

United States Army Garrison Dugway Proving Ground

Technical Exhibit 02

Paving IDIQ Contract – Roadway and Parking Specifications FY 2027 – FY
2031



2026 Standard Specifications

For Road and Bridge Construction

The Department's Deputy Director of Engineering and Operations and Director of Project Development have reviewed the Standard Specifications included herein and approved them for use with the Department's construction projects and inclusion in the Department's construction contracts on the 1st day of January, 2026.

Carmen Swanwick

04/29/2025

Carmen Swanwick, Director of Project Development

Lisa Wilson

04/29/2025

Lisa Wilson, Deputy Director of Engineering and Operations

UTAH DEPARTMENT OF TRANSPORTATION

4501 South 2700 West P.O. Box 141200 Salt Lake City, UT 84114-1200
(801) 965-4000 | contactudot@utah.gov

Introduction

These Standard Specifications are for use in Utah Department of Transportation and are for use in construction contracts at US Army Garrison Dugway Proving Ground.

The Standard Specifications are written to the Contractor. They define the Contractor's responsibility, what the Contractor is expected to provide, and the Department's expectations. All actions are to be performed by the Contractor unless otherwise noted.

A Special Provision may revise the Standard Specification for a specific contract.

Signature Sheet

The Signature Sheet on Page 2 of the 2026 Standard Specification book shows the approval for all Standard Specifications included herein for use with the Department's construction projects and inclusion in the Department's construction contracts. The signatures are made by the Department Deputy Director of Engineering and Operations and the Department Director of Project Development.

Specification Index

The Specification Index lists all 2026 Standard Specifications.

The Index also lists the date of the last approved change to each specification. The date is a weblink to the Standards Committee Meeting Agenda where a redlined version of the Specification can be viewed with an explanation of the change on the agenda Submittal Sheet. This same link can also be found in the footer of each Specification.

References

Web references within these Standards are listed at <http://www.udot.utah.gov/go/standardsreferences>.

Reference to external publications in any 2026 Standard is to the most recent version, issue, or edition in effect on the date of the *Notice to Bidders*, unless stated otherwise.

Layout

SECTION NUMBER
TITLE

PART (1, 2, or 3)

ARTICLE (such as 1.1, 1.2, 2.1, 2.2, 3.1, 3.2)

Paragraph (such as A, B, C under an article)

Example:

PART 1 GENERAL

1.1 ARTICLE

 A. Paragraph

 1. Subparagraph

 a. Subparagraph

 1) Subparagraph

 a) Subparagraph

Page Numbering

The Standard Specifications of this book each have separate numbering. There is no cumulative page numbering.

2026 Specification Index

Specification Number	Specification name	Date = Change Approved
00100	GENERAL PROVISIONS	2/18/2021
00570	DEFINITIONS	2/16/2023
01501	MOBILIZATION	11/18/2021
01554	TRAFFIC CONTROL	9/14/2023
01558	TEMPORARY PAVEMENT MARKINGS	2/17/2022
01571	TEMPORARY ENVIRONMENTAL CONTROLS	2/17/2022
01572	DUST CONTROL AND WATERING	9/14/2023
01721	SURVEY	
01741	FINAL CLEANUP	9/14/2023
01891	MOVE SIGNS	6/8/2023
01892	RECONSTRUCT CATCH BASIN, MANHOLE, AND MISCELLANEOUS BOXES	6/17/2021
02056	EMBANKMENT, BORROW, AND BACKFILL	2/20/2025
02072	GEOSYNTHETICS	2/17/2022
02075	GEOTEXTILES	9/16/2021
02216	TEST-HOLE UTILITY	2/28/2019
02221	REMOVE STRUCTURE AND OBSTRUCTION	9/14/2023
02231	SITE CLEARING AND GRUBBING	9/14/2023
02316	ROADWAY EXCAVATION	9/14/2023

02317	STRUCTURAL EXCAVATION AND BACKFILL	2/17/2022
02318	DITCH CONSTRUCTION	11/18/2021
02342	SHOTCRETE	9/16/2021
02372	GABIONS AND REVET MATTRESSES	9/16/2021
02373	RIPRAP	10/25/2018
02376	ROLLED EROSION CONTROL PRODUCTS	2/17/2022
02610	DRAINAGE PIPE	9/12/2024
02613	DRAINAGE PIPE END SECTIONS	11/18/2021
02618	DRAINAGE PIPE LINER	12/7/2023
02620	CONCRETE LINED PIPE INVERT	6/8/2023
02622	PIPE UNDERDRAINS	9/16/2021

02633	CONCRETE DRAINAGE STRUCTURES	11/18/2021
02635	GRATES, COVERS, FRAMES, TRASH RACKS AND MANHOLE STEPS	11/18/2021
02645	PRECAST CONCRETE BOX CULVERT AND THREE-SIDED STRUCTURES	9/16/2021
02701	PAVEMENT SMOOTHNESS	12/8/2022
02705	CONCRETE AND ASPHALT CUTTING	11/18/2021
02721	UNTREATED BASE COURSE (UTBC)	11/18/2021
02732	CRACK SEALING (NEW)	2/20/2025
02735	MICRO SURFACING	11/18/2021
02741	ASPHALT MIX	11/18/2021
02743	ASPHALT MIX FOR BIKE AND PEDESTRIAN PATHS	11/18/2021

02744	STONE MATRIX ASPHALT (SMA)	2/28/2019
02745	ASPHALT MATERIAL	11/18/2021
02746	HYDRATED LIME	11/18/2021
02748	PRIME COAT/TACK COAT	2/15/2024
02750	PARTIAL DEPTH REPAIR FOR CONCRETE PAVEMENT USING HOT APPLIED REPAIR MATERIALS	6/9/2022
02751	PARTIAL DEPTH REPAIR FOR CONCRETE PAVEMENT USING PORTLAND CEMENT CONCRETE	6/9/2022
02752	PORTLAND CEMENT CONCRETE PAVEMENT	11/18/2021
02753	FULL DEPTH SLAB REPLACEMENT FOR CONCRETE PAVEMENTS	6/17/2021
02754	DOWEL BAR RETROFIT	2/20/2025

02755	CONCRETE SLAB JACKING	9/14/2023
02761	LONGITUDINAL RUMBLE STRIP	12/5/2024
02765	PAVEMENT MARKING - WATERBORNE PAINT	2/17/2022
02768	PAVEMENT MARKING MATERIALS (Warranty Specification)	6/28/2018
02771	PEDESTRIAN ACCESS RAMP	2/20/2025
02776	CONCRETE FLATWORK	12/7/2023
02785	CHIP SEAL COAT	2/15/2024
02786	OPEN-GRADED SURFACE COURSE (OGSC)	9/8/2022
02787	BONDED WEARING COURSE (BWC)	6/9/2022
02821	CHAIN LINK FENCING AND GATES	6/28/2018
02825	CATTLE GUARD	12/7/2023

02841	W-BEAM GUARDRAIL	12/7/2023
02842	DELINEATORS	
02843	CRASH CUSHIONS AND BARRIER END TREATMENTS	12/7/2023
02844	CONCRETE BARRIER	12/5/2024
02845	HIGH TENSION CABLE BARRIER	12/7/2023
02890	RETROREFLECTIVE SHEETING	2/16/2023
02891	TRAFFIC SIGNS	2/20/2025
02894	PRECAST ROADSIDE SIGN FOUNDATIONS	
02911	HYDRAULIC EROSION CONTROL PRODUCTS	12/8/2022
02912	TOPSOIL	6/8/2023
02922	SEED, TURF SEED, AND TURF SOD	6/8/2023

02961	ROTOMILLING	9/8/2022
02963	PROFILE ROTOMILLING	
02981	GRINDING PAVEMENT	9/12/2024
02983	GRINDING FOR GROOVED-IN PAVEMENT MARKINGS	2/17/2022
03055	PORTLAND CEMENT CONCRETE	2/17/2022
03056	SELF-CONSOLIDATING CONCRETE (SCC)	10/25/2018
03058	RAPID STRENGTH CONCRETE (RSC)	6/17/2021
03152	CONCRETE JOINT CONTROL	12/5/2024
03211	REINFORCING STEEL AND WELDED WIRE	11/18/2021

03390	CONCRETE CURING	
03575	FLOWABLE FILL	
03605	APPROACH SLAB JACKING	9/16/2021
06055	TIMBER AND TIMBER TREATMENT	
09981	CONCRETE COATING	9/12/2024
13431	PRECAST CONCRETE FIBER OPTIC AND UTILITY VAULT	12/8/2022

SECTION 00100

GENERAL PROVISIONS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Provisions related to the interpretation of the contract.
 - 1. Includes document organization, language, and communication.

1.2 RELATED SECTIONS

- A. Section 00570: Definitions

1.3 REFERENCES

- A. UDOT Structures Design and Detailing Manual (SDDM)

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS **Not Used**

1.6 GENERAL

- A. The plans and specifications are expressed in US customary units except where a referenced document uses the International System of Units as the standard.

1.7 ORGANIZATION

- A. Specifications
 - 1. The specifications are organized by sections.
 - a. Each section defines a portion of the project requirements and is identified by a number and a title.

- b. References to specification sections use section numbers.
 - 1) Section references apply to the complete specification section consisting of the Standard Specification and Special Provision for the referenced section.
- 2. Sections up to and including Section 01499 are general specifications applicable to every contract unless specified as applicable under certain conditions.
 - a. General specifications include administrative requirements applicable to the contract and apply without reference.
 - b. General specifications apply to and govern the work of all Sections of the specifications
- 3. Sections beginning with Section 01500 through the end of the specifications are technical specifications for specific pay items.
 - a. The pay items in a contract designate the technical specifications that apply to a specific contract.
- 4. Section numbers of Special Provisions indicate the extent of the revision to the Standards.
 - a. A section number followed by “S” (for example, “01234S”), indicates a new section or a complete replacement of a Standard Specification section.
 - b. A section number followed by “M” (for example, “01234M”), indicates that part of a Standard is changed, added, or deleted.
- 5. Specification sections are written in a three-part format. Each Section contains the following primary PARTS:
 - a. PART 1 GENERAL
 - 1) Describes administrative and procedural requirements unique to the section.
 - b. PART 2 PRODUCTS
 - 1) Describes systems, assemblies, equipment, products, materials, fabrications, and mixes.
 - c. PART 3 EXECUTION
 - 1) Describes site installation or application, including preparatory actions and post-installation cleaning and protection.
 - 2) Includes site-built or site-manufactured assemblies.
- 6. Information describing the section format related to articles, paragraphs, and sub-paragraphs is shown in Figure 1.

Figure 1

Specification Layout	
SECTION NUMBER	TITLE
PART (1, 2, or 3)	
ARTICLE (such as 1.1, 1.2, 2.1, 2.2, 3.1, 3.2)	
Paragraph (such as A, B, C under an article)	
Example:	
PART 1	GENERAL
1.1	ARTICLE
A.	Paragraph
1.	Subparagraph
a.	Subparagraph
1)	Subparagraph
a)	Subparagraph

B. References

1. Reference to external publications is to the most recent version, issue, or edition in effect on the date of the *Notice to Bidders*, unless stated otherwise.
2. Reference to a specification section incorporates all applicable requirements of the referenced section.
 - a. Example: A reference in Part 2 of a section to Section 03055 for the product Class AA(AE) Concrete incorporates into the section all General, Products, and Execution requirements in Section 03055 applicable to Class AA(AE) Concrete.
 - b. Example: A reference in Part 3 of a section that refers to Section 03390 for concrete curing incorporates into the section all General, Products, and Execution requirements in Section 03390 applicable to curing the concrete elements.
 - c. Example: A reference in Part 3 of a section to Section 03310 for concrete placing and finishing incorporates into the section all General, Products, and Execution requirements in Section 03310 applicable to placing and finishing concrete.
3. References to articles and paragraphs within a section may omit the terms "Article," "Paragraph," and "Subparagraph."
 - a. Example: A reference to "this Section, 3.2 A4c" means Article 3.2, paragraph A, subparagraph 4c in the same section where the reference is made.

4. Applicable requirements in a referenced section or external publication apply the same as if they are included in their entirety in the section from which they are referenced.
 - a. The requirements of a section take precedence over referenced requirements in the case of conflict.
5. Interpret the words “shown,” “specified,” or “described” without a specified location as follows:
 - a. “Shown” means “shown in the plans.”
 - b. “Specified” means “specified in the specifications.”
 - c. “Described” means “described in the Contract.”

1.8 LANGUAGE

- A. The plans and specifications are written to the Contractor.
- B. The plans and specifications are written in the active voice where the subject is expressed or implied.
 1. Instructions written in the imperative mood are directed at the Contractor.
 - a. The subject, “the Contractor,” is implied.
 - b. Imperative mood statements are considered mandatory.
 - 1) Example: “Place the concrete.”

2. Decisions and actions by the Department or third parties are written in the indicative mood.
 - a. The subject, “the Engineer” or a third party, is expressed.
 - 1) Example: “The Engineer will take a sample.”
- C. The Contract Documents may omit modifying words such as “all” and “any” and articles such as “the” and “an,” but the fact that a modifier or an article is absent from one statement and appears in another is not intended to affect the interpretation of either statement in the interest of brevity.
- D. Refer to Section 00570 for the meaning of abbreviations or terms used in these specifications, the plans, and other contract documents.
 1. Refer to the UDOT SDDM for the meaning of abbreviations used in the structure drawings.

1.9 COMMUNICATION

- A. Submit documents and direct questions in writing to the Engineer.
- B. Orders, approvals, authorizations, and requests to the Contractor are by the Engineer unless specifically described as by another.
- C. The following items from the Department are in writing:
 1. Approvals
 2. Authorizations
 3. Decisions
 4. Notifications
 5. Orders
 6. Responses
 7. Direction
- D. The following items from the Contractor must be in writing:
 1. Assignments
 2. Notifications
 3. Proposals
 4. Reports
 5. Requests
 - a. Includes Requests For Information (RFIs), sequentially numbered
 6. Subcontracts
 7. Test results

8. Documentation and communication are deemed to be in writing.
9. Approvals and electronic signatures may be required and are deemed to be executed and binding.

1.10 REQUESTS FOR INFORMATION (RFI)

- A. Request an interpretation of the requirements in the Contract Documents if:
 1. Unable to determine from the Contract Documents the exact material, process, or system to be installed
 2. Elements of construction are required to occupy the same space (interference)
 3. An item of Work is described differently at more than one place in the Contract Documents
- B. Submission of RFIs
 1. Prepare and submit RFIs on a form completely filled in and fully legible after photocopying or scanning
 2. Give each RFI a discrete, consecutive number.
 3. Provide on each page of the RFI and each attachment to the RFI the project name, project number, date, RFI number, and a descriptive title.
 4. Sign all RFIs attesting to a good faith effort to determine from the Contract Documents the information requested for interpretation.
- C. Subcontractor-Initiated and Supplier-Initiated RFIs
 1. Subcontractors and material suppliers submit RFIs through the Contractor. The Contractor reviews subcontractor- and supplier-initiated RFIs and submits them attached to an RFI form prepared, signed, and submitted by the Contractor.
 - a. RFIs submitted directly by subcontractors or material suppliers will be returned unanswered to the Contractor.
 - b. Review all subcontractor- and supplier-initiated RFIs and take actions to resolve issues of coordination, sequencing and layout of the Work.
- D. Requested Information
 1. Provide the information required to understand and analyze the circumstances causing the RFI and to prepare a clarification or direction as to how to proceed.
 2. The RFI may be returned unanswered if the provided information is insufficient.
- E. Unacceptable Uses for RFIs
 1. To request approval of submittals

2. To request approval of substitutions
3. To request changes that involve change in Contract Time and Contract Sum
4. To request different methods of performing Work than those indicated in the Contract Drawings and Specifications

F. RFI Log

1. Prepare and maintain a log of RFIs
 - a. Furnish copies of the log showing outstanding RFIs if requested by the Engineer.

G. Review Time

1. The Engineer will return RFIs to the Contractor within 14 calendar days of receipt.

PART 2 PRODUCTS NOT USED

PART 3 EXECUTION NOT USED

END OF SECTION

SECTION 00570

DEFINITIONS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Definitions and acronyms used throughout Department Specifications.

1.2 RELATED SECTIONS

- A. Section 00777: Change Management
- B. Section 01280: Measurement

1.3 REFERENCES

- A. Code of Federal Regulations

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS **Not Used**

1.6 ACRONYMS AND ABBREVIATIONS

- A. Interpret acronyms and abbreviations used in the Contract as follows:
 - 1. AAR Association of American Railroads
 - 2. AASHTO American Association of State Highway and Transportation Officials
 - 3. ACI American Concrete Institute
 - 4. AGC Associated General Contractors
 - 5. AI Asphalt Institute
 - 6. AIA American Institute of Architects
 - 7. AISC American Institute of Steel Construction
 - 8. AISI American Iron and Steel Institute
 - 9. AMRL AASHTO Materials Reference Laboratory
 - 10. ANLA American Nursery and Landscape Association
 - 11. ANSI American National Standards Institute
 - 12. API American Petroleum Institute
 - 13. APL Approved Products List
 - 14. ARA American Railway Association
 - 15. AREA American Railway Engineering Association

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16.	AREMA	American Railway Engineering and Maintenance-of-Way Association
17.	ARTBA	American Road and Transportation Builders Association
18.	ASCE	American Society of Civil Engineers
19.	ASLA	American Society of Landscape Architects
20.	ASTM	American Society for Testing and Materials
21.	ATMS	Advanced Traffic Management System
22.	AWPA	American Wood Preservers' Association
23.	AWWA	American Water Works Association
24.	AWG	American Wire Gauge
25.	AWS	American Welding Society
26.	CCTV	Closed Circuit Television
27.	CFR	Code of Federal Regulations
28.	DBE	Disadvantaged Business Enterprise
29.	DBS	Department Business System
30.	DMS	Dynamic Message Sign
31.	DTM	Digital Terrain Model
32.	ESS	Environmental Sensor Sign
33.	EUSERC	Electric Utility Service Equipment Requirements Committee
34.	FHWA	Federal Highway Administration
35.	FSS	Federal Specifications and Standards (General Service Administration)
36.	GRC	Galvanized Rigid Conduit
37.	GSA	General Services Administration
38.	ID	Identification
39.	IMSA	International Municipal Signal Association
40.	ISO	International Organization for Standardization
41.	ITE	Institute of Traffic Engineers
42.	ITS	Intelligent Transportation System
43.	LFOT	Local Field Operations Test
44.	LRFD	Load and Resistance Factor Design
45.	MIL	Military Specifications
46.	MMF	Multi-Mode Fiber
47.	MUTCD	Utah Manual on Uniform Traffic Control Devices (This applies to all references to the MUTCD in Utah Department of Transportation 2017 Standard Specifications and Drawings, Supplemental Specifications and Drawings, Special Provisions, and Plan Sheets.) Refer to http://www.udot.utah.gov/go/standardsreferences for a link to the Utah MUTCD.
48.	NEC	National Electric Code
49.	NEMA	National Electrical Manufacturers Association
50.	NIST	National Institute of Standards and Technology

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51.	NTPEP	National Transportation Product Evaluation Program
52.	NVLAP	National Verification Laboratory Acceptance Program, (Bureau of Standards)
53.	OSHA	Occupational Safety and Health Administration
54.	OTDR	Optical Time Domain Reflectometer
55.	PM	Project Manager
56.	PCA	Portland Cement Association
57.	PDPL	Performance Data Products Listing
58.	PTI	Post-Tensioning Institute
59.	RMS	Ramp Metering Station
60.	RWIS	Roadway Weather Information System
61.	SAE	Society of Automotive Engineers
62.	SMF	Single-Mode Fiber
63.	SSPC	Society for Protective Coatings
64.	SWPPP	Storm Water Pollution Prevention Plan
65.	TMS	Traffic Monitoring Station
66.	TTQP	Transportation Technician Qualification Program
67.	UBS	UDOT Bidding System
68.	UDOT	Utah Department of Transportation
69.	UFC	Unified Facilities Criteria
70.	UFGS	Unified Facilities Guide Specification
71.	UL	Underwriters Laboratory Inc.
72.	UPDES	Utah Pollution Discharge Elimination System
73.	USASI	United States of American Standard Institute
74.	USC	United States Code
75.	VIDS	Video Image Detection System
76.	VMS	Variable Message Sign
77.	WIM	Weigh in Motion
78.	WWPA	Western Wood Products Association

1.7 TERMS

A. Interpret terms used in the Contract as follows:

51. **Act of God** – Any cataclysmic phenomenon of nature beyond the Government and Contractor's control. Weather is not considered an act of God unless it can be shown conclusively that such weather could not have been anticipated as a normal hazard of the contract.
52. **Actual Cost** – Contractor's cost to provide labor, material, equipment owned or invoiced rental, and administrative overhead necessary for the work. Actual Cost excludes profit.
53. **Advertisement** – The public announcement requesting proposals for specified work or materials.

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54. **Approve or Approval** – Formal conditional determination by the Contracting Officer's Representative (COR) that a particular matter or item is satisfactory for the project. Such determination may be based on requirements beyond those set forth in the contract and may reflect preferences by the Department. Items approved are considered to be as-if part of Department design.
55. **Authorize or Authorization** – Formal conditional determination by the COR that a particular matter or item designed or produced by others may be used. Such determination may be based on reasonable and accepted engineering principles typical for the work produced. Authorization does not relieve the Contractor from responsibility for negligent acts, errors, correctness of details, conformance to the contract, and the successful completion of the work.
56. **Award** – The Government's acceptance of a bid or proposal.
57. **Backfill** – Material used to replace, or the act of replacing material removed during construction.
58. **Bid** – Bid proposal. A proposer's written offer or proposal to perform stated work at the quoted prices.
59. **Bid Documentation** – All writings, working papers, computer printouts, charts, and data compilations containing or reflecting a proposer's information, data, or calculations used to determine the bid proposal. Bid documentation includes documents used to decide and apply:
- Equipment rates
 - Overhead rates
 - Labor rates
 - Efficiency or productivity factors
 - Arithmetic extensions
 - Subcontractor and material supplier quotations
- Bid documentation excludes any Government documents provided to the proposer used to prepare the bid proposal.
60. **Bridge** – A structure, including supports, erected over a depression or an obstruction such as water, highway, or railway, and having:
- a. A track or passageway for carrying traffic or other moving loads or utilities
 - b. A length measured along the center of roadway of more than 20 ft between under copings of abutments or extreme ends of openings for multiple boxes.
61. **Bridge Length** – The measurement taken at the centerline of the roadway between front faces of bridge abutments, spring lines of arches, or extreme ends of openings for multiple

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boxes; bridge length can also include multiple pipes, where the clear distance between openings is less than half of the smaller contiguous opening.

- 62. **Bridge Roadway Width** – The clear width measured at right angles to the longitudinal centerline of the bridge between the bottom of curbs or in the case of multiple height of curbs, between the bottoms of the lower risers or if curbs are not used, between inner faces of parapet or railing.
- 63. **Certification** – A written document or affidavit officially declaring confirmation of a statement or information as being true, accurate, or genuine.
- 64. **Calendar Day** – Every day shown on the calendar, beginning and ending at midnight.
- 65. **Certificate of Compliance** – A certification, including a signature by a person having legal authority to act for the manufacturer, stating that the product or assembly to be incorporated into the project was fabricated according to the applicable requirements.
- 66. **Certified Test Report** – A test report from the manufacturer or an independent testing laboratory, including a signature by a person having legal authority to act for the manufacturer or the independent testing laboratory stating that the test results show the product or assembly to be incorporated into the project has been sampled and tested and the samples have passed all the specified tests.
- 67. **Change Order** – An order issued by the Government that alters the original contract and is agreed to by the and Contractor.
- 68. **Construction Limits** – The established boundaries within the right-of-way or construction easements that define the construction area.
- 69. **Contract** – Written agreement between the Government and the Contractor establishing the obligations of the parties for the performance of the work prescribed. The contract includes the following, all of which constitute one instrument:
 - a. Specifications and Plans
 - b. Materials Minimum Sampling and Testing
 - c. Notice to Contractors
 - d. Bidding Schedule
 - e. Measurement and Payment
 - f. Bid Conditions Disadvantaged Business Enterprise (federal project)
 - 1. Contract DBE Goal Percentage
 - 2. DBE Bid Assurance
 - g. E.E.O. Affirmative Action Requirements (federal project)
 - h. Specific E.E.O Opportunity Responsibilities (federal project)
 - 1. Apprentice Training Hours

Definitions

- i. Equal Opportunity (state project)
 - j. Title VI Appendix A & E
 - k. FHWA – 1273 (federal project)
 - l. Cargo Preference Act (federal project)
 - m. Wage Rates (when applicable)
 - n. Notice of Award
 - o. Notice to Proceed
 - p. Authorized contract time extensions
 - q. Change orders, directives and agreements required to complete the work in an acceptable manner
70. **Contract Amount (price component)** – The summation of the products of the quantities shown in the bid schedule multiplied by the unit bid prices for the items in the Contractor’s bid proposal.
71. **Contract Bid Item** – A specific unit of work for which a price is provided in the contract.
72. **Contract Bonds** – The approved form of security, executed by the Contractor and the Contractor’s surety or sureties, guarantying complete execution of the contract, including change orders, directives, and agreements, and the payment of all legal debts pertaining to the construction of the project.
- a. **Contract Payment Bond** – The security executed by the Contractor and furnished to the Government to guaranty payment of all Contractor legal debts pertaining to the construction of the contract.
 - b. **Contract Performance Bond** – The security executed by the Contractor and furnished to the Government to guaranty completion of the work under the contract.
73. **Contract Completion** – The day, determined by the COR, when all work specified in the contract is satisfactorily completed and all obligations of the Contractor under the contract are fulfilled.
74. **Contract Time** – The substantial completion date or number of working days or calendar days allowed for substantial completion of the contract, including authorized time extensions.
75. **Contracting Officer (KO)** – The individual authorized by the Government to sign a contract with the Contractor and make changes to the contract. No Government individual other than the KO is authorized to make changes to the Contract requirements.
76. **Contracting Officer’s Representative** – The individual authorized in writing by the KO with the technical expertise to monitor Contractor performance and compliance with the Contract requirements.
77. **Contractor** – The individual or legal entity contracting with

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the Government for performance of prescribed work.

78. **Debarment** – Action taken by the Government or federal government pursuant to policies or regulations that prohibits a person or company from performing work on a public project.
79. **Delay** – An event, action, force, or factor causing work to extend beyond the specified contract time.
80. **Differing Site Conditions** – Subsurface or latent physical conditions at the project site that:
- a. Differ significantly from those indicated in the contract.
 - b. Present unknown physical conditions of an unusual nature that differ materially from those normally encountered and generally recognized as inherent in the nature of the required work.
81. **Digital Terrain Model (DTM)** – A topographic model of the bare earth that can be manipulated by computer programs. The data file contains the spatial elevation data of the terrain in a digital format.
82. **Electronic Communication** – A communication transmitted through e-mail, or other electronic means where a hard copy can be produced.
83. **Equipment** – All machinery, tools, apparatus, and the fuels, lubricants, batteries, and other supplies and parts needed to use, operate, and maintain these items for use in constructing and completing the work.
84. **Extra Work** – Work not provided for in the contract but found by the COR to be essential for the satisfactory completion of the contract within the contract's intended scope and said work is authorized by the KO prior to the work being performed.
85. **Falsework** - A temporary structure that supports structural elements of concrete, steel, masonry, or other materials during their construction or erection.
86. **Forms, Formwork** - The enclosures or panels that contain the fluid concrete and withstand the forces due to its placement and consolidation. Forms may in turn be supported on falsework.
87. **Geotextile** – Any permeable knitted, woven, or nonwoven textile material used with foundation, soil, rock, earth, or any other geotechnical engineering related material, as an integral part of a man-made project, structure, or system.
88. **Holidays**

Table 1

New Year's Day
Martin Luther King, Jr. Day
Presidents' Day
Memorial Day

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Independence Day
Juneteenth Day
Labor Day
Columbus Day
Veterans Day
Thanksgiving Day
Christmas Day

Holidays that fall on a Friday or a Saturday will recognize the preceding Thursday and the Friday or Saturday as holidays. Holidays that fall on Sunday will recognize both the Sunday and the following Monday as holidays.

89. **Inspector** – The COR or an authorized DPW Quality Assurance Specialist (QA) assigned to inspect work and materials.
90. **Liquidated Damages** – A predetermined sum assessed the Contractor. This sum is not considered a penalty, but as liquidated damages due the Department Government by reason of added cost of engineering, supervision, contract administration, and other items for extra expenditures of public funds for the Contractor's failure to perform as required.
91. **Major Contract Item** – An individual pay item, or item added by change order, having a contract value in excess of five percent of the original contract amount.
92. **Notice to Proceed** – Written notice to the Contractor to begin the contract.
93. **Pavement Structure** – The combination of subbase, base course, and surface course placed on a subgrade to support and distribute the traffic load to the roadbed.
 - a. **Surface Course** – One or more layers of a pavement structure designed to accommodate the traffic load, the top layer that resists skidding, traffic abrasion, and the disintegrating effects of climate. The top layer is sometimes called the wearing course.
 - b. **Base Course** – One or more layers of specified material and thickness placed on a subbase or a subgrade to support a surface course.
 - c. **Subbase** – One or more layers of specified material thickness placed on a subgrade to support a base course.
94. **Pay Item** – A specific unit of work where a price is provided in the contract and paid, subject to contract provisions, to the Contractor for the accepted quantities
95. **Physical Completion** – The day when all construction work required by, or incidental to, the contract is satisfactorily

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- completed, including all punch list work, final cleanup, and demobilization and the Contractor's only outstanding obligation under the contract is submitting or processing documentation.
96. **Plan Quantity** – A designer generated quantity for a portion of the work designated as the pay quantity for the contract.
97. **Plans** – Contract drawings showing the location, type, dimensions, and details of the required work including any combination of the following:
- a. **Standard Drawings** – Detailed drawings approved for repetitive use.
 - b. **Supplemental Drawings** - Approved additions and revisions to the Standard Drawings.
 - c. **Project Plans** – Drawings applicable to a specific contract or delivery order.
 - d. **Digital Delivery Documents** – Digital files classified as legal documents applicable to a specific contract
98. **Project** – The specific section of the roadway or other specific property where construction is to be performed together with all work constructed under the contract.
99. **Proposal** – A Proposer's written response to a Government request for proposals.
100. **Proposer** – An individual or individual's legal entity submitting a proposal in response to a request for proposals; also defined as an offeror in response to a request for proposals
101. **Request for Proposal** – A document used to solicit proposals including all other documents that are attached to that document or incorporated in that document by reference.
102. **Responsible Proposer** – A proposer capable in all respects of meeting all the requirements of a request for proposal; and fully performing all the requirements of the contract, including being financially solvent with sufficient financial resources to perform the work.
103. **Responsive Proposal** – A proposal conforming in all material respects to the requirements of the request for proposal.
104. **Resources** – The labor, equipment, materials, and incidentals necessary to perform work on a contract bid item or other element of work.
105. **Right-of-Way** – A general term denoting land, property, or interest acquired for or devoted to transportation purposes.
106. **Roadbed** – The graded portion of highway within top and side slopes, prepared as a foundation for the pavement structure and shoulders.
107. **Roadbed Material** – Material in cuts, embankments, and embankment foundations from the subgrade down that supports the pavement structure.

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108. **Roadside** – The areas between the outside edges of the shoulders and the right-of-way boundaries including unpaved median areas between inside shoulders of divided highways and areas within interchanges.
109. **Roadway, Road or Street** – That portion of land improved, designed, or ordinarily used for vehicular travel.
110. **Shoulder** – The portion of the roadway adjacent to the traveled way that supports base and surface courses.
111. **Site of Work** – As defined in Title 29 CFR Part 5.2 (I).
112. **Specifications** – The compilation of provisions and requirements for the performance of prescribed work, including any combination of the following:
- a. **Special Provisions** – A unique specification or a modification or revision to the standard specifications applicable to an individual contract.
 - b. **Supplemental Specifications** – Approved additions and revisions to the Standard Specifications.
 - c. **Standard Specifications** – Specifications approved for general application and repetitive use.
113. **Stabilization** – Modification of soils or aggregates by incorporating materials that increases load-bearing capacity, firmness, and resistance to weathering or displacement.
114. **Stop Work Order** – Notification from the KO to cease or hold work on a contract when the either the KO or the COR determines that work being performed is not in compliance with the contract or is being performed in an unsafe manner.
115. **Structures** – Buildings, bridges, box culverts, catch basins, cribbings, drop inlets, headwalls, manholes, overhead sign supports, retaining walls, and other similar features.
116. **Subgrade** – The top surface of a roadbed upon which the pavement structure, shoulders, and curbs are constructed.
117. **Subgrade Treatment** – Stabilization of roadbed material.
118. **Subcontractor** – An individual or legal entity to which a Contractor sublets part of the work.
119. **Substantial Completion** – Substantially complete. The day as determined by the COR when all of the following have occurred:
- a. The public, (including vehicles and pedestrians), have full and unrestricted use and benefit of the facilities both from the operational and safety standpoint.
 - b. All safety features are installed and fully functional, including, but not limited to, illumination, signing, pavement markings, all coats of striping paint, barrier, guardrail, impact attenuators, delineators, and all other

Definitions

- safety appurtenances.
- c. All remaining pay items in the contract are complete in addition to safety features. Only minor corrective work and replacement of temporary substitute facilities remains for physical completion.
 - d. The Contractor and COR mutually agree that all work remaining will be performed without lane closures, trail or sidewalk closures, and further delays, disruption, or impediment to the public.
120. **Substructure** – The system of bridge elements that support the superstructure. The substructure transfers the loads to the earth and retains material behind the supports. Substructure elements include abutments, bents, footings, piles, wingwalls, backwalls, etc.
121. **Superintendent** – The Contractor's authorized employee in responsible charge of work.
122. **Superstructure** – The system of bridge elements that spans the feature being crossed. The superstructure rests on the substructure. The superstructure includes the deck, parapets and girders, or other support elements (for example, trusses, arches, box girders).
123. **Surety** – The legal entity or individual, other than the Contractor, executing a bond furnished by the Contractor.
124. **Suspend Work** – Notification to cease or hold work on a contract when the KO determines that delaying the work would be beneficial to both the Contractor and the Department.
125. **Temporary Works** – Facilities that are generally designed by the Contractor and employed by the Contractor in the execution of the work and whose failure to perform properly could adversely affect the character of the contract work or endanger the safety of adjacent facilities, property, workers, or the public. Such facilities include but are not limited to falsework, forms and form travelers, cofferdams, shoring, water control systems, and temporary bridges.
126. **Traveled Way** – The portion of the roadway designated for the movement of vehicles, excluding shoulders and auxiliary lanes.
127. **Unsuitable Material** - Material not meeting specifications, organic materials, materials that are soft, springy or otherwise yielding, frozen lumps, soils such as peat or bog, and over-saturated silts, clays, or sands whose water content prevents appropriate compaction.
128. **Unusually Severe Weather** – Adverse weather (such as unusually heavy rain or snow, or unusually high or low temperatures) which at the time of year in which it occurred is

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unusual for the project site, and is more severe than any that might be anticipated for the project location during the period of expected performance as compared to the averages.

- 129. **User Costs** – Costs incurred, directly or indirectly, by the traveling public due to construction activities.
- 130. **Utility** – All privately, publicly, or cooperatively owned lines, facilities, and systems for producing, transmitting, or distributing communications, power, heat, gas, oil, water, waste, and storm water not connected with the highway drainage, signal systems, and other products that directly or indirectly serve the public.
- 131. **Value Engineering Change Proposal** – A change proposed by the Contractor and considered by the Government intended to result in project cost savings to contract pay items without reducing the essential functions and characteristics of the project. Refer to Section 00725.
- 132. **Well-Graded Material** – Material having an even distribution of different particle sizes. This even distribution of particles of different sizes results in a dense mass upon compaction.
- 133. **Work** – The elements, activities, and incidentals necessary to complete a project (including labor, materials, equipment, and the interim products and stages attained in the course of reaching completion), and all alterations, amendments, or extensions made by change order or other written orders of the Engineer.
- 134. **Working Day** – Any calendar day, except:
 - a. Contract designated holidays or days restricted in the contract.
 - b. Days when the Contractor is specifically required by the contract or letter from the KO to suspend operations through no fault of the Contractor.
 - c. Days when the COR determines that inclement weather or adverse conditions interfere with the progress of the work.
 - 1) When the COR determines that inclement weather prevents the Contractor from working for at least 50 percent of the normal working day.
 - 2) The day may be considered a working day exception even though conditions may improve, and the major portion of the day could be considered suitable for operations if weather stops the Contractor's crew from beginning work at the normal starting hour and the crew is released as a

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result.

- d. Submittals and notification requirements are based on a standard Monday through Thursday 4-day each 10-hour days work week.

- 135. **Working Drawings** – Drawings produced by the Contractor that supplement the contract drawings to provide information not included in the contract documents but that is required to fabricate, erect, transport, or temporarily support the structure or structural elements in the completion of the work. Working drawings do not supersede the contract drawings.

PART 2 PRODUCTS Not Used

PART 3 EXECUTION Not Used

END OF SECTION

SECTION 01501

MOBILIZATION

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Mobilization and preparatory work necessary to become ready to perform the work and final cleanup necessary to complete the work.

1.2 RELATED SECTIONS

- A. Section 01741: Final Cleanup

1.3 REFERENCES Not Used

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS Not Used

1.6 GENERAL

- A. Includes the moving of personnel, equipment, supplies and incidentals to each work site before beginning the work.
- B. Includes the establishment of offices, buildings, and other facilities necessary for the work.
- C. Includes labor and operations that must be performed before beginning other items under the Contract.
- D. Includes removal of personnel, equipment, and supplies from each work site at the completion of the work.
- E. Includes work that is not included with other items under the Contract such as cleanup and restoration of disturbed areas. Refer to Section 01741.

PART 2 PRODUCTS Not Used

PART 3 EXECUTION Not Used

END OF SECTION

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SECTION 01554

TRAFFIC CONTROL

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Temporary traffic control requirements, signs, devices, and personnel necessary to control vehicular and pedestrian traffic flow in a safe and efficient manner in construction zones.
- B. Work zone traffic control devices, arrow boards, and pilot cars.

1.2 RELATED SECTIONS

- A. Section 00555: Prosecution and Progress
- B. Section 02890: Retroreflective Sheeting

1.3 REFERENCES

- A. AASHTO Roadside Design Guide, Current Edition
- B. American National Standards Institute (ANSI)
- C. Americans with Disabilities Act
- D. ATSSA: American Traffic Safety Services Association Quality Standards for Work Zone Traffic Control Devices
- E. UDOT Flagger Training Handbook
- F. UDOT Guidelines for Crash Cushions and Barrier End Treatments
- G. International Safety Equipment Association (ISEA)
- H. NCHRP Report 350: Recommended Procedures for the Safety Performance Evaluation of Highway Features
- I. Utah Manual on Uniform Traffic Control Devices (Utah MUTCD)

- J. MASH 2016: AASHTO Manual for Assessing Safety Hardware published 2016

1.4 DEFINITIONS

- A. Peak Hours –Peak hours are 6:00 a.m. to 8:00 a.m. and 4:30 p.m. to 6:30 p.m., Monday through Thursday unless otherwise defined by the COR or in the contract documents.
- B. TPAR – Temporary Pedestrian Access Route – A temporary detectable pedestrian route that is provided when existing routes are disrupted, closed or relocated and includes accessibility features present in the existing facility.
- C. TPRS – Temporary Portable Rumble Strips – An array of three heavy rubber strips that create noise when they are run over and are used to alert drivers to the presence of road construction.
- D. Traffic Slow Down – An isolated planned event where traffic on a highway is reduced in speed to provide a gap for work to proceed.
1. Examples include the crossing of the roadway with heavy equipment or the adjustment of traffic control devices.

1.5 SUBMITTALS

- A. Traffic control plan for review before beginning work.
1. Include at least the following:
 - a. Traffic control measures for each phase of work in sufficient detail to provide for the safe and efficient movement of traffic, pedestrians, and bicycles during construction.
 - 1) Describe equipment, devices, and implementation instructions.
 - b. Work elements to be accomplished in each phase.
 - c. Details for each change to traffic control that is planned for the duration of the project.
 - d. Expected duration of each traffic control configuration.
 - e. Length of Need (LON) for temporary barriers.
 - f. Taper lengths, lane shift widths, device spacing, and sign locations for temporary and existing signs.
 - g. Removal or masking of traffic control elements that conflict with temporary traffic control measures, such as existing traffic signs, traffic signals, and markings.
 - h. Worker parking, work vehicle access, and

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equipment access.

- i. Location and hours of use for TPRS.
 - j. TPAR and devices for pedestrian access.
 - 2. Provide supporting engineering calculations.
 - 3. Provide the seal of a Professional Engineer licensed in the State of Utah on drawings and calculations.
- B. Traffic Control Inspection forms weekly for information.
- 1. Submit to the COR by COB on the Tuesday of the following week.

1.6 REQUIRED MEETING

- A. At least the following must attend a meeting before beginning construction at the time and location determined by the ~~Engineer~~COR:
- 1. Contractor's Traffic Control Designer
 - 2. Contractor's Traffic Control Maintainer

1.7 DEVICE CRASHWORTHINESS

- A. Crashworthiness requirements.
- 1. Devices manufactured on or before December 31, 2019 and that meet NCHRP 350 crashworthiness criteria may be used until December 31, 2024.
 - 2. Devices manufactured after December 31, 2019 are required to meet MASH 2016 crashworthiness criteria.
- B. Devices in the following FHWA device category are required to meet crashworthiness requirements:
- 1. Category 1 - cones, barrels and delineators
 - 2. Category 2 - barricades and sign stands
 - 3. Category 3 - barriers, crash cushions and truck mounted attenuators
- C. Devices in the following FHWA device category are not required to meet crashworthiness requirements:
- 1. Category 4 - arrow boards and portable variable message signs

1.8 TRAFFIC CONTROL REQUIREMENTS

- A. Meet the following requirements for traffic control and document them in the traffic control plan:
- 1. Meet the requirements in the TC series Standard Drawings, Utah MUTCD, and the Americans with Disabilities Act.
 - 2. Provide for the safe and efficient movement of traffic.

- a. Address expected delay with the Project Public Involvement Team or the Region Communications Manager if the project does not have a Public Involvement Team.
- 3. Provide for the safe and efficient movement of pedestrians and cyclists when existing facilities are disrupted, closed or relocated by a work zone.
 - a. Provide detectable temporary facilities that include accessibility features consistent with the features present in the existing pedestrian facility
- 4. Provide concrete barrier, crash cushions and attenuators including:
 - a. Protection as required for hazard mitigation for workers. Refer to TC 3 Series Standard Drawings.
 - b. Protect all hazards to motorists within the appropriate AASHTO clear zone including bridge parapets, barrier blunt ends, poles, and large equipment. Refer to the Department Guidelines for Crash Cushions and Barrier End Treatments for acceptable devices.
- 5. Provide temporary pavement markings and pavement marking removal.
- 6. Incorporate TPRS when there is a change in the width, alignment or number of lanes on freeways or interstates with 2 lanes in a single direction.

1.9 TRAFFIC CONTROL MAINTAINER

- A. Provide a traffic control maintainer to install, maintain and remove temporary traffic control devices according to the authorized traffic control plan.
- B. Qualifications:
 - 1. Certified by the Department or by the American Traffic Safety Services Association (ATSSA) as a Traffic Control Technician. A list of certifying agencies is available at <http://www.udot.utah.gov/go/standardsreferences>
- C. Authority
 - 1. Obtains and uses labor, equipment, and materials necessary to maintain traffic control.
 - 2. Changes traffic control operations according to the authorized traffic control plan.
- D. Responsibilities and Duties
 - 1. Oversee traffic control operations and TPAR devices.
 - 2. Be present and an active participant during the installation,

- maintenance, and removal of temporary traffic control devices.
3. Implement the authorized traffic control plan.
 4. Remain available 24 hours a day, 7 days a week and can be on- site within 30 minutes of notification.
 5. Correct deficiencies immediately upon notification from the Engineer.
 6. Manage traffic control inspections.
 - a. Document inspections on a form acceptable to the COR.
 - 1) Include at least the following:
 - a) Assessment of device quality.
 - b) Items in compliance with the authorized traffic control plan.
 - b. A person with the Government or ATSSA Traffic Control Technician (TCT) certification performs the inspections and signs the documentation.
 - 1) Persons that have passed TCT certification but do not have the required experience may also perform the inspections and sign documentation.
 - c. Inspect at least four times each day with at least one of the inspections conducted during nighttime hours when construction work is occurring:
 - 1) Before beginning of shift,
 - 2) At mid-shift,
 - 3) Half-hour after shift ends, and
 - 4) At the midpoint of the off-shift period.
 - d. Conduct traffic control inspections twice each day when construction is not occurring, but traffic control is present.
 - 1) Once during daylight hours.
 - 2) Once during nighttime hours.
 - 3) At least 8 hours between inspections.
 7. Coordinate project traffic control with emergency services and local law enforcement agencies.
 8. Monitor traffic queue lengths and adjust advanced warning signs to provide adequate warning of the actual back of queue resulting from construction activities.
 9. Adjust the TPRS as necessary to maintain proper alignment, spacing and location.
 10. Adjust the TPAR devices as necessary to maintain a detectable route that is consistent with the accessibility present in the existing pedestrian facility.

1.10 PRICE ADJUSTMENTS

- A. The Government reduces payment when traffic control is not in compliance with the authorized traffic control plan or when the Contractor fails to meet all requirements cited or referenced in this specification.
 - 1. The amount per day by which the Contractor's compensation will be reduced is calculated using the daily charge for Calendar Day in the Schedule of Liquidated Damages in Table of Section 00555 for Traffic Control divided by the number of contract days, whichever is greater.

PART 2 PRODUCTS

2.1 PILOT CAR

- A. Equip with a retroreflective sign.
 - 1. Refer to Section 02890.
 - 2. Utah MUTCD sign G20-4.
- B. Equip with at least two rotating, oscillating, or strobe lights.
 - 1. Minimum 4 inch diameter/width and minimum 6 ft mounting height.
 - 2. Yellow color.

2.2 FLAGGER EQUIPMENT AND CLOTHING

- A. Refer to the UDOT Flagger Training Handbook.
- B. Safety Clothing
 - 1. Flagger vest and hard hat – Orange, red-orange, or fluorescent version of these colors.
 - a. Wear safety apparel meeting the requirements of ANSI/ISEA "American National Standard for High-Visibility Apparel and Headwear" or equivalent revisions and labeled as meeting the current ANSI/ISEA publication year, standard performance for Class 3 risk exposure.
 - b. Hard hat with 10 square inches of white or strong yellow-green retroreflective tape placed around the base of the hard hat and visible to traffic from all directions.

2.3 TRAFFIC CONTROL SIGNING AND DEVICES

- A. Signs
 - 1. Comply with Section 02890.
 - 2. Comply with TC Series Standard Drawings.

3. Comply with SN Series Standard Drawings when using post mounted signs.
- B. Channelizing Devices
1. Comply with TC Series Standard Drawings.
 2. Comply with Section 02890.
- C. Precast Concrete Barrier
1. Comply with TC Series Standards Drawings.
 2. Use an approved construction zone attenuator or permanent style end sections, as listed in Department Guidelines for Crash Cushions & Barrier End Treatments.
 - a. Use a construction zone attenuator when approach ends of temporary precast barrier are within the maximum AASHTO clear zone.
 - 1) Use AASHTO Roadside Design Guide to determine proper clear zone distance requirements
 - 2) Refer to the CC Series Standard Drawings and manufacturer's recommendations to install crash cushions.
- D. Use properly rated truck-mounted attenuator for the pre-construction posted speed limit.
1. Test Level 2 for speeds 45 mph or less.
 2. Test Level 3 for speeds greater than 45 mph.
 3. Do not use a truck-mounted attenuator (TMA) to protect blunt end for more than 72 hours.
- E. Maintain cable barrier and anchor systems during construction.
1. Protect existing hazards when cable barrier and anchor systems are rendered inoperable by work.
 - a. Address barrier length of need for the hazard.
 2. Maintain the required tension in the cable barrier system when the cable is disconnected by installing anchor systems on each end of the disconnect.
 - a. Do not cut cable.
 3. Disconnect cable at cable splice or anchor system locations only.
 4. Install manufacturer's approved terminal compatible with existing cable system. Remove anchor system when cable barrier is reestablished.
 - a. Tension cable to manufacturer's requirements.
 - b. Remove anchor system.

2.4 ARROW BOARD

- A. Comply with all standards as specified in the Utah MUTCD, Section 6F.61 Arrow Boards.
- B. Refer to the TC Series Standard Drawings and the Utah MUTCD.

2.5 TEMPORARY PORTABLE RUMBLE STRIPS

- A. Roadquake 2 Series
Temporary Portable Rumble Strip manufactured
by: Plastic Safety Systems
2444 Baldwin Road
Cleveland, OH 44104
(800) 662-6338
- B. Space and locate TPRS as shown in TC Series Standard Drawings.

2.6 TEMPORARY WALKWAY AND RAMPS

- A. Refer to TC Series Standard Drawings and the Utah MUTCD

PART 3 EXECUTION

3.1 GENERAL

- A. Implement and maintain traffic control according to the authorized traffic control plan.
 - 1. Make changes immediately and notify the COR if traffic control changes that deviate from the authorized traffic control plan are required to make the work zone safe for workers or the traveling public.
 - 2. Update the traffic control plan with the changes and resubmit for review.
- B. Maintain traffic control as required by the authorized traffic control plan when traffic control is in place and work is not actively occurring, including overnight, weekends and holidays. Inspect according to this Section, Article 1.9.
- C. The ATSSA publication Quality Standards for Work Zone Traffic Control Devices identifies different levels of device quality. Meet the acceptable level for all traffic control devices.
 - 1. Wash or replace devices as needed to meet acceptable level.

- D. Maintain traffic control devices during and after all snow plowing operations.
 - 1. Clear snow and ice away from the following:
 - a. Traffic control devices to function as intended.
 - b. TPAR devices to function as intended.
 - c. Contractor provided TPAR features.

3.2 FLAGGING

- A. Flaggers must have a current Utah flagging certificate and must present proof of certification upon request by the COR.
 - 1. Acceptable certifications. Refer to <http://www.udot.utah.gov/go/standardsreferences> for a list of acceptable courses taught by certified instructors.

3.3 TRAFFIC CONTROL SIGNING AND DEVICES

- A. Use pre-construction posted speed limit to compute sign spacing, taper lengths, buffer zones, and construction clear zone.
 - 1. Use plastic drums or directional barricades for lane closure taper devices for speeds 50 mph and greater.
 - 2. Refer to the TC Series Standard Drawings for use of cones or tubular markers at night.
- B. Use pre-construction posted speed limit during construction to compute the tangent spacing for channelizing devices.
- C. Remove all traffic control devices from site of work that are no longer necessary for the authorized traffic control plan.
 - 1. Relocate traffic control devices from the roadway a distance twice that of the Work Clear Zone if they will be used within 24 hours of the daily work stoppage and are not required for immediate traffic control. Refer to the TC Series Standard Drawings.
 - a. Obtain written permission from property owner before storing traffic control devices on private property.
 - 2. Crashworthy traffic control devices moved with a barrel mover or similar equipment may be relocated to the hard surface shoulder within 1 ft of the outer edge of shoulder if they will be used for traffic control within 48 hours of the daily work stoppage and are not required for immediate traffic control. Refer to TC Series Standard Drawings.
 - 3. Cover post mounted signs completely with an opaque and durable covering when the signs are not applicable.

3.4 ARROW BOARD

- A. May substitute Type C units for Type B units. Refer to the TC Series Standard Drawings.
- B. Do not substitute Type B units for Type C units.
- C. Remove Arrow Board from the site of work when not needed for the control of traffic within a four-hour period.

3.5 TRAFFIC SIGNALS Not Used

- A. Use uniformed police officer when construction activities are impacting an operating signalized intersection.
- B. Use of flaggers at traffic signals permitted when the signals have been turned to red flash mode or are inoperable.
 - 1. Control each approach by separate flaggers.
 - a. Flaggers can control only two lanes of approach traffic.
 - 1) Third lane control permitted when left or right turn bays present.

3.6 CONSTRUCTION ZONE SPEED LIMIT REQUIREMENTS

- A. Obtain approval for regulatory and advisory speed reductions through the COR.
 - 1. Use speed reductions only during impacted times and areas.
 - 2. Restore pre-construction regulatory speed limit at locations where traffic is not being impacted by work activities.
 - 3. Refer to <http://www.udot.utah.gov/go/standardsreferences> for policy information.
 - 4. Refer to TC Series Standard Drawings.
 - 5. Use speed reduction only when construction activities impact traffic.
 - 6. Refer to SN Series Standard Drawings for post mounted sign requirements.

3.7 LIMITATION OF OPERATIONS

- A. Traffic Slow Down
 - 1. Notify and obtain approval from the Government a minimum of 48 hours prior to slow down.
 - 2. Use a vehicle with overhead flashing lights to conduct the slow down.
 - 3. Use the officer in the marked vehicle to slow down one or two lanes.
 - a. Use, in any combination either, contractor-supplied vehicles equipped with overhead amber flashing lights or additional officers in marked vehicles at the rate of one

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vehicle per lane thereafter for all lanes of the highway to affect the traffic slow down.

4. Additional vehicles as described in this article may be used in the traffic slow down to supplement the law enforcement vehicle when required by the officer.
5. The duration of any traffic slowdown is not to exceed five minutes or as approved by the COR.

3.8 TEMPORARY PORTABLE RUMBLE STRIPS (TPRS)

- A. Clean road surface with broom or blower to remove all gravel, sand, dust, or other debris.
- B. Assemble modular pieces into strips that match the width of the travel lane as closely as possible.
 1. Follow manufacturer's recommendations so that pieces are properly interlocked.
- C. Place TPRS perpendicular to traffic and centered in lane.
 1. Follow manufacturer's recommendations and TC series standard drawings for installation and product orientation.
 2. Do not glue, nail, or otherwise affix TPRS to the road surface.
- D. Place TPRS at the same time as other traffic control devices, prior to work taking place.
 1. Maintain the TPRS in proper condition, alignment, spacing, and location.
 - a. Set TPRS perpendicular to the travel lane.
 - 1) Adjust TPRS when any one rumble strip becomes skewed by a distance of 3 ft or more. Skew distance is the distance parallel to direction of travel between the ends of the strip.
 - 2) Adjust the TPRS if the parallel distance between the individual rumble strips decreases by 5 ft or more.
 - 3) Temporary paint marks may be placed to give reference of original locations.
 - b. Make adjustments to TPRS as often as necessary during working hours, but at least during each traffic control inspection.
 - 1) Adjustments to TPRS must be made within 30 minutes of discovery or notification of misalignment.
 - c. Remove TPRS during non-working hours.
 - d. Implement a Traffic Slow Down if necessary to enter the travel lanes to deploy or reset TPRS.

- E. Do not use TPRS during snow events, or at temperatures outside of the manufacturer's recommendations.

3.9 LANE CLOSURES

- A. Notify the COR in advance of every lane closure.
 - 1. Provide notification:
 - a. As soon as practical when the schedule, location, or need for an upcoming lane closure arises or changes.
 - 1) Submit an Outage Request.
 - 2. Include at least the following:
 - a. The route,
 - b. The beginning and ending locations of closure,
 - c. Number of lanes to be closed,
 - d. Direction of the closure,
 - e. The date and time of the beginning of closure, and
 - f. The date and time of the ending of the closure.

END OF SECTION

SECTION 01558

TEMPORARY PAVEMENT MARKINGS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Materials and procedures for installation of temporary pavement markings.

1.2 RELATED SECTIONS

- A. Section 02765: Pavement Marking – Waterborne Paint
- B. Section 02842: Delineators
- C. Section 02890: Retroreflective Sheeting

1.3 REFERENCES

- A. ASTM D 4592: Preformed Retroreflective Pavement Marking Tape for Limited-Service Life.
- B. Utah MUTCD

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Manufacturer's product data sheet and recommended installation instructions.

1.6 ACCEPTANCE

- A. Price Reductions for wet mil thickness do not apply.

PART 2 PRODUCTS

2.1 PAVEMENT MARKING PAINT AND GLASS BEADS

- A. Refer to Section 02765.

2.2 PAVEMENT MARKING TAPE

- A. Refer to ASTM D 4592, Type I (Removable).

2.3 TEMPORARY RAISED PAVEMENT MARKERS

- A. Provide plastic raised pavement markers that have reflective material with clear covers as needed for the particular application.
 - 1. Supply marker body manufactured of polyurethane plastic in color specified for required type.
 - a. Width 4 inch, height 2 inch, thickness 0.06 inch.
 - 2. Retroreflective sheeting to be at least 1 square inch and according to Section 02890 for retroreflective sheeting requirements.
 - 3. Provide clear polyvinyl chloride covers attached to marker body with heavy duty staples.
 - 4. Supply marker with a method of attaching to pavement surface.
- B. Raised Pavement Marker Types
 - 1. Type Y1 – Yellow body with yellow reflective sheeting on both sides.
 - a. Optional – Type B1, black body with yellow reflective sheeting on both sides.
 - 2. Type W1 – White body with white reflective sheeting on both sides.
 - a. Optional – Type B2, black body with white reflective sheeting on both sides.

PART 3 EXECUTION

3.1 GENERAL

- A. Apply temporary pavement markings along the entire length of any roadway surfaces opened to traffic.
 - 1. Refer to the authorized traffic control plan for the pavement markings layout.
 - a. Duplicate pre-project pavement markings layout when not shown in the traffic control plan, unless otherwise directed by the Engineer
 - 2. Use the same segment and cycle length of permanent markings for all temporary broken-line pavement markings.
- B. Use channelizing devices or other delineation as directed by the Engineer to indicate road user paths in work zones when it is not possible to provide a clear path with temporary pavement markings.

3.2 PAVEMENT MARKING PAINT AND GLASS BEADS

- A. Re-apply pavement marking paint at two week intervals to maintain markings that provide a clear path during night and twilight periods and wet pavement conditions.

3.3 PAVEMENT MARKING TAPE

- A. Apply pavement marking tape according to manufacturer's directions.
- B. Maintain or re-apply pavement marking tape to maintain markings that provide a clear path during night and twilight periods and wet pavement conditions.
- C. Inspect and replace immediately any loose, missing, or damaged pavement markings.
- D. Remove the tape immediately before paving.

3.4 TEMPORARY RAISED PAVEMENT MARKERS

- A. Attach raised pavement marker according to manufacturer's directions.
- B. Space raised pavement markers as follows:
 - 1. Solid line On 10 ft centers.
 - 2. Broken line Three on 5 ft centers spaced on a 40 ft cycle length.
- C. Inspect and replace immediately any loose, missing, or damaged markers.
- D. Remove markers immediately before paving.

END OF SECTION

SECTION 01571

TEMPORARY ENVIRONMENTAL CONTROLS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Temporary environmental controls to control erosion and prevent sediment laden runoff from leaving the construction site and areas under the Contractor's control.

1.2 RELATED SECTIONS

- A. Section 02075: Geotextiles

1.3 REFERENCES

- A. AASHTO M 288: Geotextile Specifications for Highway Applications
- B. AASHTO Construction Stormwater Field Guide
- C. ASTM D 4355: Deterioration of Geotextiles by Exposure to Light, Moisture and Heat in a Xenon Arc Type Apparatus
- D. ASTM D 4491: Water Permeability of Geotextiles by Permittivity
- E. ASTM D 4751: Determining Apparent Opening Size of a Geotextile
- F. Utah Pollutant Discharge Elimination System, Utah Construction General Permit (CGP) and UDOT Municipal Separate Storm Sewer System (MS4) Permit

1.4 DEFINITIONS

- A. Check Dam – A fiber roll or stone structure placed across a roadside ditch to temporarily protect ditch from channel erosion by slowing velocity of stormwater runoff and intercepting and trapping sediment.
- B. Disturbed Area – Areas within a construction site where existing vegetative cover, or existing stabilized areas, have been removed or altered and exposed soils are susceptible to increased erosion and sedimentation.

- C. Drop-Inlet Barrier – A barrier placed around a storm drain inlet grate, situated outside of roadway pavement condition, that is designed to intercept and trap sediment-laden runoff before entering the storm drain system.
- D. Fiber Roll – Wood excelsior, rice or wheat straw or coconut fibers rolled or bound with netting to form a tube-like structure used to intercept and trap sediment.
- E. Final Stabilization – Procedures and controls completed as the final measure to protect disturbed areas of a construction site from erosion and sedimentation until vegetation regrowth occurs to provide ultimate erosion protection.
 - 1. Includes work within areas to be vegetated such as establishing final grades, placing topsoil, incorporating seed; roughening slopes by walking track-mounted equipment up and down slopes; applying rolled and hydraulic erosion control products; and installing other landscape treatments to protect exposed soils from erosion.
 - 2. Includes work within areas intended to remain unvegetated such as placing final pavement; installing stone, gravel and other stable material that will prevent erosion of underlying soil.
- F. Gutter-Inlet Barrier – A device designed and prefabricated to secure to the top, envelop or hang below a storm drain inlet grate, situated within roadway pavement condition, that keeps sediment and debris from entering the storm drain system.
- G. Pipe-Inlet Barrier – A barrier placed at a pipe inlet that intercepts and traps sediment before entering the pipe.
- H. Sediment Trap – A small temporary excavated basin installed at low points on a construction site designed to trap sediment-laden runoff to allow sediment to settle out before leaving site.
- I. Silt Fence – A geotextile fabric fence used to intercept and trap sediment in a sheet flow situation, along the perimeter of a disturbed area.
- J. Slope Drain – A polyethylene pipe temporarily placed on a slope to collect and transport storm runoff down the face of a slope until permanent drainage facilities are installed or vegetation growth is adequate.
- K. Stabilized Construction Entrance – A layer of stone, underlined with a geotextile fabric, placed at a construction site entrance or exit used to reduce the amount of sediment or mud tracked onto adjacent paved roadways by vehicles leaving the construction site.

- L. Fiber Roll or Straw Bale Barrier – Temporary barriers installed by securing fiber rolls or straw bales end to end along perimeter of a disturbed area designed to intercept and slow sediment laden runoff before it leaves a construction site.
- M. Temporary Berm – A ridge of compacted soil with or without a shallow ditch that diverts stormwater runoff from a slope to a controlled release point.
- N. Temporary Environmental Fence – A high-visibility fence barrier used to delineate and prevent encroachment on sensitive areas.

1.5 SUBMITTALS

- A. APL Compliance Form for information required for the following products:
 - 1. Fiber Roll
 - 2. Silt Fence
 - 3. Gutter-Inlet Barrier

PART 2 PRODUCTS

2.1 TEMPORARY ENVIROMENTAL CONTROLS

- A. Fiber Roll
 - 1. Diameter (minimum weight per linear foot)
 - a. 18 inch (3 lb per linear foot)
 - b. 12 inch (2 lb per linear foot)
 - c. 9 inch (1 lb per linear foot)
 - 2. Functional Longevity – 24 months minimum (includes netting material).
 - 3. Matrix material – Wood excelsior, rice or wheat straw, and coconut fibers (coir) or in combination.
 - a. Material must be weed free.
 - 4. Netting – UV stabilized synthetic or coir material, with 1 inch maximum opening size, secured at end for matrix containment.
 - 5. Wood Stakes
 - a. 18 inch Fiber Roll – ¾ inches and 1½ inches by 3 feet minimum dimensions.
 - b. 12 inch Fiber Roll – ¾ inches and 1½ inches by 18 inch minimum dimensions.
 - c. 9 inch Fiber Roll – ¾ inches and 1½ inches 18 inch minimum dimensions.

- B. Silt Fence. Refer to EN Series Standard Drawings.
 - 1. Silt Fence Fabric – 3 foot minimum width, conforming to Table 7 of AASHTO M 288.
 - 2. Wood Post – 1½ inches by 1½ inches by 4 feet minimum dimensions.
 - 3. Fasteners – Staples, wire, cable ties, or nails sufficient to maintain fabric attachment to post.
- C. Check Dam. Refer to EN Series Standard Drawings
 - 1. Fiber Roll – 12 inch diameter, or
 - 2. Stone – Angular, well-graded 2 to 6 inch diameter.
- D. Drop-Inlet Barrier. Refer to EN Series Standard Drawings
 - 1. Fiber Roll – 18 inch diameter, or
 - 2. Silt Fence
 - a. Wooden Support Frame – 2 by 4 inch (nominal) wood studs.
- E. Gutter-Inlet Barrier
 - 1. Apparent Opening Size (ASTM D 4751) – between 20 and 40 sieve.
 - 2. UV Resistance (ASTM D 4355) – 65percent minimum.
 - 3. Flow Rate (ASTM D 4491) – 100 gpm/ft² minimum.
 - 4. Filter Material – Monofilament, woven or nonwoven geotextile.
 - 5. Provide protection to entire inlet opening.
 - 6. Types:
 - a. Above Inlet Gate
 - 1) Mount securely to the top side of the inlet grate at each corner with cable ties, wire or similar.
 - b. Inlet Cover Gate
 - 1) Sewn geotextile fabric that envelopes entire inlet grate.
 - 2) Must have built-in lifting straps or other device to allow removal of inlet grate and barrier.
 - c. Below Inlet Gate
 - 1) Mount device securely to the inlet grate or have independent frame that allows geotextile bag to hang below grate to capture runoff.
 - 2) Must be designed with a bypass feature that allows stormwater to be conveyed into the conveyance system when geotextile is filled to capacity.
 - 3) Must be able to remove from storm drain inlet and maintain device without dumping captured sediment into the storm drain system.

- F. Pipe-Inlet Barrier. Refer to EN Series Standard Drawings.
 - 1. Fiber Roll – 18 inch diameter, or
 - 2. Stone – Angular, well-graded 2 to 6 inch diameter.
- G. Temporary Berm. Refer to EN Series Standard Drawings.
 - 1. Compacted existing soil.
 - 2. Free of debris, such as trees, brush, obstructions and other objectionable material that will not allow for compaction of berm material.
- H. Temporary Environmental Fence
 - 1. Fence Fabric
 - a. Polyethylene, high-density (HDPE) and UV stabilized
 - b. Height – 4 ft minimum
 - c. Color – Orange
 - 2. Posts
 - a. Wood Post – 1½ inches by 1½ inches by 4 feet minimum dimensions.
 - b. Fasteners – Staples, wire, cable ties or nails sufficient to maintain fabric attachment to post.
- I. Sediment Trap. Refer to EN Series Standard Drawings.
 - 1. Stone – Angular, well-graded 6 to 12 inch diameter
- J. Slope Drain. Refer to EN Series Standard Drawings.
 - 1. 12 inch diameter single wall polyethylene pipe
 - 2. Polyethylene pipe end section
 - 3. Stone – Angular, well-graded 6 to 12 inch diameter
 - 4. Wood Stakes – 1½ inches by 1½ inches by 3 feet minimum dimensions.
- K. Stabilized Construction Entrance. Refer to EN Series Standard Drawings.
 - 1. Stone – Crushed aggregate, well-graded 2 to 3 inch diameter.
 - 2. Geotextile Fabric (Separation) – Refer to Section 02075.
- L. Fiber Roll Barrier. Refer to EN Series Standard Drawings.
 - 1. Fiber Roll – 18 inch diameter.
- M. Straw Bale Barrier. Refer to EN Series Standard Drawings.
 - 1. Straw Bales – Certified weed free straw bales by the Utah Department of Agriculture.
 - 2. Wood Stakes – 1½ inches by 1½ inches by 4 feet minimum dimensions.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Install appropriate controls as shown before beginning earth disturbing activities.
- B. Refer to installation procedures outlined in EN Series Standard Drawings and the AASHTO Construction Stormwater Field Guide.
- C. Install temporary environmental fence in the required locations before construction activities begin.
 - 1. Install posts at a 12 ft maximum spacing so the fence does not sag more than 2 inches between posts.
 - 2. Weave the fence over the support posts alternating every two loops and secure it to the posts with fasteners.
- D. Install Gutter-Inlet Barrier according to manufacturer's recommendations.

3.2 INSPECTION

- A. Check installed controls before and after each rain event to verify proper working function and compliance with the CGP.
- B. Replace controls that are not properly working to prevent erosion and sedimentation.

3.3 MAINTENANCE

- A. Maintain controls to function properly until surrounding disturbed areas have met final stabilization measures.
- B. Remove accumulated sediments from controls when depth reaches 50 percent of the control height or when it interferes with the performance of the control.
- C. Properly dispose of accumulated sediment.

3.4 REMOVAL

- A. Remove temporary environmental controls when surrounding disturbed areas have met final stabilization measures, except as follows:
 - 1. Do not remove perimeter controls, such as silt fence, fiber rolls or straw bales, when they protect a wetland or waterway unless the surrounding area meets final stabilization requirements identified within the CGP.
 - 2. When the COR determines that controls should remain in place.
- B. Remove temporary environmental fence and posts upon completion of construction.

END OF SECTION

SECTION 01572

DUST CONTROL AND WATERING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Requirements for dust control and pre-wetting.

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES Not Used

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS Not Used

PART 2 PRODUCTS

2.1 WATER

- A. Free of dirt, silt, and other detrimental matter in adequate quantities for dust control and watering requirements.

2.2 EQUIPMENT

- A. Water distribution system – Self-propelled, pressure distributor with a spray system, equipped with a positive shut-off control.
 - 1. Pressure pump must have the capacity to apply the whole load uniformly.
- B. Water truck – 1,000 U.S. gallons minimum capacity, with the capacity clearly and permanently marked on the tank.

PART 3 EXECUTION

3.1 APPLICATION

- A. Apply water for dust control in quantities and locations as needed to maintain environmental compliance and pre-wet base materials.
 - 1. Dust control may be required at any time.
 - 2. Do not waste water.

END OF SECTION

SECTION 01721

SURVEY

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Construction surveying, staking, measurement, and calculations essential to complete the project and properly control the entire work.
- B. Processes and procedures for implementing Machine Control Guidance (MCG) technology. MCG uses Global Positioning System (GPS) and Robotic Total Station (RTS) in conjunction with three-dimensional computer models to determine the precise location and elevation of the materials being moved.

1.2 RELATED SECTIONS

- A. Section 02765: Pavement Marking Paint

1.3 REFERENCES

- A. UDOT CADD Standards

- B. UDOT Plan Sheet Development Standards
- C. UDOT Survey & Geomatics Standards Manual
- D. Utah Administrative Code
- E. Utah Code

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Submittals must be signed and sealed by a Professional Land Surveyor licensed in the State of Utah.
- B. Survey Monuments certified supplement
 1. Refer to this Section, Article 3.12, paragraph C3 for submittal of drawings and notes.

1.6 MEASUREMENT Not Used

1.7 PAYMENT

- A. Survey work is incidental to other items of work and no separate measurement or payment will be made unless otherwise specified.

1.8 QUALITY ASSURANCE

- A. Assume responsibility for survey and control of the work and for correcting errors whether the errors are discovered during the actual survey work or in subsequent phases of the project and bears cost overruns resulting from errors.

PART 2 PRODUCTS

2.1 EQUIPMENT

- A. Tools, supplies, and stakes suitable for use in highway survey work.
- B. Stakes and hubs of sufficient length to provide a solid set in the ground with sufficient surface area above ground for necessary

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legible markings.

- C. Survey instruments and supporting equipment capable of achieving the specified tolerances.
 - 1. Calibrate survey equipment annually and check for accuracy before beginning survey work and as required throughout the duration of the project. Make the calibration certificate available upon the COR's request.

2.2 EQUIPMENT FOR GOVERNMENT USE Not Used

PART 3 EXECUTION

3.1 PREPARATION

- A. Establish construction survey points, elevations, and grades as necessary to control layout and complete the work.
 - 1. Verify control surveying and staking meet specified tolerances before beginning work.
- B. Calculate grades, elevations, offsets, and alignment data necessary for staking or setting items of work.
- C. The Government will provide, if available:
 - 1. Horizontal and vertical control points and elevation benchmarks.
 - a. Do not disturb control points and elevation benchmarks.
 - 2. Cross sections developed during design, if any
 - 3. Electronic project data
- D. Record keeping – Keep field notes, diaries, and books according to standard surveying practice.
 - 1. Make available survey records including field notebooks and forms used for the work to the Engineer upon request.
 - 2. Electronic records and reports are acceptable
- E. Contract Provision Disclaimer
 - 1. Provide a written request to the COR to obtain electronic data.
 - a. Waive claims that may result from the use of or reliance on the electronic data.
 - 2. Electronic Data is prepared by the Government for its own purposes and not for the benefit of private individuals or businesses.
 - 3. Indemnify and hold the Government harmless for damages,

costs, attorney fees, or other liabilities that might be incurred as a result of the Government's use and reliance on the Contractor's modified data.

3.2 DIRECTED SURVEY Not Used

3.3 COMPUTATIONS AND PLOTS Not Used

3.4 STAKE MAINTENANCE AND MARKINGS

- A. Provide and maintain reference/location stakes that identify stationing at least every 100 ft until the work has reached physical completion and accepted by the COR for each alignment and approved for removal.
 - 1. Provide reference/location stakes at whole station intervals such as 1032+00.
- B. Maintain staking necessary for the work until the construction has reached physical completion, accepted by the COR, and approved for removal.
 - 1. Legibly mark survey stakes with station and offset referenced to their respective control line.
 - 2. Mark slope, reference, and guard stakes with station.
 - 3. Renew illegible or damaged stakes.

3.5 SURVEY TOLERANCES

- A. Follow the guidelines set in the UDOT Survey & Geomatics Standards manual.
 - 1. Refer to <http://www.udot.utah.gov/go/standardsreferences> to view this manual.
- B. Relocate initial horizontal and vertical survey control points in conflict with construction to areas that will not be disturbed by construction operations.
 - 1. Furnish to the COR the coordinates and elevations for the relocated points before the initial points are disturbed.
- C. Protect benchmarks from construction activities.
 - 1. Position benchmarks to allow a level rod to stand vertically and squarely on the mark.
- D. Reference benchmarks to centerline and horizontal measurements.
 - 1. Compensate or correct for systematic errors, including those associated with instrument calibration
 - 2. Select the appropriate equipment and methods, and use trained personnel.

3. Use appropriate error propagation and other measurement design theory to select the proper instruments, field procedures, geometric layouts and computational procedures to control random errors.
- E. Apply appropriate procedures to verify that the allowable positional tolerance of such points is not exceeded if radial survey methods, GPS, or other acceptable technologies or procedures are used to locate or establish points on the survey.
- F. The positional tolerance may be tested by:
1. Comparing the relative location of points in a survey as measured by an independent survey of equal or higher accuracy or,
 2. The results of a minimally constrained, correctly weighted least squares adjustment of the points on the survey.
- G. Employ field procedures, instrumentation and adequate survey personnel in order to achieve a precision of 0.02 feet (or 6 mm) + 20 ppm.
1. See Table 1 for construction staking tolerances.

Table 1

CONSTRUCTION STAKING TOLERANCES		
Description	<u>Horizontal</u> <u>Vertical</u>	
	Decimals of a foot	
Box Culverts	± 0.02	± 0.02
Bridge Superstructures	± 0.02	± 0.02
Bridge Substructures	± 0.02	± 0.02
Clearing and Grubbing Stakes	± 1.00	-----
Construction Centerline Control	± 0.05	-----
Construction Centerline Station	± 0.10	-----
Curbs, Walks, and Bike Paths	± 0.03	± 0.02
Grade Stakes – Roadway Subgrade	± 0.20	± 0.05
Grade Stakes – Top of Rock	± 0.20	± 0.03
Grade Stakes – Roadway Finish	± 0.10	± 0.02
Luminaire and Signal Poles (incl.	± 0.20	± 0.20
Manholes, Inlets, and Culverts	± 0.10	± 0.03
PCC Pavement	± 0.10	± 0.02
Slope Stakes and References	± 0.30	± 0.10
Traffic Markings	± 0.20	-----
Walls – Retaining, MSE, Sound, etc.	± 0.05	± 0.05
Wetland Mitigation Control Stakes	± 0.20	± 0.20

Notes:

1. Stakes for miscellaneous items not listed above will have a horizontal and vertical tolerance of 0.20 ft, unless otherwise directed.
2. Features that are to be constructed flush to another surface should take on the same tolerance as that surface.
3. Staking tolerances for special circumstances will be discussed at the pre-survey meeting.
4. Meet the appropriate construction tolerances for the material as specified in the special provisions or standard specifications, regardless of the construction staking tolerances, specific to the work item.
5. Use bridge control once established, independent of other project controls for the duration of the bridge construction. Refer to Survey & Geomatics Standards manual for bridge control methodology. Refer to <http://www.udot.utah.gov/go/standardsreferences>.
6. Tolerances stated for each type of construction stake in this table indicates the acceptable deviation of the position of each reference point from its computed position relative to the given alignment and grade. Staked positions are generally checked using electronic stakeout reports and, if within tolerances, the stated position is accepted. Reference points may also have an accuracy relative to each other for precise measurements such as structures.

2. Survey Staking Methods:

- a. Furnish reference stakes for slope stakes and stakes used for setting items for work.
- b. Furnish the following for projects using Conventional Survey Methods:
 - 1) Establish and set slope stakes and references on both sides of centerline at cross section locations.
 - a) Place slope stakes at a maximum centerline spacing of 25 ft when the centerline curve radius is less than or equal to 500 ft.
 - b) Place slope stakes at a maximum spacing of 50 ft when the centerline curve radius is greater than 500 ft.
 - c) Place reference stakes at a maximum centerline spacing of 100 ft on tangents.
 - 2) Establish slope stakes in the field as the actual point of intersections of the design slope with the natural ground line.
 - 3) Set slope stake references outside the clearing limits.
 - 4) Include reference point and slope stake information on the reference stakes.

3. Furnish the following for projects using Machine Control Guidance Methods:

- a. Establish and set location stakes and references on one side of centerline at cross section locations.
 - 1) Place cross section stakes at a maximum spacing of 300 ft. (Cross-section stakes reference physical Survey

- points in the proposed cross-section, such as edge of pavement or top back of curb)
 - b. Place slope stakes at a maximum spacing of 300 ft.
 - c. Establish slope stakes in the field at the actual point of intersections of the design slope with the natural ground line.
 - d. Set slope stake references outside the clearing limits.
 - e. Include reference point information on the reference stakes.
 - f. Provide adequate bench marks throughout the project for construction equipment equipped with MCG to check setup and tolerances.
 - 1) Perform equipment checks at least once per day.
 - 2) Record equipment checks in a log for verification by the Engineer.
- 4. Staking limits – Reference/Location stakes can serve the purpose of the following staking requirements as long as required information for both purposes can be written on the stake.
 - a. Stake clearing limits on both sides of centerline at each established station.
 - 1) Locate the clearing limit on the ground as shown by the cut and fill limits on the plans.
 - b. Stake right of way limits, or temporary construction easement (TCE) if one exists, every 500 ft maximum on tangents, every 250 ft maximum on curves where ROW is not delineated by existing fence lines or other obvious boundaries.
 - 1) Stake Right of Way limits at right of way break/angle points along the right of way lines.
 - 2) Reduce the distance if staking distance is affected by line of sight.
 - c. Stake environmental control limits on both sides of centerline at each established station.
 - 1) Locate the environmental control limits on the ground as shown by the slope rounding contours and environmental and silt fence locations as shown on the Plans.
 - 2) Stake environmental control limits every 25 ft in environmentally sensitive areas.
 - 3) Provide staking as needed to guarantee the silt fence is located inside of right of way in standard silt fence installations where stations/locations are not called out on the environmental control plan sheets,.

I. Setting grade finishing stakes (Conventional Survey or RTK):
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1. Grade elevations and horizontal alignment:
 - a. On centerline.
 - b. On each shoulder at roadway cross section locations and between centerline and shoulder with a maximum spacing of 25 ft.
 - c. At the top of subgrade and the top of each aggregate course.
2. Locations:
 - a. Set stakes on centerline, on each normal shoulder, and on the shoulder of the turnout where turnouts are constructed.
 - b. Set hubs at the center and along the edges of parking areas.
 - c. Set stakes in ditches to be paved.
3. Maximum spacing between stakes along the alignment is 50 ft.
4. Use brushes or guard stakes at each grade finishing stake.
5. Reset grade finishing stakes as many times as necessary to construct the subgrade and each aggregate course.

J. Grade Verification (Machine Control Guidance)

1. The following procedure will only be applicable for verification of roadway layers for grade elevations and horizontal alignment.
 - a. The Department will use the Contractor provided survey equipment that is capable of verifying grade to tolerances required.
 - b. The Department will verify elevations at the following locations:
 - 1) On centerline.
 - 2) On each shoulder at roadway cross section locations and between centerline and shoulder with a maximum spacing of 25 ft.
 - 3) At the top of subgrade and the top of each aggregate course.
2. Locations:
 - a. On centerline, on each normal shoulder, and on the shoulder of the turnout where turnouts are constructed.
 - b. At the center and along the edges of parking areas.
 - c. At the top of subgrade and the top of each aggregate course.
 - d. In ditches to be paved.
3. The Government may verify and document elevations at a 300 ft maximum spacing between locations along the alignment.
 - a. The Government reserves the right to increase the spacing between grade verification locations up to, but

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not to exceed, 1000 ft if a level of confidence can be attained by the COR.

3.6 CONCRETE PAVING

- A. Develop a method of horizontal and vertical survey control for the placement of concrete pavement.
 - 1. Use laser, wire, or string line to maintain horizontal and vertical control.
 - 2. Maximum spacing, 50 ft
 - 3. Set control on both sides of roadway.
- B. 3D Paving, Machine Control Guidance, Wireless Paver
 - 1. Stagger survey control on either side of the roadway to provide a good strength of figure.
 - a. Place control points for MCG according to equipment manufacturer recommendations.
 - b. The instrument setup must obtain vertical accuracies within ± 0.02 ft of the existing control.
- C. Stake concrete joint and station stamp locations.

3.7 DRAINAGE STRUCTURES

- A. Stake drainage structures.
 - 1. Survey and record the ground profile along the centerline of the structure.
 - 2. Determine the slope catch points at inlets and outlets.
 - 3. Set reference points and record information necessary to determine structure length and end treatments.
 - 4. Stake ditches or grade to make the structure functional.
 - 5. Plot the profile along centerline of the structure to show the natural ground, the flow line, the roadway section, and the structure.
 - 6. Mark guard stakes with the following, when applicable:
 - a. Diameter, length, and type of culvert such as 18 inch x 35 ft corrugated metal pipe (cmp)
 - b. The vertical and horizontal distance from the hub to the invert at the end of the culvert or intermediate point as needed or directed
 - c. Flow line grade of the pipe
 - d. Station
 - 7. Provide a reference at a maximum spacing of 50 ft for storm sewers and waterlines. Reference inverts of pipe at manholes.

3.8 BRIDGES

Not Used

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3.9 BOX CULVERTS

- A. Set horizontal and vertical control and reference points.
 - 1. Establish and reference the centerline, back of parapet or barrier, skew, and flow line elevations at inlet, outlet, and breaks.

3.10 CURB AND GUTTER

- A. Set curb and gutter staking at 25 ft intervals on tangent and 10 ft intervals on curve radii.
- B. Set line and grade for curb and gutter within 0.02 ft of the proposed or established grade line.

3.11 GUARDRAIL

- A. Stake guardrail vertical and horizontal control at a maximum spacing of 25 ft on tangent sections and 10 ft on curves.

3.12 EXISTING SURVEY MONUMENTS

- A. Locate and reference private and public land survey monuments that may be destroyed by project construction activities before disturbing those existing monuments and under the direction of a Professional Land Surveyor licensed in the State of Utah.
- B. Complete referencing and reestablishing those existing monuments before project completion.
- C. Reference and reestablish the monuments created by the county surveyor.
 - 1. Notify the county surveyor at least 30 days before the destruction of monument.
 - 2. Coordinate the reestablishment of section corner and quarter corner monuments with the county surveyor.
 - 3. Prepare a corner record to be filed in the office of the County Surveyor's Office per Utah State Code 17-23-17.5 unless prepared by the County Surveyor's Office.
 - 4. Submit a certified supplement to the Survey Control Sheet with notes to the COR showing references to section corners and quarter corners.

3.13 RETAINING WALLS

- A. Set horizontal and vertical control and reference points based upon the projects primary and secondary control.
 - 1. Establish and reference the centerline offsets for the walls, radius points, and the beginning and ending wall locations as shown.
- B. Stake retaining wall vertical and horizontal control at no more than 25 ft on tangent sections and 10 ft on curved sections.

3.14 PAVEMENT MARKING

- A. Layout temporary and permanent pavement markings according to Section 02765.
 - 1. Place references for traffic striping no more than 150 ft apart on tangents and 50 ft on curves.

3.15 CLEANUP

- A. Remove and dispose of flagging, lath, stakes, and other staking material after the project has reached physical completion and the COR has approved removal.

3.16 UTILITIES

- A. Stake control lines as needed in cooperation with the utility operators so their facilities can be relocated to their proper final position.
- B. Stake crossings or potential points of conflict between facilities to give proper horizontal and vertical control for the relocation.
- C. Schedule this survey work with the utility companies to minimize delays and disruption of survey stakes.
- D. Replace disturbed stakes as necessary to facilitate the relocations.
- E. The Contractor is responsible for costs incurred to relocate utility more than once due to inaccurate or incomplete staking.
- F. Collect survey grade coordinate data for exposed, relocated, and new utilities during construction.
 - 1. Collect data in 1 ft intervals unless otherwise directed in the UDOT Survey & Geomatics Standards manual. Refer to <http://www.udot.utah.gov/go/standardsreferences>.

3.17 EXISTING MILEPOST SIGNS

- A. Locate existing milepost sign stations within the project limits.
- B. Reestablish locations of milepost signs before project completion if construction activities required removal of existing milepost signs.
 - 1. Reset sign location at original station of existing sign.
 - a. Exceptions
 - 1) Prior physical limitations listed in this Section, Article 3.17, paragraph A were removed during construction and no longer prevent installation of a sign at the actual mile point.
 - 2) Roadside conditions or newly constructed physical limitations would prevent reestablishment of milepost sign within 3 ft of its original station.
 - b. Contact the COR to determine how to proceed in either of these special cases.
 - 2. Establish an appropriate offset for each milepost sign to meet installation and clear zone requirements.
- C. Contact the COR to determine the preferred action for reestablishing the milepost signs where the alignment of the roadway was modified during construction to the extent that the new measured mile point locations of milepost signs were shifted more than 10 ft from their original location. New measured mile point locations of milepost signs were shifted more than 10 ft from their original location.

END OF SECTION

SECTION 01741
FINAL CLEANUP

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Final cleanup and restoration of all areas disturbed by construction.

1.2 RELATED SECTIONS

- A. Section 01455: Materials Quality Requirements

1.3 REFERENCES

- A. AASHTO Roadside Design Guide

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS Not Used

1.6 PAYMENT PROCEDURES

- A. All costs associated with Final Cleanup are incidental to other items of work and no separate measurement or payment will be made.
- B. Contractor is responsible for cleanup work outside the area disturbed by the Contractor for items of debris generated by the Contractor during performance of the work.

PART 2 PRODUCTS Not

Used PART 3

EXECUTION

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3.1 CLEANING PROCEDURES

- A. Do not damage the final pavement surface.
- B. Only use equipment with rubber tread on the final pavement surface.
- C. Clean drainage facilities of debris and obstructions caused by construction.
 - 1. Dispose of material removed.
- D. Remove or cover with fine material from roadway excavation or borrow, large rocks or boulders on fill slopes with the following exception:
 - 1. Large rocks and boulders protruding from the final graded surface six inches or less, on slopes steeper than 3:1 or beyond the clear zone.
- E. Do not undercut the slope on cut slopes.
 - 1. Remove all overhanging rocks.
 - 2. Solid ledge rock or partially buried boulders 0.33 yd³ or more may be left in place on slopes steeper than 4:1 beyond the clear zone.
- F. Clean and finish areas within right-of-way limits as follows:
 - 1. Dispose of trash and debris.
- G. Clean and finish areas within staging and office sites.

END OF SECTION

SECTION 01891

MOVE SIGNS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Materials and procedures for moving or resetting

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signs and moving or reconstructing mailbox supports.

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete

1.3 REFERENCES

- A. AASHTO M 232: Zinc Coating (Hot-Dip) on Iron and Steel Hardware
- B. ASTM A 513: Electric Resistance Welded Carbon and Alloy Steel Mechanical Tubing

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS Not Used

PART 2 PRODUCTS

2.1 CONCRETE

- A. Refer to SN Series Standard Drawings.
- B. Use an air-entrained (AE) concrete mix. Refer to Section 03055.

PART 3 EXECUTION

3.1 MOVE SIGN

- A. Maintain existing signs until construction requires removal.
 - 1. Coordinate with COR a minimum of 24 hours before removing any sign.
- B. Relocate existing signs as shown.
- C. Remove concrete from existing posts.
- D. Reset sign following SN Series Standard Drawings.
- E. Fill and compact the hole created by removing sign post.
 - 1. Compact to density of adjacent material.

END OF SECTION

SECTION 01892

RECONSTRUCT CATCH BASIN, MANHOLE, AND MISCELLANEOUS BOXES

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Lower and raise catch basin, manhole, and miscellaneous boxes such as meter, valve, and monument boxes to meet the grade of the adjacent surfaces.

1.2 RELATED SECTIONS

- A. Section 01721: Survey
- B. Section 02721: Untreated Base Course (UTBC)
- C. Section 03055: Portland Cement Concrete
- D. Section 03211: Reinforcing Steel and Welded Wire

1.3 REFERENCES

- A. AASHTO M 105: Gray Iron Castings
- B. AASHTO M 199: Precast Reinforced Concrete Manhole Sections

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Submit request to use alternative grade rings to the COR for review.
 - 1. Include manufacturer's product data sheets and installation instructions for alternative grade ring material.

PART 2 PRODUCTS

2.1 CONCRETE

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- A. Class AA(AE) – Refer to Section 03055.

2.2 REINFORCING STEEL

- A. Coated reinforcing steel. Refer to section 03211.

2.3 GRADE RINGS

- A. Gray iron
 - 1. Grade 30B or greater. Refer to AASHTO M 105.
- B. Precast concrete grade ring
 - 1. Refer to AASHTO M 199.
- C. Alternative grade ring material may be used when previously authorized by the COR.

2.4 UNTREATED BASE COURSE

- A. Refer to 02721.

PART 3 EXECUTION

3.1 RECONSTRUCT CATCH BASIN, MANHOLE, AND MISCELLANEOUS BOXES

- A. Lower impacted catch basin, manhole, and miscellaneous boxes prior to rotomilling.
- B. Remove concrete collar, walls, frame, grade ring, and riser as necessary to lower or raise the item.
- C. Protect all remaining features.
- D. Reconstruct catch basin and manhole to match the appropriate CB Series Standard Drawings.
- E. Set the top so that all corners are recessed up to 1/4 inch lower than the finished surface unless otherwise shown.
- F. Provide a concrete collar extending at least 12 inches outside the frame or flange and to a depth equal to the pavement but not less than 8 inches.

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- G. Reestablish damaged survey monuments.
 - 1. Refer to Section 01721

3.2 PAVEMENT REPAIR

- A. Place and compact untreated base course below the pavement structure.
- B. Place pavement to match the existing pavement surface.

END OF SECTION

SECTION 02056

EMBANKMENT, BORROW, AND BACKFILL

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Embankment, backfill, and bridge approach embankments.

1.2 RELATED SECTIONS

- A. Section 03575: Flowable Fill

1.3 REFERENCES

- A. AASHTO M 145: Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes
- B. AASHTO T 11: Materials Finer than 75 μm (No. 200) Sieve in Mineral Aggregates by Washing
- C. AASHTO T 27: Sieve Analysis of Fine and Coarse Aggregates
- D. AASHTO T 99: Moisture-Density Relations of Soils Using a 2.5 kg (5.5-lb) Rammer and a 305 mm (12 inch) Drop
- E. AASHTO T 180: Moisture-Density Relations of Soils Using a 4.54 kg (10-lb) Rammer and a 457 mm (18 inch) Drop

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- F. UDOT Materials Manual of Instruction
- G UDOT Minimum Sampling and Testing Requirements

1.4 DEFINITIONS

- A. Well-graded material – Material having an even distribution of different particle sizes. This even distribution of particles of different sizes results in a dense mass upon compaction.

1.5 SUBMITTALS

- A. Provide the following before delivering material to the project:
 - 1. Supplier and source of materials.
 - 2. Gradation analysis. Refer to AASHTO T 27 and T 11.
 - 3. Soil classification when applicable. Refer to AASHTO M 145.
 - 4. Maximum Dry Density and Optimum Moisture Determination
 - a. Use AASHTO T 180 Method D for A-1 soils and AASHTO T 99 Method D for all other soils.
- B. Engineering proposal for alternate materials or trench configurations for drainage pipe bedding and pipe backfill as outlined in this Section, Article 2.8 C.

1.6 ACCEPTANCE

- A. Acceptance sampling and testing is according to UDOT Minimum Sampling and Testing Requirements.
- B. Government reserves the right to select and test material from any location at the construction site.
 - 1. Establish the limits of nonconforming material sampled non- randomly and correct.
- C. Density Requirements – Acceptance is on a lot-by-lot basis.
- D. Remove nonconforming material and replace with acceptable material.

PART 2 PRODUCTS

2.1 GENERAL

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- A. Provide materials free of contamination from chemical or petroleum products for embankment and backfill placements. Materials may include recycled Portland Cement Concrete.
 - 1. Do not include asphalt pavement materials.

2.2 MATERIALS

- A. Borrow
 - 1. Classifications A-1-a through A-4. Refer to AASHTO M 145.
- B. Granular Borrow
 - 1. Classification A-1-a. Refer to AASHTO M 145.
 - 2. Non-plastic, well-graded, 3 inch maximum.
- C. Granular Backfill Borrow
 - 1. Classification A-1-a. Refer to AASHTO M 145.
 - 2. Well-graded, 2 inch maximum.
- D. Free-Draining Granular Backfill
 - 1. Meet the following gradation:

Table 1

Free Draining Granular Backfill Gradation	
Sieve Size	Percent Passing
1½ inch	90-100
1 inch	20-55
¾ inch	0-15
⅝ inch	0-5

2.3 EMBANKMENT

- A. Borrow or suitable roadway excavation materials excluding organic, frozen, or contaminated soils.

2.4 DRAINAGE PIPE BEDDING AND BACKFILL

- A. Drainage Pipe Bedding and Drainage Pipe Backfill
 - 1. Granular Backfill Borrow, Classification A-1. Refer to AASHTO M145.
 - a. Well-graded material.
 - b. Maximum aggregate size is 1½ inches for plastic pipe, 2 inches for all other pipes.
 - 2. Flowable fill. Refer to Section 03575.

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- a. Use only for drainage pipe backfill.
- B. Other materials or trench configurations for drainage pipe bedding and backfill may be used only upon previous approval of the Contractor's engineering proposal by the COR. Proposals using this option may include the use of native material or uniformly graded materials enclosed in an appropriate drainage geotextile. Any proposal must include all of the following:
 - 1. Stamped drawings and specifications signed and sealed by a Professional Engineer licensed in the state of Utah.
 - 2. Evaluation of site specific conditions and surrounding soils, including potential for migration of fines.
 - 3. A structural evaluation of the pipe support system for the proposed pipe that includes the pipe structural capacity and the depth of fill.
 - 4. Complete bedding or backfill source information including gradation, soil classification, and laboratory testing reports.

PART 3 EXECUTION

3.1 GENERAL

- A. Complete clearing, grubbing, stripping, and stockpiling topsoil before placing material.
- B. Requirements when placing material during freezing or snowy conditions:
 - 1. Do not place embankment on frozen or snow-covered areas.
 - a. Remove snow and frozen material from embankments, foundations, and borrow areas and furnish embankment material that can be compacted to the specified density.
 - 2. Do not deliver or use frozen material.
- C. Use appropriate compaction equipment adjacent to pipes, abutments, back walls, approach slabs, wing walls, retaining walls, and other structures.
 - 1. Expand the width of the trench to accommodate necessary compaction equipment.
 - 2. Compact by hand areas where compaction equipment cannot compact the soil.
- D. Over-excavate unsuitable material.

- E. Do not use unsuitable material for embankment or backfill placement.
- F. Density Requirements
 - 1. Borrow, Granular Borrow, Granular Backfill Borrow, Embankment for Bridge, and Drainage Pipe Bedding
 - a. Meet minimum density test average of 96 percent of maximum laboratory density with no single determination lower than 92 percent.
 - 1) Use AASHTO T 180 Method D for A-1 soils and AASHTO T 99 Method D for all other soils.
 - 2) Maintain appropriate moisture for compaction during processing.
 - 2. Drainage Pipe Backfill
 - a. Meet minimum density test average of 92 percent maximum laboratory density with no single determination less than 90 percent.
 - 1) Use AASHTO T 180 Method D for A-1 soils.
 - 2) Maintain appropriate moisture for compaction during processing.
 - b. Meet the pavement section material density requirement for pipes that encroach into the pavement section or use flowable fill.
 - 3. Meet 100 percent of the developed field density for material with more than 30 percent retained on the $\frac{3}{4}$ inch sieve.
 - 4. Free-Draining Granular Backfill
 - a. Meet 100 percent of the developed field density.
 - 1)

3.2 EMBANKMENT AND BORROW PLACEMENT

- A. Place roadway excavation or borrow or both in embankment section with the highest quality material in the top portion of the embankment.
- B. Scarify and compact the top 8 inches of the surface of the working platform or foundation to at least 90 percent of maximum laboratory density when the embankment height is 6 ft or less.
- C. Break and scarify all underlying concrete pavement surfaces so that pieces do not exceed 1 ft² before placing embankment over an existing concrete pavement surface that is outside the limits of removal or excavation shown.

Reconstruct Catch Basin, Manhole, and Miscellaneous Boxes

1. Remove other pavement surfaces that are not Portland Cement Concrete.
- D. Maintain Drainage
 1. Grade and maintain the roadway to ensure adequate drainage.
 2. Maintain drainage pipes and drainage ditches or provide temporary facilities when interrupting items such as irrigation systems, sewers, and under-drains.
- E. Place an initial layer to act as a working platform over soft, wet ground when approved by the COR.
 1. Density requirements do not apply to the working platform.
 2. Meet density requirements for embankment placed above the working platform.
- F. Do not place initial layer of embankment until COR inspects and accepts the working platform or foundation.
- G. Spread embankment materials uniformly in layers not exceeding 1 ft (uncompacted depth) and compact to the density requirements.
 1. Reduce the lift thickness or modify operations if tests show unsatisfactory density.
- H. Finish subgrade surface within ± 0.2 ft of line and grade.
- I. Do not use rock or broken concrete materials over 1 ft in any dimension.
- J. Distribute larger particles so space exists for placing and compacting embankment material.
- K. Do not place rocks larger than 4 inches or broken concrete within 1 ft of the subgrade surface.
- L. Do not use compacting equipment that causes shear failure in the embankment.

3.3 GRANULAR BORROW, GRANULAR BACKFILL BORROW, AND BACKFILL PLACEMENT

- A. Finish surface within ± 0.1 ft of line and grade.
- B. Compact material in maximum 6 inch layers (uncompacted depth) to the density requirement.

Reconstruct Catch Basin, Manhole, and Miscellaneous Boxes

- C. Backfill for structures such as bridges, foundations, box culverts, drains, and other structures.
 - 1. Place Embankment unless otherwise shown.

3.4 DRAINAGE PIPE FOUNDATION, BEDDING, AND BACKFILL PLACEMENT

- A. Place in 6 inch layers (uncompacted depth) and compact to the density requirement.
- B. Place uniform layers of drainage pipe backfill on both sides of the pipe and compact to the density requirement before placing successive lifts.
- C. Fully compact the haunch areas.

3.5 EMBANKMENT FOR BRIDGE PLACEMENT Not Used

3.6 FREE-DRAINING GRANULAR BACKFILL PLACEMENT

- A. Compact material in 1 ft maximum layers to the density requirement.
- B. Finish surface within ± 0.2 ft of line and grade.

END OF SECTION

SECTION 02072

GEOSYNTHETICS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Geosynthetics for subgrade stabilization and base reinforcement.

1.2 RELATED SECTIONS

- A. Section 02075: Geotextiles

1.3 REFERENCES

- A. ASTM D 374: Thickness of Solid Electrical Insulation
- B. ASTM D 4355: Deterioration of Geotextiles by Exposure to Light, Moisture and Heat in a Xenon Arc-Type Apparatus
- C. ASTM D 4759: Determining the Specification Conformance of Geosynthetics
- D. ASTM D 5818: Exposure and Retrieval of Samples to Evaluate Installation Damage of Geosynthetics
- E. ASTM D 6637: Determining Tensile Properties of Geogrids by the Single or Multi-rib Tensile Method
- F. ASTM D 7737: Individual Geogrid Junction Strength
- G. ASTM D 7748: Flexural Rigidity of Geogrids, Geotextiles and Related Products
- H. ASTM D 7864: Determining the Aperture Stability Modulus of Geogrids

1.4 DEFINITIONS

- A. Geogrid – A biaxial polymeric grid formed by a regular network of integrally connected tensile elements with apertures of sufficient size to allow interlocking with surrounding soil, rock, or earth to function primarily as reinforcement.

- B. Geosynthetic Composite System – A system of geogrid for reinforcement and confinement of the base course material along with a separation geotextile or approved alternate systems.
- C. Minimum Average Roll Value (MARV) – Reported value of a property taken as two standard deviations below the population mean based on testing and determined according to ASTM D 4759 to verify that the product meets specification requirements.

1.5 SUBMITTALS

- A. APL Compliance Form for information is required for the following products.
 - 1. Geosynthetic Composite System
 - 2. Alternative Geosynthetic Composite Systems
- B. Manufacturer's product data sheets and recommended installation instructions.

1.6 ACCEPTANCE

- A. The Department rejects material at installation if it has defects, rips, holes, flaws, deterioration, or damage incurred during manufacture, transport, handling or storage.

1.7 PACKAGING, SHIPPING AND STORING

- A. Labeling and Tagging
 - 1. Identify each package by a tag or label securely affixed to the outside of the roll on at least one end.
 - 2. Provide the following required information on the tag:
 - a. Name of the geotextile manufacturer
 - b. Brand name of the product
 - c. Width, length, and package weight
 - c. Roll number(s)
 - d. Lot number(s)
- B. Storage and Protection.
 - 1. Prevent excessive mud, wet concrete, epoxy, or other deleterious materials from coming in contact with and affixing to the geosynthetic materials.
 - 2. Store at temperatures above -20 degrees F and below 140 degrees F.
 - 3. Rolled materials may be laid flat or stood on end.
 - 4. Do not expose geosynthetic materials to direct sunlight for a period longer than recommended by the manufacturer.
 - 5. Keep geosynthetic materials dry until installation.

PART 2 PRODUCTS

2.1 GEOSYNTHETIC COMPOSITE SYSTEM

- A. Geogrids.
1. Meet the requirements of Table 1.

Table 1

Geogrid Property Requirements		
Property	Test Method	Value ³
Aperture size (in) nominal dimension	ASTM D 374	1.0 (MD) x 1.3 (XMD)
Rib Thickness (in)	ASTM D 374	0.05
Junction Thickness (in)	ASTM D 374	0.120
Junction efficiency, % of rib ultimate tensile strength ¹ (min.)	ASTM D 7737 Method A	93%
Tensile strength, 2% strain (lb/ft)	ASTM D 6637 Method B	410 (MD) x 620 (XMD)
Geosynthetic Sheet Stiffness/Modulus (lbs/in)(min.)	ASTM D 6637 Method B	2,500 (XMD) ²
Ultimate Strength (lb/ft)	ASTM D 6637 Method B	1,310 (MD) x 1,970 (XMD)
Ultraviolet resistance (%) minimum retained tensile strength after 500 hours	ASTM D 4355	70
Flexural stiffness (mg-cm)	ASTM D 7748	750,000
Aperture Stability (kg-cm/deg)	ASTM D 7864	6.5
Note:		
1. Junction efficiency is the ratio of (ASTM D7737 Junction Strength) / (ASTM D 6637 Method "A" Single Rib) x 100		
2. Sheet stiffness/modulus value = Tensile Strength @ 2% strain ÷ 0.02 ÷ 12		
3. Minimum Average Roll Values (MARVs) in accordance with ASTM 4759		
4. Machine Direction (MD) Cross Machine Direction (XMD)		
5. Independent testing shall be performed by an accredited laboratory through the Geosynthetic Accreditation Institute-Laboratory Accreditation Program.		

- B. Separation Geotextile.
1. Refer to Section 02075.

2.2 ALTERNATIVE GEOSYNTHETIC COMPOSITE SYSTEMS

- A. Alternative geosynthetic composite systems not on the APL may be used if approved.
 - 1. Approval will be based on full scale laboratory testing or field testing of the alternate system immediately adjacent to and within 50 ft of a baseline composite system using materials meeting the criteria in Article 2.1.
 - a. Test two different pavement sections with each system.
 - 1) Aggregate Base Course (minimum 6 inch), Geosynthetic composite system, subgrade (Modulus 5 to 25 ksi).
 - 2) Aggregate Base Course (minimum 6 inch), Geosynthetic composite system, Stabilizing Material (minimum 6 inch), Geosynthetic composite system, subgrade (Modulus 0.55 to 5 ksi).
 - b. Submit composite resilient modulus values from both systems.
 - c. Submit deformation values for both systems.
 - d. The alternate system performance must meet or exceed the baseline composite system performance.

PART 3 EXECUTION

3.1 INSTALL GEOSYNTHETIC COMPOSITE SYSTEM

- A. Orient the geosynthetic rolls parallel to the roadway centerline. Use widths that produce overlaps of parallel rolls at the centerline and shoulders.
- B. Geosynthetic Composite System
 - 1. Separation Geotextile.
 - a. Install the separation component under the reinforcement component if the system is two separate products.
 - b. Meet the Installation requirements of Section 02075.
 - 2. Geogrid.
 - a. Provide minimum overlap according to manufacturer's recommendations.
 - b. Install geogrid according to the guidelines provided by the manufacturer or as directed by the Engineer.
 - 3. Alternate Geosynthetic Composite System.
 - a. Follow manufacturer's recommendations.
- C. Temporarily secure the geosynthetics in place using ties, staples, pins, sand bags or backfill, or as determined by the Engineer.

- D. Place, spread, and compact granular fill material in such a manner that minimizes the development of wrinkles in or movement of the geosynthetic composite system.
- E. Do not operate vehicles over the geosynthetic composite system until at least six inches of loose fill has been placed.
 - 1. Keep the turning of vehicles to a minimum to prevent vehicles from displacing the fill and damaging the geosynthetic system.
- F. Replace any roll of geosynthetic damaged before, during, or after installation.
 - 1. Replace the affected area and overlap the geosynthetic one foot on all sides adjacent to the damaged area.
 - 2. Follow ASTM D 5818 for exposure and retrieval of damages done during installation of evaluation samples.

END OF SECTION

SECTION 02075
GEOTEXTILES

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Geotextiles

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES

- A. AASHTO M 288: Geotextile Specification for Highway Applications

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Manufacturer's product data sheet and recommended installation instructions.

1.6 PACKAGING, SHIPPING, AND STORING

- A. Protect the geotextile from direct sunlight, chemicals, mud, dirt, and debris during shipment and storage.
- B. Labeling and Tagging
 - 1. Identify each package by a tag or label securely affixed to the outside of the roll on at least one end.
 - 2. Provide the following required information on the tag:
 - a. Name of the geotextile manufacturer
 - b. Brand name of the product, width, length, and package weight of geotextile

PART 2 PRODUCTS

2.1 EROSION CONTROL GEOTEXTILE

- A. Conforming to AASHTO M 288.

2.2 DRAINAGE GEOTEXTILE

- A. Class 2 nonwoven drainage geotextile according to AASHTO M 288.

2.3 SEPARATION GEOTEXTILE

- A. Class I fabric according to AASHTO M 288 with Apparent Opening Size of 0.60 mm maximum average roll value.
 - 1. Do not use woven slit film geotextiles in or under the pavement structure.

2.4 STABILIZATION GEOTEXTILE

- A. Class I fabric according to AASHTO M 288 with Apparent Opening Size of 0.43 mm maximum average roll value.

2.5 WEED BARRIER GEOTEXTILE

- A. Woven, needle-punched, polypropylene, 5.0 oz, weed barrier geotextile designed for commercial use.

2.6 THREAD FOR SEWN SEAMS

- A. Refer to AASHTO M 288, Appendix, Seaming.

PART 3 EXECUTION

3.1 GENERAL EROSION CONTROL, DRAINAGE, SEPERATION AND STABALIZATION

- A. Place geotextile on areas that are smooth and free of projections or depressions.
 - 1. Do not drag the geotextile across the subgrade.
 - 2. Roll geotextile out as smoothly as possible in the direction of vehicle travel.
- B. Do not operate construction equipment or traffic directly on geotextile.
- C. Repair immediately if damaged
 - 1. Place patch over damaged area and extend 3 ft beyond the perimeter of the tear or damage.

- D. Place cover material as soon as possible.
 - 1. Do not leave uncovered for more than five days
 - 2. Do not tear, puncture or shift geotextile during placement.
 - 3. Use at least a 6 inch thick cover layer or twice the maximum aggregate size, whichever is thicker.
 - 4. Do not dump cover material directly on the geotextile except as a starter course.
- E. Limit construction vehicles in size and mass so rutting in the initial layer above the geotextile is not more than 3 inch deep or half the layer thickness, whichever is less.
 - 1. Do not turn vehicle wheels on the first layer.

3.2 INSTALL EROSION CONTROL GEOTEXTILE

- A. Overlap the geotextile at least 2 ft at all longitudinal and transverse joints or sew the geotextile unless otherwise specified.
- B. Place overlapped geotextile so that the upstream sheet overlaps the downstream sheets.
- C. Overlap each sheet over the next downhill sheet for placement on slopes.
- D. Anchor the geotextile using key trenches or aprons at the crests and toes of the slope.

3.3 INSTALL DRAINAGE GEOTEXTILE

- A. Place and secure geotextile to provide direct contact against the excavated surface.
- B. Overlap successive sheets of geotextile at least 1 ft in the down-gradient direction of flow.
- C. Overlap geotextile at least 1 ft at the top of the trench, where applicable.
- D. Place fill beginning with the sheets overlapping above subsequent sheets to hold geotextile in place.

3.4 INSTALL SEPARATION GEOTEXTILE

- A. Overlap the geotextile at least 1 ft at all longitudinal and transverse joints or sew the geotextile unless otherwise specified.
- B. Place fill to hold geotextile in place beginning with the sheets overlapping above subsequent sheets.
- C. Use pins 18 inches long to help secure the geotextile during installation.

3.5 INSTALL STABILIZATION GEOTEXTILE

- A. Overlap the geotextile at least 2 ft at all longitudinal and transverse joints or sew the geotextile unless otherwise specified.
- B. Overlap each sheet over the next downhill sheet for placement on slopes.
- C. Place fill to hold geotextile in place beginning with the sheets overlapping above subsequent sheets.
- D. Use pins 18 inches long to help secure the geotextile during installation.

3.6 INSTALL WEED BARRIER GEOTEXTILE

- A. Preparation
 - 1. Remove sharp objects, large stones, and undesirable vegetation.
 - 2. Cut an "X" over each plant and push geotextile under plant base if placing geotextile over an existing bed.
 - 3. Roll geotextile over soil and cut an "X" for each plant hole if placing over a new bed.
 - 4. Fold excess geotextile under and cover with specified landscaping materials.
- B. Surface Cover
 - 1. Provide at least 4 inch of cover on all areas on the geotextile.
 - 2. Increase thickness of cover material over geotextile to three times the diameter of the largest rock material if using large landscape rock.
 - 3. Do not leave any portion of geotextile exposed to direct sunlight.
- C. Repair
 - 1. Repair immediately if damaged.
 - 2. Place patch over damaged area and extend 3 ft beyond the perimeter of the tear or damage.

- D. Maintenance
 - 1. Maintain surfaces and supply additional landscape materials where necessary including areas affected by erosion.

3.7 SEW

- A. Stitch seam such that no subgrade can be seen through the seam and the edge of the geotextile will not distort or be damaged under the force of cover material placement.

END OF SECTION

SECTION 02216

TEST-HOLE UTILITY

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Procedure for test-hole operations to identify and locate utilities during construction activities.

1.2 RELATED SECTIONS

- A. Section 01721: Survey

1.3 REFERENCES

- A. CI/ASCE 38-02: Standard Guideline for the Collection and Depiction of Existing Subsurface Utility Data

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Provide records to the COR within 5 working days of test-hole operations.

PART 2 PRODUCTS Not

Used PART 3

EXECUTION

3.1 PREPARATION

- A. Identify locations of possible conflicts including excavation, pipe culvert extensions, catch basins, and other construction elements to test-hole.

- B. Request permission to test-hole from the COR at least 24 hours prior to excavation.

3.2 EXPLORATION

- A. Use vacuum excavation or other methods that protect the integrity of the utility.
- B. Perform test-hole operations to provide Quality Level A per CI/ASCE 38- 02.

3.3 RECORDS

- A. Measure and record utility location according to CI/ASCE 38-02.
 - 1. Measure and record horizontal and vertical location in project coordinate system per Standard Specification 01721.
 - 2. Measure and record depth from grade, outside diameter, orientation, material composition, and configuration of non-encased, multi- conduit systems.

3.4 BACKFILL AND RESTORATION

- A. Backfill test-hole excavation according to utility specifications when outside of roadway prism.
 - 1. Backfill according to the most restrictive of the Government's specifications if test-hole excavation is within the roadway prism.
 - 2. Backfill within 24 hours or as otherwise determined.
- B. Backfill test-hole excavation to the same depths and grades with equivalent or better materials of surrounding pavement or terrain.

END OF SECTION

SECTION 02221

REMOVE STRUCTURE AND OBSTRUCTION

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Removal and disposal of structures and obstructions that interfere with construction.
- B. Salvage of equipment or materials when specified.

1.2 RELATED SECTIONS

- A. Section 00727: Control of Work
- B. Section 01355: Environmental Compliance
- C. Section 01721: Survey
- D. Section 02056: Embankment, Borrow, and Backfill
- E. Section 02705: Pavement Cutting
- F. Section 03055: Portland Cement Concrete
- G. Section 03575: Flowable Fill

1.3 REFERENCES **Not Used**

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS **Not Used**

1.6 PROJECT SITE CONDITIONS

- A. Protect adjacent structures and utilities and their contents that are designated to remain.

PART 2 PRODUCTS

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2.1 CONCRETE

- A. Use Class A (AE) Concrete. Refer to Section 03055.

2.2 GRANULAR BORROW AND GRANULAR BACKFILL BORROW

- A. Refer to Section 02056.

2.3 FLOWABLE FILL

- A. Refer to Section 03575.

PART 3 EXECUTION

3.1 PREPARATION

- A. Review all work procedures with the COR.
- B. Coordinate utility location according to Section 00727.
 - 1. Locate and protect all active utilities.
 - 2. Before beginning work:
 - a. Notify COR
 - b. Notify owners of affected utilities
 - c. Have the utilities in the area located by Blue Stakes
- C. Restore utility services disturbed by construction operations.
- D. Disconnect water service by excavating to the corporation stop and turning it off. Disconnect the service line from the corporation stop.
- E. Remove items such as existing cesspools and leach lines.
- F. Excavate all material necessary to permit removal.

3.2 BACKFILL AND COMPACTION

- A. Fill all holes or pits resulting from removal operations with suitable material.
- B. Compact the backfilled areas to the density of the surrounding ground or as specified.

3.3 REMOVAL AND DISPOSAL

- A. Remove and dispose of material promptly using methods acceptable to the COR and according to applicable rules and regulations.
 - 1. Include excavation, removal, transportation, and disposal costs in the item of work.
- B. Remove concrete to at least 2-3 ft below the subgrade surface or finished ground lines.
- C. Dispose of removed obstructions at a site secured by the Contractor.
 - 1. Obtain required permits and provide an environmentally safe area for disposal of removed items. Refer to Section 01355.

3.4 SALVAGE

- A. Salvage designated equipment and materials.

3.5 REMOVE BUILDING, BASEMENT, AND FOUNDATION

- A. Move or demolish designated buildings including basements, foundations, sidewalks, pavement slabs, porches, fences, and outbuildings on each parcel.
- B. The Government is not responsible for vandalism or theft that occurs to the building or its contents that reduces the value of the salvage or increases the cost of removal after the award of the Contract.
- C. Break the floor into pieces not larger than 1 ft² in areas to remain in place. Remove and dispose of pieces over 1 ft².

3.6 REMOVE SEPTIC TANK AND UNDERGROUND TANK

- A. Empty and dispose of tank contents according to Section 01355.
- B. Break down and remove tank and appurtenances to at least 2 ft below the subgrade surface or finished ground lines.
- C. Break the floor into pieces not larger than 1 ft² in area.

3.7 REMOVE BURIED FUEL TANK

- A. Remove buried fuel storage tanks and dispose of tank contents according to all applicable Laws and Regulations.
- B. Do not spill fuel on subgrade.
- C. Comply with the State and local authorities having jurisdiction over fuel tank removals.

3.8 REMOVE BOX CULVERT

- A. Remove structure so that no remaining portion is closer than 3 ft to a watercourse or closer than 2 ft to the subgrade and embankment surface or within 2 ft of the finished ground lines.
- B. Remove structures that will interfere with proposed construction.
- C. Complete blasting or other removal operations of existing structure that may damage new construction before beginning the new work.

3.9 REMOVE CONCRETE HEADWALL

- A. Remove headwall where designated.

3.10 REMOVE CONCRETE SLOPE PROTECTION

- A. Remove portions of the existing slope protection and cutoff wall as shown.
- B. Mark the concrete slope protection removal limits in the presence of the Engineer.
- C. Saw cut the existing slope protection to full depth.
- D. Do not damage the portions of concrete slope protection that will remain.

3.11 REMOVE MANHOLE, CLEANOUT, DIVERSION BOX, AND CATCH BASIN

- A. Maintain satisfactory by-pass service during construction operations.
- B. Plug unused sewers with a 2 ft long concrete plug.

3.12 REMOVE PIPE AND PIPE END SECTION

- A. Remove pipe, end sections and attached appurtenances.
- B. Cut pipe to 2 ft inside the Department's right-of-way when the
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pipe extends into private property.

1. Seal the remaining pipe with a 2 ft long concrete plug.
- C. Seal openings in remaining manhole or catch basin walls with a concrete plug.

3.13 ABANDON PIPE

- A. Completely fill the pipe with flowable fill.
1. Obtain approval from the COR for alternate materials.
- B. Survey the location of exposed abandoned pipes ends. Refer to Section 01721.

3.14 REMOVE CONCRETE AND ASPHALT PAVEMENT

- A. Cut existing pavement on the designated lines with straight vertical edges free from irregularities when joining new construction to existing pavement. Refer to Section 02705.
- B. Completely remove pavement down to the underlying base course or subgrade.

3.15 REMOVE RAILROAD TRACK Not Used

3.16 OBLITERATE CONCRETE ROAD

- A. Break up pavement into pieces not larger than 1 ft² in area.
1. Scarify and cover broken concrete with at least 1 ft of suitable backfill material.
- B. Fill depressions and form rounded slopes to blend with the natural or surrounding contours.
- C. Grade materials either along the toe of an embankment or into a depression or borrow pit.
1. Cover with at least 1 ft of suitable backfill material.

3.17 REMOVE CONCRETE SIDEWALK AND CONCRETE DRIVEWAY

- A. Remove concrete to the nearest expansion joint or saw cut to provide proper grades and connections.
- B. Make concrete cuts straight, vertical to the surface, full depth, and

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free from irregularities. Refer to Section 02705.

3.18 REMOVE CONCRETE CURB, CONCRETE GUTTER, CONCRETE CURB AND GUTTER, RAISED ISLAND, AND BITUMINOUS CURB

- A. Remove concrete curb, concrete gutter, concrete curb and gutter, raised island, bituminous curb, and parts of such improvements to an existing joint or joint sawed with a vertical face.
- B. Remove material to provide proper grades and connections.

3.19 REMOVE CONCRETE BARRIER

- A. Precast
 - 1. Remove connection pins and concrete barrier including any stabilization pins.
 - 2. Dispose or salvage as required.
 - a. Contact the Engineer for delivery location and coordination for salvaged barrier.
- B. Cast-In-Place
 - 1. Saw cut the pavement at the front face of the cast-in-place barrier.
 - 2. Remove barrier and underlying pavement.

3.20 REMOVE GUARDRAIL

- A. Remove and dispose of guardrail, posts, hardware, anchor assemblies, terminal assemblies, and attached posts, signs, and delineators.
- B. Remove steel posts to at least 8 inches below the subgrade surface or finished ground lines.

3.21 REMOVE CATTLE GUARD

- A. Salvage existing traffic grill units and welded end guard units.
 - 1. Contact the Engineer for delivery location and coordination for salvaged traffic grill units and welded end guard units.
- B. Cut existing pavement as required. Refer to Section 02705.
- C. Remove existing fence as required.
- D. Remove the cattle guard to at least 2 ft below the subgrade surface or finished ground lines.

- E. Prevent livestock from entering work site from adjacent properties during removal and installation.

3.22 REMOVE FENCE

- A. Prevent people or livestock from entering work site from adjacent properties during removal and installation.
- B. Remove fence, posts, and foundations to at least 2 ft below subgrade or finished ground lines.
- C. Do not damage vegetation and ground cover during removal operations.

3.23 REMOVE TREE

- A. Remove trees as shown.
 - 1. A tree consists of stump, root, trunk, branches, and foliage.
 - 2. Multiple leaders rising from a common root will not be counted separately.
 - 3. Remove the root system to at least 2 ft below the finished ground level and within a 2 ft radius of the stump.

3.24 REMOVE UTILITY POLE

- A. Remove pole and all appurtenances.
- B. Remove foundation to at least 2 ft below subgrade or finished ground lines.

END OF SECTION

SECTION 02231

SITE CLEARING AND GRUBBING

PART 1 GENERAL

1.1 SECTION INCLUDES

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- A. Clear, grub, remove, and dispose of trees, stumps, and debris within the designated limits of the roadways, channels, easements, and other designated areas.

1.2 RELATED SECTIONS

- A. Section 01355: Environmental Compliance
- B. Section 01571: Temporary Environmental Controls
- C. Section 02221: Remove Structure and Obstruction

1.3 REFERENCES Not Used

1.4 DEFINITIONS

- A. Clear – Remove and dispose of trees, stumps, logs, limbs, sticks, vegetation, debris, and other material on the natural ground surface.
- B. Grub – Remove and dispose of roots, buried logs, debris, organic matter, and other deleterious materials under the ground surface.

1.5 SUBMITTALS Not Used

1.6 PAYMENT PROCEDURES Not Used

PART 2 PRODUCTS Not Used

PART 3 EXECUTION

3.1 PREPARATION

- A. Verify with the COR the vegetation or objects to be removed.
- B. Review work procedures with the COR.
- C. Schedule work carefully with consideration for property owners and general public.
- D. Refer to Section 01571 for temporary environmental measures.

3.2 VEGETATION REMOVAL

- A. Grub the areas 2 ft below natural ground, within the limits of clearing, of all stumps, roots, buried logs, and all other underground obstructions.
- B. Stumps, roots, and non-perishable solid objects may remain in cleared areas where the embankment is:
 - 1. 2 ft or more above the natural ground.
 - 2. At least 2 ft away outside the slope stake lines.
- C. Completely grub stumps and roots where a structure is to be constructed, piles are to be driven, or unsuitable material is to be removed.

3.3 BACKFILLING

- A. Backfill all stump holes, cuts, depressions, and other holes resulting from clearing and grubbing within areas to receive embankment.
 - 1. Compact backfilled areas to the density of the surrounding ground.
- B. Consider backfill as incidental to the work.

3.4 DISPOSAL

- A. Dispose of material. Refer to Section 01355.
- B. Do not dispose of material within the designated roadbed.
- C. Dispose of material as follows:
 - 1. Haul to the landfill.
 - 2. Use material to widen embankments and flatten embankment side slopes as approved by the COR.
 - 3. Reduce wood to chips a maximum of ½ inch thick for mulching cut and fill slopes.
 - a. Chips may be buried or distributed uniformly on the ground surface and mixed with the underlying earth so the mixtures will not sustain burning.

3.5 TREE REMOVAL

- A. Refer to Section 02221.

3.6 PROTECTION

- A. Land monuments, property markers, or official datum points
 - 1. Protect until their removal is approved.
 - 2. Reference for re-establishment before removing.
- B. Protect trees from damage to roots and branches if they are designated to remain.
- C. Protect other vegetation and objects designated to remain.

END OF SECTION

SECTION 02316
ROADWAY EXCAVATION

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Excavation of all material within designated areas.
- B. Rock excavation and removal.
- C. Placement of excavated material in embankment and other areas.

1.2 RELATED SECTIONS

- A. Section 00725: Scope of Work
- B. Section 00820: Legal Relations and Responsibility to the Public
- C. Section 01355: Environmental Compliance
- D. Section 01571: Temporary Environmental Controls
- E. Section 01721: Survey
- F. Section 02056: Embankment, Borrow, and Backfill
- G. Section 02075: Geotextiles
- H. Section 02231: Site Clearing and Grubbing
- I. Section 02705: Pavement Cutting
- J. Section 02912: Topsoil

1.3 REFERENCES

- A. National Fire Protection Association (NFPA) Codes and Standards
- B. OSHA Construction Standards

1.4 DEFINITIONS

- A. Rock – Material that cannot be excavated and removed without blasting, chipping, cutting, or ripping.

1.5 SUBMITTALS Not Used

1.6 ACCEPTANCE AND PAYMENT PROCEDURES

- A. Payment is quantified by the cubic yard.
- B. Payment includes excavation, removal, transportation, and disposal.

PART 2 PRODUCTS

2.1 MATERIALS FOR OVER-EXCAVATED AREAS

- A. Refer to Section 02056.
- B. Geotextile Fabric – Refer to Section 02075.

2.2 USE OF ON-SITE MATERIALS Not Used

2.3 EXPLOSIVES

- A. Use of explosives is not allowed.

PART 3 EXECUTION

3.1 PREPARATION AND PROTECTION

- A. Refer to Sections 00820 and 01571.
- B. Locate and protect utilities as necessary.
- C. Clear and grub within the designated area before starting excavation. Refer to Section 02231.

3.2 BLASTING MATERIAL STORAGE Not Used

3.3 TOPSOIL

- A. Remove topsoil according to Section 02912.

3.4 DEWATERING

- A. Keep excavation free from surface and ground water through all stages of construction.
 - 1. Maintain adequate drainage during all stages of construction through pumping, pipe culverts, and drainage ditches.
 - 2. Provide temporary facilities when interrupting items such as irrigation systems, sewers, and under drainages.

3.5 EXCAVATION – STANDARD PROCEDURES

- A. Finish excavation to reasonably smooth and uniform surface.
- B. Provide and maintain satisfactory access to roads, streets, and adjacent property during all phases of construction according to the Traffic Control Plan.
- C. Remove material in all cut sections to the depth shown.
 - 1. Scarify to an 8 inch depth and compact subgrade to at least 90 percent of maximum laboratory density before placing pavement section.
- D. Excavate and waste unsuitable material.
- E. Material for backfilling or finishing
 - 1. Use suitable granular material encountered in excavation to construct the top layers of embankment, finishing the roadbed, or backfill when directed by the Engineer.
 - 2. Haul the granular material directly from excavation to the final position on the roadbed when practical.

3.6 ROCK REMOVAL – NONEXPLOSIVE METHOD

- A. Excavate solid rock to the specified depth and backfill with acceptable material.
 - 1. Rock removed more than the specified depth will not be measured or paid for.
 - 2. Backfilling depth greater than the specified depth will not be measured or paid for.

3.7 ROCK REMOVAL – EXPLOSIVE METHOD Not Used

3.8 ROCK FACES

- A. Scale rock cuts of all loose rocks and fragments and leave in a neat and safe condition.

3.9 PAVEMENT

- A. Cut existing pavement on the designated lines with straight vertical edges free from irregularities when joining new construction to existing pavement. Refer to Section 02705.
- B. Excavate all pavement as shown.
- C. Dispose of pavement at the landfill or as otherwise indicated.

END OF SECTION

SECTION 02317

STRUCTURAL EXCAVATION AND BACKFILL

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Excavation for and backfilling of bridges, foundations, box culverts, pipe culverts, drains, and other structures.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill

1.3 REFERENCES

- A. AASHTO LRFD Bridge Construction Specifications

1.4 DEFINITIONS

- A. Unsuitable material – Material not meeting specifications, organic materials, frozen lumps, soils such as peat or bog, and over-saturated silts, clays, or sands whose water content prevents appropriate compaction.

1.5 SUBMITTALS

- A. Working Drawings
 - 1. Drawings for Temporary Works.
 - a. Includes shoring, bracing, and cofferdams.

1.6 PAYMENT PROCEDURES

- A. Granular Backfill Borrow and Free Draining Granular Backfill are paid for separately.

PART 2 PRODUCTS

2.1 BACKFILL FOR STRUCTURES

- A. Refer to Section 02056.

2.2 DRAINAGE PIPE BACKFILL

- A. Refer to Section 02056.

PART 3 EXECUTION

3.1 PREPARATION Not Used

3.2 EXCAVATION

- A. Structural excavation includes:
 - 1. Excavating foundations for structures.
 - 2. Control and removal of water.
 - a. Dewater the excavation if ground water is encountered.
 - b. Continue dewatering activities as necessary to prevent damage to the work.
 - 3. Installation and removal of facilities required to complete the work unless specified or allowed to remain in place.
- B. Excavate rock and other hard strata to elevations and dimensions shown.
 - 1. Clean and cut to a firm surface as shown.
 - a. Foundation cannot contain loose material.
 - 2. Clean out seams and fill with concrete, mortar, or grout.
- C. Do not over-excavate unless indicated in the contract documents or -approved by the COR.
 - 1. Over-excavate and waste material unsuitable for the structure foundation to at least 1 ft below the design elevation
 - 2. Cost of over-excavation is paid.
- D. Clean the space under structures, in channels, and adjacent areas affected by operations to prevent drift and scour.
- E. Dispose of all unused excavation material.

3.3 BACKFILL AND COMPACTION

- A. Backfill and compact excavated spaces not occupied by abutments, foundations, and other permanent work as described.
 - 1. Backfill to the surface of the surrounding ground, with a

2. sufficient allowance for settlement.
Bring backfill up uniformly on all sides of structures.

END OF SECTION

SECTION 02318

DITCH EXCAVATION

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Small ditch and surface ditch excavation.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill

1.3 REFERENCES **Not Used**

1.4 DEFINITIONS

- A. Small ditch – Ditch or channel, not V-shaped, with a bottom width less than 12.0 ft.
- B. Surface ditch – V-shaped, 1.0 ft deep minimum and 3.0 ft wide across the top, or as shown.

1.5 SUBMITTALS **Not**

Used PART 2

PRODUCTS **Not**

Used

PART 3 EXECUTION

3.1 SMALL DITCH EXCAVATION

- A. Form the ditch as shown.
- B. Place and compact excavated material in embankments. Refer to Section 02056.

3.2 SURFACE DITCH EXCAVATION

- A. Construct the ditch along the contour of the ground.
- B. Place excavated material to form a berm on the downhill side of the ditch.
- C. Shape the ditch and berm so that power-driven mowers can operate on the graded surface.

END OF SECTION

SECTION 02342

SHOTCRETE

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Shotcrete mix design, application and curing

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete

1.3 REFERENCES

- A. AASHTO T 152: Air Content of Freshly Mixed Concrete by the Pressure Method
- B. ASTM C 309: Liquid Membrane-Forming Compounds for Curing Concrete
- C. ASTM C 1604: Obtaining and Testing Drilled Cores of Shotcrete

1.4 DEFINITIONS

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- A. Dry Mix Process – Shotcrete production process in which most of the water is added at the nozzle.
- B. Nozzle Operator – Worker on shotcrete crew who manipulates the nozzle, controls consistency of dry mix shotcrete, and controls final disposition of the material.
- C. Rebound – Shotcrete that fails to adhere to the surface on which the shotcrete is being applied.
- D. Wet Mix Process – Shotcrete production process in which cement, aggregate, water and admixtures are first mixed together before introduction into the delivery hose.

1.5 SUBMITTALS

- A. Shotcrete mix design with mix proportions and test results for approval.
 - 1. The Department approves shotcrete mix designs based on preconstruction test panel results.
 - 2. Include at least the following:
 - a. Aggregate source and grading
 - b. Cement type and brand
 - c. Water source
 - d. Type and dosage of hydration stabilizing/extended-set admixtures, if used
 - e. Discharge time limit for wet mix shotcrete
 - f. Preconstruction test panel test results including at least the following:
 - 1) Sample identification, including mix design and test panel number and orientation.
 - 2) Date and time of core sample preparation including sample dimensions.
 - 3) Date and time of compression test.
 - 4) 28 day compressive strength.
 - 5) Names and signatures of people performing the compression test.
 - 6) Name of nozzle operator who produced test panel.
 - 7) Air content measured at the time of test panel creation (wet mix only).
- B. Written documentation of the shotcrete supervisor's and nozzleman's qualifications including proof of ACI Shotcrete Nozzleman certification for review.
 - 1. ACI Shotcrete Nozzleman Certification
 - 2. Experience on at least three comparable projects.
 - a. Include project names, locations and owners' contact information
- C. Production summary reports for each shift
 - 1. Submit for information within 24 hours of completing shift.
 - 2. Include at least the following:
 - a. Observations of success or problems of equipment operation, application, final product conditions, and any other relevant issues during production and application.
 - 1) Include sketches for locations as necessary.
 - b. Shotcrete batch number(s) if applicable

1.6 ACCEPTANCE

- A. The Department will perform testing according to Table 1.

Table 1

Sampling and Testing Schedule			
Property or Characteristic	Test Methods or Specifications	Frequency	Sampling Point
Air Content (Wet Mix Only)	AASHTO T 152	1 per load ¹	Prior to placement
28 day Compressive Strength	ASTM C 1604	1 set per 50 cy; at least one set per day	Production test panels ²

¹ When continuous mixing is used, sample every 10 cy.

² A single compressive strength test result is the average result from three 3-inch diameter core specimens from the same test panel tested according to ASTM C 1604 at 28 days.

- B. Shotcrete that fails to meet the strength requirements based on the following will be cause for rejection.
1. Specified 28 day minimum compressive strength test result less than 0.85 times the minimum 28 day compressive strength.
 2. A single core compressive strength test result less than 0.75 times the minimum 28 day compressive strength.
 3. The engineer may accept a 'reject' lot based on engineering analysis. The Department will apply a price reduction on rejected shotcrete allowed to stay in place.
- C. Shotcrete that has been damaged by frost, or which lacks uniformity, exhibits segregation, honeycombing, lamination with a shotcrete stage, or shows excessive cracking will be cause for rejection.

PART 2 PRODUCTS

2.1 SHOTCRETE

- A. Refer to Section 03055 for material requirements.
- B. Conform to the requirements in Table 2.

Table 2

Shotcrete Process	Air Content Range	28 day Minimum Compressive Strength ² (psi)
Dry Mix	N/A	4,000
Wet Mix	8% - 10% at the pump	4,000

² According to ASTM C 1604

- C. Use the amount of water required to produce shotcrete of suitable strength, consistency, quality, and uniformity with the minimum amount of rebound.

2.2 MIX DESIGN

- A. Design mixes to meet the requirements of this Section and project specific criteria.
- B. Preconstruction Test Panels
 - 1. Refer to this Section, Article 3.1.

2.3 CURING COMPOUND

- A. Use a curing compound that complies with ASTM C 309 Type 1-D or Type 2, Class A.

PART 3 EXECUTION

3.1 TEST PANELS

- A. Forms and Construction
 - 1. Construct wood forms for preconstruction and production test panels at least 3.5 inch thick by 2 ft by 2 ft in size.
 - 2. Construct wood forms with plywood back 0.75 inch thick minimum.
 - 3. Do not include reinforcing steel bars in the test panels
- B. Preconstruction Test Panels
 - 1. Produce test panels for each proposed mix design.
 - a. Include a separate test panel for each anticipated shooting orientation (slab, slopes, vertical, and overhead), and each proposed nozzle operator.

- b. Assign a test panel number and record specific information about the panel required for submittal.
 - 2. Notify the COR of the time and location of preconstruction test panel production so that the COR may witness the production.
- C. Production Test Panels
 - 1. Produce separate panels for each nozzle operator applying shotcrete.
 - a. Produce production test panels for every 50 cubic yards of material placed, but not less than one panel per day.
 - b. Shoot test panels in the same manner as that being used on the project, including distance from nozzle, angle and orientation.
 - c. Include a separate test panel for each mix design used by the nozzle operator and each shooting orientation that mix design was used in (slab, slopes, vertical, and overhead).
 - 2. Cure test panels in similar conditions to what is anticipated in the field and until test specimens are sampled.
- D. Compressive Strength Test
 - 1. Use a core drill to remove three, 3 inch diameter samples from the center of the test panel, at least 5 inch clear of the edges.
 - 2. Remove the test specimens at least 21 calendar days and no more than 26 calendar days age.
 - a. Coordinate timing of specimen removal, delivery and testing with the COR.
 - 3. Store cores in watertight plastic bags and label with the project number, test panel number, date of placement, and name of nozzle operator.

3.2 SURFACE PREPARATION

- A. Prepare shotcrete application surface according to the following:
 - 1. Remove loose material, mud, rebound, and other foreign matter.
 - 2. Clear any vegetation or organic material from the surface.
 - 3. Remove curing compound on previously placed shotcrete surfaces by sandblasting.
 - 4. Install depth gages on 4 ft centers longitudinally and vertically to indicate the thickness of shotcrete layers.
 - 5. Dampen natural excavated surfaces immediately before shooting.
 - a. Remove standing water before shooting.
 - 6. Achieve saturated surface dry condition on shotcrete surfaces.

3.3 APPLICATION

- A. Limitations
 - 1. Do not apply shotcrete during adverse weather including rain, snow, and high winds without adequate protection authorized by the Engineer.
 - 2. Cease placement operations when the ambient temperature drops below 50 degrees F.
- B. Use authorized nozzle operators and approved mix designs.
- C. Apply shotcrete in one or more layers.
 - 1. Apply shotcrete within discharge time limit indicated in the approved mix design.
 - 2. Make the surface of each shotcrete layer uniform and free of sags, drips, or runs.
 - 3. Limit the layer thickness to 3 inch per application with no sloughing or sagging.
 - 4. Broom or scarify the applied surface and allow the layer to harden if additional thickness is required.
 - 5. Remove laitance, loose material, and rebound.
 - a. Promptly remove rebound from the work area.
 - 6. Construction Joints
 - a. Taper construction joints at an approximately 45 degree angle from the receiving surface.
 - b. Form joints by cutting plastic shotcrete. Roughen shotcrete in the joint face while it is still plastic.

3.4 CURING

- A. Apply curing compound to shotcrete surfaces which are exposed in their final condition.
 - 1. Apply at twice the rate indicated for formed surfaces recommended by the manufacturer.
- B. Cure for at least 7 consecutive days or the time necessary to attain 70 percent of the specified 28 day minimum compressive strength, whichever period is less.
 - 1. Immediately repair damage to the curing compound film during the specified curing period by re-spraying.

END OF SECTION

SECTION 02372

GABIONS AND REVET MATTRESSES

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Rock filled gabions and revet mattresses.

1.2 RELATED SECTIONS

- A. Section 02075: Geotextiles
- B. Section 02317: Structural Excavation and Backfill

1.3 REFERENCES

- A. AASHTO T 85: Specific Gravity and Absorption of Coarse Aggregate
- B. AASHTO T 96: Resistance to Degradation of Small-Size Course Aggregate by Abrasion and Impact in the Los Angeles Machine
- C. AASHTO T 104: Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate
- D. ASTM A 975: Double-Twisted Hexagonal Mesh Gabions and Revet Mattresses (Metallic-Coated Steel Wire or Metallic-Coated Steel Wire With Poly(Vinyl Chloride) (PVC) Coating)
- E. ASTM D 7014: Assembly and Placement of Double-Twisted Wire Mesh Gabions and Revet Mattresses

1.4 DEFINITIONS

- A. Gabion – A double-twisted wire mesh enclosure uniformly partitioned into internal cells, interconnected with other similar units, and filled with rock to form flexible, permeable, monolithic structures.
- B. Revet mattress – Similar to gabions but having a larger footprint and smaller height.
- C. Rock diameter - The equivalent diameter (average width) of a given rock.

1.5 SUBMITTALS

- A. Wire enclosure manufacturer's product data sheet and installation instructions.
 - 1. Conform to ASTM D 7014 for installation instructions.
- B. The rock source, gradation, and laboratory values for the properties in Table 1.

PART 2 PRODUCTS

2.1 GABIONS AND REVET MATTRESSES

- A. Wire enclosure
 - 1. Gabion
 - a. Galvanized 8 x 10 wire mesh type, mesh opening 3.25 inch nominal x 4.50 inch nominal. Refer to ASTM A 975
 - b. PVC coated galvanized 8 x 10 wire mesh type, mesh opening 3.25 inch nominal x 4.50 inch nominal. Refer to ASTM A 975.
 - 2. Revet mattress
 - a. Galvanized 6 x 8 wire mesh type, mesh opening 2.50 inch nominal x 3.25 inch nominal. Refer to ASTM A 975.
 - b. PVC coated galvanized 6 x 8 wire mesh type, mesh opening 2.50 inch nominal x 3.25 inch nominal. Refer to ASTM A 975.
- B. Rock
 - 1. Hard, durable, resistant to weathering, and free from seams, cracks and other structural defects.
 - a. Do not use shale, mudstone, and other rock that may break into smaller pieces during handling.
 - b. Do not use concrete or asphalt rubble.
 - 2. Meet the properties listed in Table 1.

Table 1

Physical Properties		
Property	Value	Test Method
Specific Gravity	2.4 min	AASHTO T 85
Absorption	2% max	AASHTO T 85
Soundness of Aggregate using Sodium Sulfate or Magnesium Sulfate	12% max or 17.5% max	AASHTO T 104
Resistance to Degradation	45% max	AASHTO T 96

3. Provide rock diameters between 4 to 8 inches or as shown.
4. Retain 95 percent or more of the rock on a square screen having the same opening size as the wire enclosure.
 - a. Control gradation by sample measurement or by visual inspection, as determined by the Engineer.

2.2 EROSION CONTROL GEOTEXTILE

- A. Refer to Section 02075, except use a nonwoven 12 oz/yd² maximum average roll value.

PART 3 EXECUTION

3.1 PREPARATION

- A. Excavate and grade to provide a firm and uniform bearing surface. Refer to Section 02317.
- B. Install erosion control geotextile as shown.

3.2 GABION AND REVET MATTRESS INSTALLATION

- A. Follow manufacturer's installation instructions.
- B. Place each empty wire enclosure into its final position as shown.
 1. Tie or fasten to adjacent wire enclosures along all containing edges in order to form a continuously connected, monolithic structural unit.

- C. Place rock into the wire enclosure in a manner that ensures proper alignment, minimizes voids, avoids bulges, and prevents damage to the wire enclosure.
 - 1. Hand place rock on exposed vertical faces as necessary to develop a satisfactory face graduation and prevent loss of smaller rock fill through the openings.

END OF SECTION

SECTION 02373

RIPRAP

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Loose and compacted riprap.

1.2 RELATED SECTIONS

- A. Section 02075: Geotextiles
- B. Section 02317: Structural Excavation

1.3 REFERENCES

- A. AASHTO T 85: Specific Gravity and Absorption of Coarse Aggregate
- B. AASHTO T 96: Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
- C. AASHTO T 104: Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate

1.4 DEFINITIONS

- A. D_{50} - The median rock diameter within a riprap mass for which 50 percent of the rocks are smaller and 50 percent are larger.
- B. Rock diameter - The equivalent diameter (average width) of a given rock.

1.5 SUBMITTALS

- A. The rock source, gradation, and laboratory values for the properties in Table 1.

PART 2 PRODUCTS

2.1 RIPRAP

- A. Rock
1. Angular, hard, durable, resistant to weathering and free from seams, cracks and other structural defects.
 - a. Do not use shale, mudstone, or other rock that may break into smaller pieces in the process of handling and placing.
 - b. Do not use concrete or asphalt rubble.
 2. Meet the properties listed in Table 1.

Table 1

Physical Properties		
Property	Value	Test Method
Specific Gravity	2.5 min	AASHTO T 85
Absorption	2% max	AASHTO T 85
Soundness of Aggregate using Sodium Sulfate or Magnesium Sulfate	12% max or 17.5% max	AASHTO T 104
Resistance to Degradation	40% max	AASHTO T 96

3. Well graded rock throughout the riprap layer to produce a dense mass. Refer to Table 2 for riprap gradation limits.
 - a. Control gradation by sample measurement or by visual inspection, as determined by the Engineer.

Table 2

Riprap Gradation Limits	
Rock Diameter Range (ft)	Percent of Gradation Smaller Than
1.5D ₅₀ to 1.7D ₅₀	100
1.2D ₅₀ to 1.4D ₅₀	85
1.0D ₅₀ to 1.15D ₅₀	50
0.4D ₅₀ to 0.6D ₅₀	15
0.1D ₅₀	10

2.2 EROSION CONTROL GEOTEXTILE

- A. Refer to Section 02075, except use a nonwoven 12 oz/yd² minimum average roll value.

PART 3 EXECUTION

3.1 PREPARATION

- A. Excavate and grade to provide a firm and uniform bearing surface. Refer to Section 02317.
- B. Install erosion control geotextile as shown.

3.2 LOOSE RIPRAP

- A. Place rocks to provide a secure unsegregated dense mass.
 - 1. Distribute and manipulate the rocks so that the larger rocks are uniformly distributed and the smaller rocks serve to fill the spaces between the larger rocks.

3.3 COMPACTED RIPRAP

- A. Place rocks conforming to this Section, Article 3.2.
- B. Compact riprap to remove irregular surface protrusions larger than 3 inches.

END OF SECTION

SECTION 02376

ROLLED EROSION CONTROL PRODUCTS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Temporary and Permanent Rolled Erosion Control Products (RECP).

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. ASTM D 4355: Deterioration of Geotextiles by Exposure to Light, Moisture and Heat in a Xenon Arc Type Apparatus
- B. ASTM D 6459: Determination of Rolled Erosion Control Product (RECP) Performance in Protecting Hillslopes from Rainfall-Induced Erosion
- C. ASTM D 6460: Determination of Rolled Erosion Control Product (RECP) Performance in Protecting Earthen Channels from Stormwater-Induced Erosion
- D. ASTM D 6525: Measuring Nominal Thickness of Rolled Erosion Control Products
- E. ASTM D 6566: Measuring Mass per Unit Area of Turf Reinforcement Mats
- F. ASTM D 6818: Tensile Properties of Rolled Erosion Control Products

1.4 DEFINITIONS

- A. Rolled Erosion Control Products (RECP) – Degradable or non-degradable prefabricated blankets formed with natural or synthetic materials installed to reduce soil erosion and aid in vegetative establishment.
 - 1. Temporary RECP are composed of degradable natural fibers bound between two degradable natural or synthetic fiber netting to form a continuous matrix designed to provide erosion control protection up to 36 months.

2. Permanent RECP are manufactured with non-degradable synthetic fibers, filaments, nets, wire mesh or other elements into a permanent, three-dimensional matrix, and installed where established vegetation may be insufficient to protect against erosion within high flow channels, ditches or along very steep slopes.

1.5 SUBMITTALS

- A. Authorized Products List (APL) Compliance Form for information.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products in original ultra violet (UV) weather-resistant factory labeled packages showing the name of the manufacturer and product description.
 1. Protect from damage due to climatic conditions or construction operations.

PART 2 PRODUCTS

2.1 ROLLED EROSION CONTROL PRODUCTS (RECP)

- A. Meet the requirements in Table 1 for Temporary RECP.

Table 1

Requirements for Temporary Rolled Erosion Control Products				
Property	Test Method (ASTM)	RECP Type 2	RECP Type 3	RECP Type 4
Typical Functional Longevity		12 months	24 months	36 months
Slope Gradient (H:V)		≤ 3:1	≤ 2:1	≤ 1:1
Tensile Strength	D6818	≥ 75 lbs/ft	≥ 100 lbs/ft	≥ 125 lbs/ft
C Factor	D6459	≤ 0.20	≤ 0.10	≤ 0.05
Unvegetated Shear Stress	D6460	≥ 1.75 lbs/ft ²	≥ 2.0 lbs/ft ²	≥ 2.25 lbs/ft ²

- B. Meet the requirements in Table 2 for Permanent RECP.

Table 2

Requirements for Permanent Rolled Erosion Control Products		
Property	Test Method (ASTM)	Turf Reinforcement Mat (TRM)
Slope Gradient (H:V)		$\leq 0.5:1$
Vegetated Shear Stress	D6460	$\geq 10.0 \text{ lbs/ft}^2$
Material Mass per Unit Area	D6566	$\geq 8.0 \text{ oz/yd}^2$
Tensile Strength	D6818	$\geq 175 \text{ lbs/ft}$
UV Stability	D4355	$\geq 80\%$
Thickness	D6525	$\geq 0.25 \text{ inch}$

PART 3 EXECUTION

3.1 PREPARATION

- A. Complete all required grading, topsoil placement, and seeding in designated areas before installing RECP.
- B. Make soil surface stable, firm, free of rocks, roots and other obstructions.
- C. Install RECP within 24 hours after seeding.

3.2 INSTALLATION

- A. Minimize disturbance of the prepared seedbed when installing RECP.
- B. Install RECP according to manufacturer's recommendations including any requirements to anchor RECP into the soil.
- C. Unroll RECP parallel to the primary direction of flow and place it in direct contact with the soil.
 - 1. Do not stretch the product or allow it to "tent" or bridge over surface inconsistencies during installation.
- D. Within a channel, ditch or swale, install RECP to allow runoff to flow directly to the centerline of ditch, not undermining or bypassing the RECP.

- E. Place additional staples in areas such as swales, base of humps, against rock outcrops and as required achieving maximum contact between RECP and the soil.

END OF SECTION

SECTION 02610

DRAINAGE PIPE

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Pipe used to convey surface water and stormwater by gravity flow.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill
- B. Section 02317: Structural Excavation
- C. Section 03055: Portland Cement Concrete

1.3 REFERENCES

- A. AASHTO M 170: Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe
- B. AASHTO M 196: Corrugated Aluminum Pipe for Sewers and Drains
- C. AASHTO M 207: Reinforced Concrete Elliptical Culvert, Storm Drain and Sewer Pipe
- D. AASHTO M 243: Field-Applied Coating of Corrugated Metal Structural Plate for Pipe, Pipe-Arches, and Arches
- E. AASHTO M 245: Corrugated Steel Pipe, Polymer Precoated, for Sewers and Drains
- F. AASHTO M 294: Corrugated Polyethylene Pipe, 300- to 1500-mm (12- to 60-in.) Diameter
- G. AASHTO M 304: Poly (Vinyl Chloride) (PVC) Profile Wall Drain Pipe and Fittings Based on Controlled Inside Diameter
- H. AASHTO M 330: Polypropylene Pipe, 300- to 1500-mm (12- to 60-
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in.) Diameter

- I. AASHTO M 335: Steel-Reinforced Polyethylene (PE) Ribbed Pipe, 300- to 1500-mm (12- to 60-in.) Diameter
- J. AASHTO R 82: Pipe Joint Selection for Highway Culvert and Storm Drains
- K. ASTM C 443: Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
- L. ASTM C 990: Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants
- M. ASTM D 1056: Flexible Cellular Materials—Sponge or Expanded Rubber
- N. ASTM F 477: Elastomeric Seals (Gaskets) for Joining Plastic Pipe

1.4 DEFINITIONS

- A. Deflection – Deviation from a pipe’s original shape.
- B. NASSCO – National Association of Sewer Service Companies.
- C. Nominal Diameter – Inside diameter of the pipe specified.
- D. Pipe Interior Roughness:
 - 1. Corrugated – Interior surface that is formed into a series of alternating crests and valleys with a Manning’s “n” coefficient greater than 0.013.
 - 2. Smooth Lined – Interior surface that is essentially smooth, with a Manning’s “n” coefficient less than or equal to 0.013.
- E. Profile Wall Pipe – Pipe that has a smooth interior wall with an exterior that is either is ribbed, corrugated, or smooth.
 - 1. The wall may contain hollow cores.
- F. Spring Line – Location of the maximum horizontal dimension of a pipe.
- G. Thermoplastic Pipe – Pipe made from plastic materials such as HDPE, PVC, and Polypropylene.

1.5 SUBMITTALS

- A. Manufacturer’s certificate of compliance.
- B. Manufacturer’s data sheets and installation instructions.
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- C. Documentation from AASHTO NTPEP showing compliance with Table 3 for the thermoplastic pipe material and diameter supplied for information.
- D. NASSCO Pipeline Assessment certification for remote video technician for information.
- E. Pipe inspection report for each pipe for review. Include at least the following:
 - 1. The project number, date, and time of the inspection for each pipe inspection, the pipe identification used in the plan set, and type and size of pipe.
 - 2. A video recording of each pipe inspection in a digital format for remote inspection.
 - 3. Written and still image documentation of locations where alignment deviations, joint gaps, pipe damage, and any other deficiencies were observed.
 - 4. Written documentation of deflection testing and measurements.

1.6 ACCEPTANCE

- A. General
 - 1. Each pipe is accepted for payment after verifying that the following criteria have been met:
 - a. Alignment
 - b. Deflection
 - c. Joint gap
 - d. Damage
 - 2. Proposed resolutions for nonconforming pipes require the seal of a licensed Professional Engineer competent in the structural design of the pipe material being evaluated.
- B. Alignment
 - 1. Do not exceed the installation horizontal and vertical alignment tolerances in Table 1.

Table 1

Installation Alignment Tolerances		
Design Grade	Horizontal	Vertical*
		inch/100 ft
> 1%	Not to exceed the pipe manufacturer's requirements	1½
0.5% - 1%		1
< 0.5%		½

* Increase tolerance by 50 percent for pipes with silt-tight joints.

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- C. Deflection
 - 1. Do not exceed a 5 percent shape change from the nominal diameter.
 - a. Proposals for acceptance of pipes with a deflection greater than 7.5 percent will be rejected.
- D. Joint Gap
 - 1. Do not exceed the joint separation, measured along the length of pipe, in Table 2.
 - a. Repair or replace joints that show visible signs of soil or water infiltration.

Table 2

Joint Gap Tolerances	
Nominal Diameter (inches)	Separation (inches)
12 to 36	0.75
42 to 48	1.00
54 to 90	1.25
96 to 144	1.75

- E. Damage
 - 1. Pipes are to be free from:
 - a. Cuts, cracks, spalls, chips, or punctures.
 - 1) Cracks are allowable up to 0.01 inch in width for reinforced concrete pipe
 - b. Loss or delamination of coatings.
 - c. Exposed reinforcing steel.
 - d. Imperfect concrete mixing and casting such as honeycomb or open texture.

PART 2 PRODUCTS

2.1 PIPE

- A. General
 - 1. Provide pipe according to the size, material or interior roughness, and joint type specified.
 - a. Meet the material requirements in Table 3.
 - b. Meet the joint requirements of AASHTO R 82 for silt-tight and leak-resistant joints.
 - c. Determine the strength and thickness of the pipe required based on the cover over the pipe. Refer to the DG Series Standard Drawings.

- d. Maintain the same type of pipe material, joint, strength, and thickness throughout the entire length of pipe.
2. Internally label each section of pipe with the manufacturer's name or trademark, nominal diameter, and manufacture date. Include the pipe class, gauge, and coating according to the pipe material type.
 - a. Place the pipe so that the location of the label is above the spring line of the pipe.

Table 3

Pipe Material Specifications	
Interior Roughness and Material Type	
Corrugated	
Polymer Coated Corrugated Steel Pipe and Pipe Arch	AASHTO M 245
Aluminum Corrugated Pipe and Pipe Arch	AASHTO M 196
Smooth	
Profile Wall Polyethylene (HDPE) Pipe	AASHTO M 294
Profile Wall Polyvinyl Chloride (PVC) pipe	AASHTO M 304
Profile Wall Polypropylene Pipe	AASHTO M 330
Steel Reinforced Thermoplastic Ribbed Pipe	AASHTO M 335
Polymer Coated Spiral Rib Steel Pipe and Pipe Arch	AASHTO M 245
Spiral Rib Aluminum Pipe and Pipe Arch	AASHTO M 196
Reinforced Concrete Pipe	AASHTO M 170
Elliptical Reinforced Concrete Pipe	AASHTO M 207

- B. Reinforced Concrete Pipe
 1. Use a ~~Department~~ **UDOT** prequalified supplier of precast concrete products.
 2. Use Type V Cement. Refer to Section 03055.
 3. Do not cast lift holes except for circular pipe that has a nominal diameter greater than 54 inches or any elliptical pipe.
 - a. Plug and seal lift holes in a manner such that the pipe section will meet pressure requirements.
 4. Use a rubber gasket for circular reinforced concrete pipe. Refer to ASTM C 443.
 5. Use a mastic joint sealant for elliptical reinforced concrete pipe. Refer to ASTM C 990.

C. Metal Pipe – Steel and Aluminum

1. Do not allow pipes of different types of metal to contact each other.
 2. Use rubber gaskets on all connecting bands. Refer to ASTM D 1056 for flat gaskets and ASTM C 443 for O-ring gaskets.
 3. Use bands with projections (dimple bands) only in extension of existing pipes or a field cut where annular corrugations do not exist.
- D. Thermoplastic Pipe - HDPE, PVC, Polypropylene and Steel Reinforced Thermoplastic Ribbed Pipe
1. Use an AASHTO NTPEP compliant supplier of thermoplastic pipe.
 2. Do not use in permanent above ground installations.
 3. Do not use greater than 36 inch diameter for PVC pipe.
 4. Do not use greater than 60 inch diameter for HDPE, Polypropylene, and Steel Reinforced Thermoplastic Ribbed pipe.
 5. Use bell and spigot joints with an elastomeric rubber gasket. Refer to ASTM F 477.

2.2 PIPE FOUNDATION, BEDDING AND BACKFILL MATERIAL

- A. Refer to Section 02056.

2.3 ASPHALT MASTIC

- A. Refer to AASHTO M 243.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Excavation and Backfill
1. Refer to Section 02317 and DG Series Standard Drawings.
 2. Keep trenches free from water.
 3. Grade and prepare the bottom of the trench to provide a firm and uniform bearing throughout the entire length of the pipe.
 - a. Do not use blocking to bring the pipe to grade.
 - b. Shape the pipe foundation and bedding to have recesses to fit any projecting hubs or bells.
 4. Remove and relay or replace pipe that is out of alignment, not joined properly or damaged before placing backfill.
- B. Follow manufacturer's recommendations for connecting pipes and for connecting pipe to end sections, concrete headwalls, catch

basins, and similar structures.

1. Do not allow pipes of different types of metal to contact each other.
2. Spray or brush-coast aluminum pipe contacting concrete with asphalt mastic to a minimum thickness of 0.05 inch.

3.2 POST INSTALLATION PIPE INSPECTION

A. General

1. Inspect pipes after installation of pipe, placement of backfill and before placing pavement or finished grade according to Table 4 and Table 5.

Table 4

Roadway Functional Classification	Percent of Pipes to Inspect
Primary	100
Secondary and Tertiary	50

Table 5

Pipe Testing Requirements				
Pipe Size	Visual		Deflection*	
	Manual	Remote	Manual	Mandrel
≤ 48-inch dia.		X		X
> 48-inch dia.	X		X	

* Deflection testing is required for circular metal and thermoplastic pipe only.

2. The COR will determine which pipes to inspect when the percentage of pipes to inspect is less than 100 percent.
 - a. Inspect additional pipes with apparent defects and as directed by the COR.
3. Inspection is not required for pipes less than 20 ft in length.
4. Notify the COR at least 24 hours before performing an inspection.
5. Clean and flush the pipe with water immediately before the inspection.
 - a. Remove and dispose material and debris from pipes.

B. Remote Inspection

1. CCTV Equipment
 - a. Record video using a crawler mounted camera capable of panning and tilting to a 90 degree angle with the axis

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- of the pipe and rotating 360 degrees.
 - b. Center the camera head in the pipe both vertically and horizontally to allow a clear picture of the entire periphery of the pipe.
 - c. Video image must be continuously illuminated, clear, focused, and free from roll, static, or other image distortion qualities that may prevent the reviewer from evaluating the pipe's condition.
 - 1) Calibrate the video's lighting and focus to view the internal markings within the pipe.
- 2. CCTV Inspection
 - a. Remote video inspection operator must have a current NASSCO Pipeline Assessment certification.
 - b. Use the video image to observe horizontal and vertical alignment deviations, joints gaps, and pipe damage.
 - 1) Document all observations with captions in the video.
 - 2) Note each location and provide a still image in the inspection report.
 - c. Do not move the crawler through the pipe at a speed greater than 30 ft/minute.
 - 1) Stop the crawler and video the entire circumference at each joint.
 - 2) Stop the crawler and zoom when necessary to video observations.

C. Manual Inspection

- 1. Notify the COR at least 24 hours before performing an inspection.
 - a. Follow UOSH requirements for inspecting confined entry spaces.
- 2. Observe and document the following:
 - a. Deflection - Take the following measurements every 10 ft along the length of the pipe to the nearest ¼ inch:
 - 1) Vertically from the crown to invert.
 - 2) Horizontally at the spring line.
 - 3) Two measurements, each diagonally at 45 degrees to the pipe springline.
 - b. Cracks – Measure cracks using a device capable of measuring 0.01 inch.
 - c. Joint Gaps – Measure the widest separation at each joint to the nearest ¼ inch.
 - d. Damage.

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- D. Mandrel Inspection
1. Notify the COR at least 24 hours before performing an inspection.
 - a. Verify the diameter of the mandrel with a proving ring or other method according to manufacturer's recommendations.
 2. Mandrel
 - a. Contain at least nine equally spaced runners (40 degree angles).
 - b. Length not less than the diameter.
 3. Deflection test
 - a. Pull a mandrel that is 5 percent less than the pipe nominal inside diameter.
 - b. Perform the following if the mandrel does not pass:
 - 1) Record the maximum distance achieved from the inlet side.
 - 2) Remove the mandrel and continue the inspection from the outlet end of the pipe toward the inlet end. Record the maximum distance achieved from the outlet side.
 - 3) Repeat inspection with a mandrel that is 7.5 percent less than the pipe nominal inside diameter.

END OF SECTION

SECTION 02613

DRAINAGE PIPE END SECTIONS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Precast concrete and prefabricated steel end sections for drainage pipes.

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete
- B. Section 03056: Self-Consolidating Concrete (SCC)

1.3 REFERENCES

- A. AASHTO M 218: Steel Sheet, Zinc-Coated (Galvanized), for Corrugated Steel Pipe
- B. ASTM A 500: Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
- C. ASTM D 1056: Flexible Cellular Materials – Sponge or Expanded Rubber

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS **Not Used**

PART 2 PRODUCTS

2.1 CONCRETE END SECTION

- A. Concrete
 - 1. Wet cast - Class AA(AE). Refer to Section 03055.
 - 2. Self-Consolidating Concrete. Refer to Section 03056.
- B. Refer to DG Series Standard Drawings.
- C. Use a UDOT prequalified supplier of precast concrete products.

2.2 STEEL END SECTION

- A. End Section: Galvanized steel sheeting according to AASHTO M 218.
- B. Safety Bar for Safety End Section: Schedule 40 ASTM A 500 Class B steel pipe. Galvanize after fabrication according to AASHTO M 111.
- C. Smooth Tapered Sleeve, when required: Galvanized steel sheeting according to AASHTO M 218.
- D. Refer to DG Series Standard Drawings.

PART 3 EXECUTION

3.1 GENERAL

- A. Furnish precast concrete or prefabricated steel end sections compatible with the adjoining pipe.

3.2 INSTALLATION

- A. Prepare a smooth foundation free of rock and debris, capable of uniformly supporting the end section.
- B. Place the end section at the same alignment and slope as the adjoining pipe.
- C. Embed the toe plate for steel end sections.
- D. Connect the end section to the adjoining pipe according to the DG Series Standard Drawings.
 - 1. Use a $\frac{3}{8}$ inch neoprene flat gasket when connecting to a dissimilar metal. Refer to ASTM D 1056.
- E. Backfill and grade slopes to match the end section.

END OF SECTION

SECTION 02618

DRAINAGE PIPE LINER

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Drainage lining methods such as polyvinyl chloride (PVC) fold-and-form, cured in place pipe (CIPP), and segmental thermoplastic and metal pipe.

1.2 RELATED SECTIONS

- A. Section 02610: Drainage Pipe

1.3 REFERENCES

- A. AASHTO M 326: Polyethylene (PE) Liner Pipe, 300- to 1600-mm Diameter, Based on Controlled Outside Diameter
- B. AASHTO LRFD Bridge Design Specifications
- C. ASTM D 790: Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
- D. ASTM D 1784: Rigid Poly (Vinyl Chloride) (PVC) Compounds and Chlorinated Poly (Vinyl Chloride) (CPVC) Compounds
- E. ASTM D 5813: Cured-In-Place Thermosetting Resin Sewer Piping Systems
- F. ASTM F 1216: Rehabilitation of Existing Pipelines and Conduits by the Inversion and Curing of a Resin-Impregnated Tube
- G. ASTM F 1504: Folded Poly (Vinyl Chloride) (PVC) Pipe for Existing Sewer and Conduit Rehabilitation
- H. ASTM F 1743: Rehabilitation of Existing Pipelines and Conduits by Pulled-in-Place Installation of Cured-in-Place Thermosetting Resin Pipe (CIPP)
- I. ASTM F 1867: Installation of Folded/Formed Poly (Vinyl Chloride) (PVC) Pipe Type A for Existing Sewer and Conduit Rehabilitation

- J. ASTM F 1871: Folded/Formed Poly (Vinyl Chloride) Pipe Type A for Existing Sewer and Conduit Rehabilitation
- K. ASTM F 1947: Installation of Folded Poly (Vinyl Chloride) (PVC) Pipe into Existing Sewers and Conduits
- L. ASTM F 2019: Rehabilitation of Existing Pipelines and Conduits by the Pulled in Place Installation of Glass Reinforced Plastic (GRP) Cured-in-Place Thermosetting Resin Pipe (CIPP) Using the UV-Light Curing Method

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Include the following information into a single submittal for each drainage pipe to be lined, for review. Use the naming convention used in the plans for each location. Provide the seal of a Professional Engineer (PE) or Professional Structural Engineer (SE) for engineering calculations. Divide information into the sections listed below.
 - 1. Work Area
 - a. Locations and dimensions of any temporary access roads
 - b. Locations and dimensions of pipe liner assembly and insertion area “footprints” or area of disturbance.
 - c. Distance of insertion footprint from the traveled way
 - d. Amount of time the footprint will be exposed
 - e. Shoring method when excavation is proposed
 - f. Written access agreements from property owners when work area is outside Department Right of Way
 - 2. Preparation
 - a. Host pipe cleaning
 - b. Management of flows in the host pipe
 - c. Initial pipe inspection
 - 3. Installation
 - a. Method of pipe liner installation and finishing
 - b. Bulkhead and grouting details for segmental thermoplastic and metal liner
 - c. Curing method identifying required curing times, temperatures, and pressures
 - d. Any potential for producing contaminant concentrations in excess of surface water quality standards or ambient groundwater quality standards, as applicable
 - e. Control and disposal of any contaminants from materials or installation methods during and after installation
 - f. Final pipe inspection

4. Product Data
 - a. Engineering calculations and recommended size and wall thickness of pipe liner for products not specified in Section 02610.
 - 1) The host pipe is considered to be fully deteriorated and unable to carry loads.
 - 2) Meet or exceed AASHTO HL-93 or interstate alternate loading according to AASHTO LRFD Bridge Design Specifications and interim specifications for pipe liner load capacity.
 - b. Maximum allowable external grouting pressure on pipe liner and joints for segmental thermoplastic and metal liners.
5. Certificate of compliance.
6. Mix Designs
 - a. Grout Mix design
 - b. Trial batch test data

PART 2 PRODUCTS

2.1 PVC FOLD-AND-FORM LINER

- A. PVC resin manufactured from virgin materials containing no fillers.
 1. Meet or exceed the physical property requirements in ASTM F 1504 or ASTM F 1871 for the cell classification as defined in ASTM D 1784.
- B. Do not use greater than 30 inch diameter.

2.2 CIPP LINER

- A. Fabric tube consisting of one or more layers of absorbent non-woven felt fabric, felt/fiberglass, or fiberglass meeting the requirements of ASTM D 5813 and ASTM F 1216, ASTM F 1743, or ASTM F 2019.
 1. Use a CIPP liner that meets or exceeds the following physical properties outlined in Table 1.

Table 1

Minimum Physical Properties		
Property	Test Method ASTM	Value
Initial Flexural Modulus	ASTM D 790	250,000 PSI @73° F
Flexural Strength	ASTM D 790	4,500 PSI @73° F

2. Use a sufficient quantity of resin for impregnation to fill the volume of air voids in the pipe liner with additional allowance for polymerization and for any migration of resin into the cracks and joints in the host pipe.
 - a. Color the resin with a pigment compatible with the resin system as specified by the resin manufacturer.

2.3 SEGMENTAL THERMOPLASTIC LINER

- A. Solid wall polyethylene pipe. Refer to AASHTO M 326.
- B. Profile wall thermoplastic pipe. Refer to Section 02610.

2.4 SEGMENTAL METAL LINER

- A. Metal Pipe. Refer to Section 02610.

2.5 GROUT

- A. Cementitious low density grout mixture having the following characteristics.
 1. Suitable with the site conditions, construction methods, and pipe liner structural capacity.
 2. Compressive strength between 100 and 400 psi
 - a. Determine strength from trial batches at 28 days
 3. Fluidity to completely fill voids and the annular space between the host pipe and the pipe liner without segregating
 4. Fly ash not to exceed 30 percent of cementitious material
 5. Shrinkage not to exceed 1 percent by volume

PART 3 EXECUTION

3.1 GENERAL

- A. Furnish the type of drainage pipe liner shown with the following exception:
 1. Select from any of the drainage pipe liner products listed in this Section, Part 2 meeting the allowable pipe size reduction when no specific product is shown.

3.2 PREPARATION

- A. General
 - 1. Provide adequate control of water in the host pipe when necessary.
 - a. Repair damage caused as a result of flow control or lack thereof.
 - 2. Provide the necessary controls to prevent pollutants from being transported downstream.
- B. Host Pipe Cleaning
 - 1. Remove material to include sediment, rocks, and miscellaneous debris to inspection and install the pipe liner.
 - 2. Use a cleaning method and tools that will not cause damage to the pipe or surrounding roadway features.
 - a. Notify the COR when cleaning could result in damage.
 - b. Do not cause erosion of surrounding areas from cleaning operations.
 - c. Minimize sediment from being transported downstream.
 - 3. Collect and properly dispose of all material removed.
- C. Host Pipe Inspection
 - 1. Perform remote inspection for pipes with a nominal diameter less than 48 inches using a closed-circuit television (CCTV) crawler.
 - 2. Perform a visual inspection for pipes with diameter greater than 48 inches.
 - 3. Inspect the condition of the host pipe and the suitability of the pipe liner installation. Document and notify the Engineer of the following:
 - a. Collapsed or impassable sections of the pipe
 - b. Horizontal or vertical alignment deviations
 - c. Structural concerns
 - d. Loss of material surrounding the pipe exterior
 - e. Infiltration of water
 - f. Repairs required to perform the installation
- D. Drainage Pipe Liner
 - 1. Verify host pipe internal size and length before ordering pipe liner.
 - 2. PVC fold-and-form and CIPP liner dimensions
 - a. Size the pipe liner to have an outside diameter that will tightly fit the inside of the host pipe.

- b. Provide sufficient pipe liner length to continuously line the host pipe with no intermediate joints.
- 3. Segmental Thermoplastic and Metal liner dimensions
 - a. Size the pipe liner to have an outside dimension that will freely pass through the host pipe and accommodate grout placement.
 - b. Use only one type of pipe liner and joint system in any single host pipe.

3.3 INSTALLATION

- A. General
 - 1. Remove sediment and debris that might have accumulated after the initial cleaning immediately before installing the pipe liner.
 - 2. Handle and install the pipe liner in a manner that will not cause damage to the pipe liner.
 - 3. Provide a smooth finish and a tight seal between the pipe liner and the drainage structure walls at pipe penetrations.
 - 4. Restore lateral and inlet connections.
 - 5. Inspect the rehabilitated pipe upon completion and before acceptance by the Engineer.
 - a. Evaluate pipe liners that contain dry spots, wrinkles, delamination, holes, cracks, joint gaps, soft spots, blisters, or other defects.
- B. PVC fold-and-form and CIPP Liner
 - 1. Meet minimum installation requirements according to ASTM F 1867 or ASTM F 1947 for PVC fold-and form
 - 2. Meet minimum installation requirements according to ASTM F 1216 or ASTM F 1743 for CIPP.
 - 3. Follow the manufacturer's recommended curing and relaxation period required to hold the pipe liner against the host pipe.
 - a. Provide sufficient gauges, monitoring devices, and tests to determine the effectiveness of the curing.
- C. Segmental Thermoplastic and Metal Liner
 - 1. Construct a bulkhead at each end of the host pipe to fully support the fluid thrust loads exerted by the placement of grout.
 - a. Vent bulkheads to allow release of trapped air as the grout flows into the annular space.
 - 2. Provide a minimum of 24 hours for the pipe liner temperature to equalize within the host pipe before placing grout.

3. Fill the entire full length of annular space between the host pipe and the pipe liner with grout.
 - a. Provide sufficient gauges, monitoring devices, and tests to determine the effectiveness of the grout placement.
4. Do not exceed the pipe liner maximum specified grouting pressure.
 - a. Use a pumping unit with a pressure meter and valves capable of limiting grout pumping pressures to the maximum pressure recommended by the manufacturer at the end of the discharge pipe.
5. Place grout in lifts to avoid excessive buoyant forces or grouting pressures.
 - a. Allow each lift to cure sufficiently before proceeding with next lift.
6. Remove any internal braces or bands once the grout is sufficiently cured and before acceptance.

END OF SECTION

SECTION 02620

CONCRETE LINED PIPE INVERT

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Concrete lining for corrugated metal pipe invert.

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete
- B. Section 03211: Reinforcing Steel and Welded Wire
- C. Section 03310: Structural Concrete
- D. Section 03390: Concrete Curing

1.3 REFERENCES NOT USED

1.4 DEFINITIONS NOT USED

1.5 SUBMITTALS

- A. Include the following information into a single submittal for review for each drainage pipe to be lined. Use the naming convention used in the plans for each location. Divide information into the sections listed below.
 - 1. Work Area
 - a. Locations and dimensions of any temporary access roads
 - b. Locations and dimensions of work area "footprint"
 - c. Amount of time the footprint will be exposed
 - 2. Preparation
 - a. Host pipe cleaning
 - b. Management of flows in the host pipe
 - c. Initial pipe inspection
 - 3. Installation
 - a. Method of placing and finishing concrete
 - b. Curing method, identifying required curing times and temperatures

- c. Any potential for producing contaminant concentrations in excess of surface water quality standards or ambient groundwater quality standards as applicable
- d. Control and disposal of any contaminants from materials or installation methods during and after installation
- e. Final pipe inspection

PART 2 PRODUCTS

2.1 CONCRETE

- A. Concrete, Class AA(AE), Type V cement. Refer to Section 03055.
- B. Obtain approval from the COR for an alternate material.

2.2 WELDED WIRE REINFORCEMENT

- A. Galvanized welded wire reinforcement 6 x 6-W1.4 x W1.4. Refer to Section 03211.

PART 3 EXECUTION

3.1 PREPARATION

- A. General
 - 1. Complete backfill and embankment placement over the pipe before concrete lining.
 - 2. Provide the necessary controls to prevent pollutants from being transported downstream.
 - 3. Verify host pipe internal diameter and length before ordering concrete.
- B. Host Pipe Cleaning
 - 1. Remove material to include sediment, rocks, and miscellaneous debris in order that an inspection and concrete lining can be properly performed.

2. Use a cleaning method and tools that will not cause damage to the pipe or surrounding roadway features.
 - a. Notify the COR when cleaning could result in damage.
 - b. Do not cause erosion of surrounding areas from cleaning operations.
 - c. Minimize sediment from being transported downstream.
 3. Collect and properly dispose of all material removed.
- C. Host Pipe Inspection:
1. Perform a visual inspection
 2. Inspect the condition of the host pipe and the suitability of the concrete lined invert installation. Document and notify the COR of the following:
 - a. Collapsed or impassable sections of the pipe
 - b. Horizontal and vertical alignment deviations
 - c. Structural concerns
 - d. Loss of material surrounding the pipe exterior
 - e. Infiltration of water
 - f. Repairs required to perform the installation

3.2 PLACEMENT

- A. Shape and support the welded wire reinforcement to provide 1 inch clearance from the crest of the metal corrugations.
- B. Place concrete at least 4 inches above the crest of the corrugations and to a width one third of the circumference of a round pipe or the maximum span of an arch pipe.
 1. Control the thickness of the concrete by installing non-corrosive pins, nails, or other gauging devices into the crest of the corrugations, perpendicular to the surface.
- C. Finish the concrete to a floated surface finish according to Section 03310.
- D. Cure concrete according to Section 03390.
 1. Provide a minimum of 48 hours before allowing water to flow through the pipe after applying the curing compound.
- E. Provide adequate control of water in the host pipe when necessary.
 1. Repair damage caused as a result of flow control or lack thereof

END OF SECTION

SECTION 02622

PIPE UNDERDRAINS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Pipe underdrains.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill
- B. Section 02075: Geotextiles
- C. Section 02317: Structural Excavation
- D. Section 02633: Concrete Drainage Structures

1.3 REFERENCES

- A. AASHTO M 252: Corrugated Polyethylene Drainage Pipe
- B. AASHTO M 278: Class PS46 Poly (Vinyl Chloride) (PVC) Pipe
- C. AASHTO M 294: Corrugated Polyethylene Pipe, 300-to 1500-mm (12- to 60-in.) Diameter
- D. AASHTO M 304: Poly (Vinyl Chloride) (PVC) Profile Wall Drain Pipe and Fittings Based on Controlled Inside Diameter
- E. AASHTO M 330: Polypropylene Pipe, 300- to 1500-mm (12- to 60-in.) Diameter
- F. AASHTO PP 63: Pipe Joint Selection for Highway Culvert and Storm Drains
- G. AASHTO National Transportation Product Evaluation Program (NTPEP)

1.4 DEFINITIONS

- A. Profile Wall Pipe – A pipe that has a smooth interior wall with an exterior that is either is ribbed, corrugated, or smooth. The wall may contain hollow cores.

1.5 SUBMITTALS

- A. Manufacturer's product data sheets and installation instructions for information.
- B. Documentation from AASHTO NTPEP showing compliance with Table 1 for the pipe material and diameter supplied for information.

PART 2 PRODUCTS

2.1 PIPE

- A. Thermoplastic profile wall pipe having perforations around the entire circumference meeting the material requirements shown in Table 1.

Table 1

Perforated Pipe Material Specifications	
Material Type	AASHTO Specification
High Density Polyethylene (HDPE)	M 252, M 294
Polyvinyl Chloride (PVC)	M 278, M 304
Polypropylene	M 330

- B. Soil tight pipe joints according to AASHTO PP 63.
- C. Pipe fittings matching the pipe material type.

2.2 DRAINAGE GEOTEXTILE

- A. Refer to Section 02075.

2.3 FREE DRAINING GRANULAR BACKFILL

- A. Refer to Section 02056.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Excavate a trench to a width of the outside pipe diameter plus 2 ft and to a depth of 3 inch below the bottom of the pipe.
 - 1. Refer to Section 02317.
 - 2. Keep trenches free of water.
- B. Grade and prepare the bottom of the trench to provide a firm and uniform bearing surface.
- C. Place drainage geotextile within the trench as shown.
- D. Place and compact the bottom 3 inch of the trench with free draining granular backfill.
- E. Connect pipes to concrete drainage structures according to Section 02633.
 - 1. Place an end cap at the upstream end of the pipe when not connected to a structure.
- F. Place and compact the remainder of the free draining granular backfill to depth shown.

END OF SECTION

SECTION 02633

CONCRETE DRAINAGE STRUCTURES

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Materials and procedures for constructing concrete drainage structures from the CB and DB Series Standard Drawings.

1.2 RELATED SECTIONS

- A. Section 01721: Survey
- B. Section 02056: Embankment, Borrow, and Backfill
- C. Section 02635: Grates, Solid Covers, Frames, and Manhole Steps
- D. Section 03055: Portland Cement Concrete
- E. Section 03056: Self-Consolidating Concrete (SCC)
- F. Section 03152: Concrete Joint Control
- G. Section 03211: Reinforcing Steel and Welded Wire
- H. Section 03310: Structural Concrete
- I. Section 03390: Concrete Curing

1.3 REFERENCES

- A. AASHTO M 198: Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants
- B. AASHTO M 199: Precast Reinforced Concrete Manhole Sections
- C. AASHTO M 213: Preformed Expansion Joint Fillers for Concrete Paving and Structural Construction (Non-extruding and Resilient Bituminous Types)
- D. AASHTO M 235: Epoxy Resin Adhesives

- E. AASHTO M 315: Joints for Concrete Pipes and Manholes Using Rubber Gaskets
- F. AASHTO Standard Specification for Highway Bridges – current edition
- G. ASTM C 361: Reinforced Concrete Low-Head Pressure Pipe
- H. ASTM C 443: Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
- I. ASTM C 478: Precast Reinforced Concrete Manhole Sections
- J. ASTM C 857: Minimum Structural Design Loading for Underground Precast Concrete Utility Structures
- K. ASTM C 858: Underground Precast Concrete Utility Structures
- L. ASTM C 891: Installation of Underground Precast Concrete Utility Structures
- M. ASTM C 1107: Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
- N. ASTM C 1244: Concrete Sewer Manholes by the Negative Air Pressure (Vacuum) Test Prior to Backfill
- O. UDOT Quality Management Plans

1.4 DEFINITIONS

- A. Catch Basin/Drop Inlet – A structure accepting drainage from gutters or medians or other channels and discharging the water through a conduit. Refer to the CB and DB Series Standard Drawings for shape and dimensions of Standard Catch Basins.
- B. Inlet – A grated surface connection to a closed conduit such as a storm drain. A structure at the upstream end of a cross culvert. The upstream end of any structure through which water may flow.
- C. Manhole (access hole) – A circular structure for access and joining pipes. Refer to the CB Series Standard Drawings for the Standard Detail for a Manhole.

1.5 SUBMITTALS

- A. Concrete mix design for approval according to Section 03055 or 03056.
- B. Precast Structures
 - 1. Provide verification the structures are furnished by a UDOT pre-qualified precast supplier.
 - 2.
- C. Repair Procedure
 - 1. Submit to the COR for review before performing any repairs. Refer to this Section, Article 1.6, paragraph B.

1.6 ACCEPTANCE

- A. Select and construct or install cast-in-place precast drainage structures according to this Section.
 - 1. Cast in place construction.
 - 2. Precast construction.
 - a. Field verify the fit and function at structure locations before manufacturing.
 - b. The Government will not reimburse the Contractor for precast structures that do not fit existing field conditions.
- B. Repair or replace any structure that has the following:
 - 1. Fractures or cracks passing through the wall except for a single end crack that does not exceed the thickness of the precast unit.
 - 2. Defects showing improper proportioning, mixing, or molding.
 - 3. Honeycombing and open texture.
 - 4. Damaged or cracked ends that prevent joining manhole or inlets grade rings and sections.
 - 5. Any continuous crack with a surface width of 0.01 inch or more that extends more than 12 inches anywhere on the wall.
 - 6. Limit cracks or fractures for grade rings or similar structures. Refer to ASTM C 478.
- C. Precast Structures
 - 1. Furnish precast drainage structures according to the CB Series Standard Drawings.
 - a. Pre-qualify the supplier according to the UDOT Quality Management Plan – Precast/Prestressed Concrete Structures.
 - b. Furnish precast structures that are plumb and square within $\frac{1}{8}$ inch per foot so that precast adjoining elements fit.
 - c. Mark structures with date of casting and supplier

identification.

- D. Upon completing each installation and before placing backfill.
 - 1. Verify the structures and pipe connections appear watertight.
 - 2. Test according to this Section, Article 3.3 when directed by the COR.

PART 2 PRODUCTS

2.1 CONCRETE

- A. Wet cast – Class AA(AE). Refer to Section 03055.
- B. Dry cast – Submit mix design for approval.
 - 1. Minimum cement content 470 lb/yd³
 - 2. Maximum water/cementitious ratio 0.40
- C. Self-Consolidating Concrete – Refer to Section 03055.

2.2 REINFORCING STEEL AND WELDED WIRE

- A. Refer to Section 03211.

2.3 STRUCTURAL CONCRETE

- A. Refer to Section 03310.

2.4 JOINTS AND SEALERS

- A. Preformed Joint Filler. Refer to AASHTO M 213 and AASHTO M 198.

2.5 WATERSTOPS

- A. Refer to Section 03152 for materials requirements.
- B. Refer to AASHTO Standard Specification for Highway Bridges, Division II, sub-section 8.9.3.4 for installation requirements.

2.6 NON-SHRINK GROUT

- A. Use non-shrink grout according to ASTM C 1107.

2.7 FORMS

- A. Use plywood, wood, metal, glass, or a combination of these materials.

2.8 GASKETS AND JOINT SEALANTS FOR CONNECTING PRECAST SECTIONS

- A. Furnish gaskets for sealing precast sections that meet ASTM C 443 requirements.
- B. Furnish gaskets for sealing precast concrete manholes that meet AASHTO M 315.
- C. Furnish epoxy resin adhesive. Refer to AASHTO M 235.
- D. Furnish "O" Ring according to ASTM C 361 as shown in the CB Series Standard Drawings.

2.9 MANHOLE/FRAME GASKET

- A. Place $\frac{3}{4}$ inch diameter minimum extruded rope Type B flexible plastic gaskets between the manhole frame and the concrete risers according to AASHTO M 198 requirements.

2.10 JOINTING MASTIC

- A. Furnish water-resistant elastic jointing mastic of plastic bituminous materials and inert fillers that do not lose slump or plasticity when applied to a vertical metal surface and heated to 120 degrees F.
- B. Furnish joint mastic that can be applied evenly and adhere at temperature range of 40 to 120 degrees F or higher.

2.11 GRATES, SOLID COVERS, FRAMES, AND MANHOLE STEPS

- A. Refer to Section 02635.

PART 3 EXECUTION

3.1 PREPARATION

- A. Verify fit and function with field conditions before manufacturing or constructing any structure. Refer to Section 01721.

- B. Furnish structures free of voids, cracks, and with beveled corners and edges. Securely attach all inserts in the proper location.
 - 1. Prevent cold joints in the structure.
- C. Clean and prepare the mating surfaces before assembly of pipes with structure.
 - 1. Use one of the following methods to connect the pipes to the structure for precast:
 - a. Pipe boot according to pipe manufacturer specifications for pipe type.
 - b. Non-shrink grout to seal the pipe connection.
- D. Excavate the material under the box location to a minimum depth of 4 inches and backfill with suitable backfill material and compact.
 - 1. Excavate sufficiently to place and compact bedding and backfill material according to Section 02056.
 - 2. Add as needed, a sand-leveling course no greater than 2 inches in depth to the backfill material.
 - a. Excavate the area to the appropriate depth to accommodate the backfill and leveling course, when used.

3.2 INSTALLATION

- A. Manholes
 - 1. Furnish precast concrete manholes according to ASTM C 478 and use self-centering watertight joints that meet ASTM C 443. Refer to CB Series Standard Drawings.
 - a. Cure wet cast members according to Section 03390.
- B. Grade Rings/Catch Basin Grade Sections
 - 1. Furnish according to ASTM C 478 with anchor bolt-holes as shown on the CB Series Standard Drawings.
 - a. Cure wet cast members according to Section 03390.
- C. Precast Inlets and Boxes
 - 1. Furnish structures according to CB Series Standard Drawings.
 - a. Attach and secure all inserts at the place of manufacture such as wall sleeves, gaskets or piping, sumps, steps, access hatches, and any other inserts as shown on CB Series Standard Drawings.
 - 2. Manufacture structures according to applicable requirements of ASTM C 858 and as modified by this Section.
 - a. Meet AASHTO M 199 and ASTM C 857 requirements.
 - b. Cure wet cast members according to Section 03390.
 - 3. Provide sufficient lifting points for a safe installation.

- a. Locate lifting devices to avoid interference with the reinforcing steel.
- 4. Do not ship precast units until after 28-day compressive strength has been attained.
 - a. Protect the unit from any damage.
- 5. Refer to ASTM C 891. Comply with manufacturer installation guidelines.
 - a. Inspect precast drainage structures for defects before lowering into excavation.
 - b. Clean mating surfaces of all foreign materials such as dirt, mud, and stones and apply proper joint sealing material where applicable.
 - c. Assemble all joints tightly.
 - d. Use care when joining precast elements in cold weather. Do not force joints together with mechanical equipment.
 - e. Sufficiently warm all sealing materials to flow without causing damage to precast joint elements.
- 6. Furnish structures with appropriate openings for connecting pipe.
 - a. Cast or cut structure openings. Do not expose reinforcing steel or reduce reinforcing steel covering at openings.
 - b. Do not modify precast units in the field by cutting or enlarging holes or by making any other changes without the manufacturer's and Engineer's-COR's approval.
 - c. Modify precast units only according to manufacturer requirements.
- 7. Do not place precast drainage structure in excavation that has water and frozen surfaces.
- 8. Plug lift insert recesses with a 1:1 sand to cement grout mix.
 - a. Finish flush with top, bottom, or both surfaces of concrete.
- D. Cast-in-place concrete drainage structures.
 - 1. Refer to CB and DB Series Standard Drawings.
 - 2. Construct according to Section 03310.
 - 3. Cure according to Section 03390.

3.3 TESTING

- A. Conduct either a vacuum test or ex-filtration test to verify the drainage structures are watertight at the direction of the Engineer upon failure of the visual inspection referenced in this Section, Article 1.6.
 - 1. Furnish all necessary equipment and materials for repair and re-test any structures that fail any tests.
 - 2. Do not conduct vacuum and ex-filtration tests concurrently.

- B. Vacuum Test – Follow the test procedure outlined below:
1. Vacuum test precast structures after assembly and before backfilling.
 - a. Form a seal between the vacuum base and the manhole rim/precast structure cover. Secure pipe plugs to prevent movement while the vacuum is drawn.
 - b. Draw a vacuum of 10 inches of mercury (Hg). Record the time for the vacuum to drop to 9 inches.
 - c. Passing drop rates for the time to drop to 9 inches are:

<u>Diameter/Width</u>	<u>Time to Drop 1 inch Hg</u>
up to 4 ft	30 seconds
4 ft and larger	40 seconds
 - d. Make necessary repairs if the structure fails the test. Repairs and repair procedures must be acceptable to the Engineer.
 - e. Disassemble the manhole and replace the gaskets if preformed plastic gaskets are pulled out during the vacuum test.
- C. Ex-filtration Test – Follow test procedure ASTM C 1244 as modified below:
1. Plug all pipes leading into or out of the precast structure for a watertight seal.
 2. Fill precast structure with water to a level 3-4 inches below the casting rim or lid.
 3. Let the water stand for two hours before beginning the test to allow absorption into the precast structure.
 4. Place additional water to bring the water level back to 3-4 inches below the rim or lid after the two hour stabilization.
 5. Test for at least two hours and verify the leakage is less than shown on table 1.

Table 1

Precast Structure Ex-Filtration Test – Allowable Leakage						
Water Depth (measured from invert to water level)	Allowable water drop per hour					
Feet	Maximum Horizontal Internal Dimension					
	4 ft.*		5 ft *		6 ft *	
	(gals)	(inches)	(gals)	(inches)	(gals)	(inches)
2	0.8	0.32	1.0	0.40	1.2	0.48
4	1.6	0.64	2.0	0.8	2.4	0.96
6	2.4	0.96	3.0	1.21	3.6	1.45
8	3.2	1.29	4.0	1.61	4.8	1.93
10	4.0	1.61	5.0	2.01	6.0	2.42
12	4.8	1.93	6.0	2.42	7.2	2.90
14	5.6	2.25	7.0	2.82	8.4	3.38
16	6.4	2.58	8.0	3.22	9.6	3.87
18	7.2	2.90	9.0	3.63	10.8	4.35
20**	8.0	3.22	10.0	4.03	12	4.84
* Adjust volume loss proportionally for different size not shown ** Provide an engineering analysis for equivalent ex-filtration rates for depths greater than 20 feet.						

END OF SECTION

SECTION 02635

GRATES, COVERS, FRAMES, TRASH RACKS, AND MANHOLE STEPS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Grates, covers, frames, trash racks and manhole steps.

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. AASHTO M 31: Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
- B. AASHTO M 111: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
- C. AASHTO M 199: Precast Reinforced Concrete Manhole Sections
- D. AASHTO M 270: Carbon and High-Strength Low-Alloy Structural Steel Shapes, Plates, and Bars and Quenched-and-Tempered Alloy Structural Steel Plates for Bridges
- E. AASHTO M 306: Drainage, Sewer, Utility, and Related Castings
- F. ASTM A 536: Standard Specification for Ductile Iron Castings
- G. ASTM D 4101: Polypropylene Injection and Extrusion
- H. AWS D1.1: Structural Welding Code - Steel

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Shop drawings for cast iron alternative to steel grates and frames, for review.
- B. Manhole step manufacturer's product data sheets and installation instructions for information.

Grates, Covers, Frames, Trash Racks, and Manhole Steps
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PART 2 PRODUCTS

2.1 GRATES, COVERS, FRAMES, AND TRASH RACKS

- A. Refer to the DB and GF Series Standard Drawings.
 - 1. Structural Steel
 - a. AASHTO M 270, Grade 36.
 - b. Fabricate according to AWS D1.1
 - c. Hot-dip galvanize after fabrication according to AASHTO M 111.
 - 2. Gray Iron.
 - a. AASHTO M 306, Class 35B.
 - 3. Ductile Iron.
 - a. ASTM A 536 Grade 80-55-06.

2.2 MANHOLE STEPS

- A. Refer to AASHTO M 199 and GF Standard Series Drawings.
 - 1. Steel Rod
 - a. AASHTO M31, Grade 60.
 - 2. Copolymer Polypropylene Plastic
 - a. ASTM D 4101.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Provide a firm and even bearing surface on the frame so that the grate or cover will not rock under the influence of loading.
- B. Install according to manufacturer's instructions.

END OF SECTION

SECTION 02645

PRECAST CONCRETE BOX AND THREE-SIDED CULVERT STRUCTURES

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Single-cell precast concrete box culverts, multi-cell precast concrete box culverts, and precast concrete three-sided structures as shown.
- B. Precast or cast-in-place secondary elements such as cutoff walls, aprons, footings, floor slabs, headwalls, and wingwalls as shown.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill
- B. Section 02317: Structural Excavation
- C. Section 03055: Portland Cement Concrete
- D. Section 03211: Reinforcing Steel and Welded Wire
- E. Section 03310: Structural Concrete
- F. Section 03390: Concrete Curing
- G. Section 03575: Flowable Fill

1.3 REFERENCES

- A. AASHTO M 111: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
- B. AASHTO T 106: Compressive Strength of Hydraulic Cement Mortars (Using 2-in. Cube Specimens)
- C. AASHTO T 160: Length Change of Hardened Hydraulic Cement Mortar and Concrete

- D. AASHTO T 161: Resistance of Concrete to Rapid Freezing and Thawing
- E. AASHTO LRFD Bridge Design Specifications
- F. AASHTO Manual for Bridge Evaluation
- G. ASTM C 877: External Sealing Bands for Concrete Pipe, Manholes, and Precast Box Sections
- H. ASTM C 882: Bond Strength of Epoxy-Resin Systems Used With Concrete By Slant Shear
- I. ASTM C 928: Packaged, Dry, Rapid-Hardening Cementitious Materials for Concrete Repairs
- J. ASTM C 990: Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants
- K. ASTM C 1504: Manufacture of Precast Reinforced Concrete Three-Sided Structures for Culverts and Storm Drains
- L. ASTM C 1577: Precast Reinforced Concrete Monolithic Box Sections for Culverts, Storm Drains, and Sewers Designed According to AASHTO LRFD
- M. Precast/Prestressed Concrete Institute (PCI) Design Handbook
- N. UDOT Bridge Management Manual
- O. UDOT Quality Management Plan
- P. UDOT Structures Design and Detailing Manual (SDDM)

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Working Drawings
 - 1. Fabrication and installation drawings for review of all fabricated materials, and cast-in-place elements that are included in this item of work and are not detailed in the contract plans.
 - a. Include details not provided in the contract documents for the fabrication, transportation, erection, and construction of the members included in this item of work.

- b. Detail all phases of construction including layout, joint details, lifting devices, casting methods, construction placement, and details of any cast-in-place elements included in this item of work.
- c. Note proposed transportation methods.
- d. Include inventory and operating load ratings in a table on the first sheet of the drawings for structures with spans greater than 20 ft.
 - 1) Measure span length along roadway centerline.
- e. Load Rating Report and supplemental documentation according to the AASHTO Manual for Bridge Evaluation and the UDOT Bridge Management Manual when load rating is required.
- f. Provide the seal of a Professional Engineer (PE) or Professional Structural Engineer (SE) licensed in the State of Utah on the drawings and supporting engineering calculations for the following:
 - 1) Special designs of precast concrete box culverts designed by the fabricator.
 - 2) Precast concrete three-sided structures.
 - 3) Lifting devices.
 - 4) Structural members and ancillary appurtenances designed by the fabricator.
 - 5) Connections between precast and cast-in-place members and appurtenances.
 - 6) Load ratings.
 - 7) Any other means and methods that require an engineer's design.

- B. Dry Cast Concrete
 - 1. Certificate of Compliance.

- C. Structural Non-Shrink Grout
 - 1. Certificate of Compliance.

1.6 ACCEPTANCE

- A. Precast concrete box culverts and precast concrete three-sided structures may be accepted at a reduced price when the concrete strength is below that specified.
 - 1. Price adjustment pay factor according to Section 03055.

PART 2 PRODUCTS

2.1 CONCRETE

- A. Wet Cast Concrete
 - 1. Concrete Class AAA(AE) according to Section 03055.
 - a. 28-day minimum compressive strength of 5,000 psi.
- B. Dry Cast Concrete
 - 1. Minimum cement content 470 lb/yd³ of concrete.
 - 2. Meet specified 28 day minimum compressive strength.

2.2 REINFORCING STEEL AND WELDED WIRE

- A. Coated – Refer to Section 03211.

2.3 JOINT SEALANT

- A. Refer to ASTM C 990.
- B. Use a flexible butyl rubber material with a minimum cross-section of $\frac{3}{4}$ x $1\frac{1}{2}$ inches as a joint sealant for box culverts.
- C. Joint sealant furnished by culvert manufacturer.

2.4 JOINT WRAP

- A. Refer to ASTM C 877.

2.5 STRUCTURAL NON-SHRINK GROUT

- A. Use gray, non-shrink grout containing no calcium chloride or admixture containing calcium chloride or other ingredient in sufficient quantity to cause corrosion to steel reinforcement.
- B. Use quick-setting, rapid strength gain, non-shrink, and high-bond strength grout.
- C. Meet all the requirements of AASHTO T 160 with the exception that the Contractor-supplied cube molds will remain intact with a top firmly attached throughout the curing period.

- D. Mix structural non-shrink grout just before use according to the manufacturer's instructions.
- E. Refer to Table 1 for additional structural non-shrink grout requirements.

Table 1

Structural Non-Shrink Grout			
*Properties	Requirements	ASTM	AASHTO
Accelerated Weathering: Tested Medium Accepted Weight Loss	<3% White Utah Road Salt <15% @ 300 Cycles		T 161
Compressive Strength	>3,000 psi @ 24 hours >5,000 psi @ 7 days		T 106
Accepted Bond Strength	>1,000 psi @ 24 Hours	C 882 as modified by C 928 8.5	
Length Change	No expansion after 7 days		T 160

* Certified test results from an independent AASHTO accredited testing laboratory are acceptable.

2.6 LIFTING DEVICES

- A. Use lifting devices that can support the required vertical and horizontal forces with the applicable safety factors as specified in the Component Handling and Erection Bracing requirements of the PCI Design Handbook.
- B. Galvanize according to AASHTO M 111.

2.7 GROUTED SPLICE COUPLERS

- A. Refer to Section 03211.

2.8 GRANULAR BACKFILL BORROW

- A. Refer to Section 02056.

2.9 FREE DRAINING GRANULAR BACKFILL

- A. Refer to Section 02056.

2.10 FLOWABLE FILL

- A. Refer to Section 03575.

2.11 MANUFACTURE

- A. Use a UDOT Certified Concrete Precaster for precast concrete box culverts and three-sided structures according to the UDOT Quality Management Plan: Precast-Prestressed Concrete Structures.
- B. Precast Concrete Box Culverts
1. Refer to ASTM C 1577, except as modified by this Section.
 2. Provide the design of special designs for sizes and loads other than those shown in Table 1 of ASTM C 1577 if the design is not provided.
 3. Prepare special designs according to AASHTO LRFD Bridge Design Specifications, Section 12 and the SDDM.
 4. Provide 1 inch minimum concrete cover to reinforcing steel bars.
 5. Provide minimum concrete cover to reinforcement where welded wire fabric is used, equal to the greater of three times the diameter of the wire or 1 inch.
 6. Provide 2 inch minimum concrete cover to the reinforcing steel in the top of the top slab of box sections covered with less than 2 ft of fill for all types of reinforcement.
 7. Provide tongue and groove ends or equivalent to transfer shear between sections at all joints.
 8. Design and form section ends so that the sections can be laid together to make a continuous line of box sections compatible with the permissible variations in Section 11 of ASTM C 1577.
- C. Precast Concrete Three-Sided Structures
1. Refer to ASTM C 1504 with the following exceptions:
 - a. Design the structure according to AASHTO LRFD Bridge Design Specifications, Section 12.
 - b. Design for HL-93 live loading.
 2. Provide a maximum longitudinal wire spacing of 8 inches for welded wire. Provide a circumferential wire spacing not greater than 4 inches or less than 2 inches.
 3. Provide minimum concrete cover to welded wire fabric equal to the greater of three times the diameter of the wire or 1 inch.
 4. Provide 2 inch minimum concrete cover to all reinforcing steel in the top of the top slab for three-sided structures covered by less than 2 ft of fill.
- D. Project the reinforcing steel at least 12 inches out of the precast sections and square off the concrete face where precast sections join cast-in-place concrete.

Precast Concrete Box and Three-Sided Culvert Structures

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- E. Provide the number and type of lifting devices required to support the vertical and horizontal forces with the applicable safety factors as specified in the Component Handling and Erection Bracing requirements of the PCI Design Handbook.
 - 1. Use a maximum diameter of 3 inches when lifting holes are used.
 - 2. Locate holes to avoid interference with the reinforcing steel.
- F. Do not locate horizontal and vertical changes in alignment greater than 10 degrees at a joint between precast segments of a precast concrete box culvert unless otherwise approved by the ~~Engineer~~COR.
- G. Permanently mark each precast unit with date of casting and supplier identification.
- H. Concrete Curing – Refer to Section 03390.
- I. Do not use precast segments less than 5 ft in lay length.

2.12 QUALITY ASSURANCE

- A. Precast Elements
 - 1. Prevent cracking or damage of precast elements during shipping, handling, and storage.
 - 2. Replace defects and breakage of precast elements.
 - a. Members that sustain damage or surface defects during fabrication, handling, storage, hauling, or erection are subject to review or rejection.
 - b. Obtain approval before performing repairs.
 - c. Repair work must reestablish the elements' structural integrity, durability, and aesthetics to the satisfaction of the ~~Engineer~~Government.
 - d. Determine the cause when damage occurs and take corrective action.
 - e. Failure to take corrective action, leading to similar repetitive damage, can be cause for rejection of the damaged element.
 - f. Cracks that extend to the nearest reinforcement plane and fine surface cracks that do not extend to the nearest reinforcement plane but are numerous or extensive are subject to review and rejection.
 - g. Full depth cracking and breakage greater than 1 ft are cause for rejection.
 - 3. Document all test results. The quality control file will contain at least the following information:
 - a. Element identification

- b. Date and time of cast
- c. Concrete cylinder test results
- d. Quantity of used concrete and the batch printout
- e. Form-stripping date and repairs if applicable
- f. Location/number of blockouts and lifting inserts
- g. Temperature and moisture of curing period
- h. Lifting device details, requirements, and inserts

PART 3 EXECUTION

3.1 EXCAVATING, TRENCHING, BEDDING, AND BACKFILL

- A. Refer to Section 02317.
- B. Excavate to at least 6 inches below the bottom of the precast structure or as shown.
- C. Scarify, and moisture condition, and compact the top 8 inches of the excavated ground surface to at least 90 percent of the maximum dry density.
- D. Place a bedding of at least 6 inches of granular backfill borrow or as shown.
 - 1. Level and compact bedding material to provide uniform support of the structure along its entire supported width and length.
 - 2. Limit the soil gradation for granular backfill borrow used as bedding under precast members to 100 percent passing the $\frac{3}{4}$ inch sieve.
 - 3. Modify the soil gradation for free draining granular backfill used as bedding under precast members to 100 percent passing the $\frac{3}{4}$ inch sieve.
- E. Backfill structure as shown.
- F. Refer to the authorized fabrication and installation drawings for excavation, bedding, and backfill requirements where a precast three-sided structure is placed on a footing.
- G. Fill the gap between multiple single cell culverts with flowable fill as shown.

3.2 INSTALLATION

- A. Inspect precast elements for defects before lowering into trench.
- B. Repair or replace defective, damaged, or unsound precast elements.
- C. Use a trench width adequate to place and compact bedding material.
 - 1. Minimum outside width of trench is the outside width of structure plus 2 ft each side.
- D. Lay precast elements starting at the downstream end.
- E. Carefully lower precast elements into the trench with suitable equipment to prevent damage.
- F. Remove all dirt and foreign material from joints.
- G. Apply joint sealant to precast box culvert.
 - 1. Place the joint sealant material on the bottom half of the groove (bell) of the box last placed.
 - 2. Place the balance of the joint sealant material on the top half of the tongue (spigot) of the box to be set.
 - 3. Place the joint sealant material about 1 inch from the leading edge of the groove and tongue.
 - 4. Maintain the joint sealant material at 70 degrees F or greater during placement.
- H. Pull precast box culvert sections together to ½ inch joint gap spacing measured face to face of adjoining concrete surfaces with the joint gap being uniform on all sides of the precast box culvert.
 - 1. Use appropriate pulling devices to avoid misalignment and damage to box sections.
 - 2. The ~~Engineer~~COR may approve up to a ¾ inch joint gap spacing in cases where it is necessary to adjust the total length of a box culvert run.
 - 3. Reject box sections when the installation tolerance cannot be met due to casting variations.
 - 4. Prevent soil from being forced into the joint as the box sections are placed.
 - 5. Disassemble joint, check position of joint sealant, repair alignment, and re-install when adjoining elements cannot be pulled together to meet joint requirements.
- I. Do not disturb previously completed joints during laying operation.

- J. Do not lay precast elements when water is in the trench.
- K. Place precast three-sided structure sections against previous sections as tightly as possible while maintaining alignment.
 - 1. Follow manufacturer's installation recommendations.
 - 2. Do not exceed joint tolerances in the authorized fabrication and installation drawings.
- L. Apply joint wrap to the top slab and side walls at all joints.
 - 1. Clean top and sides of concrete surface at joints before placing joint wrap.
 - 2. Use a minimum width of 2 ft centered on the joint.
- M. Install precast cutoff walls, aprons, wingwalls, and headwalls, as follows:
 - 1. Establish working points, working lines, and benchmark elevations before placing elements.
 - 2. Clean bonding surfaces between elements of debris, dust, and rust before connecting elements to achieve the required bonding between protruding bars and elements.
 - 3. Place cut-off wall and apron elements as shown in the plans.
 - a. Adjust the final location of the cut-off wall or apron elements if actual joint gaps cause the final location to vary.
 - b. Adjust the height of each apron element by means of leveling devices or shims.
 - 4. Lift apron segments using lifting devices as shown on the authorized fabrication and installation working drawings.
 - 5. Set elements in the proper horizontal location. Check for proper alignment and grade within specified tolerances.
 - 6. Adjust vertical leveling devices before full release of the apron from the crane to reduce the amount of torque required to turn the bolts in the leveling devices. Check for proper grade within specified tolerances.
 - 7. Place or pump flowable fill into voids and pockets as shown.

3.3 JOINTS

- A. Mechanically connect the exterior sections of precast three-sided structures at all top joints within a minimum length of 12 ft from each end of the structure.
 - 1. Use at least four mechanical connections per joint with a maximum spacing of 10 ft.
 - 2. Galvanize all plates, shapes, and hardware.

- B. Connect precast three-sided structures to the footing, pedestal, or slab 2 ft from the outermost exterior edge of the structure at all four corners of the structure with a galvanized rigid mechanical connection.
 - 1. Locate the connection on the interior face of the segment to allow for future inspection.

3.4 LIFTING HOLES AND DEVICES

- A. Plug lifting holes and lifting device recesses with structural non-shrink grout.
 - 1. Finish flush with all concrete surfaces.

3.5 CAST-IN-PLACE CONCRETE

- A. Refer to Section 03310.

3.6 WATERPROOFING MEMBRANE

- A. Apply a waterproofing membrane to the top slab and side walls of concrete box culverts and three-sided structures for the full length of the structures.

END OF SECTION

SECTION 02701

PAVEMENT SMOOTHNESS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Testing and determination of a minimum International Roughness Index (IRI) for pavement smoothness for the following pavements:
 - 1. Hot Mix Asphalt (HMA)
 - 2. Open Graded Surface Course (OGSC)
 - 3. Bonded Wearing Course (BWC)
 - 4. Stone Matrix Asphalt (SMA)
 - 5. Micro-surfacing
 - 6. Portland Cement Concrete Pavement (PCCP)
 - 7. PCCP rehabilitation.

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1.2 RELATED SECTIONS

- A. Section 02742S: Project Specific Surfacing Requirements
- B. Section 02745: Asphalt Material
- C. Section 02981: Grinding Pavement

1.3 REFERENCES

- A. AASHTO M 328: Inertial Profiler
- B. AASHTO R 43: Quantifying Roughness of Pavements
- C. AASHTO R 54: Accepting Pavement Ride Quality When Measured Using Inertial Profiling Systems
- D. AASHTO R 57: Operating Inertial Profiling Systems
- E. ASTM E 2560: Data Format for Pavement Profile
- F. UDOT Materials Manual of Instruction

1.4 DEFINITIONS

- A. Pavement Section
 - 1. Each travel lane or median, 528 ft long. Sections include:
 - a. All traffic lanes
 - b. Medians and two way left turn lanes 8 ft and wider
 - c. Turn lanes
 - 2. Each pavement section is laid out consecutively from the start of the project.
- B. Structure Section
 - 1. Each travel lane or median, 528 ft long. Sections include:
 - a. Bridges, approach slabs, and pavement within 25 ft of the approach slab with final riding surfaces placed as part of the contract.
- C. Wheel Path
 - 1. A continuous parallel line 2.5 ft inside the lane or median lines.
- D. Mean Roughness Index (MRI)
 - 1. Average of two wheel path IRIs taken from each pavement section.
- E. Localized Roughness
 - 1. Profile deviations in a continuous 25 ft pavement as identified using the ProVAL "Smoothness Assurance" analysis, calculating IRI with a continuous short interval of 25 ft [7.62 m] and the 250- mm filter applied.

1.5 SUBMITTALS

- A. Certifications for profilers and operators, for information.
- B. Summary report of acceptance profile testing. Include at least the following:
 - 1. Pavement Section identification
 - 2. Structure Section identification
 - 3. MRI results by section
 - 4. Localized Roughness results.
 - 5. Include any electronic project files.
 - a. Flag utility locations in the acceptance profile.
 - b. Identify milepost or project stationing for the beginning of the profile collection.
 - c. Flag beginning and end of grinding locations in acceptance profile.

- C. Original raw data files compatible with the ASTM E 2560 standard. Refer to AASHTO R 43.
 - 1. Data files collected prior to beginning work for Category 2 Pavement Sections.
 - 2. Data files collected before corrective work.
 - 3. Data files collected prior to placing final riding surface.
 - 4. Data files collected for acceptance.

1.6 ACCEPTANCE

- A. Limit transverse pavement deviations to less than $\frac{3}{16}$ inch from the lower edge of a 10-ft straightedge.
- B. Limit Localized Roughness greater than 15 ft in length as specified in Section 02742S.
 - 1. Measure localized roughness for the following locations:
 - a. All traffic lanes
 - ~~b.~~ ~~Ramps~~
 - ~~e.b.~~ Medians and two way left turn lanes 8 ft and wider
 - ~~d.c.~~ Turn lanes
 - ~~e.d.~~ Shoulders wider than 4 ft
 - ~~f.e.~~ Bike lanes
 - ~~g.f.~~ Structures and approach slabs with the final riding surfaces placed as part of the contract
 - 2. Localized Roughness requirements apply to all pavements and structures regardless of length.
 - 3. Localized Roughness requirements do not apply to micro-surfacing.
 - 4. Include profile deviations from bridge decks, approach slabs and transitions, manholes, valves, and other facilities in the profile when the contract requires the adjustment, new construction or reconstruction of these facilities.
 - 5. Exclude profile deviations from bridge decks, approach slabs and transitions, manholes, valves, and other facilities in the profile when the contract does not include adjustment, new construction or reconstruction of these facilities.
 - a. Category 2 Pavement Sections: the COR may waive areas of Localized Roughness due to existing conditions.
 - 1) Provide an explanation for each location where existing conditions prevent the requirements from being met.
 - 2) Do not remove these areas from the profile for incentive/disincentive calculations.

- C. The Government applies Incentive/Disincentive for smoothness based on the profile of the final surface. MRI is evaluated for each Pavement Section and Structure Section with 250-mm filter applied using the most recent version of Profile Viewer and Analysis (ProVAL) software. Refer to AASHTO R 54 and ASTM E 2560.
1. Refer to Section 02742S for definitions of Category 1 Pavement Sections and Category 2 Pavement Sections.
 2. Include all Structure Sections regardless of length and pavements greater than 528 ft.
 3. The Government will apply incentives/disincentives before placement of a chip seal or micro-surfacing.
 4. Pavement sections requiring corrective action for localized roughness at the time of acceptance testing are not eligible for incentives.
 5. The Government applies Incentive/Disincentive for Category 1 Pavement Sections and Structure Sections based on the tables in Section 02742S
 6. The Government calculates the percent improvement for Category 2 Pavement Sections using the following formula:

$$\frac{[MRI_o - MRI_f] \times 100}{MRI_o}$$

Where:

MRI_o = MRI of original roadway surface

MRI_f = MRI of final corrected roadway surface

- a. The Government applies Incentive/Disincentive based on the tables in Section 02742S and the following:
 - 1) Sections with a $MRI_f \leq 90$ do not require corrective action.
 - 2) Disincentive does not apply to sections with an $MRI_f \leq 70$.
7. The Government calculates the percent improvement for micro-surfacing Pavement Sections according to the formula in Article 1.6.C.6 and applies Incentive/Disincentive based on the table in Section 02742S.
8. The Government will prorate incentives/disincentives for partial pavement sections.
9. The Government will apply an additional \$1000 disincentive per pavement section to:
 - a. Pavement Sections with HMA and SMA final surfaces where grinding exceeds 30 yd².
 - b. Pavement Sections with grinding on the final surface of OGSC, and BWC.

- c. The additional disincentive does not apply to the first or last 25 ft of the project.
- 10. The Government does not apply Incentives/Disincentives to:
 - a. Pavements shorter than 528 ft
 - b. Shoulders
 - c. Bike Lanes
 - d. Medians narrower than 8 ft
 - e. Horizontal curves with a centerline curvature radius less than 900 ft and areas within the superelevation transitions to these short radius curves
 - f. Horizontal tapers
 - g. Surfaces within 15 ft of bridge decks and approach slabs not constructed as part of the contract
- 11. The COR may accept a pavement section requiring corrective action for MRI at a \$1000 disincentive per pavement section.

1.7 OPERATOR QUALIFICATIONS

- A. Use a certified operator according to UDOT Materials Manual of Instruction 995.

PART 2 PRODUCTS

2.1 FLUSH COAT

- A. CQS-1h or CSS-1h emulsified asphalt. Refer to AASHTO M 208 and Section 02745.
 - 1. Select emulsion according to the time constraints required for Maintenance of Traffic (MOT) and the ability to fully cure before allowing traffic on the roadway.

2.2 PROFILING EQUIPMENT

- A. Certify equipment according to UDOT Materials Manual of Instruction 995.
 - 1. Use only the prescribed filter settings from the certification for all measurements in this Section.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

- A. Notify the COR at least two working days before performing acceptance testing for pavement smoothness.

1. Clearly define each of the pavement sections to be evaluated.
- B. Perform work necessary to prepare the pavement for testing, including
 1. Sweep and clean debris.
 - a. Do not profile if water is on roadway.
 2. Remove or relocate barricades and utility plates so profiler can maintain constant speed and alignment.
 3. Mark begin and end locations as well as any excluded areas.
- C. The Government may perform Independent Assurance testing according to AASHTO R 54 Verification Testing

3.2 MICROSURFACING

- A. Profile before and after placement of micro-surfacing.

3.3 MRI OF ORIGINAL ROADWAY SURFACE

- A. Profile existing roadway surface before beginning construction to determine MRI_o for Category 2 pavement Sections.
 - a. Collect profiles with GPS location enabled.

3.4 PAVEMENT PROFILE CORRECTION

- A. Perform preliminary profiling and roadway smoothness evaluation to identify any defects exceeding acceptance limits.
- B. Correct defects before performing acceptance testing.
 1. Category 1 Pavements: Measure and correct Localized Roughness defects in the underlying surface before placement of a final surface of OGSC, BWC, SMA, chip seal, or micro-surfacing.
 2. Correct defects across lanes, medians, and shoulders in widths terminating at a lane line, edge of pavement, or center of the lane either by grinding or by surface replacement as approved by the COR. Refer to Section 02981 for grinding requirements
 - a. Include areas not eligible for incentive/disincentive.
 3. Taper corrected areas for smooth transverse transitions and surface drainage.

3.5 ACCEPTANCE PROFILING

- A. Perform acceptance testing for smoothness after all corrective work has been performed.
 - 1. Collect longitudinal profiles in each wheel path and in the center of each paved shoulder and bike lane using UDOT certified profilers and operators in accordance with AASHTO R 54, R 57 and M 328.
 - a. Collect profiles with GPS location enabled
 - 2. Category 1 Pavements: Acceptance testing can be performed before raising manholes and valves.
 - 3. Category 1 Pavements: Acceptance testing can be performed before grooving in pavement messages.
 - 4. Profile Structure Section prior to placement of thin-bonded overlays.
- B. Seal corrected areas in BWC, OGSC, HMA, and SMA with a flush coat application.
 - 1. Apply the flush coat at an undiluted rate of 0.11 ± 0.01 gal/yd²
 - 2. Allow flush coat to cure for 24 hour prior to any profiling.

3.6 PAVEMENT THICKNESS

- A. Re-test pavements for thickness after grinding as determined by the COR.

END OF SECTION

SECTION 02721

UNTREATED BASE COURSE (UTBC)

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Production, construction, and compaction of UTBC used for pavements, shoulders, and incidental construction.

1.2 RELATED SECTIONS

- A. Section 01572: Dust Control and Watering

1.3 REFERENCES

- A. AASHTO T 11: Materials Finer than 75- μ m (No. 200) Sieve in Mineral Aggregates by Washing
- B. AASHTO T 19: Bulk Density ("Unit Weight") and Voids in Aggregate
- C. AASHTO T 27: Sieve Analysis of Fine and Coarse Aggregates
- D. AASHTO T 89: Determining the Liquid Limit of Soils
- E. AASHTO T 90: Determining the Plastic Limit and Plasticity Index of Soils
- F. AASHTO T 96: Resistance to Degradation of Small-Sized Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
- G. AASHTO T 180: Moisture-Density Relations of Soils Using a 4.54 kg (10 lb) Rammer and 457 mm (18 in) Drop
- H. AASHTO T 193: The California Bearing Ratio
- I. AASHTO T 255: Total Evaporable Moisture Content of Aggregate by Drying
- J. AASHTO T 335: Determining the Percent of Fracture in Coarse Aggregate
- K. UDOT Minimum Sampling and Testing Requirements

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Written report for approval for each aggregate class and source, a minimum of five working days before placement. Include the following:
 - 1. Aggregate suitability. Refer to this Section, Part 2.
 - 2. Name of supplier and location of source.
 - 3. Maximum Dry Density and Optimum Moisture Content and associated test result data. Refer to AASHTO T 180, Method D.
 - 4. Job mix gradation including single values for each sieve size, No. 4 and finer. The target values must be within the gradation limits of Table 2.
- B. Job-mix gradation changes
 - 1. Refer to this Section, Article 3.1.

1.6 ACCEPTANCE

- A. Sampling and testing of material is according to UDOT Minimum Sampling and Testing Requirements.
- B. Type I Placement – Pavement Section
 - 1. Use Class A aggregate, Table 1.
 - 2. The Contractor is responsible for QC testing for moisture, gradation, and laboratory density and in-place density determinations.
 - 3. The Government may takes random samples and tests for moisture, gradation, and laboratory density and in-place density determinations.
 - 4. Meet gradation limits and applicable tolerances of Table 2 for each gradation test. Evaluate each subplot separately and do not average with other sublots.
 - 5. Meet minimum density test average of 97 percent of maximum laboratory density with no test less than 94 percent.
- C. Type II Placement – Incidental includes placement for Curb, Curb and Gutter, Driveways, Pedestrian Access Ramps, Sidewalk, Waterways, Flatwork, and other items of work in the contract to which UTBC is included and not measured or paid for separately.
 - 1. Use Class A aggregate, Table 1.
 - 2. The Contractor is responsible for QC testing for moisture, gradation, and laboratory density and in-place density determinations.
 - 3. The Government may take random samples and tests for moisture, gradation, and laboratory density and performs in-place

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- density determinations.
4. Meet gradation limits and applicable tolerances of Table 2 for each gradation test. Each subplot will be evaluated separately and not averaged with other sublots.
 5. Meet minimum density test average of 95 percent of maximum laboratory density with no test less than 92 percent.
- D. Type III Placement – Shoulder
1. Use Class A or B aggregate, Table 1.
 2. Adjust moisture content before compaction.
- E. Material not meeting the gradation requirements may be allowed to remain in-place at the discretion of the COR provided density requirements are met. Additional lots may not be placed until the deficiencies are addressed and corrected.
- F. Correct material that does not meet the specified criteria by scarifying, placing additional material, re-mixing, reshaping, and re-compacting when determined by the COR.
- G. Do not place additional material on any unaccepted layer.

PART 2 PRODUCTS

2.1 AGGREGATES

- A. Well-graded, clean, hard, tough, durable, and sound mineral aggregates consisting of crushed stone, crushed gravel, or crushed slag, free of organic matter and contamination from chemical or petroleum products, according to Table 1.

Table 1

Aggregate Properties			
	Aggregate Class		
	A	B	
Dry Rodded Unit Weight	Not less than 75 lb/ft ³		AASHTO T 19
Liquid Limit/Plastic Index	Non-plastic	PI ≤ 6	AASHTO T 89 AASHTO 90
Aggregate Wear	Not to exceed 50 percent		AASHTO T 96
Gradation	Table 2		AASHTO T 11 AASHTO T 27
CBR with a 10 lb surcharge measured at 0.20 inch penetration	70% Minimum	N/A	AASHTO T 193
Two Fractured Faces	50% Min	N/A	AASHTO T 335

- B. Establish the job mix (target) gradation for the $\frac{3}{4}$ inch sieve and finer within the gradation limits.
1. The Job Mix Gradation Tolerance is the allowable deviation from the job mix (target) gradation on the applicable sieves.
 2. All other percents passing will be within the gradation limits. Refer to AASHTO T 11 and AASHTO T 27.

Table 2

Gradation Limits		
Sieve Size	Job Mix Gradation Target Band	Job Mix Gradation Tolerance
1½ inch	100	
1 inch	90 - 100	±9.0
$\frac{3}{4}$ inch	70 - 85	±9.0
$\frac{1}{2}$ inch	65 - 80	±9.0
$\frac{3}{8}$ inch	55 - 75	±9.0
No. 4	40 - 65	±7.0
No. 16	25 - 40	±5.0
No. 200	7 - 11	±3.0

Percent passing based on total aggregate (dry weight) and fine and coarse aggregate with approximately the same bulk specific gravities.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Mixing – Provide moisture content of ± 2 percent of optimum at the time of placement. Refer to AASHTO T 180, Method D and AASHTO T 255.
- B. Procedures for Changing the Job-Mix Gradation
1. Submit changes in writing 24 hours before placement for approval by the COR.
- C. Placing – Place in layers of uniform thickness and compact each layer to a thickness not to exceed a 6 inch depth. Do not place on any frozen surface. Refer to Section 01572.
- D. Finishing – Uniform line and grade with surface deviations no more than $\frac{3}{8}$ inch in 10 ft in any direction.
1. Profile Tolerance – Correct any profile deviations greater than $\frac{3}{8}$ inch.
 - a. Rework minimum of 4 inch lift to achieve homogeneous density.
 - b. Determine limits of correction based on extent of deviation.

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- c. Continue finishing until existing deviation is less than $\frac{3}{8}$ inch.
- E. Compaction – Maintain optimum moisture content $\pm$ 2 percent.
 - 1. Use appropriate compaction equipment adjacent to abutments, backwalls, approach slabs, wing walls, retaining walls, and other structures.
 - 2. Use a minimum of two passes with a roller for Type III placement or as directed by the COR.

END OF SECTION

SECTION 02732

CRACK SEALING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Procedure for sealing cracks in pavement using hot-pour crack sealant or flexible repair mastic material.

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. AASHTO T 51: Ductility of Bituminous Materials
- B. AASHTO T 228: Specific Gravity of Semi-Solid Asphalt Materials
- C. AASHTO T 300: Force Ductility of Asphalt Materials
- D. ASTM D 36: Softening Point of Bitumen (Ring-and-Ball Apparatus)
- E. ASTM D 3111: Flexibility Determination of Hot-Melt Adhesives by Mandrel Bend Test
- F. ASTM D 4402: Viscosity Determination of Asphalt at Elevated Temperatures Using a Rotational Viscometer
- G. ASTM D 5329: Sealants and Fillers, Hot-Applied, For Joints and Cracks in Asphaltic and Portland Cement Concrete Pavements
- H. ASTM D 6690: Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavement
- I. ASTM D 8260: Hot-Applied Asphalt Aggregate-Filled Mastic

1.4 DEFINITIONS

- A. Flush fill – Filling the crack so that the material is level with the pavement surface.
- B. Overband – Extending the sealant or mastic slightly beyond the crack edges to seal both sides and protect against moisture intrusion.

1.5 SUBMITTALS

- A. Provide the following for hot-pour crack sealants and flexible repair mastic materials.
 - 1. Test results demonstrating compliance with the AASHTO and ASTM properties listed in this specification for review.
 - a. Use an AASHTO-accredited independent laboratory or the Product Evaluation and Audit Solutions (PEAS) program.
 - b. Provide test results for the current product formulation performed within the past five years, or from a current PEAS evaluation report.
 - 2. Manufacturer Installation Instructions for information Material Safety Data Sheets (MSDS) for information

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Supply material pre-proportioned, pre-blended, pre-reacted and pre-packaged in lined boxes weighing no more than 30lb.
 - 1. Use a dissolvable lining that will completely melt and become part of the mastic upon subsequent re-melting.
 - 2. Deliver in the manufacturer's original sealed container.
 - a. Clearly mark each container with the manufacturer's name, trade name of mastic, batch or lot number, and recommended safe heating and application temperatures.

PART 2 PRODUCTS

2.1 HOT-POUR ASPHALT PAVEMENT CRACK SEALANT – UDOT TYPE 1

- A. Meet the requirements of ASTM D 6690 Type 1.
 - 1. Use material that does not track in normal traffic after 45 minutes from application.
 - 2. Use material that pours readily and penetrates $\frac{1}{4}$ inch and wider cracks for the entire application temperature range recommended by the manufacturer.

2.2 HOT-POUR ASPHALT PAVEMENT CRACK SEALANT – UDOT TYPE 2

- A. Meet the requirements of Table 1.
 - 1. Use material that does not track in normal traffic after 45 minutes from application.
 - 2. Use material that pours readily and penetrates $\frac{1}{4}$ inch and wider cracks for the entire application temperature range recommended by the manufacturer

Table 1

Hot-Pour Asphalt Pavement Crack Sealant – UDOT Type 2			
Application Properties			
Test Method	Property	Minimum	Maximum
ASTM D 36	Softening Point (°F)	200	
AASHTO T 51	Ductility, modified, 1cm/min, 39.2° F (4° C), cm	30	
AASHTO T 228	Specific Gravity, 60° F (15.6° C)		1.140
AASHTO T 300 (a)	Force-Ductility, lb force		4
ASTM D 4402	Viscosity, 380°F (193.3°C), SC4-27 spindle, 20 rpm, Cp		2,500
ASTM D 5329	Cone Penetration, 77° F (25° C), 150 g, 5 sec., dmm		75
ASTM D 5329	Flow 140°F (60° C), 5 hrs 75° angle, mm		1
ASTM D 5329	Resilience, 77° F (25° C), 20 sec., percent	40	
ASTM D 5329 (b) (c)	Bond, Non-Immersed	Pass	
ASTM D 5329	Asphalt Compatibility	Pass	
(a) Maximum of 4 lb force during the specified elongation of 30 cm @ 1 cm/min, 39.2° F (4° C).			
(b) Three 12.7mm Specimens pass 3 cycles at 50% ext. at 0 °F.			
(c) The development at any time during the test procedure of a crack, separation, or other opening that at any point is over 6 mm deep, in the sealant or between the sealant and concrete block shall constitute failure of the test specimen. Measure the depth of the crack, separation or other opening perpendicular to the side of the sealant showing the defect.			

2.3 HOT-POUR ASPHALT PAVEMENT CRACK SEALANT – UDOT TYPE 3

- A. Meet the requirements of Table 2.
1. Use material that does not track in normal traffic after 45 minutes from application.
 2. Use material that pours readily and penetrates 1/4 inch and wider cracks for the entire application temperature range recommended by the manufacturer

Table 2

Hot-Pour Asphalt Pavement Crack Sealant – UDOT Type 3			
Application Properties			
Test Method	Property	Minimum	Maximum
ASTM D 36	Softening Point (°F)	205	
ASTM D 113	Ductility, 77° F (25° C), cm	40	
ASTM D 3111	Flexibility, 1" (25 mm) mandrel, 90° bend, 2 sec., 0°F (-18°C)	no cracks	
ASTM D 5329	Cone Penetration, 77° F (25° C), 150 g, 5 sec., dmm		60
ASTM D 5329 (a)(b)	Bond, Non-Immersed		Pass
ASTM D 5329	Resilience, 77° F (25° C), 20 sec., percent	40	
ASTM D 5329	Asphalt Compatibility	Pass	
(a) Two out of three 25.4mm specimens pass 5 cycles at 50% ext. at 20 °F			
(b) The development at any time during the test procedure of a crack, separation, or other opening that at any point is over 6 mm deep, in the sealant or between the sealant and concrete block shall constitute failure of the test specimen. Measure the depth of the crack, separation or other opening perpendicular to the side of the sealant showing the defect.			

2.4 FLEXIBLE REPAIR MASTIC MATERIAL

- A. Meet the requirements of ASTM D 8260, Type I or Type II

2.5 HOT-POUR CRACK SEALANT FOR CONCRETE PAVEMENTS

- A. Meet the requirements of ASTM D 6690, Type II

2.6 EQUIPMENT

- A. Sealant Heating Equipment
1. Use a double boiler or circulating hot oil heat transfer unit for indirectly heating product.
 - a. Use a unit capable of constant agitation.
 - b. Do not use direct heat transfer units (tar pots).
- B. Pavement Crack Router
1. Use equipment sufficiently maneuverable to follow meandering cracks and provide a uniform and controlled routing depth.
 - a. Use bits sharp enough to create a clean cut.

- C. Compressed Air Lance or Hot Air Lance
 - 1. Use a lance capable of providing clean, oil-free compressed air at a minimum volume of 90 cubic feet per minute at a pressure of 120 lbs per square inch.

PART 3 EXECUTION

3.1 PREPARATION

- A. Use a pavement crack router to open transverse cracks between $\frac{1}{8}$ inch and $\frac{1}{2}$ inch wide.
 - 1. The COR may require routing longitudinal cracking if additional bonding and flexibility is desired.
 - 2. Route joints and to a $\frac{1}{2}$ inch wide x $\frac{1}{2}$ inch deep or $\frac{3}{4}$ inch wide x $\frac{3}{4}$ inch deep configuration.
 - 3. Remove material from each side of the crack to produce a vertical, intact surface with no loosely bonded aggregates.
- B. Use compressed air to clean crack walls and the surface around the crack.
 - 1. Use a hot air lance when surfaces are wet or when air temperature is less than 40 degrees F.
 - a. Do not use on saturated pavements.
 - b. Do not burn the surrounding pavement.
 - c. Seal cracks before pavement cools down.

3.2 ASPHALT PAVEMENT SEALANT APPLICATION

- A. Fill cracks less than or equal to $1\frac{1}{2}$ inch wide with hot-pour crack sealant.
 - 1. Follow manufacturer safe heating and application temperatures.
 - 2. Flush fill in applications prior to placement of a surface treatment or overlay.
 - a. Remove crack overfill by squeegee.
 - 3. Overband, using a sealing disc, when a surface treatment or overlay is not scheduled within one year.
- B. Fill cracks greater than $1\frac{1}{2}$ inch wide with flexible repair mastic material.
 - 1. Verify the mastic fills the crack and extends slightly beyond the edges to create a controlled overband, sealing both sides of the crack.
- C. Install material in a manner that produces an acceptably smooth riding surface.

- D. Reapply material to any crack segment where material shrinkage or insufficient material was applied in a previous pass or application.

3.3 LIMITATIONS

- A. Do not place during rain or other adverse weather conditions.
- B. Do not place if surface moisture is present.
- C. Do not place on concrete bridge decks.

END OF SECTION

SECTION 02735

MICRO-SURFACING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Products and procedures for mixing and spreading a properly proportioned homogeneous mixture of aggregate, mineral filler, additives, polymer-modified asphalt emulsion and water to an existing pavement providing a firm surface adhesion and smooth stable skid resistant texture.

1.2 RELATED SECTIONS

- A. Section 02746: Hydrated Lime

1.3 REFERENCES

- A. AASHTO M 17: Mineral Filler for Bituminous Paving Mixtures
- B. AASHTO M 85: Portland Cement
- C. AASHTO M 208: Cationic Emulsified Asphalt
- D. AASHTO T 11: Materials Finer Than 75 µm (No. 200) Sieve in Mineral Aggregate
- E. AASHTO T 27: Sieve Analysis of Fine and Coarse Aggregates

- F. AASHTO T 49: Penetration of Bituminous Materials
- G. AASHTO T 53: Softening Point of Bitumen
- H. AASHTO T 59: Testing Emulsified Asphalts
- I. AASHTO T 96: Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
- J. AASHTO T 104: Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate
- K. AASHTO T 112: Clay Lumps and Friable Particles in Aggregate

- L. AASHTO T 176: Plastic Fines in Graded Aggregates and Soils by Use of the Sand Equivalent Test
- M. AASHTO T 278: Surface Frictional Properties Using the British Pendulum Tester
- N. AASHTO T 279: Accelerated Polishing of Aggregates Using the British Wheel
- O. AASHTO T 315: Determining the Rheological Properties of Asphalt Binder Using a Dynamic Shear Rheometer (DSR)
- P. AASHTO T 316: Viscosity Determination of Asphalt Binder Using Rotational Viscometer
- Q. AASHTO PP 72-11: Recovering Residue from Emulsified Asphalt Using Low-Temperature Evaporative Techniques
- R. ASTM D 6372: Testing and Construction of Micro-Surfacing
- S. ASTM D 7405: Multiple Stress Creep and Recovery (MSCR) of Asphalt Binder Using a Dynamic Shear Rheometer
- T. International Slurry Seal Association (ISSA) Specifications and Guidelines
- U. UDOT Minimum Sampling and Testing Requirements
- V. UDOT Quality Management Plans

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Provide the mix design for review 14 days before beginning construction.
 - 1. Refer to this Section, Article 2.6.
 - 2. Provide the COR with the following for asphalt and polymer emulsion with job-mix design.
 - a. Test report that meets the requirements of this Section, Article 2.1.
 - b. Certificate of analysis and compliance from the manufacturer for each batch.
 - c. Target gradation for combined aggregate and mineral filler.
 - d. Name of the asphalt and polymer emulsion supplier.

- B. Provide test reports for mineral aggregate.
 - 1. Refer to this Section, Article 2.2.
- C. Provide Manufacturer's Certificate of Compliance for Mineral Filler.
 - 1. Refer to this Section, Article 2.3.
- D. Provide calibration documentation for each mixing unit including an individual calibration for each material at various settings that corresponds to the mixing unit metering device.
- E. Changes in the job-mix gradation.
 - 1. Submit a written request for a change in the job-mix gradation.
 - 2. Submit a new job-mix design if changes in gradation are outside the gradation band allowed by the stockpile tolerance in Table 4.

1.6 ACCEPTANCE

- A. Acceptance sampling and testing of material according to UDOT Minimum Sampling and Testing Requirements.
- B. Government will assess price adjustments based on the average daily application rate of aggregate and emulsion residue per lot.
 - 1. A lot is defined as one day's production.
 - 2. Use the computerized monitoring system from the paver for the aggregate and emulsion totals per lot.
 - 3. Obtain square yards from the Contractors Daily Report, verified by the Quality Control.
 - 4. Pay Factor = PF_{rate} (Table 1) x $PF_{residue}$ (Table 2)
- C. Application rate pay factor (PF_{rate}) is based on application rate of dry aggregate.
 - 1. Aggregate application target rate is 25 lbs/yd²

Table 1

Pay Factor Application Rate	
Average Daily Application Rate (lbs/yd ²)	PF_{rate}
25.0 or Greater	1.0
20.0 to 24.9	(Avg. Rate ÷ 25)
Less than 20.0	Reapplication with option to remove

- D. Emulsion residue pay factor (PF_{residue}) is based on the residue of the emulsion placed each lot.
1. Target Value = Residual asphalt target value for emulsion based on the mix design.
 2. Emulsion residue = Based on the Certificate of Compliance from the Emulsion Supplier.

Table 2

Pay Factor Emulsion Residue Content	
Average Daily Emulsion Residue %	PF_{residue}
Target Value less than $\pm 0.6\%$	1.0
Target Value $\pm 0.61\%$ to Target Value $\pm 1.00\%$	0.8
Greater than 1% of Target Value	Reapplication with option to remove

PART 2 PRODUCTS

2.1 EMULSIFIED ASPHALT

- A. Use a quick-set polymer-modified asphalt emulsion manufactured specifically for micro-surfacing.
1. Refer to AASHTO M 208 grade CQS-1 h (cationic) emulsified asphalt. Delete settlement and cement mixing test requirements.
 2. The supplier must ~~be on the approved list adhering~~adhere to the UDOT Quality Management Plan 508 Asphalt Emulsion.
- B. Mill or blend the polymer material into the asphalt or emulsifier solution before the emulsification process.
1. Obtain certification from the asphalt emulsion manufacturer that the emulsion contains at least 3.0 percent polymer solids based on the weight of asphalt (asphalt residual).
- C. Refer to Table 3 for Modified Emulsion Residue.

Table 3

Modified Emulsion Residue By Distillation		
Test Method	Description	Specification
AASHTO T 49	Penetration, 25° C	40-90
AASHTO T 53	Softening point	57° C Min
AASHTO T 59-modified (a)	Residue by distillation	62% Min.
AASHTO T 316	Rotational Viscosity 135° C	650 CPS Min
Modified Emulsion Residue By Evaporation, AASHTO PP 72-11 Procedure B		
AASHTO T 315 (b)	Original DSR, G* at 58° C, kPa	7 - 14
AASHTO T 315 (b)	Original DSR, Phase Angle, δ , at 58° C	75 Max.
ASTM D 7405 (b)	Multiple Stress Creep Recovery (MSCR) at 64° C, % recovery at 3.2 kPa stress level	25 Min.
(a) Modified distillation procedure – Heat emulsion residue to $177 \pm 5^\circ \text{C}$ and maintain that temperature for 20 min. Perform the distillation within 60 ± 15 min. (b) Do not reheat on completion of evaporation. Complete residue testing within 48 hr of performing the evaporation procedure. Pull small specimens from the evaporation sample for rheological testing and ball by hand using gloves that will not affect the residue such as nitrile gloves.		

- D. Formulate the polymer modified emulsified asphalt so that when the paving mixture is applied with relative humidity at not more than 50 percent and the ambient air temperature of at least 75 degrees F, the paving mixture will sufficiently cure so that traffic can be allowed in one hour.

2.2 MINERAL AGGREGATE

- A. Use 100 percent crushed mineral aggregates, clean and free from organic matter or other detrimental substances that meet the following requirements:
1. Maximum clay lumps and friable particles of 2 percent for coarse and fine aggregates. Refer to AASHTO T 112.
 2. Maximum weighted sodium sulfate soundness loss of 15 percent. Refer to AASHTO T 104.
 3. Maximum loss by abrasion of 30 percent. Refer to AASHTO T 96.
 4. Sand equivalent of 60 or greater. Refer to AASHTO T 176.
 5. Minimum polishing value of 31. Refer to AASHTO T 278 and T 279.
 - a. Perform tests on aggregate before crushing.
 - b. Do not use predominantly limestone or dolomite aggregates.

- B. Establish a job mix or target gradation within the gradation band as specified in Table 4.
1. Base the mix design on the target gradation.
 2. The percent passing each sieve must not vary by more than the stockpile tolerance and still remains within the gradation band. Refer to AASHTO T 11 and AASHTO T 27.

Table 4

Job-Mix Gradation Design Limits		
Sieve Size	Job Mix Gradation Target Band	Stockpile Tolerances
$\frac{3}{8}$	100	0
#4	70-90	±5
#8	45-70	±5
#16	28-50	±5
#30	19-34	±5
#50	12-25	±4
#100	7-18	±3
#200	5-15	±2

Percent passing based on total aggregate (dry weight) and fine and coarse aggregate with approximately the same bulk specific gravities.

2.3 MINERAL FILLER

- A. Use one of the following for Mineral Filler. Refer to AASHTO M 17.
1. Non-air entrained Type I/II portland cement. Refer to AASHTO M 85.
 2. Hydrated lime, free from lumps or foreign matter. Refer to Section 02746.
- B. Determine the amount of mineral filler needed through the laboratory mix design.
1. The amount of mineral filler used must be between 0.5 percent and 2.0 percent by the weight of dry aggregate.
 2. Adjust the percentage of mineral filler ± 0.5 percent as necessary for better consistency or to optimize set times with Engineer's approval.

2.4 WATER

- A. Use potable water free from harmful salts, reactive chemicals, and any other contaminants.

2.5 ADDITIVES

- A. Use additives as required to accelerate or retard the break-set of the micro-surfacing mix, to improve the resulting finished surface, or to increase adhesion.
 - 1. Determine the initial additive quantities from the mix design for the micro-surfacing mix or individual materials.
 - 2. Use additives compatible with the other components of the mix.

2.6 JOB-MIX DESIGN

- A. Design according to ASTM D 6372.
 - 1. Show each ingredient amount meets the following:
 - a. Residual asphalt cement content, 7.0 percent minimum by dry total weight of aggregate.
 - b. Aggregate gradation (target) within the job-mix gradation design limits in Table 4.
 - c. Mineral filler, percentage by total dry weight of aggregate.
 - d. Polymer modifier 3.0 percent minimum polymer solids based on the residual asphalt content certified by emulsion supplier.
 - 2. Identify the optimum emulsion content as a percentage of the dry weight of aggregate to meet Table 5 requirements.
 - 3. Identify additives as determined by design testing to control mix set times and cohesion.
 - a. Provide acceptable limits for additives.
 - 4. Conform to the ISSA A143 specifications listed in Table 5.
 - 5. Use the same materials and aggregate gradation to be used on the project.
 - 6. Provide a micro-surfacing mixture that can be spread in variable thickness cross-sections, ruts, scratch courses, and surfaces.

Table 5

ISSA A 143 Specifications		
ISSA Test No.	Description	Specification
ISSA TB-139	<u>Wet Cohesion</u> @ 30 Minutes Minimum (Set) @ 60 Minutes Minimum (Traffic)	12 kg-cm Minimum 20 kg-cm Minimum or Near Spin
ISSA TB-109	Excess Asphalt by LWT Sand Abrasion	50 g/ft ² Maximum (538 g/m ² Maximum)
ISSA TB-114	Wet Stripping	Pass (90% Minimum)
*ISSA TB-100	<u>Wet-Track Abrasion Loss</u> One-hour Soak Six-day Soak	50 g/ft ² (538 g/m ²) Maximum 75 g/ft ² (807 g/m ²) Maximum
ISSA TB-147	Lateral Displacement	5% Maximum
ISSA TB-144	Classification Compatibility	11 Grade Points Minimum (AAA, BAA)
ISSA TB-113	Mix Time @ 77° F (25° C)	Controllable to 120 Seconds Minimum

* Perform the wet track abrasion test under laboratory conditions as a component of the mix design process.

2.7 EQUIPMENT

- A. Use mixing equipment specifically designed and manufactured to mix and place micro-surfacing.
 1. Mix the material by an automatically sequenced, self-propelled micro-surfacing mixing machine with a continuous flow revolving multi-blade double shafted mixing unit that:
 - a. Accurately delivers and proportions the aggregate, emulsified asphalt, mineral filler, controls setting additive, and water.
 - b. Discharges the mixed product on a continuous flow basis.
 2. Use a machine with sufficient storage capacity for aggregate, emulsified asphalt, mineral filler, control additive, and water to maintain an adequate supply to the proportioning controls.
 3. Use a machine capable of self-loading materials while continuing to place micro-surfacing.
 4. Full control of the forward and reverse speed during applications of the micro-surfacing material.
 - a. Use original manufacturer designed equipment for the self-loading device, opposite side driver stations, and forward and reverse speed controls.
- B. Use a machine with a hydraulically adjustable (while applying mixture) type spreader box with a positive screed adjustment for yield control and a positive adjustment for the joint matcher.

- C. Equip the micro-surfacing spreader box with the following:
1. Ribbon flights mounted on an adjustable shaft to continually agitate and distribute the materials throughout the box.
 2. Curb bumpers and replaceable runners with at least 5 ft. long end runners.
 3. A sufficient walkway to provide access to either side of the spreader box without walking through the freshly placed material.
 - a. The equipment must provide sufficient turbulence to prevent the mix from setting in the box or causing excessive side buildup or lumps.
 4. Flexible seals in contact with the road, front and rear, to prevent the loss of the mixture from the box.
 5. A secondary strike-off located approximately 2 to 3 ft behind the primary strike-off to minimize transverse corrugations and to improve surface texture.
 - a. The secondary strike-off must have elevation and width adjustments similar to the primary strike-off and a pivot point that can be tilted for texturing or raised completely off the surface.
 - b. The secondary strike-off must have the same adjustments as the spreader box.
 6. Ability to side shift the box to compensate for variations in the pavement geometry.
 7. Capability of applying micro-surfacing mixture in variable widths up to 15 ft.
- D. Use a rut filling spreader box specifically designed to fill ruts with an average depth greater than ½ inch.
1. Ruts greater than ½ inch deep require multiple passes to restore the cross section.
- E. Use a computerized material monitoring system with integrated material control devices such that the amount of each material can be determined at any time.
1. Calibrate each material control device before each mix application and as often thereafter as determined by the Engineer.
 2. Use a monitoring system capable of recording, displaying, and printing the following information:
 - a. Individual sensor counts for emulsion, aggregate, mineral filler, water, and additive.
 - b. Aggregate, emulsion, and mineral filler output in pounds per minute.
 - c. Percentages of emulsion, mineral filler, water, and additive.
 - d. Cumulative totals of aggregate, emulsion, mineral filler, water, and additive.

- e. Scale factor for all materials.
- F. Check the daily totals from the computerized monitoring system for the aggregate and the daily total from the scalping screen weighing system are within 2 percent.
- G. Calibrate each mixing unit ~~in the presence of the Engineer~~ before construction.
 - 1. Do not use any machine until calibration has been completed.

PART 3 EXECUTION

3.1 LIMITATIONS

- A. Do not apply micro-surfacing during rain when road surface moisture is present or during other adverse weather conditions.
- B. Do not apply micro-surfacing if either the pavement or air temperature is below 50 degrees F.
- C. Do not apply micro-surfacing when the temperature is projected below 37 degrees F within 48 hours of placing micro-surfacing.
- D. Cease micro-surfacing operations when the weather or other conditions prolong opening road surface to traffic beyond two hours.
- E. Remove and replace the micro-surfacing if any of the following occurs:
 - 1. Lumping, balling, or unmixed aggregates.
 - 2. Separation of the coarse aggregate from the emulsion and fines.
 - 3. Excessive breaking of emulsion inside the spreader box.
 - 4. Streaking caused by oversized aggregate.
 - 5. Flushing or excessively rich areas appearing in the micro-surfacing after two hours from the time of placement.
 - 6. Any measurable rutting, shoving, or other evidence of premature deformation when exposed to traffic.
- F. Keep traffic off roadway surface until the micro-surfacing has cured.
 - 1. Allow for additional curing time at locations such as driveways, intersections, and where sharp turning movements may take place or where vehicles may accelerate quickly.

3.2 STOCKPILE

- A. Construct individual 500-ton stockpiles of micro-surfacing aggregates.

1. Engineer will verify stockpiles at least one and at most seven days before use.
 2. Combining, altering, or moving 500-ton stockpiles may require re-approval by the COR before use.
- B. Notify the Engineer at least seven calendar days before micro-surfacing placement in order for the initial stockpiles to be sampled and tested for acceptance.
- C. Obtain the Engineer's written acceptance of a stockpile before its use for micro-surfacing.
- D. Provide stockpile Quality Control information to the COR for every 500 tons of aggregates to include the following:
1. Aggregate gradation meeting job-mix formula tolerances according to Table 4.
 2. Sand Equivalency. Refer to AASHTO T 176.
- E. Be capable of determining aggregate moisture within 10 minutes at all times.
- F. Rework or remove material not meeting specifications from the stockpile area. Identify stockpiles that will be reworked.
- G. The Government may retest corrected material for acceptance.

3.3 PREPARATION

- A. Clean the pavement surface of dirt, sand, dust, oil, and other objectionable material immediately before applying micro-surfacing.
- B. Allow un-sealed cracks to dry thoroughly before applying micro-surfacing when using water to clean the road surface.
- C. Cover manholes, valve boxes, drop inlets, and other service utility entrances before surfacing.
- D. Protect all structures, including items such as guardrail, guideposts, concrete barriers, drains, and parapet walls.

3.4 APPLICATION

- A. Pre-wet the pavement surface as required due to local conditions by fogging ahead of the micro-surfacing box.
1. Do not create standing water on the pavement in front of the micro-surfacing box.

- B. Place micro-surfacing mix that meets the job-mix design.
 - 1. Control the ingredient proportions with metering or measuring devices on the micro-surfacing equipment.
 - a. Use readings from the metering or measuring devices to determine compliance with limits stated in the approved job-mix design.
 - 2. Limit any increase or decrease in the amount of mineral filler added to the mix during production to ± 1 percent of the job-mix design.
 - 3. Limit the set-control agent to ± 1 percent of the job-mix design.
 - 4. The emulsion used throughout the project must be the same emulsion submitted with the job-mix design.
 - 5. The emulsion content must be ± 1 percent of the job-mix design.
 - a. ~~Engineer~~COR may require a new job-mix design and re-authorization of the micro-surfacing if large disparities occur.
 - b. Calculate the percent emulsion on the daily electronic printout from the calibrated paver for emulsion and aggregate quantities.
- C. Place a mixture according to the authorized mix design and the quality control tolerances.
 - 1. Maintain quality control documentation and make available to the COR upon request or at completion of daily work.
 - 2. Randomly calculate the percent asphalt content of the mixture from the equipment computer display readings at least three times daily.
 - 3. Randomly calculate the yield of the aggregate being placed from the equipment computer display readings at least three times daily.
 - 4. Maintain a daily report and log sheet containing the following information:
 - a. Aggregate used, ton (dry)
 - b. Micro-Surfacing emulsion used, ton
 - c. Bituminous materials for tack coat used, if specified, ton
 - d. Mineral Filler used, lbs
 - e. Water used in mixture, gallons
 - f. Additive used in mixture, gallons
 - g. Surface area completed (square yards)
 - h. Surface area application rate (dry lbs aggregate per square yard)
 - i. Percentage of emulsified asphalt based on dry aggregate
- D. Pass the mineral aggregate over a vibratory scalping screen before transfer to the micro-surfacing mixing machine to remove oversize material.
 - 1. The screening unit must be capable of providing weigh tickets for each load of material.

- E. Carry a sufficient amount of micro-surfacing in all parts of the spreader box so that full width and complete coverage is obtained with no streaks or narrow spots.
 - 1. Avoid overloading the spreader box.
 - 2. Do not spray water directly into the spreader box during the application of micro-surfacing under any circumstances.
- F. Apply micro-surfacing of proper consistency at a minimum rate of 25 lbs/yd² based on the dry weight of aggregate for each lot.
 - 1. Provide to the Engineer the square yards for each lot.
 - 2. Calculate the aggregate application rate using the daily tickets from the screening unit and daily electronic printout from the calibrated paver.
- G. Apply micro-surfacing for rut filling.
 - 1. Apply micro-surfacing as a scratch-coat pass using a steel or high density strike-off when required to fill ruts less than ½ inch deep or as directed by the COR.
 - 2. Make multiple passes with the rut filling spreader box for ruts greater than ½ inch deep.
 - 3. Allow 24-hour cure time after filling ruts, before placing additional micro-surfacing layer.
- H. Do not apply water to freshly placed micro-surfacing.

3.5 TEST STRIP

- A. Construct a test strip at least 500 ft long on the roadway before initial placement.
 - 1. The material must achieve initial set within 30 minutes and the surface shows no signs of distress when exposed to traffic after curing for 2 hours.
- B. Make necessary adjustments if test strip does not perform as required.
 - 1. Repeat the test strip process.
 - 2. The ~~Engineer~~COR may require a new job-mix design if failures indicate an ingredient problem.

3.6 FINISHING DETAILS

- A. The depth of each micro-surfacing course must not exceed twice the maximum aggregate size.
 - 1. Not required when using a rut filling spreader box.

- B. Longitudinal and transverse joints
 1. Construct longitudinal joints within 6 inches of the lane lines where possible.
 2. The overlap of micro-surfacing at any joint must not exceed 6 inches.
 3. Repair the joints if any of the following conditions exist:
 - a. Build-up of material at the joints.
 - b. Uncovered areas at the joints.
 - c. Longitudinal and transverse joints with more than $\frac{1}{4}$ inch vertical space between the surface and a 4 ft straightedge placed perpendicular to the joint.
 4. The edges of the micro-surfacing must follow the centerline, lane lines, shoulder lines, and curb lines.
 5. Repair the edges if they vary more than 6 inches.
 6. Use methods approved by the Engineer to correct deficiencies.
 - a. The repaired surfaces must be dense with a uniform texture.
 - b. Repair sections with surface irregularities the same width as the existing pass of micro-surfacing.
 - c. Small areas of patching are not permitted.
- C. Finished micro-surfacing must have a uniform texture free of scratches, tears, and other surface irregularities.
 1. Repair the surface if any of the following deficiencies exist:
 - a. More than one surface irregularity $\frac{1}{4}$ inch or wider and 10 ft or longer in any 100 ft section.
 - b. More than three surface irregularities $\frac{1}{2}$ inch or wider and more than 6 inches long in any 100 ft section.
 - c. Any surface irregularity 1 inch or wider and more than 4 inches long.
 - d. Any tire track damage to the fresh micro-surfacing.
 - e. Slick spots or any area of bleeding (surface flushing).
- D. Place micro-surfacing adjacent to concrete pavements or concrete curb and gutter with a straight longitudinal edge.
 1. Do not allow over-lap in these areas.
- E. Maintain neat construction lines at all locations.
- F. Place micro-surfacing at side streets and intersections out to right-of-way line including around radii where applicable.

- G. Use hand squeegees to spread micro-surfacing in areas where micro-surfacing machine cannot operate.
 - 1. Lightly dampen areas before mix placement.
 - 2. Provide complete and uniform coverage.
 - 3. Avoid unsightly appearance from handwork.
 - 4. Use the same type of finish in hand worked areas as applied by the spreader box.
- H. Use construction paper or comparable products so that beginning and ending joint lines from each construction pass are straight and neat.

END OF SECTION

SECTION 02741

ASPHALT MIX

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. A surface course of one or more layers of HMA comprised of aggregate, asphalt binder, hydrated lime, and other additives.
- B. Option to incorporate Reclaimed Asphalt Pavement (RAP) materials into HMA pavement.

1.2 RELATED SECTIONS

- A. Section 01456: Materials Dispute Resolution
- B. Section 02701: Pavement Smoothness
- C. Section 02742S: Project Specific Surfacing Requirements
- D. Section 02745: Asphalt Material
- E. Section 02746: Hydrated Lime
- F. Section 02748: Prime Coat/Tack Coat

1.3 REFERENCES

- A. AASHTO M 323: Superpave Volumetric Mix Design
- B. AASHTO R 35: Superpave Volumetric Design for Hot-Mix Asphalt (HMA)
- C. AASHTO T 19: Bulk Density ("Unit Weight") and Voids in Aggregate
- D. AASHTO T 89: Determining the Liquid Limit of Soils
- E. AASHTO T 90: Determining the Plastic Limit and Plasticity Index of Soils
- F. AASHTO T 96: Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine

- G. AASHTO T 104: Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate
- H. AASHTO T 112: Clay Lumps and Friable Particles in Aggregate
- I. AASHTO T 176: Plastic Fines in Graded Aggregates and Soils by Use of the Sand Equivalent Test
- J. AASHTO T 195: Determining Degree of Particle Coating of Asphalt Mixtures
- K. AASHTO T 209: Theoretical Maximum Specific Gravity and Density of Hot Mix Asphalt (HMA)
- L. AASHTO T 255: Total Evaporable Moisture Content of Aggregate by Drying
- M. AASHTO T 304: Uncompacted Void Content of Fine Aggregate
- N. AASHTO T 335: Determining the Percentage of Fracture in Coarse Aggregate
- O. UDOT Materials Manual of Instruction
- P. UDOT Minimum Sampling and Testing Requirements
- Q. UDOT Quality Management Plans

1.4 DEFINITIONS

- A. Longitudinal Joint – Any new asphalt lift abutting an existing paving lift, exceeding 200 feet in length and excluding intersections. This includes joints created by echelon paving and new asphalt placed against a milled asphalt edge.
- B. Lot – The number of tons of HMA placed in a Production Day.
- C. Minor Target Change – A change from the verified mix design gradation target on a maximum of two sieves with the following limitations.
 - 1. The maximum change in the target gradation on the #8 or any coarser sieve is limited to 3 percent passing per sieve.
 - 2. The maximum change in the target gradation on the #16 or #50 sieves is 2 percent passing per sieve.
 - 3. The maximum change in the target gradation on the #200 sieve is 0.5 percent passing.

- 4. No target change may violate the mix design requirements in this section.
- D. Overband – an 8 inch protective asphalt coating sealing the longitudinal joint of final riding surface, as proposed by the contractor and approved by the Engineer
- E. Production Day – A 24 hour period in which HMA is being placed.
- F. RAP – Recycled Asphalt Pavement. Crushed or milled asphalt materials that have been removed from pavements.
- G. Thin Overlay Pavement – An overlay where the sum of the thickness of the HMA lifts is less than two inches.

1.5 SUBMITTALS

- A. Mix design for approval at least ~~10-14~~ working days before paving according to the UDOT Materials Manual of Instruction 960.
- B. Changes in job mix design
 - 1. Submit a written request for any proposed change in the job-mix gradation.
 - a. Allow at least 12 hours for approval before incorporating a minor target change into production.
 - b. Allow at least six working days for verification and approval of any other change.
 - 2. Include documentation supporting correlation between suggested target changes and mix design volumetric requirements. Department acceptance test results or Contractor QC test data or both are acceptable.
 - 3. Submit samples according to the UDOT Materials Manual of Instruction 960 for a volumetric mix design verification for anything other than approved minor target changes. This includes changes in the aggregate source, asphalt binder source, or asphalt binder grade.
- C. Corrective action plan for approval according to this Section, Article 3.3, paragraph B and Article 3.4, paragraph A4b.
- D. Mat joint layout plan to the COR for review at least 14 calendar days before placement.

1.6 ACCEPTANCE

- A. Acceptance sampling and testing of material is according to UDOT Minimum Sampling and Testing Requirements.
- B. Gradation and asphalt binder content
 - 1. The COR evaluates a lot on the test results of four samples with the following exceptions:
 - a. Compute incentive/disincentive using the test results from three samples if only three samples can be taken for the production day.
 - b. Combine test results with the next day of production if at least three random samples cannot be taken. Take one sample for each 500 tons, or portion thereof, from the following day's production.
 - c. Add the lot to the previous day's production for the final day's production if three random samples cannot be taken.
 - d. The lot may be increased to include up to three production days when agreed upon in advance by both the Contractor and the COR when less than 900 tons are anticipated per production day.
 - 2. Evaluate the lot using the number of tests "n" in Table 3.
- C. Density and Thickness
 - 1. Obtain cores from the mat and longitudinal joint within two contract days after the pavement is placed. Refer to UDOT Materials Manual of Instruction 984.
 - a. Move transversely to a point 1 ft from the edge of the pavement for in-place mat density if the random location for coring falls within 1 ft of the edge of the overall pavement section (outer part of shoulders).
 - b. Fill core holes with HMA or high AC content cold mix and compact.
 - c. Begin testing the cores within 24 hours for density acceptance.
 - 2. Density Requirements
 - a. The target for in-place density for the mat is 93.5 percent of Theoretical Maximum Specific Gravity except for thin overlay pavements.
 - b. The target for in-place density for the longitudinal joint is 91.5 percent of the Theoretical Maximum Specific Gravity (Gmm).

- 1) Use the average of both Gmm's if two different sides or values exist for the cores taken.
- c. The target for in-place density is 92.5 percent of Theoretical Maximum Specific Gravity for Thin overlay pavement projects.
 - 1) Do not take longitudinal joint cores for thin overlay pavement.
- d. Use the average of the Theoretical Maximum Specific Gravity tests for each lot.
- e. Acceptance for in-place mat and longitudinal joint density may be based on establishing a rolling pattern for items such as bridge decks, utility work, traffic signals, detours, lane leveling, driveways, other handwork, or small projects with plan quantities less than 500 tons.
3. Thickness Requirements are based only on mat cores. The thickness requirement may be waived when matching up to existing pavement, curb and gutter for Pavement in or next to intersections.
 - a. The Government accepts a lot for thickness when:
 - 1) The average thickness is not more than $\frac{1}{2}$ inch greater or $\frac{1}{4}$ inch less than the total design thickness specified.
 - 2) No individual subplot shows a deficient thickness of more than $\frac{3}{8}$ inch.
 - b. Excess Thickness – The COR may allow excess thickness to remain in place or may order its removal.
 - 1) The Government pays for 50 percent of the mix for material in excess of the $+\frac{1}{2}$ inch tolerance when excess thickness is allowed to remain in place.
 - c. Deficient Thickness – Place additional material where lots or sublots are deficient in thickness.
 - 1) The Government pays for 50 percent of the mix for additional material over specified thickness necessary to achieve minimum lift thickness.
 - 2) Minimum compacted lift is 3 times the nominal maximum aggregate size.
 - d. Thickness tolerances established above do not apply to leveling courses. Check final surfaces in stage construction.
 - e. Check thickness regularly with a depth probe and take corrective action as necessary for thin overlay pavement.
4. Longitudinal Joint
 - a. The edge of a new asphalt mat may be removed for

the purpose of meeting longitudinal joint density requirements.

- 1) The material wasted is still included in the payment.
- 2) Up to 3 inches for a confined edge is allowed.
- 3) Up to 6 inches for an unconfined edge is allowed.

- D. The Government applies one Incentive/Disincentive for the lowest dollar value for Gradation/Asphalt Content, one Incentive/Disincentive for In- Place Mat Density, and one Incentive/Disincentive for Longitudinal Joint Density. The COR computes Incentives/Disincentives as follows for each lot. Refer to Section 02701 for smoothness requirements.
1. Compute incentive/disincentive for Gradation/Asphalt Binder and In-place Mat Density and Longitudinal Joint Density according to Table 1.
 2. Base the incentive/disincentive on Percent within Limit (PT) computation using Tables 2, 3, and 4.
 3. Use lowest single value combined for gradation (each of the sieves) and asphalt binder content for calculating the gradation/asphalt binder content incentive/disincentive.
 4. Use Tables 2, 3, and 4 to determine PT for in-place Mat Density and Longitudinal Joint Density.
 5. Meet PT of 88 or greater for in-place mat density or the Department does not pay incentives on gradation/asphalt binder content.
 6. The Government pays/assesses the longitudinal joint density incentive/disincentive per ton of HMA placed adjacent to, and on the hot side of the longitudinal joint for each lift:
 - a. The incentive/disincentive will be calculated from the average of the core densities taken from all abutting joints if the HMA mat has a longitudinal joint on more than one side.
 7. The following work is not eligible for incentive:
 - a. Items such as utility work, traffic signals, detours, lane leveling, and driveways.
 - b. Small projects with plan quantities of HMA less than 500 tons.
- E. The Government rejects lots:
1. According to Table 1.
 2. The COR may accept a reject lot. Refer to Section 01456.
 - a. A price reduction of 35 percent of the pay item or \$20 per ton, whichever is greater, will be assessed.
 - b. The lot will not be eligible for any incentive.
- F. The COR may elect to accept material on visual inspection for work

such as utility work, traffic signals, detours, lane leveling, and driveways, other hand work, or small projects with plan quantities less than 500 tons.

1. Lots accepted on visual inspection are not eligible for Incentive/Disincentive.
2. The Government reserves the option of conducting any additional acceptance tests necessary to determine that the material and workmanship meets the project requirements.
3. Acceptance for mat density and longitudinal joint density may be based on establishing and maintaining a roller pattern to obtain maximum density without over-stressing the pavement.

Table 1

Incentive/Disincentive for Asphalt Binder Content, and Mat Density	
PT Based on Min. Four Samples	Incentive/Disincentive (Dollars/Ton)
>99	2.00
96-99	1.50
92-95	1.00
88-91	0.00
84-87	-0.26
80-83	-0.60
76-79	-0.93
72-75	-1.27
68-71	-1.60
64-67	-1.93
60-63	-2.27
<60	Reject
Incentive/Disincentive for Gradation	
PT Based on Min. Four Samples	Incentive/Disincentive (Dollars/Ton)
>99	2.00
96-99	1.50
92-95	1.00
88-91	0.00
84-87	-0.26
80-83	-0.60
76-79	-0.93
72-75	-1.27
68-71	-1.60
64-67	-1.93
60-63	-2.27
56-59	-5.00
52-55	-10.00
<52	Reject
Incentive/Disincentive for Longitudinal Joint Density	
PT Based on Min Four Samples	Incentive/Disincentive (Dollars/Ton)
>99	2.00
96-99	1.50
92-95	1.00
88-91	0.00
84-87	-0.26
80-83	-0.60

76-79	-0.93
72-75	-1.27
68-71	-1.60
64-67	-1.93
60-63	-2.27
56-59	-2.60
52-55	-5.00
<52	The \$5 penalty and Overband Longitudinal Joint if Final Surface Lift

Table 2

Upper and Lower Limit Determination	
Parameter	UL and LL
¾ inch sieve for 1 inch HMA ½ inch sieve for ¾ inch HMA ⅜ inch sieve for ½ inch HMA No. 4 sieve for ⅜ inch HMA	Target Value ± 6.0%
No. 8 sieve	Target Value ± 5.0%
No.50 sieve	Target Value ± 3.0%
No. 200 sieve	Target Value ± 2.0%
Asphalt Binder Content	Target Value ± 0.35%
Mat Density	Lower Limit Target Value - 2.0% Upper Limit Target Value + 4.0%
Longitudinal Joint Density	Lower Limit Target Value - 2.0% Upper Limit Target Value + 6.0%

Table 3

Enter table in the appropriate “number of tests” column and round down to the nearest value.

Quality Index Values (QU or QL) for Estimating Percent Within Limits										
PU or PL	n=3	n=4	n=5	n=6	n=7	n=8	n=10	n=12	n=15	n=20
100	1.16	1.50	1.75	1.91	2.06	2.15	2.29	2.35	2.47	2.56
99	1.16	1.47	1.68	1.79	1.89	1.95	2.04	2.09	2.14	2.19
98	1.15	1.44	1.61	1.70	1.77	1.80	1.86	1.89	1.93	1.97
97	1.15	1.41	1.55	1.62	1.67	1.69	1.74	1.77	1.80	1.82
96	1.15	1.38	1.49	1.55	1.59	1.61	1.64	1.66	1.69	1.70
95	1.14	1.35	1.45	1.49	1.52	1.54	1.56	1.57	1.59	1.61
94	1.13	1.32	1.40	1.44	1.46	1.47	1.49	1.50	1.51	1.53
93	1.12	1.29	1.36	1.38	1.40	1.41	1.43	1.43	1.44	1.46
92	1.11	1.26	1.31	1.33	1.35	1.36	1.37	1.37	1.38	1.39

91	1.10	1.23	1.27	1.29	1.30	1.31	1.32	1.32	1.32	1.33
90	1.09	1.20	1.23	1.24	1.25	1.25	1.26	1.26	1.27	1.27
89	1.08	1.17	1.20	1.21	1.21	1.21	1.21	1.21	1.22	1.22
Table 3 Continued										
PU/PL	n=3	n=4	n=5	n=6	n=7	n=8	n=10	n=12	n=15	n=20
88	1.07	1.14	1.16	1.17	1.17	1.17	1.17	1.17	1.17	1.17
87	1.06	1.11	1.12	1.12	1.12	1.13	1.13	1.13	1.13	1.13
86	1.05	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08
85	1.03	1.05	1.05	1.05	1.05	1.04	1.04	1.04	1.04	1.04
84	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00	1.00
83	1.00	0.99	0.98	0.97	0.97	0.96	0.96	0.96	0.96	0.96
82	0.98	0.96	0.95	0.94	0.94	0.93	0.93	0.92	0.92	0.92
81	0.96	0.93	0.92	0.91	0.90	0.90	0.89	0.89	0.89	0.88
80	0.94	0.90	0.88	0.87	0.86	0.86	0.85	0.85	0.85	0.85
79	0.92	0.87	0.85	0.84	0.83	0.83	0.82	0.82	0.82	0.81
78	0.89	0.84	0.82	0.81	0.80	0.79	0.79	0.78	0.78	0.78
77	0.87	0.81	0.79	0.78	0.77	0.76	0.76	0.75	0.75	0.75
76	0.84	0.78	0.76	0.75	0.74	0.73	0.72	0.72	0.72	0.72
75	0.82	0.75	0.73	0.72	0.71	0.70	0.69	0.69	0.69	0.68
74	0.79	0.72	0.70	0.68	0.67	0.67	0.66	0.66	0.66	0.65
73	0.77	0.69	0.67	0.65	0.64	0.64	0.62	0.62	0.62	0.62
72	0.74	0.66	0.64	0.62	0.61	0.61	0.60	0.59	0.59	0.59
71	0.71	0.63	0.60	0.59	0.58	0.58	0.57	0.56	0.56	0.56
70	0.68	0.60	0.58	0.56	0.55	0.55	0.54	0.54	0.54	0.53
69	0.65	0.57	0.55	0.54	0.53	0.52	0.51	0.51	0.51	0.50
68	0.62	0.54	0.52	0.51	0.50	0.50	0.48	0.48	0.48	0.48
67	0.59	0.51	0.49	0.48	0.47	0.47	0.46	0.45	0.45	0.45
66	0.56	0.48	0.46	0.45	0.44	0.44	0.43	0.42	0.42	0.42
65	0.53	0.45	0.43	0.42	0.41	0.41	0.40	0.40	0.40	0.39
64	0.49	0.42	0.40	0.39	0.38	0.38	0.37	0.37	0.37	0.37
63	0.46	0.39	0.37	0.36	0.35	0.35	0.35	0.34	0.34	0.34
62	0.43	0.36	0.34	0.33	0.33	0.33	0.32	0.31	0.31	0.31
61	0.39	0.33	0.31	0.30	0.30	0.30	0.29	0.29	0.29	0.28
60	0.36	0.30	0.28	0.27	0.26	0.26	0.25	0.25	0.25	0.25
59	0.32	0.27	0.25	0.25	0.24	0.24	0.24	0.23	0.23	0.23
58	0.29	0.24	0.23	0.22	0.21	0.21	0.21	0.21	0.21	0.20
57	0.25	0.21	0.20	0.19	0.19	0.19	0.18	0.18	0.18	0.18
56	0.22	0.18	0.17	0.16	0.16	0.16	0.16	0.16	0.15	0.15
55	0.18	0.15	0.14	0.14	0.13	0.13	0.13	0.13	0.13	0.13
54	0.14	0.12	0.11	0.11	0.11	0.11	0.10	0.10	0.10	0.10
53	0.11	0.09	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
52	0.07	0.06	0.06	0.05	0.05	0.05	0.05	0.05	0.05	0.05
51	0.04	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 4

Definitions, Abbreviations, and Formulas for Acceptance	
Term	Explanation
Target Value (TV)	The target values for gradation and asphalt binder content are given in the Contractor's volumetric mix design. See this Section, article 1.6 for density target values.
Average (AVE)	The sum of the lot's test results for a measured characteristic divided by the number of test results—the arithmetic mean.
Sample Standard Deviations	The square root of the value formed by summing the squared difference between the individual test results of a measured characteristic and AVE, divided by the number of test results minus one.
Upper Limit (UL)	The value above the TV of each measured characteristic that defines the upper limit of acceptable production. (Table 2)
Lower Limit (LL)	The value below the TV of each measured characteristic that defines the lower limit of acceptable production (Table 2)
Upper Quality Index (QU)	$QU = (UL - AVE)/s$
Lower Quality Index (QL)	$QL = (AVE - LL)/s$
Percentage of Lot Within UL (PU)	Determined by entering Table 3 with QU.
Percentage of Lot Within LL (PL)	Determined by entering Table 3 with QL.
Total Percentage of Lot Within UL and LL (PT)	$PT = (PU + PL) - 100$
Incentive/Disincentive	Determined by entering Table 1 with PT or PL.

All values for AVE, s, QU, and QL will be calculated to at least four decimal places and carried through all further calculations. Rounding to lower accuracy is not allowed.

1.7 DISPUTE RESOLUTION

- A. Refer to Section 01456 when disputing the validity of the Department's acceptance tests.

PART 2 PRODUCTS

2.1 ASPHALT BINDER

- A. Project Specific Surfacing Requirements – Refer to Section 02742S.
- B. Asphalt Material – Refer to Section 02745.

2.2 AGGREGATE

- A. Crusher processed virgin aggregate material consisting of crushed stone, gravel, or slag.
- B. Refer to Table 5 to determine the suitability of the aggregate.
 - 1. Coarse aggregates
 - a. Retained on No. 4 sieve
 - 2. Fine aggregates
 - a. Clean, hard grained, and angular
 - b. Passing the No. 4 sieve

Table 5

Aggregate Properties – HMA			
Test Method	Test No.	75 Design Gyration and Greater	Less Than 75 Design Gyration
One Fractured Face	AASHTO T 335	95% minimum	85% min (1 inch and $\frac{3}{4}$ inch) 90% min ($\frac{1}{2}$ inch and $\frac{3}{8}$ inch)
Two Fractured Face	AASHTO T 335	90% minimum	80% min (1 inch and $\frac{3}{4}$ inch) 90% min ($\frac{1}{2}$ inch and $\frac{3}{8}$ inch)
Fine Aggregate Angularity	AASHTO T 304	45 minimum	45 minimum
Flakiness Index	UDOT MOI 933 (Based on $\frac{3}{8}$ inch sieve and above)	17% maximum	17% maximum
L.A. Wear	AASHTO T 96	35% maximum	40% maximum
Sand Equivalent	AASHTO T 176 (Pre-wet method)	60 minimum	45 minimum
Plasticity Index	AASHTO T 89 and T 90	0	0
Unit Weight	AASHTO T 19	minimum 75 lb/cu ft	minimum 75 lb/cu ft
Soundness (sodium sulfate)	AASHTO T 104	16% maximum loss with five cycles	16% maximum loss with five cycles
Clay Lumps and Friable Particles	AASHTO T 112	2% maximum	2% maximum
Natural Fines	N/A	0%	10% maximum

- C. Meet gradation requirements in Table 6.

Table 6

Aggregate Gradations (Percent Passing by Dry Weight of Aggregate)					
Sieve Size		1 inch	¾ inch	½ inch	⅜ inch
Control Sieves	1½ inch	100.0			
	1 inch	90.0 - 100.0	100.0		
	¾ inch	<90	90.0 - 100.0	100.0	
	½ inch		<90	90.0 – 100.0	100.0
	⅜ inch			<90	90.0 - 100.0
	No. 4				< 90
	No. 8	19.0 - 45.0	23.0 - 49.0	28.0 - 58.0	32.0 - 67.0
	No. 200	1.0 - 7.0	2.0 - 8.0	2.0 – 10.0	2.0 – 10.0

2.3 HYDRATED LIME

- A. Meet the requirements of Section 02746.

2.4 RECLAIMED ASPHALT PAVEMENT (RAP) (OPTIONAL)

- A. Do not adjust the asphalt binder grade if the lower end is already a PG XX-34.
- B. Do not adjust the asphalt binder grade when RAP content is not more than 15 percent by total weight of the hot mix and RAP asphalt binder content is not more than 15 percent of the total asphalt binder content by weight.
- C. Adjust asphalt binder grade according to AASHTO M 323 when RAP asphalt binder content is between 15 to 25 percent of the asphalt binder weight.
1. Select one grade softer than the grade specified. Do not adjust the asphalt binder grade if the lower end is already a PG XX-34.
 2. Provide test reports indicating that the PG grade and quantity of the recovered asphalt binder is consistent throughout the stockpile.
 3. Limit RAP to 25 percent of the total weight of the hot mix and RAP binder to 25 percent of the total binder.
- D. RAP aggregate is required to meet Table 5 with exception of Sand Equivalent. Refer to AASHTO T 176.

2.5 VOLUMETRIC MIX DESIGN

- A. Perform Superpave Volumetric Mix Design according to UDOT Materials Manual of Instruction 960 and the following:
1. Incorporate hydrated lime into all designs. Refer to Section 02746.
 2. Comply with Table 7 and Table 8.
- B. Obtain Government approval for the mix design. Refer to the UDOT Materials Manual of Instruction 960.
1. Do not begin paving until approved.
- C. Mix Design Changes
1. The Government may allow up to two minor target changes per project without penalty to contractor. The ~~Department~~ Government charges \$1,000 for each additional minor target change.
 2. The Government allows up to two volumetric mix design verifications, (including field verifications), per project. The ~~Department~~ Government charges \$3,000 for each additional laboratory or field verification required. This includes all laboratory or field volumetric mix design verifications required due to contractor initiated target changes.
 3. The COR will review each change and provide written notice of approval or rejection of each mix design change.

Table 7

Volumetric Design Gyration				
20 Years Design ESALS (Million)	Compaction Parameters			Voids Filled with Asphalt (VFA) (%)
	N _{initial} /% of G _{mm} *	N _{design} /% of G _{mm} *	N _{max} /% of G _{mm} *	
0.3	6 / ≤ 91.5	50 / 96.5	75 / ≤ 98	70 – 80 **
0.3 to < 3	7 / ≤ 90.5	75 / 96.5	115 / ≤ 98	70 – 80
3 to < 30	8 / ≤ 89	100 / 96.5	160 / ≤ 98	70 – 80
≥ 30	9 / ≤ 89	125 / 96.5	205 / ≤ 98	70 – 80

* G_{mm}: Theoretical maximum specific gravity of mix. Refer to AASHTO T 209.

** Use 67 percent for the lower limit VFA for 1-inch nominal maximum size mixture.

Table 8

Volumetric Design Requirements	
HMA design mixing and compaction temperatures	Provided by the Engineer

Dust Proportion Range	0.6 - 1.40
Voids in Mineral Aggregate (VMA) at N _{design} AASHTO R 35.9.2 using G _{sb} Oven Dry. Equation based on percent of total mix.	12.0% - 13.0% for 1 inch 13.0% - 14.0% for ¾ inch 14.0% - 15.0% for ½ inch 15.0% - 16.0% for ⅜ inch
Hamburg Wheel Tracker UDOT MOI 990	75 Design Gyrations and Greater Maximum 10 mm impression at 20,000 passes. Less than 75 Design Gyrations Maximum 10 mm impression at 10,000 passes

2.6 PRIME COAT/TACK COAT

- A. Refer to Section 02748.

PART 3 EXECUTION

3.1 HMA

- A. Dry aggregate to an average moisture content of not more than 0.2 percent by weight.
1. May be verified by AASHTO T 255.
 2. Adjust burners to avoid damage or soot contamination of the aggregate.
- B. Treat aggregate with hydrated lime. Refer to Section 02746.
1. Method A or B
 2. The Government applies a deduction for mix produced by a non- certified supplier.
 - a. The deduction is applied according to the UDOT Quality Management Plan 514 Hot-Mix Asphalt.
- C. Coat with asphalt binder 100 percent of the particles passing and 98 percent of the particles retained on the No. 4 sieve.
1. May be verified by AASHTO T 195.
 2. Discontinue operation and make necessary corrections if material is not properly coated.
- D. Maintain temperature of the HMA between the limits identified on the Volumetric Mix Design Verification Letter for mixing and compacting.
1. The Government rejects materials heated over the identified limits.

2. Remove all material rejected by the Department for overheating.
- E. Minimum compacted lift thickness is 3 times the nominal maximum aggregate size.

3.2 HMA PLANT

- A. Provide
1. Positive means to determine the moisture content of aggregate.
 2. Positive means to sample all material components.
 3. Sensors to measure the temperature of the HMA at discharge.
 4. The ability to maintain mix discharge temperature according to the mix design.
- B. Asphalt Binder Storage Tanks
1. Provide a positive means for separating and identifying asphalt grades when multiple products are used in mix production.
 2. Provide a positive means of sampling the asphalt binder.

3.3 CEASE PRODUCTION

- A. Cease production when any two out of three consecutive lots meet one of the following criteria:
1. A net disincentive
 2. Air voids at N_{des} averaged for each lot are less than 2.5 or greater than 4.75 percent
 3. VMA at N_{des} averaged for each lot are not within Target Value $\pm$ 1.25 percent
- B. Submit a corrective action plan for approval before production continues indicating the changes in production procedures that will be implemented to correct the deficiencies.
1. Address the specific issues contributing to the cease production.
 2. Obtain approval of the revised plan before production continues.
- C. The COR may require a new mix design.
- D. The COR may require Hamburg Wheel-Track testing for up to 5 lots after the cease production order.
1. Sample randomly from behind the paver for up to 5 lots after the cease production order.
 2. Failure to meet the requirements of Table 8 results in rejection of the lot.

3.4 LABORATORY CORRELATION

- A. Perform split-sample, paired t -testing with the Department based on project quality control testing using Department LQP qualified lab.
 - 1. Perform split-sample, paired t analysis on all mix acceptance tests and tests related to volumetric properties.
 - 2. Perform paired t analysis as defined in the UDOT Materials Manual of Instruction, Appendix C.
 - 3. Continue paired t -testing until at least two consecutive production days meet $\alpha = 0.05$ for a two tailed distribution.
 - 4. Resolve discrepancies in lab results within the first five production days.
 - a. Cease production if the requirements for two consecutive days of the first five days cannot be met.
 - b. Submit a corrective action plan to the COR before production continues indicating the changes in procedures that will be implemented to correct the deficiencies.
 - 1) Make paired t test results available within 24 hours of sampling.

3.5 SURFACE PREPARATION

- A. Locate, reference, and protect all utility covers, monuments, curb and gutter, and other components affected by the paving operations.
- B. Remove all moisture, dirt, sand, leaves, and other objectionable material from the prepared surface before placing the tack coat and mix.
- C. Complete spot leveling before placing pavement courses.
 - 1. Place, spread, and compact leveling mix on portions of the existing surface.
 - 2. Fill and compact any localized potholes more than 1 inch deep.
 - 3. Allow compacted mix to cool sufficiently to below 150 degrees F to provide a stable structural platform before placing additional lifts of HMA.
- D. Apply tack coat to all paved surfaces before applying a leveling course or pavement lift as required in Section 02748.
- E. Allow sufficient cure time for prime coat/tack coat before placing HMA.

3.6 SURFACE PLACEMENT

- A. Provide a compactable sloped edge adjacent to the next lane to be paved when full-width or Echelon paving is impractical and more than one pass is required.
 - 1. Coat edge with tack coat according to Section 02748 at a residual rate of 0.05 gal/yd².
 - 2. Echelon paving is the preferred method for constructing a longitudinal joint.
 - 3. Refer to Section 01554 and DD and TC Series Standard Drawings for pavement edge slope required to safely maintain traffic.
- B. Adjust the production of the mixing plant and material delivery until a steady paver speed is maintained.
- C. Construct straight joints, offset longitudinal joints 6 to 12 inches in succeeding courses.
 - 1. Construct all joints within 1 ft of the centerline, the lane line or the center of the lane.
 - 2. Tack the longitudinal edge before placing the adjacent pass.
- D. Offset transverse construction joints at least 6 ft longitudinally.
- E. Do not allow construction vehicles, general traffic, or rollers to pass over the uncompacted end or edge of freshly placed mix until the mat temperature drops to a point where damage or differential compaction will not occur.
- F. Taper the end of a course subjected to traffic at approximately 50:1 (horizontal to vertical).
 - 1. Remove the portion of the pass that contains the tapered end before placing fresh mix.
 - 2. Tack the contact surfaces before fresh mix is placed against the compacted mix.
- G. Use a motor grader, spreader box, or other approved spreading methods for projects under 180 yd², irregular areas, or for miscellaneous construction such as detours, sidewalks, and leveling courses.

3.7 COMPACTION

- A. Use a small compactor or vibratory roller at structures in addition to normal rolling.
- B. Operate in a transverse direction next to the back wall and approach slab.

3.8 LIMITATIONS

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- A. Do not place on frozen base or during adverse climatic conditions such as precipitation or when roadway surface is icy or wet.
- B. Use a release agent that does not dissolve asphalt and is satisfactory to the COR for all equipment and hand tools used to mix, haul, and place the HMA.
- C. Place HMA from April 15 through October 15, and when the air temperature in the shade and the roadway surface temperature is above 50 degrees F.
 - 1. The Government determines if it is feasible to place HMA outside these dates and temperature limits.
 - 2. Obtain authorization from the COR before paving outside these requirements.

END OF SECTION

SECTION 02743

ASPHALT MIX FOR BIKE AND PEDESTRIAN PATHS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Products and procedures for constructing a surface course of one or more layers of asphalt mix comprised of aggregate, asphalt binder, lime, and other additives for bike and pedestrian paths.
- B. Mix materials at a central mixing plant.

1.2 RELATED SECTIONS

- A. Section 02741: Asphalt Mix
- B. Section 02745: Asphalt Material
- C. Section 02746: Hydrated Lime
- D. Section 02748: Prime Coat/Tack Coat

1.3 REFERENCES

- A. AASHTO T 11: Materials Finer Than 75 μm (No. 200) Sieve In Mineral Aggregates By Washing
- B. AASHTO T 19: Bulk Density (Unit Weight) and Voids in Aggregate
- C. AASHTO T 27: Sieve Analysis of Fine and Coarse Aggregates
- D. AASHTO T 30: Mechanical Analysis of Extracted Aggregate
- E. AASHTO T 89: Determining the Liquid Limit of Soils
- F. AASHTO T 90: Determining the Plastic Limit and Plasticity Index of Soils
- G. AASHTO T 96: Resistance to Abrasion of Small Size Coarse Aggregate by Use of the Los Angeles Machine
- H. AASHTO T 104: Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate

- I. AASHTO T 112: Clay Lumps and Friable Particles in Aggregate
- J. AASHTO T 176: Plastic Fines in Graded Aggregates and Soils by Use of the Sand Equivalent Test
- K. AASHTO T 209: Theoretical Maximum Specific Gravity and Density of Asphalt Mixtures
- L. AASHTO T 304: Uncompacted Void Content of Fine Aggregate
- M. AASHTO T 308: Determining the Asphalt Binder Content of Asphalt Mixtures by the Ignition Method
- N. AASHTO T 335: Determining the Percentage of Fracture in Coarse Aggregate
- O. UDOT Materials Manual of Instruction (MOI)

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Mix design for verification and approval before paving according to UDOT Materials Manual of Instruction Section 960.
- B. Certify, for information, that the mix meets the gradation requirements of Table 2 and that the asphalt binder requirements from the Job Mix Design. Refer to AASHTO T 30 and T 308.

PART 2 PRODUCTS

2.1 ASPHALT MATERIALS

- A. Use the following asphalt materials:
 - 1. Asphalt Binder – Performance Graded (PG) Binder.
 - 2. Flush Coat – CQS-1, CSS-1 or SS-1.

2.2 AGGREGATE

- A. Refer to Section 02745 and Section 02741 excluding Aggregate Properties Table.
- B. Use Table 1 following for Aggregate Properties.

Table 1

Aggregate Properties		
Properties	Test Method	Test Requirements
One Fractured Face	AASHTO T 335	N/A
Two Fractured Face	AASHTO T 335	90% Min.
Fine Aggregate Angularity	AASHTO T 304	45 Min.
Flakiness Index	MOI 933 Based on $\frac{3}{8}$ inch and above	17% Max.
L.A. Wear	AASHTO T 96	35% Max.
Sand Equivalent	AASHTO T 176, alternate method 2, pre-wet method (test the sample in the wet condition).	45 Min.
Plasticity Index	AASHTO T 89 and T 90	0
Unit Weight	AASHTO T 19	75 lb/ft ³ Min.
Soundness Loss	AASHTO T 104 – Sodium Sulfate	16% Max. loss with five cycles
Deleterious Materials	AASHTO T 112	2% Max.
Natural Fines	---	10% Max.

C. Meet gradation in Table 2.

Table 2

Aggregate Gradations Percent Passing by Dry Weight of Aggregate AASHTO T 11 and AASHTO T 27	
Control Sieve	Percent
$\frac{1}{2}$ inch	100.0
$\frac{3}{8}$ inch	90.0 - 100.0
# 4	< 90.0
# 8	32.0 - 67.0
#200	2.0 - 10.0

2.3 HYDRATED LIME

- A. Use at least one percent hydrated lime.
- B. Refer to Section 02746.

2.4 VOLUMETRIC DESIGN REQUIREMENTS

- A. Perform Superpave Volumetric Mix Design according to UDOT Materials Manual of Instruction 960 and the following:
 - 1. Incorporate hydrated lime into all designs. Refer to Section 02746.
 - 2. Comply with Table 3.
- B. Verify the Volumetric Mix Design. Refer to UDOT Materials Manual of Instruction 960.
 - 1. Do not begin paving until verification is complete.
 - 2. The COR may accept the Volumetric Mix Design from data submitted with the proposed mix design or from a previous mix design.
 - 3. The Government reserves the right to verify any mix design submitted.

Table 3

Superpave Volumetric Mix Design		
Compaction Stage	Number of Gyration (N)	% of G _{mm} (AASHTO T 209)
Initial	5	≤ 91.5
Design	50	≥ 98.5
Maximum	75	≤ 100.0

PART 3 EXECUTION

3.1 ASPHALT MIX

- A. Refer to Section 02741, Part 3.

3.2 SURFACE PREPARATION

- A. Refer to Section 02741, Part 3.
- B. Refer to Section 02748.

3.3 COMPACTION

- A. Establish a rolling pattern to obtain maximum density without over-stressing the pavement.

3.4 ASPHALT EMULSION

- A. Apply the flush coat at a uniform rate of 0.10 gal/yd² undiluted emulsion or 0.15 gal/yd² 2:1 diluted emulsion.
 - 1. Diluted emulsion 2:1 represents two parts undiluted emulsion and one part water.

END OF SECTION

SECTION 02744

STONE MATRIX ASPHALT (SMA)

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. A surface course of one or more layers of fiber stabilized SMA comprised of aggregate, asphalt binder, hydrated lime, and other additives.

1.2 RELATED SECTIONS

- A. Section 02701: Pavement Smoothness
- B. Section 02742S: Project Specific Surfacing Requirements
- C. Section 02745: Asphalt Material
- D. Section 02746: Hydrated Lime
- E. Section 02748: Prime Coat/Tack Coat

1.3 REFERENCES

- A. AASHTO M 325: Stone Matrix Asphalt (SMA)
- B. AASHTO R 46: Designing Stone Matrix Asphalt (SMA)
- C. AASHTO R 67: Sampling Asphalt Mixtures after Compaction (Obtaining Cores)
- D. AASHTO T 11: Materials Finer Than 75 µm (No. 200) Sieve in Mineral

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Aggregates by Washing

- E. AASHTO T 19: Bulk Density ("Unit Weight") and Voids in Aggregate
- F. AASHTO T 27: Sieve Analysis of Fine and Coarse Aggregates
- G. AASTHO T 85: Specific Gravity and Absorption of Coarse Aggregate
- H. AASHTO T 89: Determining the Liquid Limit of Soils
- I. AASHTO T 90: Determining the Plastic Limit and Plasticity Index of Soils

- J. AASHTO T 96: Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact of the Los Angeles Machine
- K. AASHTO T 104: Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate
- L. AASHTO T 112: Clay Lumps and Friable Particles in Aggregate
- M. AASHTO T 166: Bulk Specific Gravity (G_{mb}) of Compacted Hot Mix Asphalt (HMA) Using Saturated-Surface Dry Specimens
- N. AASHTO T 176: Plastic Fines in Graded Aggregates and Soils by Use of the Sand Equivalent Test
- O. AASHTO T 195: Determining Degree of Particle Coating of Asphalt Mixtures
- P. AASHTO T 209: Theoretical Maximum Specific Gravity (G_{mm}) and Density of Hot Mix Asphalt (HMA)
- Q. AASHTO T 255: Total Evaporable Moisture Content of Aggregate by Drying
- R. AASHTO T 278: Surface Frictional Properties Using the British Pendulum Tester
- S. AAHTO T 279: Accelerated Polishing of Aggregates Using the British Wheel
- T. AASHTO T 304: Uncompacted Void Content of Fine Aggregate
- U. AASHTO T 305: Determination of Draindown Characteristics in Uncompacted Asphalt Mixtures
- V. AASHTO T 335: Determining the Percentage of Fracture in Coarse Aggregate
- W. ASTM C 612: Standard Specification for Mineral Fiber Block and Board Thermal Insulation
- X. UDOT Materials Manual of Instruction.
- Y. UDOT Minimum Sampling and Testing Requirements
- Z. UDOT Quality Management Plans
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1.4 DEFINITIONS

- A. Lot – The number of tons of SMA placed in a Production Day.
- B. Minor Target Change – A change from the verified mix design gradation target on a maximum of two sieves with the following limitations.
 - 1. The maximum change in the target gradation on the No. 8 or any coarser sieve is limited to 3 percent passing per sieve.
 - 2. The maximum change in the target gradation on the No. 50 sieves is 2 percent passing per sieve.
 - 3. The maximum change in the target gradation on the No. 200 sieve is 0.5 percent passing.
 - 4. No target change may violate the mix design requirements in this Section.
- C. Production Day – A 24 hour period in which SMA is being placed.

1.5 SUBMITTALS

- A. Mix design for approval at least working days before paving according to the UDOT Materials Manual of Instruction 962.
- B. Changes in job mix design
 - 1. Submit a written request for a proposed change in the job-mix gradation.
 - a. Allow at least 12 hours for approval before incorporating a minor target change into production.
 - b. Allow at least six working days for verification and approval of any other change.
 - 2. Include documentation supporting correlation between suggested target changes and mix design volumetric requirements.
 - a. Government acceptance test results or Contractor QC test data or both are acceptable.
 - 3. Submit samples according to the UDOT Materials Manual of Instruction 962 for a volumetric mix design verification for anything other than approved minor target change.
 - a. This includes changes in the aggregate source, asphalt binder source, or asphalt binder grade.
- C. Corrective action plan according to this Section, Article 3.3.
- D. Refer to this Section, Article 1.7 for laboratory correlation submittals.
- E. Mat joint layout plan to the COR for review at least 14 calendar days before placement.

1.6 ACCEPTANCE

- A. Acceptance sampling and testing of material is according to UDOT Minimum Sampling and Testing Requirements.
- B. Gradation and asphalt binder content
 - 1. Evaluate a lot on the test results of four samples with the following exceptions:
 - a. Compute incentive/disincentive using the test results from three samples if only three samples can be taken for the production day.
 - b. Combine test results with the next day of production if at least three random samples cannot be taken. Take one sample for each 500 tons, or portion thereof, from the following day's production.
 - c. Add the lot to the previous day's production for the final day's production if three random samples cannot be taken.
 - d. The lot may be increased to include up to three production days when agreed upon in advance by both the Contractor and the Engineer-COR when less than 900 tons are anticipated per production day.
 - 2. Evaluate the lot using the number of tests "n" in Table 3.
 - 3. Increase sample sizes to accommodate paired-T testing as required.
- C. Density and Thickness
 - 1. Obtain cores within two contract days after the pavement is placed. Refer to AASHTO R 67.
 - a. Mark coring location for in-place density.
 - b. Move transversely to a point 1 ft from the edge of the pavement for in-place density if the random location for coring falls within 1 ft of the edge of the overall pavement section (outer part of shoulders).
 - c. Fill core holes with SMA mix and compact.
 - d. Begin testing the cores within 24 hours for density acceptance.
 - 2. Density Requirements
 - a. The target for in-place density is 94.0 percent of Theoretical Maximum Specific Gravity
 - b. Use the average of the Theoretical Maximum Specific Gravity tests for each lot.

3. Thickness Requirements
 - a. The Government accepts a lot for thickness when:
 - 1) The average thickness is not more than $\frac{1}{2}$ inch greater or $\frac{1}{4}$ inch less than the total design thickness specified.
 - 2) No individual subplot shows a deficient thickness of more than $\frac{3}{8}$ inch.
 - b. Excess Thickness – The COR may allow excess thickness to remain in place or may order its removal.
 - 1) The Government pays for 50 percent of the mix for material in excess of the $+\frac{1}{2}$ inch tolerance when excess thickness is allowed to remain in place.
 - 2) Remove and replace the entire depth of the course if it is necessary to remove portions of the course.
 - c. Deficient Thickness – Place additional material where lots or sublots are deficient in thickness.
 - 1) The Government pays for 50 percent of the mix for additional material over specified thickness necessary to achieve minimum lift thickness.
 - 2) Minimum compacted lift is 3 times the nominal maximum aggregate size.
 - d. Thickness tolerances established above do not apply to leveling courses.
 - 1) Check final surfaces in stage construction.
 - e. Thickness acceptance for thin overlay pavement consists of checking thickness regularly with a depth probe during placement and taking corrective action as necessary.

- D. The Government applies incentive/disincentive for Gradation/Asphalt Content, and In-Place Density. The COR computes Incentive/Disincentive for each lot.
 1. Compute incentive/disincentive for Gradation/Asphalt Binder and In-place Density according to Table 1.
 2. Base the incentive/disincentive on Percent within Limit (PT) computation using Tables 2, 3, and 4.
 3. Use lowest single value combined for gradation (each of the sieves) and asphalt binder content for calculating the gradation/asphalt binder content incentive/disincentive.
 4. Use Tables 2, 3, and 4 to determine PT for in-place density.
 5. Meet PT of 88 or greater for in-place density or the Department does not pay incentives on gradation/asphalt binder content.

6. Meet control requirements of Table 2 for VMA and VCA.
 - a. VCA_{MIX} is less than VCA_{DRC} or the Department does not pay incentive for gradation/asphalt binder content.
 7. The following work is not eligible for incentive:
 - a. Items such as utility work, traffic signals, detours, lane leveling, and driveways.
 - b. Small projects with plan quantities of HMA less than 500 tons.
- E. The Government applies incentive/disincentive for smoothness according to Section 02701.
1. Refer to Section 02701 for smoothness requirements.
- F. The Government rejects lots:
1. If the PT for any individual gradation measurement listed in Table 3 is less than 60 percent.
 2. If the PT for asphalt binder content or density is less than 52 percent.
 3. The COR may accept a reject lot. Refer to Section 01456.
 - a. A price reduction of 35 percent of the pay item or \$30 per ton, whichever is greater, will be assessed.
 - b. The lot will not be eligible for any incentive.
- G. The COR may elect to accept material on visual inspection for work such as utility work, traffic signals, detours, lane leveling, and driveways, other hand work, or small projects with plan quantities less than 500 tons.
1. Lots accepted on visual inspection are not eligible for Incentive/Disincentive.
 2. The Government reserves the option of conducting acceptance tests necessary to determine the material and workmanship meets the project requirements.
 3. Acceptance for density may be based on establishing and maintaining a roller pattern to obtain maximum density without over-stressing the pavement.

Table 1

Incentive/Disincentive for Asphalt Binder Content, and Density	
PT Based on Min. Four Samples	Incentive/Disincentive (Dollars/Ton)
>99	2.50
96-99	2.00
92-95	1.25
88-91	0.00
84-87	-0.26
80-83	-0.60
76-79	-0.93
72-75	-1.27
68-71	-1.60
64-67	-1.93
60-63	-2.27
<60	Reject
Incentive/Disincentive for Gradation	
PT Based on Min. Four Samples	Incentive/Disincentive (Dollars/Ton)
>99	2.50
96-99	2.00
92-95	1.25
88-91	0.00
84-87	-0.26
80-83	-0.60
76-79	-0.93
72-75	-1.27
68-71	-1.60
64-67	-1.93
60-63	-2.27
56-59	-5.00
52-55	-10.00
<52	Reject

Table 2

Production Control for VMA and VCA_{MIX}		
VMA Average Value, x, (%) Minimum of three Samples	VCA_{MIX} Job-Mix Design	Action
$X \geq 16.5$ and $X \leq 18.5$	$VCA_{MIX} < VCA_{DRC}$	Continue Paving
$X < 16.5$ or $X > 18.5$	$0 < VCA_{MIX} - VCA_{DRC} \leq 0.5\%$	Stop Production until a corrective action plan is approved.
$X < 15.5$ or $X > 19.5$	$0.5 < VCA_{MIX} - VCA_{DRC}$	Stop Production and resubmit Mix Design

Table 3

Upper and Lower Limit Determination	
Parameter	UL and LL
³ / ₈ " sieve for 1/2" SMA	Target Value ± 5.0%
#4 sieve	Target Value ± 4.0%
# 8 sieve	Target Value ± 3.0%
# 50 sieve	Target Value ± 3.0%
# 200 sieve	Target Value ± 2.0%
Asphalt Binder Content	Target Value ± 0.35%
Density	Lower Limit: Target Value - 2.0% Upper Limit: Target Value + 3.5%

Enter table in the appropriate sample size column and round down to the nearest value.

Table 4

Quality Index Values (QU or QL) for Estimating Percent Within Limits										
PU or PL	n=3	n=4	n=5	n=6	n=7	n=8	n=10	n=12	n=15	n=20
100	1.16	1.50	1.75	1.91	2.06	2.15	2.29	2.35	2.47	2.56
99	1.16	1.47	1.68	1.79	1.89	1.95	2.04	2.09	2.14	2.19
98	1.15	1.44	1.61	1.70	1.77	1.80	1.86	1.89	1.93	1.97
97	1.15	1.41	1.55	1.62	1.67	1.69	1.74	1.77	1.80	1.82
96	1.15	1.38	1.49	1.55	1.59	1.61	1.64	1.66	1.69	1.70
95	1.14	1.35	1.45	1.49	1.52	1.54	1.56	1.57	1.59	1.61
94	1.13	1.32	1.40	1.44	1.46	1.47	1.49	1.50	1.51	1.53
93	1.12	1.29	1.36	1.38	1.40	1.41	1.43	1.43	1.44	1.46
92	1.11	1.26	1.31	1.33	1.35	1.36	1.37	1.37	1.38	1.39
91	1.10	1.23	1.27	1.29	1.30	1.31	1.32	1.32	1.32	1.33
90	1.09	1.20	1.23	1.24	1.25	1.25	1.26	1.26	1.27	1.27
89	1.08	1.17	1.20	1.21	1.21	1.21	1.21	1.21	1.22	1.22
88	1.07	1.14	1.16	1.17	1.17	1.17	1.17	1.17	1.17	1.17
87	1.06	1.11	1.12	1.12	1.12	1.13	1.13	1.13	1.13	1.13
86	1.05	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08
85	1.03	1.05	1.05	1.05	1.05	1.04	1.04	1.04	1.04	1.04
84	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00	1.00
83	1.00	0.99	0.98	0.97	0.97	0.96	0.96	0.96	0.96	0.96
82	0.98	0.96	0.95	0.94	0.94	0.93	0.93	0.92	0.92	0.92
81	0.96	0.93	0.92	0.91	0.90	0.90	0.89	0.89	0.89	0.88
80	0.94	0.90	0.88	0.87	0.86	0.86	0.85	0.85	0.85	0.85
79	0.92	0.87	0.85	0.84	0.83	0.83	0.82	0.82	0.82	0.81
78	0.89	0.84	0.82	0.81	0.80	0.79	0.79	0.78	0.78	0.78

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Table 4 Continued										
PU/PL	n=3	n=4	n=5	n=6	n=7	n=8	n=10	n=12	n=15	n=20
77	0.87	0.81	0.79	0.78	0.77	0.76	0.76	0.75	0.75	0.75
76	0.84	0.78	0.76	0.75	0.74	0.73	0.72	0.72	0.72	0.72
75	0.82	0.75	0.73	0.72	0.71	0.70	0.69	0.69	0.69	0.68
74	0.79	0.72	0.70	0.68	0.67	0.67	0.66	0.66	0.66	0.65
73	0.77	0.69	0.67	0.65	0.64	0.64	0.62	0.62	0.62	0.62
72	0.74	0.66	0.64	0.62	0.61	0.61	0.60	0.59	0.59	0.59
71	0.71	0.63	0.60	0.59	0.58	0.58	0.57	0.56	0.56	0.56
70	0.68	0.60	0.58	0.56	0.55	0.55	0.54	0.54	0.54	0.53
69	0.65	0.57	0.55	0.54	0.53	0.52	0.51	0.51	0.51	0.50
68	0.62	0.54	0.52	0.51	0.50	0.50	0.48	0.48	0.48	0.48
67	0.59	0.51	0.49	0.48	0.47	0.47	0.46	0.45	0.45	0.45
66	0.56	0.48	0.46	0.45	0.44	0.44	0.43	0.42	0.42	0.42
65	0.53	0.45	0.43	0.42	0.41	0.41	0.40	0.40	0.40	0.39
64	0.49	0.42	0.40	0.39	0.38	0.38	0.37	0.37	0.37	0.37
63	0.46	0.39	0.37	0.36	0.35	0.35	0.35	0.34	0.34	0.34
62	0.43	0.36	0.34	0.33	0.33	0.33	0.32	0.31	0.31	0.31
61	0.39	0.33	0.31	0.30	0.30	0.30	0.29	0.29	0.29	0.28
60	0.36	0.30	0.28	0.27	0.26	0.26	0.25	0.25	0.25	0.25
59	0.32	0.27	0.25	0.25	0.24	0.24	0.24	0.23	0.23	0.23
58	0.29	0.24	0.23	0.22	0.21	0.21	0.21	0.21	0.21	0.20
57	0.25	0.21	0.20	0.19	0.19	0.19	0.18	0.18	0.18	0.18
56	0.22	0.18	0.17	0.16	0.16	0.16	0.16	0.16	0.15	0.15
55	0.18	0.15	0.14	0.14	0.13	0.13	0.13	0.13	0.13	0.13
54	0.14	0.12	0.11	0.11	0.11	0.11	0.10	0.10	0.10	0.10
53	0.11	0.09	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
52	0.07	0.06	0.06	0.05	0.05	0.05	0.05	0.05	0.05	0.05
51	0.04	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 5

Definitions, Abbreviations, and Formulas for Acceptance	
Term	Explanation
Target Value (TV)	The target values for gradation and asphalt binder content are given in the CONTRACTOR's mix design. The target value for density is 94.0 percent of maximum (Rice) density.
Average (AVE)	The sum of the lot's test results for a measured characteristic divided by the number of test results, the arithmetic mean.
Standard Deviation (s)	The square root of the value formed by summing the squared difference between the individual test results of a measured characteristic and AVE, divided by the number of test results minus one. This statement does not limit the methods of calculations of s; other methods that obtain the same value may be used.
Upper Limit (UL)	The value above the TV of each measured characteristic that defines the upper limit of acceptable production. (Table 3)
Lower Limit (LL)	The value below the TV of each measured characteristic that defines the lower limit of acceptable production (Table 3)
Upper Quality Index (QU)	$QU = (UL - AVE)/s$
Lower Quality Index (QL)	$QL = (AVE - LL)/s$
Percentage of Lot Within UL (PU)	Determined by entering Table 4 with QU.
Percentage of Lot Within LL (PL)	Determined by entering Table 4 with QL.
Total Percentage of Lot (PL) Within UL and LL (PT)	$PT = (PU + PL) - 100$
Incentive/Disincentive	Determined by entering Table 1 with PT or PL.

Values for AVE, s, QU, and QL will be calculated to a minimum two decimal place accuracy that will be carried through all further calculations. Rounding to lower accuracy is not allowed.

1.7 LABORATORY CORRELATION

- A. Perform the following to be eligible for dispute resolution:
 - 1. Perform split-sample, paired *t*-testing based on project quality control testing using UDOT LQP qualified lab.
 - a. Perform split-sample, paired-*t* analysis on all mix acceptance tests related to volumetric properties and the following background testing:
 - 1) Maximum Theoretical Specific Gravity of Mix (G_{mm}), AASHTO T 209

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- 2) Bulk Specific Gravity of Mix (G_{mb}), AASHTO T 166
 - 3) Bulk Specific Gravity of Coarse Aggregates (G_{sb}), AASHTO T 85
 - b. Continue until attaining successful paired t-test results, meeting $\alpha = 0.05$, for a minimum of two consecutive production days (UDOT Materials Manual of Instruction, Part 8, Appendix C).
 - c. The COR may require that all QC testing data be received before disclosure of the QA testing results. This applies to paired t-test and all subsequent QA/QC testing data.
- B. Submit a detailed report showing tabular summaries of daily test data, paired *t*-testing calculations and any corrections made to account for failed comparisons.
 - C. Submit summary before submitting engineering analysis for dispute resolution.

1.8 DISPUTE RESOLUTION

- A. Refer to Section 01456 when disputing the validity of the Department's acceptance tests.

PART 2 PRODUCTS

2.1 ASPHALT BINDER

- A. Project Specific Surfacing Requirements – Refer to Section 02742S.
- B. Asphalt Material – Refer to Section 02745 and Quality Management Plan 509: Asphalt Binder
- C. Use the minimum asphalt binder content specified in Table 10 unless called out differently in 02742S for the project.

2.2 AGGREGATE

- A. Use crusher processed virgin aggregate material consisting of crushed stone, gravel, or slag.
- B. Refer to Table 6, to determine the suitability of the aggregate.
 - 1. Coarse aggregates:
 - a. Retained on No. 4 sieve. AASHTO T 27
 - 2. Fine aggregates:
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- a. Clean, hard grained, and angular.
- b. Passing the No. 4 sieve. AASHTO T 27

Table 6

Test Method	Test No.	Specification
One Fractured Face	AASHTO T 335	100% min.
Two Fractured Face	AASHTO T 335	90% min.
Fine Aggregate Angularity	AASHTO T 304	45 min.
Flakiness Index	UDOT MOI (Based on 3/8 inch and above)	25% max.
L.A. Wear	AASHTO T 96	28% max.
Sand Equivalent	AASHTO T 176 (Pre-wet method)	60 min.
Plasticity Index (Does not apply to Mineral Filler)	AASHTO T 89 and T 90	0 max.
Unit Weight	AASHTO T 19	75 lb/cu. ft. min.
Polishing*	AASHTO T 278 and T 279	31 min.
Soundness (sodium sulfate)	AASHTO T 104	10% max. loss with five cycles
Clay Lumps and Friable Particles	AASHTO T 112	2% max
Natural Fines	N/A	0 max.
* The Department has the right to waive this requirement if the aggregates have proven acceptable through successful past performance as determined by the Engineer.		

- C. Meet the gradation requirements in Table 7. (AASHTO T 11, AASHTO T 27)

Table 7

Control Sieve Size	½ inch	¾ inch
¾ inch	100	100
½ inch	90 - 100	100
¾ inch	45 - 78	90 - 100
No. 4	20 - 28	26 - 50
No. 8	16 - 24	20 - 28
No. 16	13 - 21	13 - 21
No. 30	12 - 18	12 - 18
No. 50	12 - 15	12 - 15
No. 200	8 - 10	8 - 10

2.3 ADDITIVES / STABILIZERS

- A. Hydrated Lime: Meet the requirements of Section 02746.
- B. Stabilizing additive: Made from virgin basalt, diabase, slag, or cellulose treated with a cationic sizing agent to enhance disbursement of the fiber as well as increase adhesion of the fiber surface with the asphalt binder and control drain-down. Meet AASHTO M 325.
1. Mineral Fiber
 - a. Dosage rate between 0.3 percent to 0.6 percent, by weight of the total mix.
 - b. Average fiber length 0.25 inches, maximum
 - c. Average Fiber thickness 0.0002 inches, maximum
 - d. Shot content (ASTM C 612)
Passing No. 60 sieve 90 – 100 percent
Passing No. 230 sieve 65 – 100 percent
 2. Cellulose Fiber
 - a. Dosage rate for cellulose is 0.2 percent to 0.4 percent by

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- weight of total mix.
- b. Using Alpine sieve analysis, fiber length of 0.25 inches maximum passing the No. 100 sieve 70 percent (± 10 percent).
- c. Using a mesh screen analysis, fibers will pass
No. 20 sieve 85 percent (± 10 percent)
No. 40 sieve 65 percent (± 10 percent)
No. 140 sieve 30 percent ± 0 percent)
- d. Ash content will be 18 percent (± 5 percent)
- e. PH will be 7.5 (± 1.0)
- f. Oil absorption will be 5.0 (± 1.0 percent)
- g. Moisture content will be <5 percent by weight of cellulose

C. Mineral Filler:

Consists of finely divided mineral matter such as rock dust, slag dust, hydrated lime, hydraulic cement, fly ash, or other suitable mineral matter. Free flowing and free of lumps.

1. Meet the following requirements:

Table 8

Sieve Size	Percent Passing
No. 30	100
No. 50	95 – 100
No. 200	55 – 100
No. 450	40 max.

2. No organic impurities
3. Plasticity Index less than 4 (not appropriate for hydrated lime and hydraulic cement)

2.4 JOB-MIX DESIGN

A. Perform Stone Matrix Asphalt Mix Design according to UDOT Manual of Instruction 962.

1. Submit for verification at least 10 working days before beginning paving.
2. Do not begin paving until verification is complete.

B. Meet the following requirements:

Table 9

Mix Design Compaction Parameters	
Design Gyration	% of G_{mm}^*
100	96.5

* G_{mm} : Maximum theoretical specific gravity of mix.

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Table 10

Minimum Asphalt Binder Content	
Combined Aggregate Bulk Specific Gravity Including Lime G_{sb}	Minimum Asphalt Binder Content %*
2.375 - 2.424	6.8
2.425 - 2.474	6.7
2.475 - 2.524	6.6
2.525 - 2.574	6.5
2.575 - 2.624	6.3
2.625 - 2.674	6.2
2.675 - 2.724	6.1
> 2.724	6.0
* Percent of total mix.	

Table 11

Mix Design Requirements	
SMA design mixing and compaction temperatures	Provided by the Engineer
Voids in Mineral Aggregate (VMA) AASHTO R 46, using G_{sb} . Equation based on percent of total mix.	17.0% min.
Voids In Course Aggregate (Stone Matrix Asphalt Mix Design)	$VCA_{MIX} < VCA_{DRC}$
Hamburg Wheel Tracker	< 10.00 mm at 20,000 Cycles.
Draindown (AASHTO T 305)	0.30 max.

- C. Mortar must meet the following. Refer to UDOT Materials Manual of Instruction 962.
1. Unaged DSR $G^*/\sin \delta \geq 5$ kPa
 2. RTFO aged DSR $G^*/\sin \delta \geq 11$ kPa
 3. PAV aged BBR Stiffness ≤ 1500 Mpa

2.5 CONTRACTOR INITIATED CHANGES IN STONE MATRIX ASPHALT MIX DESIGN

- A. Submit requests in writing to the COR at least 12 hours before incorporating changes into production.
- B. Submit a field volumetric mix design for all target changes.
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1. Include documentation supporting correlation between suggested target changes and mix design volumetric requirements. Government acceptance or Contractor QC testing data is acceptable.
 2. Field volumetric mix design verification consists of three sets of two gyratory specimens run at the new target gradation, asphalt binder content, or both.
 - a. Previous acceptance tests are acceptable for field verification.
 3. The COR provides written concurrence of the verified field volumetric mix design if the field volumetric mix design meets the volumetric requirements.
 4. Submit a new laboratory volumetric mix design from a laboratory qualified by UDOT Central Materials if the field volumetric mix verification does not meet the volumetric requirements.
 - a. Allow at least 7 working days for verification.
 5. The Government may allow up to two minor target changes per project without penalty to the Contractor.
 - a. The Government charges \$1,000 for each additional minor target change.
 6. The Government may perform up to two independent volumetric mix design verifications at no cost to the Contractor.
 - a. The Government charges \$3,000 for each additional laboratory or field verification required including all laboratory or field volumetric mix design verifications required due to contractor initiated target changes.
- C. Submit a new laboratory volumetric mix design if changes occur in the aggregate source, asphalt binder source, or grade.
- D. Do not make changes to production mix until request is reviewed and verified by the COR.

PART 3 EXECUTION

3.1 SMA

- A. Dry aggregate to an average moisture content of not more than 0.2 percent by weight.
1. May be verified by AASHTO T 255.
 2. Adjust burners to avoid damage or soot contamination of the aggregate.

- B. Treat aggregate with hydrated lime. Refer to Section 02746.
 - 1. Use method A or B.
 - 2. The Government applies a deduction for mix produced by a non- certified supplier to cover the costs of inspection.
 - 3. The deduction is applied according to the UDOT Quality Management Plan 514 Hot-Mix Asphalt.
- C. Coat with asphalt binder 100 percent of the particles passing and 98 percent of the particles retained on the No. 4 sieve.
 - 1. May be verified by AASHTO T 195.
 - 2. Discontinue operation and make necessary corrections if material is not properly coated.
- D. Maintain temperature of the SMA between identified compaction limits as defined on Volumetric Mix Design Verification Letter.
 - 1. The Government rejects materials heated over the identified limits.
 - 2. Remove all material rejected by the Government for overheating.

3.2 SMA PLANT

- A. Provide the following:
 - 1. Positive means to determine the moisture content of aggregate on a daily basis.
 - 2. Positive means to sample all material components.
 - 3. Sensors to measure the temperature of the SMA at discharge.
 - 4. The ability to maintain discharge temperature of the mix according to the mix design.
- B. Asphalt Binder Storage Tanks:
 - 1. Provide positive means of determining the quantity of material in the tank at any time.
 - 2. Provide positive means of sampling the asphalt binder from the tanks.
- C. Fiber Supply System:
 - 1. Provide a separate proportioning device interlocked with the aggregate feed or weigh system to maintain correct proportions and uniform distribution for all rates of production and batch sizes.
 - 2. Provide flow indicators or sensing devices interlocked with plant controls.
 - 3. Uniformly distribute fibers in aggregate before injecting the asphalt into the mixer. (8 to 12 seconds)
 - 4. Do not allow the fiber to become entrained in the exhaust system of the plant.

- D. Mineral Filler Supply System:
 - 1. Provide a separate proportioning device interlocked with the aggregate feed or weigh system to maintain correct proportions and uniform distribution for all rates of production and batch sizes.
 - 2. Provide flow indicators or sensing devices interlocked with plant controls.
 - 3. Uniformly distribute filler in aggregate before injecting asphalt into the mixer.
 - 4. Do not allow the filler to become entrained in the exhaust system of the plant.

3.3 CEASE PRODUCTION

- A. Cease production when any two out of three consecutive lots meet one of the following criteria:
 - 1. A net disincentive.
 - 2. Air voids averaged for each lot are less than 2.5 or greater than 4.5 percent.
- B. Refer to Table 2 of this section.
- C. Submit a corrective action plan to the COR before production continues indicating the changes in production procedures that will be implemented to correct the deficiencies.
 - 1. Address the specific issues contributing to the cease production directive.
 - 2. The COR must approve the revised plan before production continues.
- D. The COR may require a new mix design.

3.4 SURFACE PREPARATION

- A. Locate, reference, and protect all utility covers, monuments, curb and gutter, and other components affected by the paving operations.
- B. Remove all moisture, dirt, sand, leaves, and other objectionable material from the prepared surface before placing the mix.
- C. Allow sufficient cure time for prime coat/tack coat before placing SMA. Refer to Section 02748.

3.5 SURFACE PLACEMENT

- A. Provide a 3:1 (horizontal to vertical) sloped edge adjacent to the next lane to be paved when full-width or echelon paving is impractical and more than one pass is required.
- B. Construct the longitudinal joint to within 6 inches of the lane lines or at the center of the lane at the direction of the COR, but never in a wheel path.
 - 1. Core and test all longitudinal joints for compaction according to the specification if the lift is 2 or more inches thick.
 - 2. Verify all edges of the adjacent areas to through lanes have straight and uniform longitudinal lines and neat vertical edges.
 - 3. Fill core holes with SMA mix and compact.
- C. Adjust the production of the mixing plant and material delivery until a steady paver speed is maintained.
- D. Do not allow construction vehicles, general traffic, or rollers to pass over the uncompacted end or edge of freshly placed mix until the mat temperature drops to a point where damage or differential compaction will not occur.
- E. Taper the end of a course subjected to traffic at approximately 50:1 (horizontal to vertical).
 - 1. Make a transverse joint by saw or wheel cutting and removing the portion of the pass that contains the tapered end.
 - 2. Tack the contact surfaces before fresh mix is placed against the compacted mix.
- F. Use a Material Transfer Vehicle (MTV) to apply all courses of SMA. Use an MTV that internally performs additional mixing of the SMA mix and then deposits material into the paver at a uniform temperature and consistency.
 - 1. Use other approved means to deposit material into the paver when an MTV is impractical, placements such as utility work, traffic signals, detours, lane leveling, and driveways, side street tie-ins, other hand work, or small projects with plan quantities less than 500 tons.

3.6 COMPACTION

- A. Use a small compactor or vibratory roller in addition to normal rolling at structures.
- B. Operate in a transverse direction next to the back wall and approach slab.
- C. Use a 9 ton (minimum) roller.

- D. Roll surface immediately after placement staying as close as possible to the lay-down machine and assuring proper mix design placement temperatures.
 - 1. Minimize the use of vibratory rollers.
- E. Do not use full Pneumatic tire rollers.
- F. Discontinue vibration if aggregate breakdown occurs or if bleeding occurs.

3.7 LIMITATIONS

- A. Do not place SMA on frozen base or subbase.
- B. Use a release agent that does not dissolve asphalt and is satisfactory to the Engineer for all equipment and hand tools used to mix, haul, and place the SMA.
- C. Do not place SMA during adverse climatic conditions such as precipitation or when roadway surface is icy or wet.
- D. Place SMA from April 15, to October 15, and when the air temperature in the shade and the roadway surface temperature are above 50 degrees F.
 - 1. The Government determines if it is feasible to place SMA outside these dates and temperature.
 - 2. Obtain authorization from the COR before paving outside these limits.

END OF SECTION

SECTION 02745

ASPHALT MATERIAL

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Asphalt materials binders, emulsions, and crack sealant for pavements.

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES

- A. AASHTO M 81: Cutback Asphalt (Rapid-Curing Type)
- B. AASHTO M 82: Cutback Asphalt (Medium-Curing Type)
- C. AASHTO M 140: Emulsified Asphalt
- D. AASHTO M 208: Cationic Emulsified Asphalt
- E. AASHTO M 226: Viscosity Graded Asphalt Cement
- F. AASHTO M 282: Joint Sealants, Hot-Poured, Elastomeric-Type, for Portland Cement Concrete Pavements
- G. AASHTO M 320: Performance Graded Asphalt Binder
- H. AASHTO R 28: Accelerated Aging of Asphalt Binder Using a Pressurized Aging Vessel (PAV)
- I. AASHTO T 44: Solubility of Bituminous Materials
- J. AASHTO T 48: Flash and Fire Points by Cleveland Open Cup
- K. AASHTO T 49: Penetration of Bituminous Materials
- L. AASHTO T 50: Float Test for Bituminous Materials
- M. AASHTO T 51: Ductility of Bituminous Materials
- N. AASHTO T 59: Emulsified Asphalt

- O. AASHTO T 201: Kinematic Viscosity of Asphalts (Bitumens)
- P. AASHTO T 228: Specific Gravity of Semi-Solid Asphalt Materials
- Q. AASHTO T 240: Effect of Heat and Air on a Moving Film of Asphalt Binder (Rolling Thin-Film Oven Test)
- R. AASHTO T 300: Force Ductility of Asphalt Materials
- S. AASHTO T 301: Elastic Recovery Test of Asphalt Materials by Means of a Ductilometer
- T. AASHTO T 313: Determining the Flexural Creep Stiffness of Asphalt Binder Using the Bending Beam Rheometer (BBR)
- U. AASHTO T 315: Determining the Rheological Properties of Asphalt Binder Using a Dynamic Shear Rheometer (DSR)
- V. AASHTO T 316: Viscosity Determination of Asphalt Binder Using Rotational Viscometer
- W. ASTM D 2006: Method for Characteristic Groups in Rubber Extender and Processing Oils by the Precipitation Method.
- X. ASTM D 2007: Characteristic Groups in Rubber Extender and Processing Oils and Other Petroleum Derived Oils by the Clay Gel Absorption Chromatographic Method
- Y. ASTM D 2026: Cutback Asphalt (Slow Curing Type)
- Z. ASTM D 4402: Viscosity Determination of Asphalt at Elevated Temperatures Using a Rotational Viscometer
- AA. ASTM D 5329: Sealants and Fillers, Hot-Applied, For Joints and Cracks in Asphaltic and Portland Cement Concrete Pavements
- BB. ASTM D 5801: Toughness and Tenacity of Bituminous Materials
- CC. ASTM D 7643: Delta Tc
- DD. California Test Methods
- EE. UDOT Materials Manual of Instruction
- FF. UDOT Minimum Sampling and Testing Requirements (MS&TR)

1.4 DEFINITIONS

- A. Binder lot – Specific to one approved binder, for one mix and for one particular project. A binder lot starts on a Saturday and ends on a Friday.
- B. Compliance Limit – The limit for acceptance without price reductions, accounting for testing variability, for the listed properties.
- C. Composite Price Reduction – The sum of price reductions for all individual properties associated with a single asphalt binder sample.
- D. Rejection Limit – The limit for acceptance with price reductions. Material not meeting these limits will be rejected and removed and replaced.

1.5 SUBMITTALS

- A. A vendor prepared bill of lading for information showing the following for each material shipment:
 - 1. Type and grade of material
 - 2. Type and amount of additives used, if applicable
 - 3. Destination
 - 4. Consignee's name
 - 5. Date of Shipment
 - 6. Railroad car or truck identification
 - 7. Project number
 - 8. Loading temperature
 - 9. Net weight in tons or net gallons corrected to 60 degrees F, when requested
 - 10. Specific gravity
 - 11. Bill of lading number
 - 12. Manufacturer of asphalt material

1.6 ACCEPTANCE

- A. The Government accepts, rejects, or applies price reductions to all mix lots for performance-graded asphalt binder (PGAB) lots according to the compliance and rejection limits of Table 1, the Minimum Sampling and Testing Requirements, UDOT Quality Management Plan 509, and the associated mix specification.
 - 1. Rejected mix will be covered according to the associated mix specification including removal or if left in place the associated disincentive dollar amount for the rejected mix.

2. For material not meeting the compliance limit, but within the rejection limit, the price reduction applied to the mix pay item price is calculated as follows:

$$\text{Price Reduction (percent)} = 25 \times \frac{(\text{Compliance Limit} - \text{Test Result})}{(\text{Compliance Limit} - \text{Rejection Limit})}$$

3. The composite price reduction will be calculated by summing the price reduction for each parameter if more than one parameter lies outside specification limits.
 - a. The mix material will be rejected if the composite price reduction exceeds 25 percent.
- B. The maximum composite price reduction from the tested samples is applied to the mix in the binder lot if multiple samples from the binder lot are tested.
- C. Adjacent samples in the preceding, current, or subsequent lots may be tested to assess the extent and limit the amount of mix to be rejected if a binder lot is rejected.

Table 1

Specification, Compliance, and Rejection Limits for Performance-Graded Asphalt Binders (Applied to the Mix)			
Property	Specification Limit	Compliance Limit	Rejection Limit
G*/sinδ of the original PGAB at high grade temp, (kPa) Rule of 86 or lower	1.00 Min	0.84 Min	0.70 Min
G* of the original PGAB at high grade temp, (kPa)	1.30 Min	1.25 Min	1.11 Min
δ (phase angle) of the original PGAB at high grade temperature, (degrees C). Rule of 92 Binders Rule of 98 Binders	74.0 Max 71.0 Max	75.0 Max 72.0 Max	77.0 Max 74.0 Max
G*/sinδ of the RTFO Residue, (kPa)	2.20 Min	1.87 Min	1.53 Min
Elastic Recovery of RTFO Residue, (%) Rule of 92 Binders Rule of 98 Binders Rule of 104 and 110 Binders	80 Min 85 Min 90 Min	75 Min 80 Min 85 Min	65 Min 70 Min 75 Min
G*/sinδ of the PAV Residue, (kPa)	5000 Max	5250 Max	5700 Max
Stiffness of the PAV Residue at the specified low grade temperature +10°C, (MPa)	300 Max	311 Max	600 Max
Slope (m-value) of the Creep Curve at the specified low grade temperature +10°C	0.300 Min	0.295 Min	0.266 Min
Delta Tc of PAV Residue from additional BBR test (use two BBR tests). Rule of 92, 98, 104 and 110 Binders	-1.0 Min.	-2.0 Min.	-3.0 Min.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Each shipment of asphalt material must:
 - 1. Be uniform in appearance and consistency.
 - 2. Show no foaming when heated to the specified loading temperature.
- B. Do not supply shipments contaminated with other asphalt grades or materials.

PART 2 PRODUCTS

2.1 PERFORMANCE GRADED ASPHALT BINDER (PGAB)

- A. Use a qualified supplier according to the UDOT Quality Management Plan 509, Asphalt Binder.
- B. Refer to AASHTO M 320 for all PGABs having algebraic differences less than 92 degrees between the high and low design temperatures.
- C. Refer to AASHTO M 320 modified by Tables 2, 3, 4, 5, 6, 7, 8 and 9 for all PGABs having algebraic differences equal to or greater than 92 degrees between the high and low design temperatures.

Table 2

PG58-34

Original Binder		
Dynamic Shear Rheometer, AASHTO T 315	@ 58° C, G*, kPa	1.30 Min.
	@ 58° C, phase angle, degrees	74.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135° C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
RTFO Residue, AASHTO T 240		
Dynamic Shear Rheometer, AASHTO T 315	@ 58° C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	80 Min.
PAV Residue, 20 hours, 2.10 MPa, 100° C, AASHTO R 28		
Dynamic Shear Rheometer, AASHTO T 315	@ 16° C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -24° C, S, MPa	300 Max. 150 Min.
	@ -24° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -30° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

Table 3

PG64-28		
<u>Original Binder</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 64° C, G*, kPa	1.30 Min.
	@ 64° C, phase angle, degrees	74.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135° C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
<u>RTFO Residue, AASHTO T 240</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 64° C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	80 Min.
<u>PAV Residue, 20 hours, 2.10 MPa, 100 °C, AASHTO R 28</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 22° C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -18° C, S, MPa	300 Max. 150 Min.
	@ -18° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -24° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

Table 4

PG64-34		
<u>Original Binder</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 64° C, G*, kPa	1.30 Min.
	@ 64° C, phase angle, degrees	71.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135° C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
<u>RTFO Residue, AASHTO T-240</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 64° C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	85 Min.
<u>PAV Residue, 20 hours, 2.10 MPa, 100 °C, AASHTO R 28</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 19° C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -24° C, S, MPa	300 Max. 150 Min.
	@ -24° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -30° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

Table 5

PG70-22		
<u>Original Binder</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 70° C, G*, kPa	1.30 Min.
	@ 70° C, phase angle, degrees	74.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135° C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
<u>RTFO Residue, AASHTO T 240</u>		
Dynamic Shear Rheometer, AASHTO T 315	@70°C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	80 Min.
<u>PAV Residue, 20 hours, 2.10 MPa, 100 °C, AASHTO R 28</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 28° C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -12° C, S, MPa	300 Max. 150 Min.
	@ -12° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -18° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

Table 6

PG70-28		
<u>Original Binder</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 70° C, G*, kPa	1.30 Min.
	@ 70° C, phase angle, degrees	71.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135° C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
<u>RTFO Residue, AASHTO T 240</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 70° C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	85 Min.
<u>PAV Residue, 20 hours, 2.10 MPa, 100 °C, AASHTO R 28</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 25° C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -18° C, S, MPa	300 Max. 150 Min.
	@ -18° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -24° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

Table 7

PG70-34

<u>Original Binder</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 70° C, G*, kPa	1.30 Min.
	@ 70° C, phase angle, degrees	71.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135 °C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
<u>RTFO Residue, AASHTO T 240</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 70° C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	90 Min.
<u>PAV Residue, 20 hours, 2.10 MPa, 100 °C, AASHTO R 28</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 22° C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -24° C, S, MPa	300 Max.
		150 Min.
	@ -24° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -30° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

Table 8

PG76-22

<u>Original Binder</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 76° C, G*, kPa	1.30 Min.
	@ 76° C, phase angle, degrees	71.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135° C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
<u>RTFO Residue, AASHTO T 240</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 76° C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	85 Min.
<u>PAV Residue, 20 hours, 2.10 MPa, 100 °C, AASHTO R 28</u>		
Dynamic Shear Rheometer, AASHTO T 315	@ 31°C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -12° C, S, MPa	300 Max.
		150 Min.
	@ -12° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -18° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

Table 9

PG76-28

Original Binder		
Dynamic Shear Rheometer, AASHTO T 315	@ 76° C, G*, kPa	1.30 Min.
	@ 76° C, phase angle, degrees	71.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135° C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
RTFO Residue, AASHTO T 240		
Dynamic Shear Rheometer, AASHTO T 315	@ 76° C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	90 Min.
PAV Residue, 20 hours, 2.10 MPa, 100 °C, AASHTO R 28		
Dynamic Shear Rheometer, AASHTO T 315	@ 28° C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -18° C, S, MPa	300 Max. 150 Min.
	@ -18° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -24° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

Table 10

PG76-34 Highly Modified

Original Binder		
Dynamic Shear Rheometer, AASHTO T 315	@ 76° C, G*, kPa	1.30 Min.
	@ 76° C, phase angle, degrees	70.0 Max.
Rotational Viscometer, AASHTO T 316	@ 135° C, Pa.s	3 Max.
Flash Point, AASHTO T 48	°C	260 Min.
RTFO Residue, AASHTO T 240		
Dynamic Shear Rheometer, AASHTO T 315	@ 76° C, G*/sinδ, kPa	2.20 Min.
Elastic Recovery, AASHTO T 301 mod (a)	%	90 Min.
PAV Residue, 20 hours, 2.10 MPa, 100 °C, AASHTO R 28		
Dynamic Shear Rheometer, AASHTO T 315	@ 25° C, kPa	5,000 Max.
Bending Beam Rheometer, AASHTO T 313	@ -24° C, S, MPa	300 Max. 150 Min.
	@ -24° C, m-value	0.300 Min.
Delta Tc from additional BBR test, ASTM D 7643	@ -30° C	-1.0 Min.
(a) Modify paragraph 4.5 as follows: Stop the ductilometer after 20 cm has been reached and within 2 seconds. Sever the specimen at its center with a pair of scissors.		

2.2 ASPHALTIC CEMENT, LIQUID ASPHALTS, AND REJUVENATING AGENTS

- A. Refer to AASHTO M 226, Table 2 with the following modifications:
1. Delete and replace ductility at 77 degrees F (25 degrees C) with ductility at 39.2 degrees F (4 degrees C) using the values specified below.

AC - 2.5
50+

AC - 5
25+

AC - 10
15+

AC - 20
5+

- B. Cationic and Anionic Emulsified Asphalt
1. All standard Slow Setting (SS, CSS), Quick Setting (QS, CQS) Medium Setting (MS, CMS), and Rapid Setting (RS, CRS) grades including all High-Float designations (HF).
 2. Supply under the Approved Supplier Certification System (ASC).
 3. Refer to and meet AASHTO M 208 and M 140.
- C. Meet the requirements of one of these tables:
1. Table 10 – Cationic Rapid Setting Emulsified Polymerized Asphalt (CRS-2P)
 2. Table 11 – Latex Modified Cationic Rapid Setting Emulsified Asphalt (LMCRS-2)
 3. Table 12 – Cationic Medium Setting Emulsified Asphalt (CMS-2S)
 4. Table 13 – High Float Medium Setting Emulsified Asphalt (HFMS-2)
 5. Table 14– High Float Medium Setting Emulsified Polymerized Asphalt (HFMS-2P)
 6. Table 15 – High Float Medium Setting Emulsified Polymerized Asphalt (HFMS-2SP)
 7. Table 16 – High Float Rapid Setting Emulsified Polymerized Asphalt (HFRS-2P).
 8. Table 17 –Cationic Rapid Setting Emulsified Asphalt (CRS-2A, B)
- D. Curing Cut-Back Asphalt
1. Refer to specification ASTM D 2026 for slow curing (SC).
 2. Refer to specification AASHTO M 82 for medium curing (MC).
 3. Refer to specification AASHTO M 81 for rapid curing (RC).
- E. Meet the requirements for Emulsified Asphalt Pavement Rejuvenating Agent:
1. Table 18 – Type A
 2. Table 19 – Type B
 3. Table 20 – Type B Modified
 4. Table 21– Type C
 5. Table 22 – Type D

Table 11

Cationic Rapid Setting Emulsified Polymerized Asphalt (CRS-2P)			
Tests	Test Method	Min.	Max.
Emulsion			
Viscosity , SF, 140° F (60° C), s (Project-site Acceptance/Rejection Limits)	AASHTO T 59	100	400
Settlement (a) 5 days, percent	AASHTO T 59		5
Storage Stability Test (b) 1 d, 24 h, percent	AASHTO T 59		
Demulsibility (c) 35 ml, 0.8% sodium dioctyl Sulfosuccinate, percent	AASHTO T 59	40	
Particle Charge Test	AASHTO T 59	Positive	
Sieve Test, percent	AASHTO T 59		0.10
Distillation			
Oil distillate, by volume of emulsion, percent			0
Residue (d), percent		68	
Residue from Distillation Test			
Penetration, 77° F (25° C), 100 g, 5 s, dmm	AASHTO T 49	80	150
Ductility, 39.2° F (4° C), 5 cm/min, cm	AASHTO T 51	35	
Toughness, lb-in	ASTM D 5801	75	
Tenacity, lb-in	ASTM D 5801	50	
Solubility in trichloroethylene, percent	AASHTO T 44	97.5	
(a) The test requirement for settlement may be waived when the emulsified asphalt is used in less than a five-day time unless the purchaser requires the test. (b) The 24-hour (1-day) storage stability test may be used instead of the five day settlement test. (c) The demulsibility test is made within 30 days from date of shipment. (d) Distillation is determined by AASHTO T 59 with modifications to include a $350 \pm 5^\circ \text{F}$ ($177 \pm 3^\circ \text{C}$) maximum temperature to be held for 15 minutes. Modify the asphalt cement before emulsification.			

Table 12

Latex Modified Cationic Rapid Setting Emulsified Asphalt (LMCRS-2)			
Tests	Test Method	Min.	Max.
Emulsion			
Viscosity, SF, 122° F (50° C), s (Project Site Acceptance/Rejection Limits)	AASHTO T 59	140	400
Settlement (a) 5 days, percent	AASHTO T 59		5
Storage Stability Test (b) 1 d, 24 h, percent	AASHTO T 59		1
Demulsibility (c) 35 ml, 0.8% sodium Dioctyl Sulfosuccinate, percent	AASHTO T 59	40	
Particle Charge Test	AASHTO T 59	Positive	
Sieve Test, percent	AASHTO T 59		0.3
Distillation			
Oil distillate, by volume of emulsion, percent			0
Residue (d), percent		65	
Residue from Distillation Test			
Penetration, 77° F (25° C), 100 g, 5 s, dmm	AASHTO T 49	40	200
Torsional Recovery (e)		18	
(a) The test requirement for settlement may be waived when the emulsified asphalt is used in less than a five-day time unless the purchaser requires the test. (b) May use the 24-hour (1-day) storage stability test instead of the five-day settlement test. (c) Make the demulsibility test within 30 days from date of shipment. (d) Determine distillation by AASHTO T 59, with modifications to include a $350 \pm 5^{\circ}\text{F}$ ($177 \pm 3^{\circ}\text{C}$) maximum temperature to be held for 15 minutes. (e) CA 332 (California Test Method)			
Co-mill latex and asphalt during emulsification			

Table 13

Cationic Medium Setting Emulsified Asphalt (CMS-2S)		
Tests	Test Method	Specification
Emulsion		
Viscosity, SF, 122° F (50° C), s	AASHTO T 59	50 - 450
Percent residue	AASHTO T 59	60 min
Storage Stability Test, 1d, 24h, percent	AASHTO T 59	1 max
Sieve, percent	AASHTO T 59	0.10 max
Particle charge	AASHTO T 59	Positive
Oil Distillate, percent by volume of emulsion	AASHTO T 59	5-15
Residue		
Penetration, 77° F (25° C), 100g, 5 sec, dmm	AASHTO T 59	100-250
Solubility, percent	AASHTO T 59	97.5 min.

Table 14

High Float Medium Setting Emulsified Asphalt (HFMS-2)			
Tests	Test Method	Min.	Max.
Emulsion			
Viscosity, SF, 122° F (50° C), s (Project Site Acceptance/Rejection Limits	AASHTO T 59	70	300
Storage Stability Test, 1d, 24 h, percent	AASHTO T 59		1.0
Sieve Test , percent	AASHTO T 59		0.1
Distillation			
Oil Distillate, by volume of emulsion, percent	AASHTO T 59	NA	NA
Residue, percent	AASHTO T 59	65	
Residue from Distillation Test			
Penetration, 77° F (25° C), 100g, 5 s, dmm	AASHTO T 49	50	200
Float Test, 140° F (60° C), s	AASHTO T 50	1,200	
Solubility in Trichloroethylene, percent	AASHTO T 44	97.5	
Ductility, 77° F (25° C) 5cm/min, cm	AASHTO T 51	40	

Table 15

High Float Medium Setting Emulsified Polymerized Asphalt (HFMS-2P) (a)			
Tests	Test method	Min.	Max.
Emulsion			
Viscosity, SF, 122° F (50° C), s (Project Site Acceptance/Rejection Limits)	AASHTO T 59	100	450
Storage Stability Test, 1 d, 24 h, percent	AASHTO T 59		1.0
Sieve Test, percent	AASHTO T 59		0.1
Distillation			
Oil distillate, by volume of emulsion, percent	AASHTO T 59		7
Residue (b), percent	AASHTO T 59	65	
Residue from Distillation Test			
Penetration, 77° F (25° C), 100 g, 5 s, dmm	AASHTO T 49	70	300
Float Test, 140° F (60° C), s	AASHTO T 50	1,200	
Solubility in trichloroethylene, percent	AASHTO T 44	97.5	
Elastic Recovery, 77° F (25° C), percent	AASHTO T 301	50	
(a) Supply an HFMS-2P (anionic, polymerized, high-float) as an emulsified blend of polymerized asphalt cement, water, and emulsifiers. Polymerize the asphalt cement with at least 3.0% polymer by weight of the asphalt cement before emulsification. The emulsion must be smooth and homogeneous throughout with no white, milky separation, pumpable, and suitable for application through a distributor after standing undisturbed for at least 24 hours. (b) Determine the distillation by AASHTO T 59, with modifications to include a 350 ± 5° F (177 ± 3° C) maximum temperature to be held for 15 minutes.			

Table 16

High Float Medium Setting Emulsified Polymerized Asphalt (HFMS-2SP) (a)			
Tests	Test method	Min.	Max.
Emulsion			
Viscosity, SF, 122° F (50° C), s (Project Site Acceptance/Rejection Limits)	AASHTO T 59	50	450
Storage Stability Test, 1 d, 24 h, percent	AASHTO T 59		1
Sieve Test, percent	AASHTO T 59		0.1
Distillation			
Oil distillate, by volume of emulsion, percent	AASHTO T 59		7
Residue (b), percent	AASHTO T 59	65	
Residue from Distillation Test			
Penetration, 77° F (25° C), 100 g, 5 s, dmm	AASHTO T 49	150	300
Float Test, 140°F (60°C), s	AASHTO T 50	1200	
Solubility in trichloroethylene, percent	AASHTO T 44	97.5	
Elongation Recovery(c), 77° F (25° C), percent	AASHTO T 301	50	
(a) Supply an HFMS-2SP (anionic, polymerized, high-float) as an emulsified blend of polymerized asphalt cement, water, and emulsifiers. Polymerize the asphalt cement with at least 3.0% polymer by weight of the asphalt cement before emulsification. The emulsion must be smooth and homogeneous throughout with no white, milky separation, pumpable, and suitable for application through a distributor after standing undisturbed for at least 24 hours. (b) Determine the distillation by AASHTO T 59, with modifications to include a $350 \pm 5^\circ \text{F}$ ($177 \pm 3^\circ \text{C}$) maximum temperature to be held for 15 minutes. (c) Report only when penetration is greater than 300 dmm.			

Table 17

High Float Rapid Setting Emulsified Polymerized Asphalt (HFRS-2P) (a)			
Tests	Test method	Min.	Max.
Emulsion			
Viscosity, SF @ 122° F (50° C), s (Project Site Acceptance/Rejection Limits)	AASHTO T 59	50	450
Storage Stability Test (b) 1 d, 24 h, percent	AASHTO T 59		1
Demulsibility 0.02 N Ca Cl ₂ , percent	AASHTO T 59	40	
Sieve Test, percent	AASHTO T 59		0.1
Distillation			
Oil distillate, by volume of emulsion, percent	AASHTO T 59		3
Residue (b), percent	AASHTO T 59	65	
Residue from Distillation Test			
Penetration, 77° F (25° C), 100 g, 5 s, dmm	AASHTO T 49	70	150
Float Test, 140° F (60° C), s	AASHTO T 50	1,200	
Solubility in trichloroethylene, percent	AASHTO T 44	97.5	
Elastic Recovery, 77° F (25° C), percent	AASHTO T 301	58	
(a) Supply an HFMS-2SP (anionic, polymerized, high-float) as an emulsified blend of polymerized asphalt cement, water, and emulsifiers. Polymerize the asphalt cement with at least 3.0% polymer by weight of the asphalt cement before emulsification. The emulsion must be smooth and homogeneous throughout with no white, milky separation, pumpable, and suitable for application through a distributor after standing undisturbed for at least 24 hours.			
(b) Determine the distillation by AASHTO T 59, with modifications to include a 350 ± 5°F (177±3°C) maximum temperature to be held for 15 minutes.			

Table 18

Cationic Rapid Setting Emulsified Asphalt (CRS-2A,B)			
Tests	Test Method	Min	Max
Emulsion			
Viscosity, SF, 122° F (50° C), s (Project Site Rejection/Acceptance Limits)	AASHTO T 59	140	400
Storage stability test, 24 h, percent	AASHTO T 59		1
Demulsibility, 35 mL 0.8 percent Sodium Dioctyl Sulfosuccinate, percent	AASHTO T 59	40	
Particle charge test	AASHTO T 59	Positive	
Sieve test, percent	AASHTO T 59		0.10
Distillation			
Oil distillate, by volume of emulsion, percent	AASHTO T 59		0
Residue, percent	AASHTO T 59	65	
Use PG58-22 and PG64-22 as base asphalt cement for CRS-2A, B, respectively. Specification for high temperature performance – original and RTFO G*/sinδ within 3° C of grade.			

Table 19

Emulsified Asphalt Pavement Rejuvenating Agent Concentrate Type A		
Property	Test Method	Limits
Viscosity, SF, 77° F (25° C), s	AASHTO T 59	15 Min 40 Max
Residue , percent W (a)	AASHTO T 59	60 Min. 65 Max.
Miscibility Test (b)	AASHTO T 59	No Coagulation
Sieve Test, percent W (c)	AASHTO T 59	0.20 Max.
5-day Settlement, percent W	AASHTO T 59	5.0 Max.
Particle Charge	AASHTO T 59	Positive
Light Transmittance , %	UDOT MOI 8-973	30 Max.
Cement Mixing	AASHTO T 59	2 Max.
Residue from Distillation (a)		
Viscosity, 140° F (60° C), mm ² /s	ASTM D 4402	150 - 300
Flash Point, COC, ° F (° C)	AASHTO T 48	385 Min.
Asphaltenes, percent W	ASTM D 2006	0.4 Min. 0.75 Max.
Maltene Distribution Ratio (PC + A ₁)/(S + A ₂) (d)	ASTM D 2006	0.3 Min. 0.6 Max
Saturated Hydrocarbons, S (d)	ASTM D 2006	21 Min. 28 Max.
PC/S Ratio (d)	ASTM D 2006	1.5 Min.
(a) AASHTO T 59, Evaporation Test, modified as follows: Heat a 50 gram sample to 300°F until foaming ceases, then cool immediately and calculate results. (b) AASHTO T 59, modified as follows: Use a 0.02 Normal Calcium Chloride solution in place of distilled water. (c) AASHTO T 59, modified as follows: Use distilled water in place of a two percent sodium oleate solution. (d) Chemical composition by ASTM Method D-2006-70: PC= Polar Compounds, A ₁ = First Acidaffins A ₂ = Second Acidaffins, S = Saturated Hydrocarbons		

Table 20

Emulsified Asphalt Pavement Rejuvenating Agent Concentrate Type B		
Tests	Test Method	Limits
Viscosity, SF, 77° F (25° C), s	AASHTO T 59	25 - 150
Residue, percent W	AASHTO T 59 (mod) (a)	62 Min.
Sieve Test, percent W	AASHTO T 59	0.10 Max.
5-day Settlement	AASHTO T 59	5.0 Max.
Particle Charge	AASHTO T 59	Positive
Pumping Stability (b)		Pass
Residue from Distillation (a)		
Viscosity @ 140° F (60° C), mm ² /s	AASHTO T 201	2,500 - 7,500
Solubility in 1,1,1 Trichloroethylene, percent	AASHTO T 44	98 Min.
Flash Point, COC	AASHTO T 48	204° C, Min.
Asphaltenes, percent W	ASTM D 2007	15 Max.
Saturates, percent W	ASTM D 2007	30 Max.
Aromatics, percent W	ASTM D 2007	25 Min.
Polar Compounds, percent W	ASTM D 2007	25 Min.
(a) Determine the distillation by AASHTO T 59 with modifications to include a 300 ± 5° F (149 ± 3° C) maximum temperature to be held for 15 minutes.		
(b) Test pumping stability by pumping 475 ml of Type B diluted 1 part concentrate to 1 part water, at 77° F (25° C) through a ¼ inch gear pump operating at 1750 rpm for 10 minutes with no significant separation or coagulation in pumped material.		
Type B – an emulsified blend of lube oil or lube oil extract and petroleum asphalt.		

Table 21

Emulsified Asphalt Pavement Rejuvenating Agent Concentrate Type B Modified		
Property	Test Method	Limits
Viscosity, SF, 77° F (25° C), s	AASHTO T 59	50 - 200
Residue(a), percent W	AASHTO T 59	62 Min.
Sieve Test, percent W	AASHTO T 59	0.20 Max.
5-day Settlement, percent W	AASHTO T 59	5.0 Max.
Particle Charge	AASHTO T 59	Positive
Pumping Stability (b)		Pass
Residue from Distillation (a)		
Viscosity (c) 275° F (135° C), cP	ASTM D 4402	150 - 300
Penetration, 77° F (25° C), dmm	AASHTO T 49	180 Min.
Solubility in 1,1,1 Trichloroethylene, percent	AASHTO T 44	98 Min.
Flash Point, COC, ° F (° C)	AASHTO T 48	400(204) Min.
Asphaltenes, percent W	ASTM D 2007	20 - 40
Saturates, percent % W	ASTM D 2007	20 Max.
Polar Compounds, percent W	ASTM D 2007	25 Min.
Aromatics, percent W	ASTM D 2007	20 Min.
PC/S Ratio	ASTM D 2007	1.5 Min.
(a) Determine the distillation by AASHTO T 59 with modifications to include a 300±5°F (149 ± 3° C) maximum temperature to be held for 15 minutes. (b) Pumping stability is tested by pumping 475 ml of Type B diluted 1 part concentrate to 1 part water, at 77° F (25° C) through a ¼ inch gear pump operating at 1750 rpm for 10 minutes with no significant separation or coagulation in pumped material. (c) Brookfield Thermocel Apparatus-LV model. ≥ 50 rpm with a #21 spindle, 7.1 g residue, at > 10 torque		
As required by the Asphalt Emulsion Quality Management Plan 508, UDOT Minimum Sampling and Testing Requirements, the supplier certifies that the base stock contains at least 15% by weight of Gilsonite Ore. Use the HCL precipitation method as a qualitative test to detect the presence of Gilsonite.		

Table 22

Emulsified Asphalt Pavement Rejuvenating Agent Concentrate Type C		
Property	Test Method	Limits
Viscosity, SF, 77° F (25° C), s	AASHTO T 59	10 - 100
Residue (a), percent W (Type C supplied ready to use 1:1 or 2:1.	AASHTO T 59	30 Min. 1:1 40 Min. 2:1
Sieve Test, percent W (b)		0.10 Max.
5-day Settlement, percent W	AASHTO T 59	5.0 Max.
Particle Charge	AASHTO T 59	Positive
pH (May be used if particle charge test is inconclusive)		2.0 - 7.0
Pumping Stability (c)		Pass
Tests of Residue from Distillation (a)		
Viscosity, 275° F (135° C), mm ² /s	AASHTO T 201	475 - 1,500
Solubility in 1,1,1 Trichloroethylene, percent	AASHTO T 44	97.5 Min.
RTFO mass loss, percent W	AASHTO T 240	2.5 Max.
Specific Gravity	AASHTO T 228	0.98 Min.
Flash Point, COC	AASHTO T 48	232° C, Min.
Asphaltenes, percent W	ASTM D 2007	25 Min., 45 Max.
Saturates, percent W	ASTM D 2007	10 Max.
Polar Compounds, percent W	ASTM D 2007	30 Min.
Aromatics, percent W	ASTM D 2007	15 Min.
(a) Determine the distillation by AASHTO T 59 with modifications to include a 300 ± 5° F (149 ± 3° C) maximum temperature to be held for 15 minutes. (b) Test method identical to AASHTO T 59 except that distilled water is used in place of 2% sodium oleate solution. (c) Test pumping stability by pumping 475 ml of Type diluted 1 part concentrate to 1 part water, at 77° F (25° C) through a ¼ inch gear pump operating at 1750 rpm for 10 minutes with no significant separation or coagulation in pumped material.		
As required by the Asphalt Emulsion Quality Management Plan Section 508 , UDOT Minimum Sampling and Testing Requirements,, the supplier certifies that the base stock contains at least 10% by weight of Gilsonite ore. Use the HCL precipitation method as a qualitative test to detect the presence of Gilsonite.		

Table 23

Emulsified Asphalt Pavement Rejuvenating Agent Concentrate Type D		
Property	Test Method	Limits
Viscosity, SF, 77° F (25° C), s	AASHTO T 59	30 - 90
Residue, (b) percent W	AASHTO T 59	65
Sieve Test, percent W	AASHTO T 59	0.10 Max.
pH		2.0 - 5.0
Residue from Distillation (b)		
Viscosity, 140° F (60° C), cm ² /s	AASHTO T 201	300 - 1200
Viscosity, 275° F (135° C), mm ² /s	AASHTO T 201	300 Min.
Modified Torsional Recovery (a) percent	CA 332 (Mod)	40 Min.
Toughness, 77° F (25° C), in-lb	ASTM D 5801	8 Min.
Tenacity, 77° F (25° C), in-lb	ASTM D 5801	5.3 Min.
Asphaltenes, percent W	ASTM D 2007	16 Max.
Saturates, percent W	ASTM D 2007	20 Max.
(a) Torsional recovery measurement to include first 30 seconds.		
(b) Determine the distillation by AASHTO T 59 with modifications to include a 300 ± 5° F (149 ± 3° C) maximum temperature to be held for 15 minutes.		

2.3 HOT-POUR CRACK SEALANT FOR ASPHALT PAVEMENT

- A. Combine a homogenous blend of materials to produce a sealant according to properties and tests in Table 23.
- B. Packaging and Marking – Supply sealant pre-blended, pre-reacted, and pre-packaged in lined boxes weighing no more than 30 lb.
 1. Use a dissolvable lining that will completely melt and become part of the sealant upon subsequent re-melting.
 2. Deliver the sealant in the manufacturer's original sealed container.
 - a. Clearly mark each container with the manufacturer's name, trade name of sealant, batch or lot number, and recommended safe heating and application temperatures.

Table 24

Hot-Pour Asphalt Pavement Crack Sealant			
Application Properties			
Workability	Pour readily and penetrate 0.25 inch and wider cracks for the entire application temperature range recommended by the manufacturer.		
Curing	No tracking caused by normal traffic after 45 minutes from application.		
Asphalt Compatibility ASTM D 5329, Section 14.	No failure in adhesion. No formation of an oily ooze at the interface between the sealant and the asphalt pavement or softening or other harmful effects on the asphalt pavement.		
Material Handling	Follow the manufacturer's safe heating and application temperatures.		
Test Method	Property	Minimum	Maximum
AASHTO T 51	Ductility, modified, 1cm/min, 39.2° F (4° C), cm	30	
UDOT method 967	Cold Temperature Flexibility	no cracks	
AASHTO T 300 (a)	Force-Ductility, lb force		4
ASTM D 5329	Flow 140°F (60° C), 5 hrs 75° angle, mm		3
AASHTO M 282 (b)	Tensile-Adhesion, modified	300%	
AASHTO T 228	Specific Gravity, 60° F (15.6° C)		1.140
ASTM D 5329	Cone Penetration, 77° F (25° C), 150 g, 5 sec., dmm		90
ASTM D 5329	Resilience, 77° F (25° C), 20 sec., percent	30	
ASTM D 4402	Viscosity, 380°F (193.3°C), SC4-27 spindle, 20 rpm, Cp		2,500
ASTM D 5329	Bond, Non-Immersed as specified in AASHTO M 282		Pass
(a)	Maximum of 4 lb force during the specified elongation of 30 cm @ 1 cm/min, 39.2° F (4° C).		
(b)	Delete Bond, Non-Immersed modification in AASHTO M 282. Perform tensile-adhesion test according to ASTM D 5329.		

PART 3 EXECUTION Not Used

END OF SECTION

SECTION 02746

HYDRATED LIME

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Products and procedures for incorporating hydrated lime into all asphalt mixes.

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES

- A. AASHTO M 303: Lime for Asphalt Mixtures
- B. ASTM C 110: Physical Testing of Quicklime, Hydrated Lime, and Limestone
- C. ASTM C 1097: Hydrated Lime for Use in Asphalt Cement or Bituminous Paving Mixtures
- D. ASTM C 1602: Mixing Water Used in the Production of Hydraulic Cement Concrete
- E. UDOT Materials Manual of Instruction
- F. UDOT Quality Management Plans

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS Not Used

1.6 QUALITY ASSURANCE

- A. Use a supplier of Hydrated Lime that is pre-qualified through UDOT Quality Management Plan 510.
- B. Comply with UDOT Quality Management Plan 514, Asphalt Mix.

PART 2 PRODUCTS

2.1 HYDRATED LIME

- A. Hydrated Lime – Refer to AASHTO M 303, Type I.
 - 1. Refer to ASTM C 1097, Physical Requirements.
 - 2. Use test method ASTM C 110.

2.2 WATER

- A. Use potable water or water according to ASTM C 1602.

PART 3 EXECUTION

3.1 APPLICATION

- A. Add hydrated lime to all asphalt pavement mixes.
 - 1. Add lime as determined in the approved mix design.
 - 2. Base the amount of hydrated lime used on the dry weight of the virgin aggregate.
 - 3. Use either Method A or B, unless Method B is specified.
 - 4. Use Method A only when the mix is produced by a Certified Asphalt Mix Supplier. Refer to the UDOT Quality Management Plan 514 Asphalt Mix.

- B. Method A – Dynamic Feed Lime Slurry
 - 1. Lime Slurry –At least three parts water to one part lime by weight.
 - a. Adjust the amount of added water to meet the 3:1 lime slurry to account for the moisture in the stockpile.
 - b. Use lime slurry with a minimum of one part water by weight of lime.
 - 2. Add at least 1 percent lime by weight of the virgin aggregate.
 - 3. Deliver lime slurry to the twin shaft pugmill for mixing with aggregate.
 - a. Use a virgin aggregate/lime mixture that contains at least 3 percent water by weight of the virgin aggregate after the pugmill.
 - 4. Adjust quantity (percent) of lime as necessary, based on results of Hamburg Wheel Tracker test.
 - 5. Verify that Lime Slurry equipment is operating at all times.
 - a. The Engineer may require Method B, marination of the aggregate/hydrated lime mixture in the stockpile if the Asphalt Mix is supplied without hydrated lime slurry treatment.

- C. Method B – Lime and Aggregate Stockpile Marination
1. Method B is an option for a Certified Asphalt Mix Supplier.
 2. Method B is required for all Asphalt Mix Suppliers that are not certified according to the UDOT Quality Management Plan 514 Asphalt Mix.
 - a. The Government may inspect the marination process when the Asphalt Mix supplier is not certified at the time of production.
 3. The Government applies a deduction for mix produced by a non-certified supplier to cover the cost of the inspection.
 - a. The deduction is according to the UDOT Quality Management Plan 514 Asphalt Mix.
 4. Provide sufficient free moisture to thoroughly wet the aggregate and activate the lime before introducing hydrated lime.
 - a. The aggregate/lime mixture will contain at least 3 percent water by weight of the virgin aggregate.
 5. Add at least 1½ percent lime by weight of the virgin aggregate.
 6. Thoroughly mix wet aggregate/lime mixture in a twin shaft pugmill.
 7. Marinate the aggregate/lime mixture in the stockpile for at least 48 hours.
 8. Adjust quantity (percent) of lime as necessary, based on results of Hamburg Wheel Tracker test, UDOT Materials Manual of Instruction 990.
 9. Use the wet cured aggregate within 60 days except as otherwise provided in the UDOT Quality Management Plan 514 Asphalt Mix.
- D. Mixing Methods A and B – Provide a horizontal twin shaft pugmill.
1. Adjust mixing paddles in the pugmill so that the aggregate being discharged is completely coated by the lime slurry.
 2. Do not allow volume of material in the pugmill to extend above the vertical position of the blade tips.

3.2 QUALITY CONTROL

- A. Tolerance Controls
1. Tolerance lime weight vessel static calibration ± 1.5 percent
 2. Dynamic delivery calibration ± 1.5 percent
 3. Inlet flow meter ± 2 percent
 4. Discharge flow meter ± 1.5 percent

END OF SECTION

SECTION 02748

PRIME COAT/TACK COAT

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Materials and procedures for applying prime and tack coat.

1.2 RELATED SECTIONS

- A. Section 02745: Asphalt Material

1.3 REFERENCES

- A. AASHTO M 82: Cutback Asphalt (Medium-Curing Type)
- B. AASHTO M 208: Cationic Emulsified Asphalts
- C. AASHTO T 27: Sieve Analysis of Fine and Coarse Aggregates
- D. AASHTO T 201: Kinematic Viscosity of Asphalts (Bitumens)
- E. ASTM D 5095: Determination of the Nonvolatile Content in Silanes, Siloxanes and Silane-Siloxane Blends Used in Masonry Water Repellent Treatments

1.4 DEFINITIONS

- A. Emulsified Asphalt – A group of asphalt products using water and soap (emulsifier) to reduce viscosity. These products consist of approximately 60 percent asphalt and 40 percent emulsifier.
 - 1. Undiluted - No additional water has been added to the emulsion
 - 2. 1:1 Dilute - One part emulsion to 1 part water
 - 3. 2:1 Dilute – Two parts emulsion to 1 part water
- B. Prime coat – Liquid asphalt applied to a prepared subgrade.
- C. Tack coat – Emulsified asphalt to the existing surface or new pavement surface and intermediate lifts.

1.5 SUBMITTALS

- A. Material invoice or bill of lading for information.
- B. Prime coat/tack coat work plan for review.
 - 1. Include quantities, rate of application, temperatures and areas to be treated.
 - 2. Refer to this Section, Article 3.2 for application rates.
- C. Request to dilute emulsified asphalt for approval.

PART 2 PRODUCTS

2.1 PRIME COAT

- A. MC-70 or MC-250. Refer to AASHTO M 82.
 - 1. Do not dilute.

2.2 TACK COAT

- A. CQS-1h or CSS-1h emulsified asphalt. Refer to AASHTO M 208 and Section 02745.
 - 1. Use specified emulsion or select emulsion according to the time constraints required for Maintenance of Traffic (MOT) and the ability to fully cure before allowing traffic on the tack coat.
 - a. A diluted product may be used to better control distribution when the residual application rate is small (0.03 gal/yd²).
 - b. Dilute at terminal only.
 - c. Do not change dilution before obtaining approval from the Engineer.
 - 2. A cationic organosilane additive with a minimum nonvolatile content of 38 percent may be used. Refer to ASTM D5095
 - a. Use a dosage from 0.7 percent to 1.0 percent by weight of the emulsion.
 - b. Modify the emulsified asphalt requirements to meet a maximum cement mixing test of 10 percent.

2.3 BLOTTER MATERIAL

- A. Granular material meeting Table 1. Refer to AASHTO T 27.

Table 1

Granular Materials	
Sieve Size	Percent Passing
No. 4	90 to 100
No. 10	25 to 80
No. 200	0 to 15

PART 3 EXECUTION

3.1 SURFACE PREPARATION

- A. Prime Coat
 - 1. Verify the base material meets all contract requirements prior to placement.
 - 2. Clean the surface of all materials that prevent bonding to the existing surface.
- B. Tack Coat
 - 1. Clean the surface of all materials that prevent the tack coat from bonding to the existing surface such as mud, dirt, and leaves.

3.2 APPLICATION

- A. Use a pressure distributor to apply the asphalt in a uniform, continuous spread.
 - 1. Keep the viscosity between 50 and 100 centistokes. Refer to AASHTO T 201.
- B. Apply prime coat according to the authorized work plan to prepared grade at an application rate of 0.5 gal/yd²
 - 1. Spread blotter material if the prime coat does not penetrate.
 - a. Use the quantities required to absorb the excess asphalt.
 - 2. Keep construction traffic off the prime coat until it has cured and dried.
- C. Apply tack coat according to the authorized work plan between all lifts of Asphalt Mix and to all surfaces, including vertical that will come in contact with Asphalt Mix.
 - 1. Meet the residual application rate shown in Table 2.
 - a. Undiluted, 1:1 Dilute and 2:1 Dilute application rates shown are based on 60 percent residual asphalt.

- b. Adjust application rate if emulsion is not 60 percent residual asphalt to achieve required residual rate.
 - c. Adjust application rates according to field conditions.
- 2. Cover all tacked surface areas with surfacing materials the same day the tack coat is applied.

Table 2

Application Rate (gal/yd²)				
Existing Pavement Condition	Residual	Undiluted	1:1 Dilute	2:1 Dilute
New Asphalt Mix	0.03	0.05	0.10	0.08
Oxidized Asphalt Mix	0.05	0.09	0.18	0.13
Milled Asphalt Mix	0.07	0.12	0.24	0.18
Milled PCCP	0.07	0.12	0.24	0.18
PCCP	0.05	0.09	0.18	0.13

- D. Do not apply prime coat or tack coat:
 - 1. On a wet surface or where surface conditions prevent proper adhesion.
 - 2. When the surface temperature is below 50 degrees F.
 - 3. When weather conditions prevent proper adhesion.
- E. Protect all structures including items such as guardrails and guide posts from being spattered or marred.

END OF SECTION

SECTION 02750

PARTIAL DEPTH REPAIR FOR CONCRETE PAVEMENT USING HOT APPLIED REPAIR MATERIALS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Partial depth repair of spalled or delaminated concrete with hot-applied flexible polymer modified patch material in the upper one third to one half of the pavement.
- B. Placing hot-applied flexible polymer modified patch material to bring existing repairs flush with the pavement

1.2 RELATED SECTIONS

- A. Section 02753: Full Depth Slab Replacement for Concrete Pavements

1.3 REFERENCES Not Used

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. APL Compliance Form for information for hot-applied flexible polymer modified repair material.
- B. Manufacturer's product data, specifications, and recommended installation instructions.
 - 1. Include contact information for technical representative who will be on site. Refer to this Section, Paragraph 3.1, subparagraph A1.

PART 2 PRODUCTS

2.1 HOT-APPLIED FLEXIBLE POLYMER MODIFIED REPAIR MATERIAL

- A. Use repair material that will provide an impermeable, voidless mass that is solid at ambient temperatures consisting of resin, polymers, graded fillers, aggregates, and chopped fiberglass fibers; up to 3 percent recycled tire rubber, ½ percent to 1 percent steel fibers 0.5 mm to 1 mm in diameter meeting the requirements of table 1.
1. Supply material in a powder form, factory blended and bagged in meltable bags.
- B. Use bulking aggregate approved by the manufacturer..

Table 1

FINISHED PRODUCT PROPERTIES	Test Method	REQUIREMENT
Color		Grey
Tensile Strain	TEX-548-C	35% minimum @ 2"/minute
Cone Flow	TEX 549-C	6% maximum
Aggregate Settlement	TEX-551-C	3% maximum
Flexibility / Mandrel	TEX-550-C	Good or Better (no tearing at bend point)
Resilience	TEX-547-C	50% minimum
Recommended Application Temp.		300°F- 350°F
Specific Gravity	ASTM D2726	1.8-2.0

- C. Provide topping stone as recommended by the manufacturer.

2.2 EQUIPMENT

- A. Provide horizontal mixing unit with thermostatic controls.

PART 3 EXECUTION

3.1 IDENTIFICATION AND PREPARATION OF REPAIR AREA

- A. Establish and mark circular, rectangular or triangular repair areas extending at least 1 inch beyond a spalled area or 6 inch beyond the failed area.
 - 1. Limit each repair location to a maximum of 9 square feet.
 - a. Repair the pavement using full depth panel replacement when repair exceeds 9 square feet. Refer to Section 02753.
 - 2. Limits for corner breaks
 - a. Use a triangular repair when only one adjacent panel has a corner break.
 - b. Use a circular repair when two or more adjacent panels have corner breaks.
 - 1) Center repair over intersection of the joints.
 - 2) Limit repairs to 1 foot diameter
 - 3) Use triangular repair when repair is greater than 1 foot diameter.
- B. Remove concrete from repair area.
 - 1. Saw cut or core drill a minimum of 1/2 inch in depth around exterior of repair area.
 - 2. Remove material with 30 pound class or smaller hand jackhammers.
- C. Completely remove all loose particles, crack seal, oil, dust, moisture, and all other contaminants.
 - 1. Use compressed air at a minimum of 100 psi to clean all surfaces.
 - 2. Remove exposed reinforcing steel before placing the patch material.
- D. Prime all surfaces of new repair areas using primer recommended by the manufacturer to prevent moisture intrusion.
- E. Heat surface of existing patching material per manufacture recommendations.
 - 1. Replace existing patches heated above the recommended temperatures at no additional cost to the Department.

3.2 MATERIAL PLACEMENT

- A. Follow manufacturer's specifications for material preparation and placement.
 - 1. Have manufacturer's representative on site for the first day of material placement or as determined by the Engineer.
- B. Place bulking aggregate evenly throughout the patch to within $\frac{1}{4}$ inch of the surface for repairs that are 1 inch in depth or greater.
 - 1. Heat and dry bulking aggregate in a vented barrel mixer at 300 degrees F immediately before placement.
- C. Place and finish hot-applied flexible polymer modified repair material flush with existing surface.
- D. Top with manufacturer's recommended topping stone to refusal at the gel stage.
- E. Repair patches with surface profiles that vary from the existing roadway profile by more than $\frac{1}{8}$ inch.
 - 1. Remove and replace or grind patches with surface profiles greater than $\frac{1}{8}$ inch above the surrounding pavement surface.
 - 2. Apply additional patch material to patches with surface profiles greater than $\frac{1}{8}$ inch below the surrounding pavement surface.

3.3 PROTECTION

- A. Do not open to traffic until patching material has cooled or hardened sufficiently to support traffic loads per manufacturer's recommendations.
- B. Protect all placements according to the authorized traffic control plan.
- C. Remove and replace any patches that are rejected before substantial completion due to the following:
 - 1. Unsatisfactory or improper workmanship by the Contractor, including patches with surface profiles that vary from the existing roadway profile by more than $\frac{1}{8}$ inch.
 - a. Patches with a profile higher than $\frac{1}{8}$ inch from the existing roadway profile may be ground to meet existing profile instead of removal and replacement.
 - 2. Failures due to damage by the operation or public traffic.

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3.4 CLEAN PAVEMENT

- A. Sweep roadway and shoulders of all debris before opening to traffic. Prevent damage to the patches from cleaning equipment.

END OF SECTION

SECTION 02751

PARTIAL DEPTH REPAIR FOR CONCRETE PAVEMENT USING PORTLAND CEMENT CONCRETE

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Partial Depth repair of spalled or delaminated concrete in the upper one third to one half of the pavement.

1.2 RELATED SECTIONS

- A. Section 02752: Portland Cement Concrete Pavement
- B. Section 03055: Portland Cement Concrete
- C. Section 03390: Concrete Curing

1.3 REFERENCES

- A. AASHTO M 235: Epoxy Resin Adhesives.
- B. ASTM D 6690: Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Request to use alternative patching material to the Engineer for approval. Include the following:
 - 1. Manufacturer's product data sheets and recommended installation instructions.
 - 2. Contact information for technical representative who will be on site. Refer to this Section, Paragraph 3.4.C

PART 2 PRODUCTS

2.1 REPAIR MATERIAL

- A. Use Class AA(AE). Refer to Section 03055.
 - 1. Alternative patching material may be used when approved.

2.2 CONCRETE CURING COMPOUND

- A. Refer to Section 03390.

2.3 HOT APPLIED JOINT FILLER

- A. Refer to ASTM D 6690, Type II

PART 3 EXECUTION

3.1 IDENTIFICATION AND PREPARATON OF REPAIR AREA

- A. The Engineer will establish and mark rectangular or circular repair areas, extending at least 2 inch beyond a spalled area or 6 inch beyond the failed area.
- B. Remove concrete from repair area
 - 1. Saw cut or core drill a minimum of 2 inch in depth around the exterior of the repair area.
 - 2. Remove material with 30 pound class or smaller hand jackhammers.
- C. Remove all loose particles, oil, dust, and all other contaminants.
 - 1. Use sandblasting or waterblasting with at least 2,000 psi.
- D. Remove all cleaning grit before placement.
- E. Sandblast clean or remove partially exposed reinforcing steel before placing the patch material.

3.2 PREPARE EXISTING JOINTS FOR PLACEMENT

- A. Maintain a width equal to the opening in all working joints and cracks within or adjacent to the patch by inserting a removable material such as cardboard, polyfoam, or fiberboard before placing the mix.
 - 1. Use a material of uniform size and thickness.
- B. Place the joint forming material 1 inch below and 3 inches laterally beyond the patch boundary.
 - 1. Do not widen the existing joint to provide for or facilitate placement of the removable joint material.

3.3 MATERIAL PLACEMENT

- A. Prepare the existing surface by applying a bonding agent of cementitious grout designed for bonding concrete products.
 - 1. Prepare grout by combining sand and Portland Cement in a 1:1 ratio by volume and enough water to produce a creamy consistency.
 - a. Apply with a brush to bottom and sides of repair area.
- B. Place patching material before bonding agent dries.
- C. Do not begin placement of alternative patching materials before having a manufacturer representative on site. Manufacturer's representative will address the following requirements:
 - a. On site mixing permissibility and constraints
 - b. Surface preparation of the repair area and appropriate bonding agent
 - c. Consolidation requirements
 - d. Pot life of mix
 - e. Environmental conditions and limitations
- D. Remove and replace patch if any of the following occurs:
 - 1. Visible cracking
 - 2. Shrinkage, including the breaking of bond between patch and pavement
 - 3. Pop-outs including pieces of pavement greater than ½ inch diameter.
 - 4. Patch surface profiles that are more than 1/8 inch below the existing roadway surface.

- E. Grind patches with surface profiles greater than 1/8 inch above the surrounding pavement surface to match the surrounding pavement surface.

3.4 FINISH

- A. Finish the patched surface to within 1/8 inch of the existing pavement.

3.5 CURE AND PROTECT

- A. Cure the patched surface immediately after finishing operations according to Section 03390 with the following change:
 - 1. Uniformly spray the surface at a rate of 50 ft²/gal.
- B. Do not open to traffic until patching material has achieved required strength.
- C. Protect all placements according to the authorized traffic control plan.
- D. Fill overcuts flush to the pavement surface with a repair epoxy.
- E. Remove joint forming material and fill all affected joints with hot pour material. Refer to ASTM D 6690

3.6 CLEAN PAVEMENT

- A. Sweep roadway and shoulders of all debris before opening to traffic. Prevent damage to the patches from cleaning equipment.

3.7 LIMITATIONS

- A. Refer to Section 02752.

END OF SECTION

SECTION 02752

PORTLAND CEMENT CONCRETE PAVEMENT

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Portland Cement Concrete Pavement.

1.2 RELATED SECTIONS

- A. Section 02701: Pavement Smoothness
- B. Section 02742S: Project Specific Surfacing Requirements
- C. Section 03055: Portland Cement Concrete
- D. Section 03152: Concrete Joint Control
- E. Section 03211: Reinforcing Steel and Welded Wire

1.3 REFERENCES

- A. AASHTO M 31: Deformed and Plain Carbon and Low-Alloy Steel Bars for Concrete Reinforcement
- B. AASHTO M 157: Ready-Mixed Concrete
- C. AASHTO T 22: Compressive Strength of Cylindrical Concrete Specimens
- D. AASHTO T 325: Estimating the Strength of Concrete in Transportation Construction by Maturity Tests
- E. ASTM C 309: Liquid Membrane-Forming Compounds for Curing Concrete
- F. ASTM E 3013: Evaluating Concrete Pavement Dowel Bar Alignment Using Magnetic Pulse Induction
- G. American Concrete Institute (ACI) Standards
- H. UDOT Minimum Sampling and Testing Requirements
- I. UDOT Quality Management Plan 506

1.4 DEFINITIONS

- A. Panel – Area of pavement within a traffic lane or shoulder bound by two transverse joints.
- B. Production Day – A 24 hour period in which Portland Cement Concrete Pavement is being placed.

1.5 SUBMITTALS

- A. Placing and finishing plan for review before concrete placement.
 - 1. Include at least the following:
 - a. Production methods
 - b. Handling and placing
 - c. Protection and curing
 - d. Cylinder storage device. Refer to Section 03055.
 - e. Cold weather protection plan
 - f. Hot weather protection plan
- B. Manufacturer's product data sheets and installation instructions for dowel bar bond breaking compound for information before concrete placement.
- C. Verification that the batch plant meets the requirements of the UDOT Quality Management Plan for Ready-Mix Concrete for information.
- D. Pavement joint layout plan for approval before concrete placement.
- E. Copy of steel verification equipment calibration for review.
 - 1. Calibration date must be within 7 days before the first paving day.
- F. Individual batch tickets, with each load of concrete, to the COR at the job site for information.
- G. Gradation quality control test results for information.
- H. Gradation quality control corrective action plans for review when required.

- I. Manufacture's product data sheets and installation instructions for material to be used to fill core holes for review.
- J. Reinforcing steel placement verification test results for review. Include the following information:
 - 1. Test location, identified by station and offset.
 - 2. Vertical and horizontal rotational alignment. Identify steel orientation and amount of misalignment.
 - 3. The length of embedment on each side of the joint.
 - 4. Depth of the bar, at each end, from the pavement surface.
 - 5. Paving date of panel and the date of verification testing.
 - 6. Name of individual conducting verification testing and name of Department representative witnessing testing.

1.6 ACCEPTANCE

- A. The point of acceptance for air content is the in-place pavement.
 - 1. Sampling and testing will be performed at the point of placement with necessary adjustments applied based on correlation testing. The Quality Control will:
 - a. Perform the air test on concrete placed in front of the paver and again from the same load in the finished pavement. Record any change in air content.
 - b. Adjust acceptance limits for air content according to results of correlation testing to provide final in-place air content that meets the requirements specified in Section 03055.
 - c. Perform two correlation tests per paver for each day of paving operations.
 - 2. The COR will mark for later removal any areas determined to be out of conformance with the contract.
 - a. Coordinate the timing for removal of reject material with the COR
- B. Government will assess price adjustments for strength, thickness, and surface smoothness separately .
- C. Thickness
 - 1. Use cores obtained for compressive strength testing.
 - 2. COR determines the acceptability and pay factors for deficient thickness areas using Table 1.

Table 1

Price Reductions for Deficient Thickness	
Deficient Thickness (DT) (inches)	Pay Factor
$0 < DT \leq \frac{1}{8}$	1.00
$\frac{1}{8} < DT \leq \frac{1}{4}$	0.90
$\frac{1}{4} < DT \leq \frac{1}{2}$	0.75
$\frac{1}{2} < DT \leq \frac{3}{4}$	0.60
$DT > \frac{3}{4}$	Reject

- a. The COR may accept pavement deficient by more than $\frac{3}{4}$ inch at 50 percent pay or require removal and replacement.
 - b. Take two additional cores for any deficient core (one on each side) where the thickness varies by $\frac{1}{8}$ inch or more. Locate the additional core midway between the deficient core and each of the adjacent cores.
 - c. The COR graphs the deficient areas by plotting new cores and the original cores to define deficient areas, assuming the following:
 - 1) The graph represents the thickness of the pavement.
 - 2) The thickness varies linearly along the pavement length from core depth to core depth.
 - 3) The pavement is a constant depth in the transverse direction.
3. The COR may require verification of thickness after grinding for smoothness.
- D. Compressive Strength Acceptance
1. Obtain cores using a 6 inch inside diameter diamond impregnated bit. Drill cores within $\frac{1}{8}$ inch of perpendicular to the pavement surface in all directions.
 - a. The Engineer marks core locations in areas free from reinforcing steel.
 - b. Perform all coring at least 14 days and at most 21 days after the pavement has been placed.
 - 1) Immediately deliver cores to the Engineer.
 - 2) Provide a protective container for transporting cores.
 - 3) After receiving the cores, the Department will moist cure the cores until performing the 28 day compressive strength testing.

- 4) Disincentives will be assessed if coring occurs after 21 days according to the following schedule:
 - a) \$5,000 for the first placement and increasing by \$5,000 per placement thereafter.
 2. Compressive strength is determined on each core, following AASHTO T 22.
 - a. The Government will accept material according to Section 03055.
 3. Hand-placement areas will be considered separately. Refer to minimum sampling and testing requirements for Section 03055.
 4. The Engineer notifies the Contractor, within three calendar days of determining the 28 day compressive strength, if any test is below specifications.
 - a. The Contractor has the option to contest the compressive strength results.
 - b. The Contractor obtains one set of three cores within 3 ft of the location where the contested core was obtained, if contesting the results.
 - 1) Retesting by a UDOT qualified lab must be completed within 40 calendar days after placement.
 - 2) Compressive strength is the average of the results of the three cores or the lowest of the individual cores if any one core differs by 10 percent or more from the average of three and payment will be based on Section 03055.
 - 3) A correction factor will not be applied to the compressive strength results.
 - 4) Results of retesting replaces the results of the original test. Retesting results are final.
- E. Smoothness
 1. Evaluate according to Section 02701.
- F. Reinforcing Steel Acceptance
 1. The Government will reject individual dowel bars based on the following criteria:
 - a. Horizontal rotational misalignment greater than 1.5 inch measured perpendicular to the joint, calculated using the full length of the bar.
 - b. Vertical rotational misalignment greater than 1.5 inch measured perpendicular to the surface, calculated using the full length of the bar.
 - c. Any bar not embedded at least 6 inch on each side of the joint in tangent sections.

- d. Any bar not embedded at least 4 inch on each side of the joint in horizontal curves.
 - e. Any bar not entirely within the vertical middle third of the pavement.
 - 2. The Government will reject individual tie bars based on the following criteria:
 - a. Any bar not embedded at least $\frac{1}{3}$ of the bar length on each side of the joint.
 - b. Any bar not entirely within the middle third of the pavement.
 - 3. Remove and replace full panels where the reinforcing steel meets the dowel or tie bar rejection criteria. The Engineer may accept rejected panels at 50 percent of the original unit bid price.
 - a. Saw cut the reinforcing steel identified as noncompliant for panels accepted at the reduced price.
 - 1) Do not saw cut compliant reinforcing steel.
 - 2) A maximum of 3 consecutive transverse joints per subplot may be saw cut and the associated panels accepted at the reduced price.
 - 3) Panels with reinforcing steel within 3 inch of the pavement surface are not eligible to be left in place.
 - 4. The COR may verify the location of the dowel bars visually before placement when baskets are used.
- G. Quantity adjustment when overlaying existing pavement surfaces not constructed with the current project.
- 1. Adjust quantity when accepted batched volume overruns or under runs neat-line volume.
 - a. The COR and Contractor determine batched volume at time of placement.
 - b. Batched volume is the total batched material adjusted to design yield minus rejected or wasted material.
 - 2. Adjust quantity before any price adjustment for non-conforming material.
 - 3. Determine overrun/underrun quantity by the following formula:

$$QA = 0.5 \left(\frac{V_a - V_n}{V_n} \right) Q_m$$

QA	=	Adjusted quantity in yd ²
V _a	=	Batched Volume
V _n	=	Neat-line Volume
Q _m	=	Measured quantity in yd ²

1.7 QUALITY CONTROL

- A. Gradation
 - 1. Perform quality control testing of the combined aggregate gradation. Sample and test within the first 100 cubic yards of production, after shutdowns of 1 hour or more and randomly thereafter for each 800 cubic yards.
 - a. Sample and test the blended aggregate when aggregates are blended before charging the mixing drum.
 - b. Sample and test each aggregate and calculate the combined gradation when aggregates are batched individually.
 - c. Provide test results to the COR Daily.
 - 2. Immediately notify the COR if a test does not meet the gradation requirement in Section 03055 and make corrections to bring the aggregate gradation within the requirements.
 - a. Suspend production and provide a written corrective action plan to the Engineer for review if two or more consecutive test results fail to meet the requirements.
 - b. Do not resume production until the corrective action plan is authorized.
 - 3. QC testing is not required for hand placements.
- B. Provide nondestructive verification of the reinforcing steel location within the hardened concrete within 5 days after each production day's paving. Meet the requirements of ASTM E 3013.
 - 1. Verify dowel bar placement the full length of every third transverse joint.
 - a. Test the full length of every transverse joint between compliant locations if non-compliant dowel bars are identified.
 - 2. Randomly verify tie bar placements for one third of the bars placed.
 - a. Test every tie bar between compliant locations if non-compliant tie bar placement is identified.

PART 2 PRODUCTS

2.1 CONCRETE

- A. Use Class AA(P). Refer to Section 03055.
 - 1. Class AA(AE) may be used for hand placements. Refer to Section 03055.

2.2 CONCRETE CURING COMPOUND

- A. Refer to ASTM C 309, Type 2, Class A.
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2.3 JOINT SEALERS

- A. Use hot poured joint sealant meeting Section 03152.

2.4 REINFORCING STEEL

- A. Tie Bars
 - 1. Use deformed epoxy coated reinforcing steel. Refer to Section 03211.
- B. Dowel Bars
 - 1. Use plain epoxy coated reinforcing steel. Refer to Section 03211.
 - 2. Basket assemblies must meet the following requirements:
 - a. Use a U-shape leg for the assembly frame.
 - b. Use a minimum 0.306 inch diameter wire.
 - c. Epoxy coating is not required.

2.5 BATCH PLANT

- A. Use a prequalified supplier according to UDOT Quality Management Plan 506, Ready Mix Concrete.

2.6 VEHICLES FOR HAULING

- A. End dump trucks with watertight beds and end gates and rounded corners.
- B. Agitator trucks with open tops.
- C. Transit mixers that conform to AASHTO M 157.
- D. Do not use bottom or belly dump units.

2.7 SLIP FORM PAVER

- A. Self-propelled machine with no fluid leaks equipped with automatic line and grade control capability.
- B. Capabilities
 - 1. Spreading the dumped concrete uniformly across the grade by an auger or a traveling strike-off device.
 - 2. Vibrating, tamping, striking-off, and shaping the concrete to the desired line grade and thickness in one continuous pass.

- C. Vibrator Minimum Requirements
 - 1. Eccentric Diameter 1 $\frac{7}{8}$ inch
 - 2. Frequency 7,000 to 9,000 vibrations per minute
 - 3. Spacing 18 inch maximum, mounted longitudinally
- D. Mount vibrators to provide adequate concrete consolidation.
 - 1. Do not interfere with dowel bar basket assemblies.
- E. Use trailing forms long enough to leave a smooth, straight, vertical edge.

2.8 CURING APPLICATION EQUIPMENT

- A. Use curing compound application machine with a fully atomizing type power spray and a wind protection hood.

PART 3 EXECUTION

3.1 LINE AND GRADE CONTROL

- A. Use a system that limits deviations in the pavement surface to $\frac{1}{8}$ inch in 10 feet and deviations in the vertical edge of the pavement to $\frac{1}{4}$ inch and maintains specified pavement thickness.
- B. Slip Form Paving
 - 1. Establish the necessary stakes for line and grade control over existing surfaces and provide the elevation control benchmarks.
 - 2. Equip machinery with a control system that automatically controls concrete placement to the specified longitudinal grades.
- C. Formed Paving
 - 1. Keep forms free from warps, bends, kinks, and equal in depth to the specified pavement edge.
 - 2. Tightly join form sections by an interlocking joint free of vertical and horizontal movement.
 - 3. Stop paving operations if the side forms do not meet or hold line and grade.
- D. Profile grind existing adjacent pavement before concrete placement.

3.2 BATCH AND MIX

- A. Separate and stockpile coarse aggregate in two sizes for 2 ½ inch, 2 inch and 1½ inch gradations with the separation being made on the 1 inch sieve for 2 ½ inch and 2 inch gradations and the ¾ inch sieve for 1 ½" gradations.
- B. Mixing – Operate the drum at manufacturer's recommended speed.
 - 1. Conduct mixing efficiency tests at the start of concrete placement and evaluate as specified in AASHTO M 157, Annex A-1.
 - 2. Increase mixing time in 10-second increments if necessary until the mixer efficiency evaluation is passed.
 - a. Correct poor mixing efficiency.
 - 3. Do not allow buildup of cement or mortar on the mixer drums and blades.
 - 4. Mix for at least 30 seconds after the last addition of water or cement
- C. Maintain a minimum mixing time of 80 seconds at the manufacturer's recommended mixing speed after all materials are in the drum.
 - 1. Include actual mix time on each computerized batch ticket as a part of the information automatically recorded.
- D. Do not add water to the mix after acceptance testing.

3.3 PLACE CONCRETE

- A. Keep the base surface moistened 500 ft in front of the paver without allowing areas of standing water.
- B. Discharge and place the mixed concrete within the time frame listed below after introducing the mixing water to the cement and aggregates.
 - 1. Concrete not placed within the following time period may be rejected.
 - a. Non-agitating Haul Equipment 45 minutes
 - b. Agitating Haul Equipment 75 minutes
 - 2. The COR may authorize alternate time periods based on concrete and ambient temperatures.
- C. Do not interfere with dowel basket assemblies if redistributing the concrete in front of the paver.
 - 1. Minimize redistributing the concrete in front of the paver.
 - 2. The COR may require retesting the concrete if redistribution occurs.

- D. Consolidate concrete using vibrators.
 - 1. Position vibrators on finishing equipment ahead of strike-off auger or final screed.
 - 2. Vibrate, screed, and mechanically tamp the spread concrete.
 - a. Thoroughly vibrate adjacent to and along the faces of the forms.
 - 1) Stop vibrating and tamping elements when the forward movement of the paver stops.
 - 3. Monitor the operation of vibrators.
 - a. Check each vibrator at the beginning and ending of each day.
 - b. Repair or replace vibrators as necessary.
 - c. Stop paving operations immediately if any indication of malfunction occurs.
 - d. Resume operations only after repairing or replacing the vibrator.
 - 4. Use hand-operated vibrators on a regular pattern, not to exceed 12 inches in each direction, for hand placements.
- E. Do not add water to the surface for finishing.
 - 1. Paving operations may be shut down and the concrete rejected if water is added.
 - 2. Use evaporation retarders only after obtaining written approval by the COR
- F. Provide protection for existing surfaces when placing concrete in an adjoining longitudinal section (companion placement).
 - 1. Do not place concrete for companion placement before existing concrete has achieved at least a compressive strength of 2,500 psi, determined by Maturity Method, AASHTO T 325, or cast cylinders.

3.4 HANDLE AND PLACE REINFORCING STEEL

- A. Keep reinforcing steel clean, free from damage, and free from distortion.
- B. Place tie bars in the middle third of the slab.
 - 1. Refer to PV Series Standard Drawings.
 - 2. Place in the direction of paving and parallel to the slab surface.
 - 3. Place by using automatic bar inserters, bar supports, through forms, or drilled and epoxied after concrete has hardened.
 - a. Do not manually insert tie bars while concrete is still plastic if using a paver.
 - 4. Maintain at least 6 inch clearance between tie bars and dowel bars.
 - a. Eliminate tie bars that interfere with dowel bars.
 - b. Do not adjust dowel bar locations.

- C. Place dowel bars in the middle third of the slab depth, parallel to the centerline and surface of the slab.

3.5 FINISH

- A. Finish the surface smooth and true to grade by machine float immediately after placing concrete.
 - 1. Finish at a rate equal to the progress of the paving operation.
- B. Shut down the mixing operation until the situation is resolved if preliminary finishing is delayed more than 30 minutes after initial screeding.
- C. Fill honeycomb areas in the vertical edge of the pavement with mortar.
- D. Texture the pavement by burlap drag unless the final texture includes artificial turf drag.
 - 1. Use at least three plies of wet burlap and drag parallel to the centerline without tearing.
 - 2. Complete the drag finish with one pass.
 - 3. Spray water directly on the final burlap drag only in the quantity necessary to keep the burlap wet.

3.6 PAVEMENT TEXTURE

- A. Refer to Section 02742S for pavement texture type.
- B. Longitudinal Tining
 - 1. Produce grooves of approximately $\frac{1}{8}$ inch wide by $\frac{1}{8}$ inch deep spaced $\frac{3}{4}$ inch apart and parallel to the longitudinal joint.
 - 2. Keep tining devices clean and free from encrusted mortar and debris to provide uniform groove dimensions.
 - 3. Time tining so that the grooves do not close up.
- C. Diamond Grinding
 - 1. Produce resultant surface in a parallel, corrugated type texture.
 - a. Allowable width of grooves is $\frac{1}{16}$ inch to $\frac{1}{4}$ inch.
 - b. Allowable distance between grooves is $\frac{1}{16}$ inch to $\frac{1}{4}$ inch.
 - c. Maximum allowable height of ridges is approximately $\frac{1}{16}$ inch.
 - 2. Maintain cross slope drainage.
- D. Transverse Tining
 - 1. Produce randomly spaced $\frac{1}{2}$ inch to $\frac{3}{4}$ inch transverse grooves approximately $\frac{1}{8}$ inch deep.

- E. Artificial Turf Drag
 - 1. Drag artificial turf along the pavement in the direction of paving.
 - 2. Use plastic turf that is wide enough to cover the entire pavement width and produce a uniform texture with corrugations $\frac{1}{16}$ inch to $\frac{1}{8}$ inch deep.
 - 3. Use turf with a blade density of 7,200 blades/ft² and each blade at least $\frac{3}{4}$ inch long.
 - 4. Continuously monitor the texturing operation.
 - 5. Weight the turf if necessary to produce an acceptable texture using a uniformly distributed load.
 - 6. Remove buildup of cementitious or other materials that may produce an uneven or unacceptable texture.
- F. Grinding may be required to correct unacceptable texturing.
- G. Rumblestrips
 - 1. Install by diamond grinding when required.
 - 2. Rotomilling is not an acceptable method of installation.

3.7 CURE

- A. Apply concrete curing compound to the entire pavement surface and exposed edges immediately after completing finishing operations.
 - 1. Apply the concrete curing compound in two approximately equal applications.
 - 2. Apply the second application in the opposite longitudinal direction as the first at a combined application rate equal to 100 ft²/gal.
 - 3. Allow at least 30 minutes between applications.
 - 4. Hand spray small and irregular areas and areas inaccessible to mechanical spraying equipment.
- B. Stop the paving operations until the problem is resolved if the concrete curing compound application behind the paving machine is delayed.
 - 1. Use fogging to keep the pavement moist until the concrete curing compound application resumes.
 - a. Use fogging equipment with compressed air misters that atomize the water and produce a very fine mist and not a spray.
 - b. Use equipment that allows for adjusting the rate of fogging depending on the conditions that are present.
 - c. Maintain misters at least 5 ft above the concrete surface and aimed in a direction above horizontal.
 - d. Do not use fogging to apply excess water to the concrete surface to aid finishing.
 - e. Do not affect the water/cement ratio of the concrete.

- f. Discontinue fogging when a fine coating of water or sheen is visible on the concrete surface.
2. Prevent damage to the pavement surface texture.

3.8 PROTECTION

- A. Protect pavement against damage and marring.
 1. Construct crossings to bridge the concrete when necessary.
- B. Do not allow hauling equipment or traffic on the pavement until 100 percent of the specified 28 day minimum compressive strength has been achieved. Verify strength by one of the following:
 1. Maturity method – AASHTO T 325
 2. Cast cylinders
- C. Do not permit hauling equipment or traffic on the pavement before all sawed joints are sealed.

3.9 JOINTS

- A. Construct contact joints, sawed joints, and transverse expansion joints as shown in the plans and approved joint layout.
- B. Keep the joint faces at right angles to the top pavement surface with longitudinal joints parallel to the centerline and coinciding with the traffic lane lines.
- C. Longitudinal Contact Joints
 1. Do not allow the finished surface across longitudinal contact joints to deviate from a straight line by more than $\frac{1}{4}$ inch in 10 ft when tested with a straight edge.
 2. Cease operations until specified tolerances are achieved if the edge slump requirements are not satisfied within 200 ft.
 3. Repair the edge by the following procedures if the edge slump exceeds the specified $\frac{1}{4}$ inch in 10 ft before placing adjacent concrete:
 - a. Saw off the slumped edge to the full thickness with a diamond saw.
 - b. Drill holes in the sawed edge and epoxy in new tie bars.
 - c. Profile grinding may be considered as approved by the Engineer.
 4. Cut and replace or straighten bent tie bars and re-coat with epoxy paint at the bend point before placing concrete in the adjacent lane as determined by the Engineer.

- D. Transverse Contact Joints
 - 1. Construct transverse contact joints normal to the centerline without keyways on the vertical face.
 - 2. Form joints with bars placed through the form or saw joints then drill and epoxy bars.
- E. Longitudinal and Transverse Sawed Joints
 - 1. Single cut all transverse and longitudinal joints. Refer to PV Series Standard Drawings.
 - 2. Saw joints before uncontrolled cracking occurs.
 - 3. Conduct continuous sawing operations during both day and night regardless of weather conditions.
 - 4. Provide lighting during nighttime sawing.
 - 5. Thoroughly clean and dry joints before placing sealant.
 - a. Clean the joint using at least 100 psi air.
 - b. Equip air compressors with operating oil and water traps.
 - 6. Fill the joints to within ¼ inch of the finished surface with joint sealer.
 - a. Remove and replace joint sealer of overfilled joints.
 - 7. Match joints in adjacent lanes to form a continuous line across the pavement width including the concrete shoulders.

3.10 FILL PAVEMENT CORE HOLES

- A. Fill the core holes with the same concrete mix or other product authorized by the Engineer within 24 hours after coring.
- B. Verify the holes are cleaned with no standing water before filling with concrete.
- C. Consolidate the concrete by rodding or vibrating.
- D. Strike off level with the pavement surface and texture.
- E. Protect concrete in core holes from any damage for at least 48 hrs.

3.11 DEFECTIVE PANELS

- A. The COR determines defective panels before substantial completion.
- B. Repair or replace cracked, damaged or otherwise defective panels.
 - 1. Complete repairs before acceptance testing for smoothness.
 - 2. Remove and replace panels when multiple full depth cracks separate the panel into three or more pieces.

3. Use methods that do not disturb or damage adjacent panels.
4. Remove and replace panel with any full depth transverse crack within 4 ft or less of a transverse sawed joint.
5. Drill and epoxy tie-bars/dowel bars as required into existing pavement.
6. Match the profile and texture of existing pavement.
7. Repair any crack connecting joints appearing before substantial completion.
 - a. Rout to a 1 inch depth by $\frac{1}{8}$ inch width and seal with silicone joint sealer. Refer to Section 03152.
8. Leave tight random cracks less than $\frac{1}{64}$ inch wide undisturbed.

3.12 LIMITATIONS – GENERAL

- A. Meet limitations of Section 03055 except as modified below.
 1. Precipitation
 - a. Cease operation when rain is threatening.
 - b. Remove, replace, or repair any pavement damaged by rain or hail as determined by the Engineer.
 - 1) Pavement is considered damaged when rain or hail leaves noticeable texture on the surface.
 - 2) Do not finish rainwater or hail into the concrete surface.
 2. Surface Evaporation
 - a. Limitations apply at any time of the year when any combination of air temperature, relative humidity, and wind velocity have the potential to impair the quality of fresh or hardened concrete or otherwise result in abnormal properties.
 - b. Prepare materials required for evaporation control measures and have them available on site.
 - c. Initiate evaporation control measures when concrete and air temperatures, relative humidity of the air, and wind velocity have the capacity to evaporate water from a free water surface at a rate equal to or greater than 0.2 lb/ft²/hr.
 - 1) Determine the evaporation rate of surface moisture by the Menzel Formula. Refer to ACI 305.1.
- B. Mix Consistency
 1. Cease operations upon having four consecutive batches rejected due to noncompliant air content of the mix.
 2. Do not resume paving operations until the issues have been resolved to the satisfaction of the Engineer.

- C. Verification of Reinforcing Steel
 - 1. Cease paving operations upon identifying 5 percent or more of the transverse joints with non-compliant steel reinforcement.
 - 2. Operations may not resume until the issues have been resolved to the satisfaction of the Engineer.

3.13 LIMITATIONS – COLD WEATHER

- A. Cold weather limitations apply when the temperature is forecast to fall below 40 degrees F within 14 calendar days of placement.
- B. Comply with the following requirements for placing concrete in cold weather:
 - 1. Do not use chemical additives in the concrete to prevent freezing.
 - 2. Provide all necessary cold weather protection for in-place concrete using items such as covers, insulation, and heat.
 - 3. Do not place concrete in contact with frozen surfaces.
 - 4. Adequately vent combustion-type heaters that produce carbon monoxide.
 - 5. Protect the concrete from freezing when the ambient air temperature is 36 degrees F or below until a compressive strength of 3,500 psi has been achieved, determined by either:
 - a. Maturity method – AASHTO T 325
 - b. Field cure cylinders
 - 6. Maintain moist conditions to avoid loss of moisture from the concrete due to heat applied
 - 7. Limit the drop in temperature next to the concrete surfaces when removing heat to 20 degrees F during any 12 hour period until the surface temperature of the concrete reaches that of the atmosphere.
 - 8. Measure, monitor, and record the concrete surface temperature at the frequency in the approved cold weather protection plan.
 - a. Provide the data to the Department within 24 hours of measurement.
 - 9. Paving may occur when base surface temperature is above 35 degrees F in the shade and the ambient air temperature is above 45 degrees F.
 - 10. Remove and replace concrete when any portion of the panel has frozen.
 - 11. Do not use material containing frost or lumps.

- C. Heating Aggregate and Water
 - 1. Provide and operate heating devices when heated aggregates are required.
 - 2. Use aggregates free of ice.
 - 3. Heat aggregates uniformly, avoid overheating or developing hot spots.
 - 4. Use either steam or dry heat.

3.14 LIMITATIONS – HOT WEATHER

- A. Comply with approved hot weather protection plan when ambient air temperature is 85 degrees F to 100 degrees F in the shade.
- B. Discontinue paving when ambient air temperature exceeds 100 degrees F in the shade.

END OF SECTION

SECTION 02753

FULL DEPTH SLAB REPLACEMENT FOR CONCRETE PAVEMENTS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Remove full panel or partial panel of existing pavement.
- B. Clean, grade, and reconsolidate base.
- C. Install dowels and tie bars.
- D. Place and cure concrete.

1.2 RELATED SECTIONS

- A. Section 02752: Portland Cement Concrete Pavement
- B. Section 03055: Portland Cement Concrete
- C. Section 03058: Rapid Strength Concrete (RSC)
- D. Section 03390: Concrete Curing

1.3 REFERENCES

- A. AASHTO M 235: Epoxy Resin Adhesives
- B. ASTM C 309: Liquid Membrane-Forming Compounds for Curing Concrete
- C. ASTM C 1074: Estimating Concrete Strength by the Maturity Method

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Authorized Products List (APL) Compliance Form for Epoxy Resin Adhesive.
- B. Sample of the end caps for approval before use on the project.

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PART 2 PRODUCTS

2.1 FULL DEPTH SLAB REPLACEMENT WITH PORTLAND CEMENT CONCRETE (PCC)

- A. Use AA(AE) concrete. Refer to Section 03055
 - 1. Meet a 28 day flexural strength of 650 psi verified through trial batch.
 - 2. Use high range water reducers such as Super Plasticizers.

2.2 FULL DEPTH SLAB REPLACEMENT WITH RSC

- A. Use RSC. Refer to Section 03058.

2.3 CONCRETE CURING COMPOUND

- A. Refer to ASTM C 309, Type 2, Class A for repairs with PCC
- B. Refer to ASTM C 309 Type 2, Class B for repairs with RSC

2.4 REINFORCING STEEL

- A. Refer to Section 02752.

2.5 EPOXY RESIN ADHESIVE

- A. Use an epoxy resin adhesive according to AASHTO M 235. Select Type and Class for the application.

PART 3 EXECUTION

3.1 PREPARATION

- A. Remove full panel of damaged existing pavement.
 - 1. Determine the extent and dimensions of the panel removal as shown or as directed by the Engineer. Refer to PV Series Standard Drawings.
 - 2. Complete the removal and make full depth cuts around the perimeter of the rectangular section to be removed.
 - a. Minimize saw overcuts.
 - 3. Remove panels by lift-out method.
 - a. Use chains and lift pins to facilitate removal and minimize disturbance of the base material.

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4. Repair damage to adjacent slabs and underlying base courses caused by removal operations.
 5. Remove all loose particles of existing concrete before placing new concrete.
- B. Grade and compact the existing base.
1. Use the largest piece of vibratory or impact compaction equipment practical.
 2. Continue compaction until additional work does not improve the measured density
- C. Form any side that does not have an adjacent panel.
1. Form to match existing panels, providing a vertical edge.
- D. Place Dowel and Tie Bars
1. Place bars in locations according to PV Series Standard Drawings. Use tie bars or smooth dowels where indicated on standard drawings.
 2. Stockpile bars and keep them clean and free from damage.
 3. Drill holes mid-depth of the slab without causing damage to the remaining pavement section and meet bar placement tolerances.
 4. Drill multiple holes simultaneously with drills held horizontally in a rigid frame.
 - a. Prevent drill bits from wandering.
 5. Clean holes of dust, grease, and other contaminants.
 6. Inject epoxy resin adhesive into the back of the drilled hole.
 - a. Provide sufficient quantity of bonding compound to completely fill the void between the bar and the outer limits of the drilled hole.
 - b. Rotate one full revolution while inserting bar.
 - c. Use retention rings to prevent the bonding compound from flowing out of the hole.
 7. Align bars horizontally and vertically to meet requirements of the Standard Drawings and allow them to stabilize before mix placement.
 8. Repair any bar coating damage with appropriate repair material.
 9. Place tight fitting end caps made of non-metallic materials that allow ¼ inch movement on protruding dowels used at expansion joints.
 10. Coat protruding portion of dowel bar with lubricant consisting of paraffin wax, lithium grease, or other semi-solid, inert lubricant approved by the COR.
 11. Remove and replace loose bars before placing concrete mix.

- E. Prepare panel joints for concrete placement.
 - 1. Maintain existing pavement joint layout.
 - 2. Place a bond breaker approved by the COR on the existing pavement edges that become transverse or longitudinal panel joints.
 - 3. Saw cut joint on the same line if repairs straddle an existing joint line. Saw according to Section 02752.

3.2 PLACE CONCRETE

- A. Place concrete according to Section 02752 for panels placed with PCC.
- B. Place concrete according to Section 03058 for panels replaced with RSC

3.3 FINISH CONCRETE

- A. Finish panels to $\pm \frac{1}{8}$ inch of existing profile.
 - 1. Correct panel profiles in excess of $\frac{1}{8}$ inch higher than the existing pavement profile by surface grinding or removal and replacement.
 - 2. Correct panel profiles in excess of $\frac{1}{8}$ inch lower than the existing pavement profile by removal and replacement of the panel.
 - 3. Make necessary corrections to the concrete finish.
- B. Do not tool joints that are to be saw cut and sealed.
- C. Texture the surface to match the existing pavement.

3.4 CURE, PROTECT AND SAW CUT CONCRETE

- A. Cure the concrete pavement.
 - 1. Refer to Section 02752 when using PCC
 - 2. Refer to Section 03058 when using RSC.
- B. Protect the individual placements with approved barricades.
- C. Saw cut all previously existing joints to original dimensions.
- D. Fill all sawing overcuts with repair epoxy approved by the Engineer.

3.5 DEFECTIVE PAVEMENT PANELS

- A. Remove and replace defective panels. Refer to Section 02752.

3.6 LIMITATIONS

- A. Do not open to traffic until 3000 psi strength is reached. Verified by either:
 - 1. Maturity Method – Refer to ASTM C 1074
 - 2. Cast cylinders when using PCC
 - 3. Concrete Compressive Strength of 3500 psi is required in cold weather.

END OF SECTION

SECTION 02754

DOWEL BAR RETROFIT

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Installation of coated dowel bars across existing transverse joints and cracks.

1.2 RELATED SECTIONS

- A. Section 03152: Concrete Joint Control
- B. Section 03211: Reinforcing Steel and Welded Wire

1.3 REFERENCES

- A. AASHTO T 22: Compressive Strength of Cylindrical Concrete Specimens
- B. ASTM C 157: Length Change of Hardened Hydraulic-Cement Mortar and Concrete
- C. ASTM C 309: Liquid Membrane-Forming Compounds for Curing Concrete
- D. ASTM C 882: Bond Strength of Epoxy-Resin Systems Used With Concrete By Slant Shear
- E. ASTM C 928: Packaged, Dry, Rapid-Hardening Cementitious Materials for Concrete Repairs
- F. UDOT Quality Management Plan 506

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Manufacturer's data product sheet and recommended installation instructions for information.
- B. Volumetric Mixer annual inspection and calibration records according to the Quality Management Plan 506 for information.

- C. The names and contact information of at least two individuals who have experience on previous projects in all aspects of the dowel bar retrofit work for information.
 - 1. Provide a written summary of each individual's experience.
 - 2. One of these individuals must be present on the project site during all working hours to oversee dowel bar retrofit work.

1.6 QUALITY CONTROL

- A. Provide a technical representative from the patching material product manufacturer for consultation.
 - 1. The representative must be onsite during test section activities.
 - 2. The Government reserves the right to require the technical representative to be onsite at any time during product installation.

PART 2 PRODUCTS

2.1 MATERIALS

- A. Dowel Bars with Bond Breaking Compound
 - 1. Use 1 ½ inch x 18 inch plain epoxy coated reinforcing steel. Refer to Section 03211.
 - 2. Use a bond breaking compound applied at the manufacturing facility.
- B. Chair Devices
 - 1. Refer to bar support requirements in Section 03211.
 - 2. Size to provide a clearance of at least ½ inch between the bottom of the bar and the supporting surface.
- C. End Caps
 - 1. Provide tight fitting, non-metallic end caps for 1 ½ inch diameter dowel bar, manufactured for dowel bar retrofit applications.
- D. Caulking Filler
 - 1. Use a commercial caulk capable of bonding to concrete.
- E. Patching Material
 - 1. Use prepackaged concrete patching material with an aggregate extension conforming to the manufacturer's recommendations.
 - 2. Use mix with 100 percent of washed aggregate passing the ¾ inch screen.
 - 3. Select patching material that meets the performance criteria listed in Table 1.

Table 1

Patching Material		
Compressive Strength	Test Method	Value
At 3 hours	AASHTO T 22	3000 psi
At 24 hours	AASHTO T 22	5000 psi
Shrinkage in 14 days	ASTM C 157	0.04 percent, max
Bond Strength	ASTM C 882 as modified by ASTM C 928, Section 8.5	1000 psi, minimum @ 24 hours

- F. Foam Core Board
 - 1. Capable of maintaining the joint or rack.
 - 2. At least $\frac{3}{8}$ inch thick.
 - 3. Made of rigid Styrofoam or closed cell foam with poster board material or plastic faced material on each side.
- G. Curing Compound
 - 1. ASTM C 309, Type 1D Class A

2.2 EQUIPMENT

- A. Jackhammers
 - 1. 30 lbs or smaller.
- B. Gang Mounted Saws
 - 1. Capable of cutting at least 3 slots per wheel path simultaneously.
 - 2. Capable of vacuuming water and paste residue from surface immediately after sawing.
- C. Volumetric Mixers
 - 1. Meet the requirements of the Quality Management Plan 506 for volumetric mixers if used.
- D. Sandblaster or Water Blasting Equipment
 - 1. Water blasting equipment must be capable of producing at least 3000 psi at the nozzle.

PART 3 EXECUTION

3.1 CONSTRUCTION

- A. Test Section
 - 1. Provide a test section consisting of at least 24 complete dowel bar retrofits at a location determined by the Engineer before beginning production work.
 - a. Take one 6 inch diameter full-depth core at three separate locations after completing the test section to assess the installation.
 - 1) Inspect the cores in the presence of the COR.
 - 2) Verify proper dowel placement and that there are no voids around the bar circumference.
 - 3) Repair core holes using the approved patching material.
 - 4) Inspect cured patching material for cracking.
 - b. Begin production work after obtaining the COR's concurrence.
 - 2. Additional cores may be required on subsequent production days as determined by the COR.
- B. Saw cut the dowel slot into pavement as shown in the PV Series Standard Drawings.
- C. Use a jackhammer for concrete removal in cut slot areas.
 - 1. Produce a flat level surface on the slot bottom
- D. Sand or water blast to clean all exposed surfaces and cracks removing slurry and loose concrete.
 - 1. Demonstrate acceptable results if using water blasting.
- E. Clean up and properly dispose of all residues from the saw, jackhammer, sand or water blasting process, and other sources.
 - 1. Re-clean patching areas if work is delayed.
- F. Place caulking filler in existing joint or crack to prevent intrusion of patching material.
 - 1. Dry and clean surface receiving caulk.

- G. Place the foam core board on the dowel bar in line with the transverse joint or crack.
 - 1. Fit the foam core board tightly around the dowel bar and to the bottom and edges of the slot.
 - 2. Adjust the height of the foam core board by cutting so it does not extend above the pavement surface.
- H. Place bars so that the bars do not extend more than 11 inch past the centerline of the slot.
- I. Thoroughly dampen concrete surfaces of the slot immediately before filling with patching material.
 - 1. Prevent standing water in the slot.
 - 2. Remove all excess water with compressed air.
- J. Fill the slot with patching material.
 - 1. Consolidate the material in the slot and around the dowel bar with an appropriate size vibrator.
 - 2. Place patching material no higher than ¼ inch above adjacent pavement before grinding.
- K. Protect the retrofit from traffic until the patching material achieves a compressive strength of 3,000 psi
- L. Begin pavement grinding within 10 working days of placing dowel bar retrofit patching materials.
 - 1. Grind patching material flush with the adjacent pavement.
- M. Remove foam core board as needed to a depth of 2 inch below surrounding surface and reseal. Refer to Section 03152.

END OF SECTION

SECTION 02755

CONCRETE SLAB JACKING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Injection of a product to raise and support a failed section of concrete pavement back to grade for a suitable ride.

1.2 RELATED SECTIONS

- A. Section 02705: Concrete and Asphalt Cutting

1.3 REFERENCES

- A. ASTM C 1107: Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
- B. ASTM D 790: Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
- C. ASTM D 1621: Compressive Properties of Rigid Cellular Plastics
- D. ASTM D 1622: Apparent Density of Rigid Cellular Plastics

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Copies of invoices from mix suppliers for information.
- B. Manufacturer's product data sheet and recommended installation instructions.
- C. A warranty letter guaranteeing all materials and workmanship for a period of one year.
 - 1. The guarantee period starts on the date of Substantial Completion.
 - 2. Include in the letter:
 - a. State Project Designation
 - b. State Project Name
 - c. Contractor and Installer Name

3. Defects (performance failures) include:
 - a. Re-settling of treated areas beyond 1/8 inch of finished grade profile
4. The guarantee covers 100 percent of the concrete slab jacking costs.
5. Replace by re-injection or slab replacement any material that fails during the warranty period.

1.8 PAYMENT PROCEDURES

- A. Liquid quantities of non-shrink grout measured using a calibrated meter paid for at the contract unit price per cubic yard.
- B. Liquid quantities of High-Density Polyurethane measured using methods approved by Engineer paid for at the contract unit price per pound.

PART 2 PRODUCTS

2.1 HIGH DENSITY POLYURETHANE

- A. Use water based formulation of expanding high-density polyurethane that sets to full compressive strength within 15 minutes after injection to raise slabs.

Table 1

High-Density Polyurethane		
PROPERTY	ASTM	LIMITS
Density	D 1622	Min 3.8 lb/ft ³ - Max 4.3 lb/ft ³
Tensile Strength	D 790*	Min. 80 psi
Elongation	N/A	Max. 5.1%
Compressive Strength at Yield Point	D 1621	Min. 60 psi

*Use the value of flexural strength or flexural yield as tensile strength.

2.2 NON-SHRINK GROUT

- A. Use packaged dry, hydraulic-cement grout (non-shrink) according to ASTM C 1107.

2.3 EQUIPMENT

- A. Equipment capable of supplying a homogenous product at the appropriate rate.

- B. Certified scales or measuring devices to measure delivered product and to proportion product components.
- C. Concrete drill or saw capable of producing circular holes of adequate size for the application type.
- D. Elevation measuring devices with an accuracy of $\frac{1}{8}$ inch.
- E. Concrete saw capable of cutting joints between failed and non-failed slabs.

PART 3 EXECUTION

3.1 PREPARATION

- A. Establish a finish target profile of pavement using elevation measuring device or string lines.
- B. Saw cut joints between failed and non-failed slabs as necessary to prevent damage to non-failed slabs. Refer to Section 02705.

3.2 INSTALLATION

- A. Drill holes as necessary in roadway slab.
- B. Inject product to evenly raise slab to finished grade profile.
- C. Fill injection holes with at least 4 inches of non-shrink grout.
- D. Final grade after jacking must be within $\pm \frac{1}{8}$ inch of finished grade profile.

END OF SECTION

SECTION 02761

LONGITUDINAL RUMBLE STRIP

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Conventional longitudinal rumble strip
- B. Sinusoidal longitudinal rumble strip

1.2 RELATED SECTIONS

- A. Section 01558: Temporary Pavement Markings
- B. Section 02742S: Project Specific Surfacing Requirements

1.3 REFERENCES **Not Used**

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS **Not Used**

1.6 ACCEPTANCE

- A. The Department will reduce the price paid per foot of longitudinal rumble strip not installed according to PV series Standard Drawings, and according to Tables 1 and 2.

Table 1

Difference from Specified Depth	Pay Factor
0%-20%	1
> 20%-33%	0.75
> 33%-50%	0.50
> 50%	Remove and Replace

Table 2

Offset from Control Line	Pay Factor
0-2 inch	1
> 2-4 inch	0.50
> 4 inch	Remove and Replace

- C. Remove and replace longitudinal rumble strips by milling 2 inch deep and a minimum of 4 ft wide with no new joints in wheel paths, leaving no pavement section less than 4 ft wide.

PART 2 PRODUCTS

2.1 RUMBLE STRIP SEAL

- A. Flush Coat, Chip Seal Coat, or Asphalt Polymer Treatment (APT). Refer to Section 02742S.

2.2 EQUIPMENT

- A. Install longitudinal rumble strip with equipment using a rotary type cutting head capable of obtaining the required groove width, depth and length in a single pass while moving in the same direction as the traffic flow.
 - 1. Use equipment with cutting heads having an independent suspension from the power unit to allow the head to self align with the slope of the shoulder.
 - 2. Use cutting heads that provide a smooth surface, approximately $\frac{1}{16}$ inch between peaks or valleys.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Install longitudinal rumble strip as shown to the dimensions and spacing in PV Series Standard Drawings.
 - 1. Establish lane widths from plan set or existing roadway if not shown, before installation of longitudinal rumble strip.
 - 2. Chip Seal Coat Application Timing
 - a. Install sinusoidal rumble strip a minimum of 30 days after placement of Chip Seal Coat.
 - b. Install conventional longitudinal rumble strip prior to placement of Chip Seal Coat.

- B. Alignment Control
 - 1. Establish a control line for placement of rumble strip using a dribble/drip line, temporary pavement marking material, or permanent application of pavement marking material according to Section 01558 in its permanent configuration and location before the installation of longitudinal rumble strip.
 - a. Establish control points at intervals no greater than 100 ft on tangents and no greater than 50 ft on curves.
- C. Construct a 120 ft long test section each production day, or when the equipment is moved to another location, to demonstrate that the equipment, personnel, and methods of operation are capable of producing acceptable results without damaging the pavement.
- D. Clean the surface of dirt, sand, dust, and other objectionable material before applying rumble strip seal.
- E. Apply rumble strip seal on asphalt surfaces after longitudinal rumble strip has been installed in the roadway.
 - 1. Refer to Section 02742S for rumble strip seal speed and application method to achieve uniform application rate and to prevent shadow areas of thinner application depths.
 - a. Apply rumble strip seal continuously not stopping for gaps in grinds.
 - 2. Apply rumble strip seal a minimum of 6 inch wider than the grind on each side.
- F. Do not apply rumble strip seal on Portland Cement Concrete Pavement (PCCP) surfaces.
- G. Remove resulting debris before opening the adjacent lane to traffic.
 - 1. Dispose of material according to Federal, State, and Local regulations.

END OF SECTION

SECTION 02765

PAVEMENT MARKING PAINT

PART 1 GENERAL

Pavement Marking Paint
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August 2020
Latest Revision: [January 1, 2017](#)

1.1 SECTION INCLUDES

- A. Furnish acrylic water-based pavement marking paint. Refer to this Section, Article 2.2 for resin requirement.
- B. Apply to hot mix asphalt or portland cement concrete pavement as longitudinal lines, transverse markings, contrast lines, and other related markings.
- C. Remove pavement markings.

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES

- A. AASHTO M 247: Glass Beads Used in Traffic Paints
- B. ASTM D 562: Consistency of Paints Measuring Krebs Unit (KU) Viscosity Using a Stormer-Type Viscometer
- C. ASTM D 1155: Roundness of Glass Spheres
- D. ASTM D 1475: Density of Liquid Coatings, Inks, and Related Products
- E. ASTM D 1644: Nonvolatile Content of Varnishes
- F. ASTM D 2205: Selection of Tests for Traffic Paints
- G. ASTM D 2743: Traffic Paint Vehicle Solids by Spectroscopy and Gas Chromatography
- H. ASTM D 2805: Hiding Power of Paints by Reflectometry
- I. ASTM D 3723: Pigment Content of Water-Emulsion Paints by Low-Temperature Ashing
- J. ASTM D 3960: Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings
- K. ASTM D 5381: X-Ray Fluorescence (XRF) Spectroscopy of Pigments and Extenders
- L. ASTM E 1347: Color and Color-Difference Measurement by Tristimulus (Filter) Colorimetry
- M. Environmental Protection Agency Testing Methods

- N. Federal Specification
- O. Federal Standards
- P. Manual on Uniform Traffic Control Devices (MUTCD)
- Q. UDOT Materials Manual of Instruction
- R. UDOT Minimum Sampling and Testing Requirements
- S. UDOT Quality Management Plans

1.4 DEFINITIONS

- A. Longitudinal Markings – pavement markings that are generally placed parallel and adjacent to the flow of traffic such as lane lines, center lines, edge lines, channelizing lines, and others.
- B. Transverse Markings – pavement markings that are generally placed perpendicular and across the flow of traffic such as shoulder markings; word, symbol, and arrow markings; stop lines; crosswalk lines; speed measurement markings; parking space markings; and others.
- C. Long-term stationary – work that occupies a location more than 3 days.

1.5 SUBMITTALS

- A. Documentation of the manufacturer and production batch identification for the paint used.

1.6 ACCEPTANCE

- A. Provide fixtures such as ball valves, gate valves, or others on paint truck for the purposes of obtaining field samples.
- B. The Government will:
 - 1. Accept pavement marking paint from qualified manufacturer supplied samples.
 - 2. Sample from the applicator's yard, at the Department's discretion for acceptance.
- C. Stop all agitation before sample is drawn.
- D. The Quality Control will:
 - 1. Visually inspect longitudinal lines and transverse markings to verify Pavement Marking Paint

- compliance with the required dimensions.
 - 2. Inspect at the end of each production day or more frequently as required.
 - 3. Verify quantities applied by either of the following methods:
 - a. Measuring both paint and bead tanks before and after application.
 - b. Witnessing the meter readings before and after application.
 - 1) A printout of meter readings instead of witnessing may be accepted at the COR's discretion.
 - 4. Sample in the field according to the UDOT Quality Management Plan 513, Pavement Marking Paint and the UDOT Minimum Sampling and Testing Requirements.
- E. Repaint any line or legend failing to meet bead application rates and dimensional requirements.
- 1. Do not remove earlier application.
- F. Price Reductions
- 1. Price reductions for pavement markings installed below the specified wet mil thickness are outlined in Table 1.

Table 1

Price Reduction for Wet Mil Thickness	
	Pay Factor
Less than 2 percent below the specified mil thickness	1.00
Less than 10 percent below the Specified wet mil thickness	0.75
Less than 15 percent below the Specified wet mil thickness	0.50
15 percent or more below the Specified wet mil thickness	0.00 *

* Repaint pavement markings at no cost to the Department. Do not remove earlier application.

2. The Government will apply price reductions for applied pavement markings that do not meet the requirements of Table 4 when the batch is sampled at the applicator's yard for quality verification according to UDOT Quality Management Plan 513, Pavement Marking Paint.
 - a. Apply the lowest pay factor from Tables 2 and 3.
3. The Department will require repainting for pavement markings more than five percentage points below total solids, pigment, or non-volatile vehicle properties from Table 4 when the project is sampled for quality verification according to Minimum Sampling and Testing Requirements.

Table 2

Price Reduction for Total Solids, Pigment and Non-Volatile Vehicle	
	Pay Factor
Less than 0.5 percent below or above the specified percentage	1.00
Less than 1.0 percent below the specified percentage	0.95
Less than 2.0 percent below the specified percentage	0.85
2.0 percent or more below the specified percentage	0.00 *

* Repaint pavement markings at no cost to the Department. Do not remove earlier application.

Table 3

Price Reductions for remaining requirements of Table 4	
	Pay Factor
Less than 0.2 percent deficient	1.00
Less than 1 percent deficient	0.90
Less than 2 percent deficient	0.80
Less than 3 percent deficient	0.70
Less than 4 percent deficient	0.60
Less than 5 percent deficient	0.50
5 percent or more below specified quantitative requirements	0.00 *

* Repaint pavement markings at no cost to the Department. Do not remove earlier application.

PART 2 PRODUCTS

2.1 PAINT

- A. Meet the requirements for Acrylic Water Based Paint specified in Table 4.

Table 4

Paint Requirements				
Property	White	Yellow	Black	Test
Pigment – Percent by weight, minimum	62.0	62.0	62.0	ASTM D 3723
Total Solids – Percent by weight, minimum	77.0	77.0	77.0	ASTM D 1644
Nonvolatile vehicle – Percent by weight vehicle, minimum*	43.0	43.0	43.0	ASTM D 3723 ASTM D 1644
Viscosity, KU @ 77 degrees F	80 – 95	80 – 95	80 – 95	ASTM D 562
Density, lb/gal, minimum	14.0	14.0	14.0	ASTM D 1475
Volatile Organic Content (VOC) – g/L, maximum	100	100	100	ASTM D 3960
Titanium Dioxide Content, lb/gal	1.0 min	0.2 max	N/A	ASTM D 5381
Color Definition	37875	33538	N/A	Federal Standard 595B
Directional Reflectance Minimum	90.0	50.0	N/A	ASTM E 1347
Dry Opacity – Minimum (5 mils wet)	0.95	0.95	N/A	ASTM D 2805

* Binder – 100 percent acrylic cross-linking polymer, by weight, as determined by infrared analysis and other chemical analysis available to the Department. Refer to ASTM D 2205.

- B. No-Pick-Up Time
 - 1. Paint may not smear or track three minutes after application to the roadway using standard application equipment, at the mil thickness required, and with an ambient shaded temperature of at least 50 degrees F.
- C. Additional Requirements
 - 1. Free of lead, chromium, or other related heavy metals. Refer to ASTM D 5381.
 - 2. Refer to ASTM D 2743 and ASTM D 5381 for tests used to verify paint samples meet ASTM requirements.

2.2 GLASS SPHERES (BEADS) USED IN PAVEMENT MARKING PAINT

- A. Heavy metal concentration: Manufacturer must provide a certificate of Pavement Marking Paint
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compliance stating that all beads contain no more than the amounts listed for the following materials as determined by testing performed according to EPA test methods 3052 and 6010C.

1. Other suitable x-ray fluorescence spectrometry analysis methods may be used to screen samples of glass spheres for arsenic, antimony and lead content.

Table 5

Heavy Metal Materials	
Material	Level (ppm, total)
Arsenic	200
Antimony	200
Lead	200

- B. Longitudinal Lines – Refer to AASHTO M 247, Specific Properties, with the following exceptions:

1. Gradation:

Table 6

Gradation	
Sieve Size	Accumulated Percent Passing
No. 18	65 – 80
No. 30	30 – 50
No. 50	0 – 5

2. Coating – Dual coating for optimum adhesion and embedment.
3. Roundness – 80 percent true spheres below the number 30 sieve. Refer to ASTM D 1155
4. Color/Clarity – Colorless/clear and free of carbon residue.
5. Refractive Index – Minimum 1.51 by oil immersion method.
6. Air Inclusions – Less than 5 percent by visual inspection.
7. Hardness – Beads above the number 30 sieve exhibit an average hardness of C70.5 when measured using the Rockwell C scale method and using a minimum sample of 100 beads.
8. Crushing Strength – Beads above the number 30 sieve exhibit an average crushing strength of 60,000 psi when measured by the L/D^2 method and with a minimum sample of 100 beads.
9. Chemical Resistance – Beads resistant to hydrochloric acid, water, calcium chloride, and sodium sulfide. TT-B Federal Specification 1325C sections 4.3.6 to 4.3.9.

- C. Transverse Markings – Refer to AASHTO M 247, Specific Properties, with the following exceptions:

1. Gradation:

Table 7

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August 2020

Latest Revision: [January 1, 2017](#)

Gradation	
Sieve Size	Accumulated Percent Passing
No. 20	90 – 95
No. 30	45 – 70
No. 50	5 – 25
No. 80	0 – 5

2. Coating – Dual coating for optimum adhesion and embedment.
 3. Roundness – The glass beads will have at least 75 percent true spheres.
 4. Refractive index – Minimum 1.51 by oil immersion method.
 5. Air Inclusions – Less than 10 percent by visual inspection.
 6. Have at least 80 percent true spheres.
- D. Beads used in Temporary Pavement Markings. Meet the above or AASHTO M 247 Type II uniform gradation.

PART 3 EXECUTION

3.1 PREPARATION

- A. Line Control
1. Establish control points at 100 ft intervals on tangent and at 50 ft intervals on curves.
 2. Maintain the line within 2 inches of the established control points and mark the roadway between control points as needed.
 - a. Remove paint that is not placed within tolerance of the established control points and replace. Refer to this Section, Article 3.4.
 - b. Maintain the line dimension within 10 percent of the width and length dimensions defined in Standard Drawings.
- B. Remove dirt, loose aggregate, curing compounds, and other foreign material and follow manufacturer's recommendations for surface preparation.

3.2 APPLICATION

- A. Use Qualified Applicators as identified in UDOT Quality Management Plan 513, Pavement Marking Paint.
- B. Apply Pavement marking paint at the following wet mil thickness:

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1. 20-25 wet mils for all longitudinal markings.
2. **Approximate** application rate for required mil thickness requirements:
 - a. 4 inch Solid Line – From 190 to 240 ft/gal
 - b. 4 inch Broken Line – From 760 to 960 ft/gal
 - c. 8 inch Solid Line – From 95 to 120 ft/gal. Use the following calculation to determine wet mil thickness if approximation is outside the range for the desired line type.

Calculation – Determine wet mil thickness

$$4 \text{ inch Solid Line – Wet mils} = \frac{4812.516 \text{ ft}^3/\text{gal mil/ft}}{X \text{ ft/gal}}$$

$$4 \text{ inch Broken Line – Wet mils} = \frac{19250.064 \text{ ft}^3/\text{gal mil/ft}}{X \text{ ft/gal}}$$

$$8 \text{ inch Solid Line – Wet mils} = \frac{2406.258 \text{ ft}^3/\text{gal mil/ft}}{X \text{ ft/gal}}$$

Where:

X = application rate. (Meter readings or dipping tanks).

- C. Refer to Table 1 for price reduction of pavement markings that are less than required wet mils in thickness.
- D. No additional payment for pavement markings placed in excess of required wet mils in thickness or exceeding dimensional requirements outlined in this Section, Article 3.2 paragraph B.
- E. Glass Sphere (Beads) – Apply at least 8 lb/gal of paint, the full length and width of line and pavement markings.
 1. Calibrate bead guns and measure bead distribution according to UDOT Materials Manual of Instruction 932, Procedure for Sampling and Accepting Pavement Marking Paint and Beads.
 2. Do not apply glass beads to contrast lines (black paint).
- F. Begin striping operations no later than 24 hours, or the next business day if a weekend, after acceptance of the finished pavement surface by the COR.
 1. Apply two applications on new bituminous surfaces.
- G. Apply lines and pavement markings only when the air and pavement temperature are:
 1. 50 degrees F and rising for Acrylic Water Based Paint.

- a. Non-grooved lines and markings applied at temperatures below 50 degrees F are temporary and must be repainted, when temperature conditions are met.
 - 1) Do not remove earlier application.
 - b. Grooved lines and markings applied below 50 degrees F must be removed and reapplied when temperature conditions are met.
- H. Comply with TC Series Standard Drawings.

3.3 CONTRACTOR QUALITY CONTROL

- A. Adhere to the requirements of UDOT Quality Management Plan 513, Pavement Marking Paint

3.4 REMOVE PAVEMENT MARKINGS

- A. Use equipment specifically designed for removal of pavement marking material.
- B. Use one of these removal methods
 - 1. High pressure water spray
 - 2. Sand blasting
 - 3. Shot blasting
- C. Do not use grinding without approval from the COR.
- D. Do not eliminate or obscure existing striping, instead of removal, by covering with black paint or any other covering.
 - 1. The COR may approve use of black paint or other obscuring material prior to installation for work durations shorter than “long term stationary” as defined in this Section, Article 1.4 and in the Temporary Traffic Control section of the MUTCD.

END OF SECTION

SECTION 02768

PAVEMENT MARKING MATERIALS

(Warranty Specification)

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Pavement Marking Tape, Pavement Marking Epoxy, Methyl-Methacrylate, and Thermoplastic legends.

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. Code of Federal Regulations, Title 40, Chapter 1, Subchapter C, part 59

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. A warranty letter stating that the manufacturer guarantees the product against material and installation defects incurred under traffic for the service life.
 - 1. The guarantee period starts on the date of Physical Completion.
 - 2. Include in the letter:
 - a. State Project Designation
 - b. State Project Name
 - c. Quantities
 - d. Surface type
 - e. Material type
 - 3. Defects (performance failures) include:
 - a. Loss of retro-reflectivity
 - b. Loss of presence.
 - 4. The guarantee covers 100 percent of the pavement marking materials and installation costs.
 - 5. Removal and replacement of the pavement marking for failed segment.
 - 6. The Government will notify the manufacturer of defects to be repaired during the guarantee period.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. According to manufacturer's recommendations.

PART 2 PRODUCTS

2.1 PAVEMENT MARKING TAPE

- A. Preformed adhesive tape with a raised profile for longitudinal lines on all pavement surfaces.
- B. Preformed adhesive tape with a flat or raised profile for legends and symbols on all pavement surfaces.
- C. Minimum service life for the following applications under all traffic volumes and wear conditions:

Table 1

TYPE	ELEVATION (ft above mean sea level)	INSTALLATION METHOD	MIN. SERVICE LIFE
Longitudinal lines	0 to 5499	Hot inlay	48 months
Longitudinal lines	0 to 5499	Grooved	72 months
Longitudinal lines	5500 or more	Grooved	48 months
Legends and symbols	All	All	24 months

- D. Performance measures for retro-reflectivity and presence.
1. Minimum retro-reflectivity
 - a. White longitudinal lines 125 millicandelas
 - b. Yellow longitudinal lines 125 millicandelas
 - c. Legends and symbols 125 millicandelas
 2. Minimum presence level – 90 percent of the longitudinal line on any 1,000 ft segment or 90 percent of the legend and symbol must be present.
 3. Failure to meet any of the specified performance measures on at least 90 percent of the longitudinal line in any 1,000 ft segment or 90 percent of a legend or symbol is considered a complete failure of that marking and requires complete replacement by the manufacturer.

2.2 PAVEMENT MARKING EPOXY

- A. Use 100 percent lead free, two-component (resin and hardener) epoxy system for longitudinal lines, legends, and symbols for all pavement surfaces.
- B. Minimum service life for the following applications under all traffic volumes and wear conditions:
 - 1. Type 1 - Fast dry
 - a. Longitudinal – Center, skip, and edge lines 24 months
 - b. Legends and symbols 12 months
 - 2. Type 2 - Slow dry
 - a. Longitudinal – Center, skip, and edge lines 48 months
 - b. Legends and symbols 24 months
- C. Performance measures for retro-reflectivity and presence.
 - 1. Minimum retro-reflectivity
 - a. White longitudinal lines 125 millicandelas
 - b. Yellow longitudinal lines 125 millicandelas
 - c. Legends and symbols 125 millicandelas
 - 2. Minimum presence level – 90 percent of the longitudinal line on any 1,000 ft segment or 90 percent of the legend and symbol must be present.
 - 3. Failure to meet any of the specified performance measures on at least 90 percent of the longitudinal line in any 1,000 ft segment or 90 percent of a legend or symbol is considered a complete failure of that marking and requires complete replacement by the manufacturer.
- D. Use beads according to manufacturer's recommendations.

2.3 METHYL METHACRYLATE

- A. Use a two-component pavement marking system compliant with Federal and State VOC regulations used for longitudinal lines, legends, and symbols and for use on all pavement surfaces.
 - B. Minimum surface life for the following applications under all traffic volumes and wear conditions:
 - 1. Longitudinal lines 48 months
 - 2. Legends and symbols 18 months
 - C. Performance Measures for retro-reflectivity and presence.
 - 1. Minimum retro-reflectivity
 - a. White Longitudinal markings 125 millicandelas
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- b. Yellow Longitudinal markings 125 millicandelas
 - c. Legends and Symbols 125 millicandelas
 - 2. Minimum presence level – 90 percent of the longitudinal line on any 1,000 ft segment or 90 percent of the legend and symbol must be present.
 - 3. Failure to meet any of the specified performance measures on at least 90 percent of the longitudinal line in any 1,000 ft segment or 90 percent of the legend or symbol is considered a complete marking failure and requires complete replacement by the manufacturer.
- D. Use beads according to manufacturer's recommendations.

2.4 PREFORMED THERMOPLASTIC HEAT FUSED PAVEMENT MARKING MATERIALS

- A. Use for legends and symbols only, on all pavement surfaces.
- B. Heat-fused preformed and sprayed pavement marking materials.
- C. Minimum service life for legends and symbols under all traffic volumes and wear conditions is 24 months.
- D. Performance measures for retro-reflectivity and presence.
 - 1. Minimum level of retro-reflectivity – 125 millicandelas
 - 2. Minimum presence – 90 percent of the each legend or symbol must be present.
 - 3. Failure to meet any of the specified performance measures on at least 90 percent of the legend or symbol is considered a complete failure of that legend or symbol and requires complete replacement by the manufacturer.

2.5 HOT MELT THERMOPLASTIC PAVEMENT MARKING MATERIALS

- A. Use for longitudinal lines only, on all pavement surfaces.
- B. Minimum service life for long lines – 24 months.
 - 1. Applies under all traffic volumes and wear conditions.
- C. Performance measures for retro-reflectivity and presence.
 - 1. Minimum retro-reflectivity – 125 millicandelas.
 - 2. Minimum presence – 90 percent of each legend or symbol present.
 - 3. Failure to meet any of the specified performance measures on at least 90 percent of the legend or symbol is considered a complete failure and requires complete replacement by the manufacturer.

PART 3 EXECUTION

3.1 PREPARATION

- A. Prepare pavement surface according to manufacturer's recommendations.

3.2 APPLICATION

- A. Apply pavement marking materials according to manufacturer's recommendations.

END OF SECTION

SECTION 02771

PEDESTRIAN ACCESS RAMP

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Pedestrian access ramp.

1.2 RELATED SECTIONS

- A. Section 02221: Remove Structure and Obstruction
- B. Section 02705: Concrete and Asphalt Cutting
- C. Section 02721: Untreated Base Course (UTBC)
- D. Section 02741: Asphalt Mix
- E. Section 02752: Portland Cement Concrete Pavement
- F. Section 02776: Concrete Flatwork
- G. Section 03055: Portland Cement Concrete
- H. Section 03390: Concrete Curing

1.3 REFERENCES

- A. AASHTO M 306: Drainage, Sewer, Utility, and Related Castings
- B. ASTM A 48: Gray Iron Castings

1.4 DEFINITIONS

- A. Pedestrian Access Ramp – the elements contained in the PA Series Standard Drawings providing an accessible pedestrian route from the sidewalk to the crosswalk, including the pedestrian elements of raised islands.

1.5 SUBMITTALS

- A. Certified Installer's Department ADA Pedestrian Ramp Training Certificate for information.
- B. Authorized Products List (APL) Compliance Form for information for detectable warning surface.

1.6 CERTIFIED INSTALLER REQUIREMENTS

- A. Certified Installer must:
 - 1. Be employed by the Contractor or Subcontractor.
 - 2. Have successfully completed UDOT's Department's ADA Pedestrian Ramp Training Course within the last 3 years. Refer to <http://www.udot.utah.gov/go/standardsreferences>.
 - 3. Be on project site during forming, installation, and inspection of pedestrian access ramp.

PART 2 PRODUCTS

2.1 CONCRETE

- A. Class AA(AE). Refer to Section 03055.
- B. Portland Cement Concrete Pavement. Refer to Section 02752.
- C. Concrete Flatwork. Refer to Section 02776

2.2 ASPHALT MIX

- A. Refer to Section 02741.

2.3 UNTREATED BASE COURSE

- A. Refer to Section 02721.

2.4 DETECTABLE WARNING SURFACE

- A. Use In-line truncated dome pattern that meets the requirements of the PA Series Standard Drawings.
- B. Provide a color that contrasts visually with the adjoining surfaces, either light-on-dark or dark-on-light.

- C. Minimize number of panels used
 - 1. Acceptable modular panel sizes are 2 ft x 2 ft, 2 ft x 3 ft, 2 ft x 4 ft, 2 ft x 5 ft, or curved panels.
- D. Acceptable products for installation:
 - 1. Polymer Composite Panel
 - a. Epoxy polymer composition
 - b. Homogenous integral color, UV stable, slip resistant, non-glare finish.
 - c. Use for new construction and retrofit construction.
 - 2. Precast Concrete Panel
 - a. Fiber reinforced high strength concrete composition
 - b. Homogeneous integral color, UV stable, slip resistant.
 - c. Use for new construction only.
 - 3. Gray Iron Casting Panel
 - a. ASTM A 48 Class 35B, as specified in AASHTO M 306, iron composition with embedment anchors
 - b. Reasonably smooth, uniform quality, slip resistant, weathered iron finish, cleaned by shot blasting, free from sand holes, burned-on sand, gas holes, cracks, and other surface defects.
 - c. Use for new construction only.

2.5 FORMS

- A. Use wood, metal, reinforced fiberglass, or plastic forms free of warps or bends and of sufficient strength to prevent deflection during the placement of concrete.
 - 1. Transition smoothly from curves to straight sections.
 - a. Keep forms in curves free of flat sections and sharp bends.
 - 2. Anchor securely in place.
 - 3. Clean the inside surfaces of dirt, concrete, and foreign material before concrete placement.

PART 3 EXECUTION

3.1 GENERAL

- A. Refer to PA Series Standard Drawings.

3.2 RECONSTRUCT PEDESTRIAN ACCESS RAMP

- A. Remove concrete and asphalt pavement, concrete sidewalk, and concrete curb and gutter as required. Refer to Section 02221.

- B. Reconstruct Pedestrian Access Ramp as required.

3.3 DETECTABLE WARNING SURFACE

- A. Install Detectable Warning Surface according to the manufacturer's product data sheets and installation instructions.
 - 1. Polymer Composite Detectable Warning Surface Panel Installation
 - a. Install cast-in-place detectable warning surface panel directly into the finished concrete surface.
 - 1) Provide a smooth transition between the panel and the surrounding concrete surface.
 - b. Install surface-applied detectable warning surface panel directly on the existing concrete surface.
 - 1) Use mechanical fasteners to secure the panel to the existing surface.
 - 2) Caulk a smooth transition bead along beveled panel edge and surrounding concrete surface.
 - 2. Precast Concrete Detectable Warning Surface Panel, and Gray Iron Casting Detectable Warning Surface Panel Installation
 - a. Install using cast-in-place method.
 - 1) Provide a smooth transition between the panel and the surrounding concrete surface.
- B. Replace Detectable Warning surface
 - 1. Remove existing surface-applied detectable warning surface without damaging the concrete surface.
 - 2. Clean concrete surface and install new detectable warning surface according to the manufacturer's product data sheets and installation instructions.

END OF SECTION

SECTION 02776

CONCRETE FLATWORK

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Concrete flatwork items such as sidewalk, plowable end section, median filler, curb, gutter, concrete lined ditch, and driveway.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill
- B. Section 02721: Untreated Base Course (UTBC)
- C. Section 03055: Portland Cement Concrete
- D. Section 03152: Concrete Joint Control
- E. Section 03211: Reinforcing Steel and Welded Wire
- F. Section 03390: Concrete Curing

1.3 REFERENCES

- A. ASTM C 979: Pigments for Integrally Colored Concrete

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. 1 ft² sample of colored concrete representative of the color to the Engineer for approval.
- B. Manufacturer's product data sheets and mixing instructions for concrete colorant.

1.6 ACCEPTANCE

- A. Concrete flatwork may be accepted at a reduced price when the concrete strength is below that specified.
 - 1. Refer to Section 03055 for pay factor adjustment.

PART 2 PRODUCTS

2.1 CONCRETE

- A. Class AA(AE). Refer to Section 03055.
- B. Colorant – High purity, chemically inert, unfading, and alkali-fast synthetic pigment coloring material according to ASTM C 979.

2.2 PREMOLDED JOINT FILLER

- A. Refer to Section 03152.

2.3 UNTREATED BASE COURSE

- A. Refer to Section 02721.

2.4 REINFORCING STEEL

- A. Coated reinforcing steel. Refer to Section 03211.

PART 3 EXECUTION

3.1 GENERAL

- A. Construct as shown in GW Series Standard Drawings and the plans.

3.2 PREPARATION

- A. Construct subgrade to plan elevations. Refer to Section 02056.
- B. Place and compact UTBC per Section 02721.
- C. Forms
 - 1. Use wood, metal, reinforced fiberglass, or plastic forms free of warps or bends and of sufficient strength to prevent deflection during the placement of concrete.
 - a. Transition smoothly from curves to straight sections.
 - 1) Keep forms in curves free of flat sections and sharp bends.
 - b. Anchor securely in place.
 - c. Clean the inside surface of all dirt, concrete, and foreign material before concrete placement.
 - 2. May use slip form machines.

- D. Concrete flatwork coloring when required.
 - 1. Match concrete color to the approved samples.
 - 2. Thoroughly mix color pigment in the concrete before placing.

3.3 PLACE AND FINISH CONCRETE

- A. Dampen the subgrade just before concrete placement.
- B. Hand methods of strike-off and consolidation are permitted.
- C. Finish the surface smooth with a non-steel concrete finishing float.
 - 1. Do not add water to the surface of the concrete.
 - 2. Remove form marks and irregularities.
- D. Round edges to a ½ inch radius.
- E. Brush exposed surfaces to a transverse broom finish except as follows:
 - 1. Provide a parallel broom finish for gutters.

3.4 EXPANSION AND CONTRACTION JOINTS

- A. Place joints perpendicular to the subgrade and as shown.
- B. Contraction Joints
 - 1. Space the joints 10 ft apart.
 - 2. Use ⅛ inch to ⅜ inch thick steel plate templates.
 - a. Remove the templates as soon as the concrete initially sets.
 - 3. Cut joint 1½ inch deep when using slip form method to place the concrete.
- C. Expansion Joints
 - 1. Use ½ inch thick premolded expansion joint filler.
 - 2. Place an expansion joint every 30 ft with the following exception not to place:
 - a. Slip form method to place concrete.
 - b. Gutters.
 - 3. Place joint filler between the sidewalk or median filler and the curb or adjacent pavement, sidewalk, driveway pavement, or structure.

3.5 CONCRETE CURING

- A. Refer to Section 03390.

END OF SECTION
Concrete Flatwork
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SECTION 02785

CHIP SEAL COAT

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Materials and procedures for applying emulsified asphalt, followed with an application of, either a standard chip seal cover material or lightweight chip seal cover material and bituminous flush coat.
- B. Cover materials.

1.2 RELATED SECTIONS

- A. Section 01554: Traffic Control
- B. Section 01558: Temporary Pavement Markings
- C. Section 02742S: Project Specific Surfacing Requirements
- D. Section 02745: Asphalt Material
- E. Section 02765: Pavement Marking Paint

1.3 REFERENCES

- A. AASHTO T 11: Materials Finer Than 75 μm (No. 200) Sieve in Mineral Aggregates by Washing
- B. AASHTO T 19: Unit Weight and Voids in Aggregate
- C. AASHTO T 27: Sieve Analysis of Fine and Coarse Aggregates
- D. AASHTO T 96: Resistance to Abrasion of Small Size Coarse Aggregate by Use of the Los Angeles Machine
- E. AASHTO T 104: Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
- F. AASHTO T 278: Surface Frictional Properties Using the British

Pendulum Tester

- G. AASHTO T 279: Accelerated Polishing of Aggregates Using the British Wheel
- H. AASHTO T 335: Determining the Percentage of Fracture in Coarse Aggregate
- I. UDOT Materials Manual of Instruction
- J. UDOT Minimum Sampling and Testing Requirements
- K. UDOT Quality Management Plan

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Test reports for information that the cover material and emulsion meets requirements of this Section, Part 2.
- B. Verification that the asphalt/polymer supplier adheres to UDOT Quality Management Plan for Asphalt Emulsion 508.
- C. All documentation verifying asphalt application rates, chip application, and other calibration verification for applied materials during the chip seal operations on a daily basis, or as requested by the Engineer.
- D. Vendor's bill of lading upon delivery and certify the material was diluted according to this Section, Part 2.

1.6 ACCEPTANCE

- A. Acceptance sampling and testing of material is according to UDOT Minimum Sampling and Testing Requirements.

PART 2 PRODUCTS

2.1 CATIONIC EMULSIONS

- A. CRS-2A according to Section 02745.
- B. CRS-2P according to Section 02745.
- C. LMCRS-2 according to Section 02745.

2.2 HIGH FLOAT EMULSIONS

- A. HFRS-2P according to Section 02745.
- B. HFMS-2 according to Section 02745.
- C. HFMS-2P according to Section 02745.

2.3 FLUSH COAT

- A. Use the emulsion as designated in Special Provision 02742S, diluted two parts concentrate to one part water by the manufacturer.

2.4 COVER MATERIAL

- A. Use crusher processed virgin aggregate consisting of natural stone, gravel, or slag according to Table 1.
 - 1. Use crusher-processed rotary-kiln lightweight expanded shale chips meeting the requirements of Table 1 for Lightweight Chip Seal Coat.

Table 1

Chip Seal Cover Material Properties			
Test	Test Method	Standard Chip Seal Type I & II	Lightweight Chip Seal
*Unit Weight	AASHTO T 19	100 lb/ft ³ , max	60 lb/ft ³ , max
One Fractured Face	AASHTO T 335	95% minimum	
Two Fractured Faces	AASHTO T 335	90% minimum	
*LA wear	AASHTO T 96	30% maximum	30% maximum
Soundness	AASHTO T 104	10% maximum	10% maximum
Flakiness Index	Materials MOI 8-933	17 maximum	25 maximum
*Stripping	Materials MOI 8-945	10% maximum	10% maximum
*Polishing	AASHTO T 278, T 279	31 minimum	31 minimum
* The Department has the right to waive this requirement if the aggregates have proven acceptable through successful past performance as determined by the Engineer.			

- B. Meet gradation limits in Table 2. Refer to AASHTO T 27 and T 11.

Table 2

Gradation Limits			
Sieve Size	Percent Passing		
	Type I	Type II	Lightweight
½ in		100 - 98	100 - 90
¾ in	100	69 - 91	55 - 80
No. 4	0 - 15	0 - 11	0 - 5
No. 8		0 - 6	0 - 3
No. 200	0 - 1	0 - 1.5	0 - 1

2.5 BLOTTER MATERIAL

- A. Refer to Section 02748.

2.6 EQUIPMENT

- A. Use distributor trucks according to the following requirements:
1. Tachometer, pressure gauges, accurate volume measuring devices or a calibrated tank, and a thermometer for measuring temperatures of the tank contents.
 2. Insulated tanks capable of storing the binder at temperatures that allow the binder to remain consistent with the appropriate viscosity for proper application rates.
 - a. Use tanks equipped with baffles to prevent pressure surges resulting from the asphalt sloshing in the tank when starting and stopping.
 - b. Use trucks equipped with devices to provide for accurate and rapid correlation and control of the amount of bituminous material being applied with that of the truck or distributor gauges.
 3. Constant volume circulation pumps and heaters to maintain a pressurized system so binder will be uniformly heated.
 - a. Circulation pump must spray a constant volume for the entire length of the spray bar for each application.
 4. Spray bar and nozzles designed to provide an appropriate fan width to provide uniform transverse distribution without corrugation or streaking.
 - a. Adjust the spray bar height to provide uniform distribution of binder across the application width and triple lapping of the binder on the pavement surface.
 - b. Use a fully circulating spray bar with a positive shutoff valve.
 5. Computerized rate control system allowing the operator to

- control all distributor operations from the cab to include:
 - a. Pressure regulation of the material application and automatic rate control adjustment to the unit ground speed.
 - 1) Hydrostatic system capable of maintaining a tolerance of ± 0.03 gal/yd².
 - b. Spray bar height and width adjustment and shut off of individual spray bar sections.
- B. Use a self-propelled aggregate (chip) spreader specifically designed and manufactured for chip seal operations, equipped with the following:
 - 1. Computerized controls that will apply a uniform, even layer of aggregate across the full width of the binder and adjust output to the unit ground speed.
 - a. Use gates adjustable to drop the correct amount of aggregate plus or minus 1 lb/yd².
 - 2. Variable width spreader with hydraulic control extension and adjustable discharge gates.
 - 3. Spreading hopper with a minimum capacity to cover a full lane of travel plus 1 ft/pass.
 - 4. Spinner broadcast type of aggregate spreader not allowed.
- C. Use sufficient number of dump trucks to circumvent any interruption in the supply of chips to the spreader.
 - 1. Use tandem axle dump trucks or larger or conveyor discharge trucks to minimize the number of hook-ups.
 - 2. Use dump trucks with matching hitches and compatible with the aggregate spreader to provide smooth hook-ups and to minimize any spillage when loading the hopper.
 - 3. Use trucks in good mechanical condition and that do not leak.
 - a. Use truck tires that do not pick up binder or aggregate when driving on the new surface.
- D. Use at least three articulating type pneumatic rollers for rolling operations.
 - 1. Use rollers weighing between 8 tons minimum and 12 tons maximum with a minimum width of 6 ft.
 - 2. Use rollers with pneumatic tires of equal size diameter and having treads satisfactory to the Engineer.
 - 3. Inflate tires so that the entire roller width area is compacted by either the rear-axle tires or the front-axle tires.
 - a. Inflate tires to 90 lb/in², or lower as approved by the Engineer.
 - 1) Maintain tire pressure within 5 lb/in².
- E. Sweeping Equipment

1. Use rotary brooms with nylon or steel bristles or pickup or vacuum brooms for pavement cleaning or brooming operations.
 - a. Keep downward pressure to a minimum.
 - b. Use water as requested by the Engineer if excessive dust is generated during sweeping operations.
 - c. Use pickup or vacuum sweepers in urban areas where aggregate accumulates in gutters or where removal is required from the edge of the shoulder.
 - d. Do not dislodge embedded aggregate when brooming chip sealed roadway.
- F. Blotter Material Equipment
 1. Apply blotter material using a truck mounted spinner broadcast spreader.

PART 3 EXECUTION

3.1 PREPARATION

- A. Clean the road surface of all dirt, sand, dust, and other objectionable material.
- B. Protect all structures including but not limited to guardrail, guideposts, concrete barriers, all drains, and parapet walls.
- C. Cover manholes, valve boxes, drop inlets, and other service utility entrances before placing any chip seal coat.
- D. Stockpile blotter material at least 0.25 lb/yd² according to Section 02748 at a site within 20 minutes delivery time of each road section being chip sealed.
 1. Have application equipment on site before beginning chip seal work.
 - a. Stockpiling of blotter material may be waived upon COR approval if blotter material can be obtained and ready to spread within 20 minutes of a road section being chip sealed.
 - b. Equipment to spread blotter material is subject to inspection and approval by the COR.

3.2 LIMITATIONS

- A. Complete all work between May 15, and August 31.

- B. Do not place chip seal coat if surface moisture is present.
- C. Place seal coat when:
 - 1. Pavement temperature is between 70 and 136 degrees F.
 - 2. Air temperature is between 50 and 110 degrees F.
 - 3. Forecasted temperature is not expected to be below 40 degrees F within 3 days after placement.
- D. Do not apply any bituminous asphalt after 6:00 p.m. if temperatures in this Section, Article 3.2, paragraph C cannot be maintained throughout all night time hours.
- E. Do not open to traffic the same day chip seal coat is placed on Interstate routes.
 - 1. Sweep and open to traffic no earlier than 6:00 a.m. the day following placement of cover material.
- F. Apply bituminous flush coat material after receiving approval from the Engineer but no earlier than 6 days after application of the cover material.
 - 1. Apply bituminous flush coat material when the air temperature in the shade is 50 degrees F and rising and the pavement temperature is 70 degrees F and rising.
 - 2. Do not apply bituminous flush coat material during fog, rain, or other adverse conditions.

3.3 COVER MATERIAL STOCKPILE

- A. Construct on a clean base to minimize contamination.
- B. Construct individual 500 ton stockpiles of aggregates. Construct to facilitate uniform dampening. Avoid excess moisture.
 - 1. COR authorizes stockpiles at least one and a maximum of seven days before use.
 - 2. Combining, altering, or moving 500 ton stockpiles may require re- authorization by the COR before use.
- C. Notify the COR at least seven calendar days before placement in order for the initial stockpiles to be sampled and tested for acceptance.
- D. Obtain the COR's authorization of a stockpile before use.
- E. Rework or remove material not meeting specifications from the stockpile area. Identify stockpiles that will be reworked.

- F. Retest corrected material for acceptance.

3.4 TEMPORARY PAVEMENT MARKINGS

- A. Apply raised pavement markers according to Section 01558.
- B. Apply temporary pavement marking paint according to Section 02765 as determined by the COR.

3.5 ASPHALT MATERIAL/COVER MATERIAL APPLICATION

- A. Apply asphalt material at a rate sufficient to obtain 50 percent chip embedment before the rolling operation and 70 percent chip embedment after rolling operation.
1. Adjust application rates throughout the project depending on existing conditions.
- B. Apply the asphalt emulsion at a minimum temperature of 145 degrees F.
- C. Do not apply asphalt material if material does not spray through the distributor in a uniform way and remain in place on the roadway.
- D. Place building paper adjacent to the transverse construction joint before starting each spraying operation.
1. Maintain the control valve to act instantaneously both at start-up and cut-off.
- E. Locate longitudinal joints within 6 inches of the traffic lane line location.
1. Construct meet lines with no skip or voids between adjacent passes.
 2. Do not place a double thickness of cover material.
- F. Calibrate the spreader at the beginning of each day and as often as necessary to comply with Table 3.
1. Maintain a distance of less than 150 ft between the distributor and the chip spreader.
 2. Maintain the chip spreader speed so that chips do not bounce or roll during application.

Table 3

	Approximate Spread Rates	
	Unit Weight lbs/ft ³	Application Rate lbs/yd ²
Lightweight Chip Seal	40-45	11.8
	45-50	12.5
	50-60	13.2

Chip Seal	60 – 65	17.0
	65 – 70	18.4
	70 – 75	19.8
	75 – 80	20.7
	80 – 85	22.1
	85 – 90	23.5
	90 – 95	24.9
	95 – 100	25.8

3.6 SURFACE ROLLING

- A. Use at least three pneumatic-tire rollers in a longitudinal direction to roll surface after the cover material has been spread.
- B. Roll at least three passes to seat the cover material.
 - 1. A pass is defined as traveling in one direction only.
- C. Control bleeding with blotter material.
- D. Set the roller speed to prevent bouncing or skidding.
 - 1. Do not exceed 5 mph.
 - 2. Reduce roller speeds during directional changes to prevent surface tearing.
- E. Synchronize the speed of the distributor and chip spreader with that of the rolling operation.
 - 1. Begin initial rolling, consisting of one complete coverage, immediately behind the chip spreader.
 - 2. Begin secondary rolling, consisting of second and third coverage, immediately after completing initial rolling.
 - 3. Synchronize all operations to keep rolling operations within 2,500 feet of the ongoing chip seal application.
- F. Sweep excess cover material off the roadway after the emulsion has set.
 - 1. Remove excess cover material to the satisfaction of the COR before opening the roadway to traffic.
- G. Repair all damage to the seal coat before opening the roadway to traffic.

3.7 BITUMINOUS FLUSH COAT APPLICATION

- A. Clean the surface of all dirt, sand, dust, loose chips, and other objectionable material before applying bituminous flush coat.
- B. Apply the bituminous flush coat at a rate of 0.11, $\pm$ 0.01 gal/yd².
 - 1. Keep traffic off the flushed surface until the bituminous material has set sufficiently to prevent tracking or pick-up.

3.8 TRAFFIC CONTROL

- A. Refer to Section 01554.

3.9 PAVEMENT MARKING PAINT

- A. Allow at least 7 calendar days after completing flush coat before applying permanent pavement markings. Refer to Section 02765.

END OF SECTION

SECTION 02786

OPEN-GRADED SURFACE COURSE (OGSC)

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Materials and procedures for constructing OGSC

1.2 RELATED SECTIONS

- A. Section 01456 Materials Dispute Resolution
- B. Section 02701: Pavement Smoothness
- C. Section 02745: Asphalt Material
- D. Section 02746: Hydrated Lime
- E. Section 02748: Prime Coat/Tack Coat

1.3 REFERENCES

- A. AASHTO T 30: Mechanical Analysis of Extracted Aggregate
- B. AASHTO T 89: Determining the Liquid Limit of Soils
- C. AASHTO T 90: Determining the Plastic Limit and Plasticity Index of Soils
- D. AASHTO T 96: Resistance to Degradation of Small Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
- E. AASHTO T 104: Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate
- F. AASHTO T 112: Clay Lumps and Friable Particle in Aggregate
- G. AASHTO T 176: Plastic Fines in Graded Aggregates and Soils by Use of the Sand Equivalent Test
- H. AASHTO T 278: Surface Frictional Properties Using the British Pendulum Tester

- I. AASHTO T 279: Accelerated Polishing of Aggregates Using the British Wheel
- J. AASHTO T 304: Uncompacted Void Content of Fine Aggregate
- K. AASHTO T 308: Determining the Asphalt Binder Content of Asphalt Mixtures by the Ignition Method
- L. AASHTO T 335: Determining the Percentage of Fracture in Coarse Aggregate
- M. UDOT Materials Manual of Instruction
- N. UDOT Minimum Sampling and Testing Requirements

1.4 DEFINITIONS

- A. Production Day – A 24-hour period in which OGSC is being placed.

1.5 SUBMITTALS

- A. Job mix gradation for approval
 - 1. Materials and documentation according to Materials Manual of Instruction, Section 954.
 - 2. Aggregate suitability test results.
- B. Changes in Job Mix Gradation
 - 1. Written request for a change in a job-mix gradation for approval.
 - 2. Allow the COR five working days to review and respond to the changes and to readjust the quantity of asphalt binder to be used.

1.6 ACCEPTANCE

- A. Acceptance sampling and testing of material is according to UDOT Minimum Sampling and Testing Requirements.
- B. Gradation and asphalt binder content
 - 1. The COR evaluates a lot on the test results of four or more samples, except when only three samples can be taken.
 - 2. Evaluate the lot using the number of tests “n” in Table 3.
 - 3. The COR informs the Contractor of the time and place of sampling not more than 15 minutes before sampling.
 - 4. Increase sample sizes to accommodate validation or third-party testing as required.

- C. The Government applies one Incentive/Disincentive for the lowest dollar value for Gradation/Asphalt Content. The Government computes the Incentives/Disincentives as follows for each lot
 - 1. Compute incentive/disincentive for Gradation/Asphalt Binder according to Table 1.
 - 2. Base the incentive/disincentive on Percent within Limit (PT) computation using Tables 2, 3, and 4.
 - 3. Use lowest single PT value combined for gradation (each of the sieves) and asphalt binder content for calculating the gradation/asphalt binder content incentive/disincentive.
- D. Thickness
 - 1. Verify the thickness with a depth probe and take corrective action if necessary.
 - a. Minimum thickness – Plan depth minus $\frac{1}{4}$ inch.
- E. Smoothness
 - 1. The Government applies incentive/disincentive for smoothness according to Section 02701.
 - 2. Refer to Section 02701 for smoothness requirements.
- F. The Government rejects lots
 - 1. If the PT for any individual gradation measurement is less than 52 percent as shown in Table 1.
 - 2. If the PT for asphalt binder content is less than 60 percent as shown in Table 1.
 - 3. The Engineer may accept a reject or non-conforming lot. Refer to Section 01456.
 - a. A price reduction of 35 percent for the OGSC pay item and a price reduction of 35 percent for the binder pay item will be assessed.
 - b. The lot will not be eligible for any incentive.

Table 1

Incentive/Disincentive for Asphalt Binder Content	
PT Based on Min. Four Samples	Incentive/Disincentive (Dollars/Ton)
>99	2.00
96-99	1.50
92-95	1.00
88-91	0.00
84-87	-0.26
80-83	-0.60
76-79	-0.93
72-75	-1.27
68-71	-1.60
64-67	-1.93
60-63	-2.27
<60	Reject

Incentive/Disincentive for Gradation	
PT Based on Min. Four Samples	Incentive/Disincentive (Dollars/Ton)
>99	2.00
96-99	1.50
92-95	1.00
88-91	0.00
84-87	-0.26
80-83	-0.60
76-79	-0.93
72-75	-1.27
68-71	-1.60
64-67	-1.93
60-63	-2.27
56-59	-5.00
52-55	-10.00
<52	Reject

Table 2

Upper and Lower Limit Determination	
Parameter	UL and LL
Asphalt Binder Content	Target Value $\pm$ 0.35%
$\frac{3}{8}$ inch sieve	Target Value $\pm$ 6.0%
No. 4 sieve	Target Value $\pm$ 6.0%
No. 8 sieve	Target Value $\pm$ 5.0%
No. 200 sieve	Target Value $\pm$ 2.0%

Table 3

Quality Index Values for Estimating Percent Within Limits										
PU/PL	n=3	n=4	n=5	n=6	n=7	n=8	n=10	n=12	N=15	n=20
100	1.16	1.50	1.75	1.91	2.06	2.15	2.29	2.35	2.47	2.56
99	1.16	1.47	1.68	1.79	1.89	1.95	2.04	2.09	2.14	2.19
98	1.15	1.44	1.61	1.70	1.77	1.80	1.86	1.89	1.93	1.97
97	1.15	1.41	1.55	1.62	1.67	1.69	1.74	1.77	1.80	1.82
96	1.15	1.38	1.49	1.55	1.59	1.61	1.64	1.66	1.69	1.70
95	1.14	1.35	1.45	1.49	1.52	1.54	1.56	1.57	1.59	1.61
94	1.13	1.32	1.40	1.44	1.46	1.47	1.49	1.50	1.51	1.53
93	1.12	1.29	1.36	1.38	1.40	1.41	1.43	1.43	1.44	1.46
92	1.11	1.26	1.31	1.33	1.35	1.36	1.37	1.37	1.38	1.39
91	1.10	1.23	1.27	1.29	1.30	1.31	1.32	1.32	1.32	1.33
90	1.09	1.20	1.23	1.24	1.25	1.25	1.26	1.26	1.27	1.27
89	1.08	1.17	1.20	1.21	1.21	1.21	1.21	1.21	1.22	1.22
88	1.07	1.14	1.16	1.17	1.17	1.17	1.17	1.17	1.17	1.17
87	1.06	1.11	1.12	1.12	1.12	1.13	1.13	1.13	1.13	1.13
86	1.05	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08
85	1.03	1.05	1.05	1.05	1.05	1.04	1.04	1.04	1.04	1.04
84	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00	1.00
83	1.00	0.99	0.98	0.97	0.97	0.96	0.96	0.96	0.96	0.96
82	0.98	0.96	0.95	0.94	0.94	0.93	0.93	0.92	0.92	0.92
81	0.96	0.93	0.92	0.91	0.90	0.90	0.89	0.89	0.89	0.88
80	0.94	0.90	0.88	0.87	0.86	0.86	0.85	0.85	0.85	0.85
79	0.92	0.87	0.85	0.84	0.83	0.83	0.82	0.82	0.82	0.81
78	0.89	0.84	0.82	0.81	0.80	0.79	0.79	0.78	0.78	0.78
77	0.87	0.81	0.79	0.78	0.77	0.76	0.76	0.75	0.75	0.75
76	0.84	0.78	0.76	0.75	0.74	0.73	0.72	0.72	0.72	0.72
75	0.82	0.75	0.73	0.72	0.71	0.70	0.69	0.69	0.69	0.68
74	0.79	0.72	0.70	0.68	0.67	0.67	0.66	0.66	0.66	0.65
73	0.77	0.69	0.67	0.65	0.64	0.64	0.62	0.62	0.62	0.62
72	0.74	0.66	0.64	0.62	0.61	0.61	0.60	0.59	0.59	0.59
71	0.71	0.63	0.60	0.59	0.58	0.58	0.57	0.56	0.56	0.56
70	0.68	0.60	0.58	0.56	0.55	0.55	0.54	0.54	0.54	0.53
69	0.65	0.57	0.55	0.54	0.53	0.52	0.51	0.51	0.51	0.50
68	0.62	0.54	0.52	0.51	0.50	0.50	0.48	0.48	0.48	0.48
67	0.59	0.51	0.49	0.48	0.47	0.47	0.46	0.45	0.45	0.45
66	0.56	0.48	0.46	0.45	0.44	0.44	0.43	0.42	0.42	0.42
65	0.53	0.45	0.43	0.42	0.41	0.41	0.40	0.40	0.40	0.39
64	0.49	0.42	0.40	0.39	0.38	0.38	0.37	0.37	0.37	0.37
63	0.46	0.39	0.37	0.36	0.35	0.35	0.35	0.34	0.34	0.34
62	0.43	0.36	0.34	0.33	0.33	0.33	0.32	0.31	0.31	0.31
61	0.39	0.33	0.31	0.30	0.30	0.30	0.29	0.29	0.29	0.28
60	0.36	0.30	0.28	0.27	0.26	0.26	0.25	0.25	0.25	0.25
<60	≤ 0.35	≤ 0.29	≤ 0.27	≤ 0.26	≤ 0.25	≤ 0.25	≤ 0.24	≤ 0.24	≤ 0.24	≤ 0.24

Enter table in the appropriate “number of tests” column and round down to the nearest value.

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Table 4

Definitions, Abbreviations, and Formulas for Acceptance	
Term	Explanation
Target Value (TV)	The target values for gradation and asphalt binder content.
Average (AVE)	The sum of the lot's test results for a measured characteristic divided by the number of test results, the arithmetic mean.
Sample Standard Deviations	The square root of the value formed by summing the squared difference between the individual test results of a measured characteristic and AVE divided by the number of test results minus one.
Upper Limit (UL)	The value above the TV of each measured characteristic that defines the upper limit of acceptable production. (Table 2)
Lower Limit (LL)	The value below the TV of each measured characteristic that defines the lower limit of acceptable production. (Table 2)
Upper Quality Index (QU)	$QU = (UL - AVE)/s$
Lower Quality Index (QL)	$QL = (AVE - LL)/s$
Percentage of Lot Within UL (PU)	Determined by entering Table 4 with QU.
Percentage of Lot Within LL (PL)	Determined by entering Table 4 with QL.
Total Percentage of Lot (PL) Within UL and LL (PT)	$PT = (PU + PL) - 100$
Incentive/Disincentive	Determined by entering Table 3 with PT or PL.

All values for AVE, s, QU, and QL will be calculated to at least a two decimal places and will be carried through all further calculations. Rounding to lower accuracy is not allowed.

1.7 DISPUTE RESOLUTION

- A. Refer to Section 01456 when disputing the validity of the Department's acceptance tests.

PART 2 PRODUCTS

2.1 ASPHALT BINDER

- A. Refer to Section 02745.

2.2 HYDRATED LIME

- A. Refer to Section 02746.

2.3 AGGREGATE MATERIALS

- A. Crusher processed virgin aggregate material consisting of crushed stone, gravel, or slag.
- B. Meet the following requirements, including Table 5.
 - 1. Coarse aggregate
 - a. Retained on No. 4 sieve.
 - 2. Fine aggregate
 - a. Clean, hard grained, and angular.
 - b. Passing the No. 4 sieve.

Table 5

Aggregate Properties		
Properties	Test Method	Test Requirement
One Fractured Face	AASHTO T 335	95 percent min.
Two Fractured Face	AASHTO T 335	90 percent min.
Fine Aggregate Angularity	AASHTO T 304	45 min.
L.A. Wear	AASHTO T 96	30% max.
Sand Equivalent	AASHTO T 176	60 min.
Plasticity Index	AASHTO T 89 and T 90	0
Polish Test	AASHTO T 278 & T 279	31 min.
Soundness (sodium sulfate)	AASHTO T 104	12% max. loss with five cycles
Clay Lumps and Friable Particles	AASHTO T 112	2% max.
Natural Fines	None	None

2.4 JOB MIX

- A. Obtain Engineer's review for job mix gradation.
 - 1. Show definite single values for the percentage of aggregate passing each sieve based on the dry weight of aggregate.
 - 2. Stay within the single value gradation limits of Table 6.
 - 3. Incorporate hydrated lime by dry weight of aggregate into all mixtures. Refer to Section 02746.

Table 6

Aggregate Gradation	
Percent Passing by Dry Weight of Aggregate	
Sieve Size	Percent
½ inch	100
¾ inch	90 - 100
# 4	35 - 45
# 8	14 - 20
# 200	2 - 4

- B. Binder Content
 - 1. The Engineer determines the binder content.

2.5 TACK COAT

- A. Refer to Section 02748

PART 3 EXECUTION

3.1 MIXING

- A. Treat the aggregate with hydrated lime. Refer to Section 02746.
- B. Mix until all particles are coated with the asphalt binder.

3.2 SURFACE PLACEMENT

- A. Apply tack coat. Refer to Section 02748.
- B. Maintain a steady paver speed.
- C. Roll sufficiently to seat without fracturing aggregate.
- D. Bring all passes up even transversely at the end of each working day.
- E. Construct longitudinal joints within 6 inches of lane lines.
- F. Remove slick spots as determined by the Engineer.

3.3 LIMITATIONS

- A. Place between May 1 and September 15.
 - 1. Obtain authorization from the Engineer before placing OGSC after September 15.
- B. Place when the air temperature in the shade and the pavement surface temperature are above 60 degrees F.
- C. Do not place if surface moisture is present.
- D. Do not place during rain or other adverse weather conditions.

END OF SECTION

SECTION 02787

BONDED WEARING COURSE (BWC)

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Materials and procedures for the placement of a Bonded Wearing Course (BWC) and associated Polymer-Modified Emulsion Membrane.

1.2 RELATED SECTIONS

- A. Section 01456: Materials Dispute Resolution
- B. Section 02701: Pavement Smoothness
- C. Section 02741: Hot Mix Asphalt (HMA)
- D. Section 02742S: Project Specific Surfacing Requirements
- E. Section 02745: Asphalt Material
- F. Section 02746: Hydrated Lime

1.3 REFERENCES

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- A. AASHTO T 11: Materials Finer Than 75- μ m (No. 200) Sieve in Mineral Aggregates by Washing
- B. AASHTO T 27: Sieve Analysis of Fine and Coarse Aggregates
- C. AASHTO T 44: Solubility of Bituminous Materials
- D. AASHTO T 49: Penetration of Bituminous Materials
- E. AASHTO T 59: Emulsified Asphalt
- F. AASHTO T 89: Determining the Liquid Limit of Soils
- G. AASHTO T 90: Determining the Plastic Limit and Plasticity Index of Soils
- H. AASHTO T 96: Resistance to Degradation of Small Size Coarse Aggregate by Abrasion and Impact the Los Angeles Machine
- I. AASHTO T 104: Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate
- J. AASHTO T 112: Clay Lumps and Friable Particle in Aggregate
- K. AASHTO T 176: Plastic Fines in Graded Aggregates and Soils by Use of the Sand Equivalent Test
- L. AASHTO T 278: Surface Frictional Properties Using the British Pendulum Tester
- M. AASHTO T 279: Accelerated Polishing of Aggregates Using the British Wheel
- N. AASHTO T 283: Resistance of Compacted Asphalt Mixture to Moisture Induced Damage
- O. AASHTO T 301: Elastic Recovery Test of Asphalt Materials by Means of a Ductilometer
- P. AASHTO T 304: Uncompacted Void Content of Fine Aggregate
- Q. AASHTO T 305: Determination of Drain down Characteristics in Uncompacted Asphalt Mixtures
- R. AASHTO T 312: Preparing and Determining the Density of Asphalt Mixture Specimens by Means of the Superpave Gyratory Compactor

- S. AASHTO T 327: Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus
- T. AASHTO T 330: The Qualitative Detection of Harmful Clays of the Smectite Group in Aggregates Using Methylene Blue Aggregates
- U. AASHTO T 335: Determining the Percentage of Fracture in Coarse Aggregate
- V. ASTM D 2995: Estimating Application Rate and Residual Application Rate of Bituminous Distributors
- W. Asphalt Institute Standards
- X. UDOT Materials Manual of Instruction
- Y. UDOT Minimum Sampling and Testing Requirements
- Z. UDOT Quality Management Plans

1.4 DEFINITIONS

- A. Bonded Wearing Course (BWC) – A spray-paver applied surface treatment consisting of a warm Polymer-Modified Emulsion Membrane and an ultra-thin, gap-graded hot mix asphalt overlay. It is typically less than 1 inch thick. The hot mix asphalt surface course is placed immediately on the emulsion. A BWC is a homogeneous wearing surface and pavement seal that can be opened to traffic immediately after cooling.
- B. Polymer-Modified Emulsion Membrane (PMEM) – A warm, modified emulsion applied to the pavement to seal the underlying pavement and provide a strong bond to the surface course.
- C. Production Day – A 24 hour period in which BWC is being placed.

1.5 SUBMITTALS

- A. Mix design data to the COR for review at least 14 calendar days before paving. Do not pave until mix design is authorized for use.
 - 1. Include all information on selection of design aggregate structure showing the target values of percent passing on all sieves listed in Table 9.
 - 2. Indicate the design asphalt binder content and the estimated asphalt binder film thickness.
 - 3. Provide test results verifying that proposed aggregate meets the requirements of Table 8.

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4. Indicate the target job mix gradation and supply QC data for the target selection.
 - a. Use these target values for price adjustments.
- B. Lottman test data using Method A lime slurry for all sources. Refer to this Section, Article 2.5.
 1. Submit both A and B lime slurry data for new sources having no historic data.
 2. Meet minimum Tensile Strength Ratio (TSR) of 80 percent.
- C. Draindown test results using AASHTO T 305.
- D. Joint layout plan to the COR for review at least 14 calendar days before placement.
- E. Daily totals – Refer to this Section, Article 3.6, paragraph Q.
 1. Clearly show placement of the longitudinal joints with respect to final pavement markings and meeting the requirements of this Section, Article 3.6, paragraph I.

1.6 ACCEPTANCE

- A. Acceptance sampling and testing of material is according to UDOT Minimum Sampling and Testing Requirements.
- B. A lot equals the number of square yards placed during each production day.
 1. The lot may be increased to include up to three production days approved when daily production rates are anticipated to be less than 15,000 yd².
- C. Incentive / Disincentive for Asphalt Binder Content and Gradation
 1. The COR evaluates a lot on the test results of four samples with the following exceptions:
 - a. Compute Incentive/Disincentive using the test results from three samples if only three samples can be taken for the production day.
 - b. Combine test results with the next day of production if at least three random samples cannot be taken. Take one sample for each 15,000 yd², or portion thereof, from the following day's production. Evaluate with the appropriate number of tests "n" in Table 4.
 - c. Add the lot to the previous day's production for the final day's production if three random samples cannot be taken.
 - d. Evaluate with the appropriate number of tests "n" in Table 4.
 2. Inform the COR of the time and place of sampling not less

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- than 24 hours before the sampling.
- a. Samples taken will be sufficiently large to allow for repeat testing according to Section 01456.
 - b. Retained material for repeat testing will be immediately stored until the represented lots are accepted.
3. Compute Incentive/Disincentive for binder content per lot based on Table 1 using the single test result with the largest deviation from the target.
 4. Incentive/Disincentive for gradation is based on Percent within Limits (PT) computation using Table 2, 3, 4, and 5.
 5. The Government will reject the lot if the PT is less than 60 percent.
 6. Any lot rejected based on gradation or binder content will not be eligible for any incentive.
- D. Reject Material
1. Remove the rejected material from the project within 72 hours and replace it with an acceptable material if the request is denied.
 2. Agree on removal time period if rotomilling is required.
 3. Government will deduct 40 percent of the bid price if a rejected lot is allowed to remain in place.
- E. Thickness
1. Verify the thickness with a depth probe and take corrective action if necessary.
 - a. Minimum thickness – Plan depth
- F. Acceptance / Disincentive for Polymer Modified Emulsion Membrane (PMEM) application rate
1. Verify the application rate of the PMEM for each lot and apply pay adjustment based on Table 6.
 - a. A lot equals the number of square yards placed during each production day.
 - b. Use the weight of PMEM provided on the weight ticket to convert from tons to gallons.

Table 1

Incentive/Disincentive	
Binder Content	Pay Adjustment (dollars per yd² of BWC)
Within $\pm 0.30\%$ of target	0.05
Between $\pm 0.31\%$ and $\pm 0.45\%$ of target	0.00
Between $\pm 0.46\%$ $\pm 0.60\%$ of target	-0.10
Greater than $\pm 0.61\%$	Reject

Table 2

Gradation Upper and Lower Limit Determination	
Parameter	UL and LL
½ inch sieve for BWC Type C ¾ inch sieve for BWC Type B	Target Value ± 6.0 percent
# 4 sieve	Target Value ± 6.0 percent
# 8 sieve	Target Value ± 5.0 percent
# 50 sieve	Target Value ± 3.0 percent
# 200 sieve	Target Value ± 2.0 percent

Table 3

Incentive/Disincentive for Gradation	
PT Based on Min. Four Samples	Pay Adjustment (dollars per yd² of BWC)
> 99	0.06
96-99	0.04
92-95	0.02
88-91	0.00
84-87	-0.01
80-83	-0.02
76-79	-0.04
72-75	-0.05
68-71	-0.06
64-67	-0.07
60-63	-0.09
<60	Reject

Table 4

Quality Index Values (QU or QL) for Estimating Percent Within Limits										
PU or PL	n=3	n=4	n=5	n=6	n=7	n=8	n=10	n=12	n=15	n=20
100	1.16	1.50	1.75	1.91	2.06	2.15	2.29	2.35	2.47	2.56
99	1.16	1.47	1.68	1.79	1.89	1.95	2.04	2.09	2.14	2.19
98	1.15	1.44	1.61	1.70	1.77	1.80	1.86	1.89	1.93	1.97
97	1.15	1.41	1.55	1.62	1.67	1.69	1.74	1.77	1.80	1.82
96	1.15	1.38	1.49	1.55	1.59	1.61	1.64	1.66	1.69	1.70
95	1.14	1.35	1.45	1.49	1.52	1.54	1.56	1.57	1.59	1.61
94	1.13	1.32	1.40	1.44	1.46	1.47	1.49	1.50	1.51	1.53
93	1.12	1.29	1.36	1.38	1.40	1.41	1.43	1.43	1.44	1.46
92	1.11	1.26	1.31	1.33	1.35	1.36	1.37	1.37	1.38	1.39
91	1.10	1.23	1.27	1.29	1.30	1.31	1.32	1.32	1.32	1.33
90	1.09	1.20	1.23	1.24	1.25	1.25	1.26	1.26	1.27	1.27
89	1.08	1.17	1.20	1.21	1.21	1.21	1.21	1.21	1.22	1.22
88	1.07	1.14	1.16	1.17	1.17	1.17	1.17	1.17	1.17	1.17
87	1.06	1.11	1.12	1.12	1.12	1.13	1.13	1.13	1.13	1.13
86	1.05	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08
85	1.03	1.05	1.05	1.05	1.05	1.04	1.04	1.04	1.04	1.04
84	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00	1.00
83	1.00	0.99	0.98	0.97	0.97	0.96	0.96	0.96	0.96	0.96
82	0.98	0.96	0.95	0.94	0.94	0.93	0.93	0.92	0.92	0.92
81	0.96	0.93	0.92	0.91	0.90	0.90	0.89	0.89	0.89	0.88
80	0.94	0.90	0.88	0.87	0.86	0.86	0.85	0.85	0.85	0.85
79	0.92	0.87	0.85	0.84	0.83	0.83	0.82	0.82	0.82	0.81
78	0.89	0.84	0.82	0.81	0.80	0.79	0.79	0.78	0.78	0.78
77	0.87	0.81	0.79	0.78	0.77	0.76	0.76	0.75	0.75	0.75
76	0.84	0.78	0.76	0.75	0.74	0.73	0.72	0.72	0.72	0.72
75	0.82	0.75	0.73	0.72	0.71	0.70	0.69	0.69	0.69	0.68
74	0.79	0.72	0.70	0.68	0.67	0.67	0.66	0.66	0.66	0.65
73	0.77	0.69	0.67	0.65	0.64	0.64	0.62	0.62	0.62	0.62
72	0.74	0.66	0.64	0.62	0.61	0.61	0.60	0.59	0.59	0.59
71	0.71	0.63	0.60	0.59	0.58	0.58	0.57	0.56	0.56	0.56
70	0.68	0.60	0.58	0.56	0.55	0.55	0.54	0.54	0.54	0.53
69	0.65	0.57	0.55	0.54	0.53	0.52	0.51	0.51	0.51	0.50
68	0.62	0.54	0.52	0.51	0.50	0.50	0.48	0.48	0.48	0.48
67	0.59	0.51	0.49	0.48	0.47	0.47	0.46	0.45	0.45	0.45
66	0.56	0.48	0.46	0.45	0.44	0.44	0.43	0.42	0.42	0.42
65	0.53	0.45	0.43	0.42	0.41	0.41	0.40	0.40	0.40	0.39
64	0.49	0.42	0.40	0.39	0.38	0.38	0.37	0.37	0.37	0.37
63	0.46	0.39	0.37	0.36	0.35	0.35	0.35	0.34	0.34	0.34
62	0.43	0.36	0.34	0.33	0.33	0.33	0.32	0.31	0.31	0.31
61	0.39	0.33	0.31	0.30	0.30	0.30	0.29	0.29	0.29	0.28
60	0.36	0.30	0.28	0.27	0.26	0.26	0.25	0.25	0.25	0.25
<60	≤0.35	≤0.29	≤0.27	≤0.26	≤0.25	≤0.25	≤0.24	≤0.24	≤0.24	≤0.24

Enter table in the appropriate sample size column and round down to the nearest value.

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Table 5

Definitions, Abbreviations, and Formulas for Acceptance	
Term	Explanation
Target Value (TV)	The target values for gradation and asphalt binder content are given in the Contractor's volumetric mix design.
Average (AVE)	The sum of the lot's test results for a measured characteristic divided by the number of test results, the arithmetic mean.
Standard Deviation (s)	The square root of the value formed by summing the squared difference between the individual test results of a measured characteristic and AVE divided by the number of test results minus one. This statement does not limit the methods of calculations of s. Other methods that obtain the same value may be used.
Upper Limit (UL)	The value above the TV of each measured characteristic that defines the upper limit of acceptable production. (Table 2)
Lower Limit (LL)	The value below the TV of each measured characteristic that defines the lower limit of acceptable production (Table 2)
Upper Quality Index (QU)	$QU = (UL - AVE)/s$
Lower Quality Index (QL)	$QL = (AVE - LL)/s$
Percentage of Lot Within UL (PU)	Determined by entering Table 4 with QU
Percentage of Lot Within LL (PL)	Determined by entering Table 4 with QL
Total Percentage of Lot (PL) Within UL and LL (PT)	$PT = (PU + PL) - 100$
Incentive/Disincentive	Determined by entering Table 1 and 3 with PT or PL.

Calculate all values for AVE, s, QU, and QL to two decimal place accuracy and carried through all further calculations. Rounding to lower accuracy is not allowed.

Table 6

PMEM Pay Adjustment	
Actual Field PMEM Rate (Gallons/Square Yard)	Pay Adjustment (dollars per yd² of BWC)
Within Specified Rate ± 0.03	0
Within Specified Rate ± 0.03 to ± 0.06	-2.00
Not Within Specified Rate ± 0.06	Remove and Replace

PART 2 PRODUCTS

2.1 ASPHALT BINDER

- A. Project Specific Surfacing Requirements. Refer to Special Provision 02742S.
- B. Refer to Section 02745.

2.2 POLYMER MODIFIED EMULSION MEMBRANE (PMEM)

- A. Project Specific Surfacing Requirements. Refer to Special Provision 02742S.
- B. Refer to Section 02745.
- C. Supply PMEM from a pre-qualified supplier under the provisions outlined in UDOT Quality Management Plan 508, Asphalt Emulsion.
- D. Meet the requirements of Table 7.

Table 7

Polymer Modified Emulsion Membrane (a)			
Tests	AASHTO Test Method	Min.	Max.
Emulsion			
Viscosity, SFS, 122° F (50° C), sec	T 59	20	100
Storage Stability Test, 1 d, 24 h, percent	T 59		1
Sieve Test (b), percent	T 59		0.1
Demulsibility, % 35 ml 0.8% sodium dioctyl sulfosuccinate	T 59	40	
Distillation			
Oil Distillate, by volume of emulsion, percent			2
Residue (c), percent		63	
Residue from Distillation Test			
Penetration, 77° F (25° C), 100 g, 5s, dmm	T 49	60	150
Elastic Recovery (d), percent	T 301	55	
Solubility			
Solubility in Organic Solvent, percent	T 44	97.5	

- (a) The emulsion will be smooth and homogeneous throughout with no white, milky separation, pumpable and suitable for application through spray bars after standing undisturbed for 24 hours. Modify before emulsification.
- (b) The sieve test may be waived if successful application of the material has been achieved in the field.
- (c) Distillation is determined by AASHTO T 59 with modifications to include a $350 \pm 5^\circ \text{ F}$ ($177 \pm 3^\circ \text{ C}$) maximum temperature to be held for a period of 15 minutes.
- (d) Modify paragraph 4.5 of AASHTO T 301 as follows: Stop the ductilometer and within two seconds, sever the specimen at its center with a pair of scissors after 20 cm has been reached.

2.3 HYDRATED LIME

- A. Refer to Section 02746.

2.4 AGGREGATE MATERIALS

- A. Crusher processed virgin aggregate material consisting of crushed stone, gravel, or slag.
- B. Determine the suitability of the aggregate according to Table 8.
 - 1. Coarse aggregate
 - a. Retained on # 4 sieve
 - 2. Fine aggregate
 - a. Clean, hard grained, and angular
 - b. Passing the # 4 sieve

Table 8

Aggregate Properties		
Properties	Test Method	Test Requirement
One Fractured Face	AASHTO T 335	95% min.
Two Fractured Face	AASHTO T 335	90% min.
Fine Aggregate Angularity	AASHTO T 304	45% min.
Flakiness Index	UDOT MOI 933 (Based on $\frac{3}{8}$ inch sieve and above)	17% max.
L.A. Wear	AASHTO T 96	30% max.
Micro-Deval Loss	AASHTO T 327	18% max. loss
Sand Equivalent	AASHTO T 176	60 min.
Plasticity Index	AASHTO T 89 and T 90	0
Polish Test	AASHTO T 278 and T 279	31 min.
Soundness (sodium sulfate)	AASHTO T 104	12% max. loss with five cycles
Clay Lumps and Friable Particles	AASHTO T 112	2% max.
Natural Fines	None	None

Methylene Blue	AASHTO T 330	10% max.
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- C. Meet gradation requirements in Table 9.

Table 9

Aggregate Gradation Percent Passing by Dry Weight of Aggregate AASHTO T 11, T 27			
Sieve Size	Percent		
	Type A	Type B	Type C
¾ inch	100	100	100
½ inch	100	100	75 - 100
⅜ inch	100	75 - 100	55 - 80
# 4	40 - 55	22 - 38	22 - 38
# 8	22 - 32	19 - 32	19 - 32
#16	15 - 25	15 - 25	15 - 25
#30	10 - 18	10 - 18	10 - 18
#50	8 - 13	8 - 13	8 - 13
#100	6 - 10	6 - 10	6 - 10
# 200	4 - 7	4 - 7	4 - 7

2.5 JOB MIX DESIGN

- A. Meet the following requirements:
1. Establish optimum binder content, minimum 5.0 percent by weight of mixture, corresponding to an estimated asphalt film thickness of at least 10 microns based on effective binder content and aggregate surface area calculated according to Asphalt Institute MS-2 Table 6.1.
 2. Verify draindown at optimum binder content is less than 0.10 percent when tested according to AASHTO T 305.
 3. Moisture susceptibility (Lottman), AASHTO T 283, with the following modifications:
 - a. Compact samples according to AASHTO T 312 using a 4 inch diameter mold and 30 gyrations.
 - b. Use mix quantity necessary to obtain compacted samples 2.5 ± 0.05 inch high.
 - c. Further test compacted samples regardless of air void levels achieved after 30 gyrations.
 - d. Apply vacuum to samples to be conditioned for 1 minute and proceed without calculating percent saturation.
 4. Use a laboratory accredited by AASHTO Materials Reference Laboratory (AMRL) for all testing.

- B. Use a pre-qualified asphalt binder supplier.
- C. Use a pre-qualified PMEM supplier.
- D. Use mixing and compaction temperatures supplied by the Engineer.
- E. The Department's Region Materials Lab will verify the Mix Design.

PART 3 EXECUTION

3.1 PRE-PAVING MEETING

- A. Arrange for Contractor's supervisory personnel, crew, and the testing laboratory personnel to meet with a representative of the Government at a mutually agreed time within two weeks before starting the BWC process to discuss methods of accomplishing all phases of the work.
 - 1. Arrange for a representative of the mix design and emulsion supplier to be present.

3.2 MIXING

- A. Mix as specified in Section 02741.
 - 1. The mineral aggregate coating will be considered satisfactory when all particles are coated.

3.3 PAVER CHARACTERISTICS

- A. Use a self-priming paver designed and built for applying the BWC with the following characteristics:
 - 1. Equipped with a receiving hopper, asphalt emulsion storage tank, system for measuring the PMEM volume applied, spray bar, and a heated, variable-width screed.
 - 2. Capable of spraying the PMEM, applying the hot mix asphalt overlay, and leveling the surface of the mat in one pass.
 - 3. Capable of placing the hot mix asphalt within five seconds after the application of the PMEM.
 - 4. Capable of paving at a controlled speed from 30 to 90 feet per minute.
 - a. Make no contact between the wheel or other part of the paving machine and the PMEM before applying the hot mix asphalt.

- B. Use a paver with a screed that is able to crown the pavement at the Bonded Wearing Course (BWC)

center and have vertically adjusted extensions to accommodate the desired pavement profile.

3.4 MATERIAL TRANSFER VEHICLE CHARACTERISTICS

- A. Use Material Transfer Vehicle (MTV) with the following characteristics:
 - 1. Able to remix the BWC to eliminate truck end segregation, minimize temperature segregation, and deliver a uniform BWC to the paver.
 - 2. Self-propelled machine totally independent of the paver.
 - 3. High capacity truck unloading system to receive BWC from the haul units.
 - 4. Minimum 25 ton on-board BWC surge capacity to minimize paver start/stops and maximize trucking efficiency.
 - 5. Equipped with a pivoting paver loading conveyor.
 - a. Able to swing to either side to allow off-lane paving.
 - 6. Paver hopper
 - a. Equipped with a hopper insert with a minimum capacity of 7 tons.
 - b. Hopper insert with mass flow design to deliver remixed BWC directly to the paver conveyor system.
- B. Deposit the BWC material directly from haul trucks into the MTV without contacting the pavement.

3.5 SURFACE PREPARATION

- A. Protect and cover manhole covers, drains, grates, catch basins, and other utility structures with plastic or building felt before paving.
 - 1. Clearly reference each location.
- B. Remove thermoplastic traffic markings.
- C. Remove symbols, characters, or other markings greater than ¼ inch above the existing pavement.
- D. Allow at least two weeks for any crack sealant to cure before placing BWC.
- E. Thoroughly clean the pavement surface to be overlaid of deleterious material, giving specific attention to accumulated mud and debris.
 - 1. Use pressurized water or vacuum systems, if necessary, to obtain a clean surface.

3.6 BWC PLACEMENT

- A. Provide a clean surface before and during paving operations.
 - 1. Immediately remove any loose material in front of the paver.
- B. Spray the PMEM with a metered mechanical pressure spray bar at a temperature of 120 to 180 degrees F at the specified application rate.
 - 1. Overlap the vertical edge with the outside nozzle 0.5 to 1.5 inches when paving next to a vertical edge to guarantee proper application of the PMEM.
 - 2. Verify the application of the PMEM is uniform over the entire surface to be paved.
- C. Calibrate the Spray Paver for application of PMEM according to ASTM D 2995 or an approved method by the Engineer before start of BWC production.
 - 1. Recalibrate Spray Paver before the startup of nonconsecutive placements of BWC material.
 - 2. Calibrate machine at 13 ft width and 17 ft width.
 - a. Verify spread rates for right and left sides are uniform, meet the specified application rates, and do not vary more than 20 percent between right and left side pads.
 - 3. Calibrate the machine on-site or within BWC mix haul distance and in the presence of the Engineer.
 - 4. Maintain a copy of the manufacturer application settings with the spray paving equipment.
 - a. Include the nozzle size and target shot rate settings.
- D. Apply hot BWC at a temperature of 290 to 330 degrees F immediately over the PMEM application.
- E. Place hot BWC over the full width of the PMEM with a heated screed.
- F. Maintain a steady spray-paver speed.
- G. Roll BWC material sufficiently to seat without fracturing mix aggregate.
- H. Use steel-wheeled, 2-axle tandem rollers, each weighing at least 10 tons.
 - 1. Use well-maintained rollers equipped with functioning water system and scrapers to prevent adhesion of the fresh mix onto the roller drums.
 - 2. Supply adequate roller units to guarantee the rolling will be accomplished promptly following the placement of the

- material.
- 3. Use a release agent (added to the water system) as needed to prevent adhesion of the fresh mix to the roller drum and wheels.
- 4. Use static mode for rolling.
- 5. Do not allow the rollers to remain stationary on the freshly placed material.
- I. Construct longitudinal joints within 6 inches of lane lines or at the center of lane, but never in a wheel path.
- J. Construct a transverse joint when the paving operation stops for more than 30 minutes.
- K. Do not overlap or hot-lap BWC.
- L. Pave through lanes after paving adjacent:
 - 1. Shoulders
 - 2. Tapers
 - 3. Transitions
 - 4. Road connections
 - 5. Private drives
 - 6. Curve widenings
 - 7. Chain control lanes
 - 8. Turnouts
 - 9. Left turn pockets
 - 10. Median borders
 - 11. On and off ramps
- M. Verify all edges of adjacent areas to through lanes that have been paved with BWC have straight, uniform, longitudinal lines, and neat vertical edges to guarantee surface is neat in appearance after through lanes are paved.
- N. Do not reintroduce previously applied BWC into the paving process over PMEM.
- O. Stop paving and remove and dispose of the contaminated material if PMEM is spilled into the spray-paver hopper.
- P. Meet the requirements of Section 02701 for smoothness.
- Q. Report daily totals of placement materials to the Engineer and include the following:
 - 1. Tonnage of BWC material placed with weight tickets

2. Gallons of PMEM applied with weight tickets
3. Square yards for the production day
4. Gallons per square yard of PMEM placed
5. Pounds per square yard of BWC material placed

3.7 LIMITATIONS

- A. Place between May 1, and September 15, when both the air temperature in the shade and the pavement surface temperature are above 60 degrees F and rising.
- B. Do not place if the Engineer determines excessive moisture is present in the pavement structure.
- C. Do not place during rain, when the surface is wet, or during other adverse weather conditions.
- D. Do not open the new pavement to traffic until the rolling operation is complete and the material has cooled below 185 degrees F.

3.8 LABORATORY CORRELATION

- A. Perform split-sample, paired t-testing with the Department based on project quality control testing using Department LQP qualified lab.
 1. Perform split-sample, paired t analysis on all mix acceptance tests and tests related to volumetric properties.
 2. Perform paired t-analysis as defined in the UDOT Materials Manual of Instruction, Appendix C.
 3. Continue paired t-testing until at least two consecutive production days meet $\alpha = 0.05$ for a two tailed distribution.
 4. Resolve discrepancies in lab results within the first five production days.
 - a. Cease production if the requirements for two consecutive days of the first five days cannot be met.
 - b. Submit a corrective action plan to the Engineer before production continues indicating the changes in procedures that will be implemented to correct the deficiencies.
 - 1) Both Contractor and Department labs must make paired t test results available within 24 hours of sampling.

3.9 DISPUTE RESOLUTION

- A. Refer to Section 01456 when disputing the validity of the Department's acceptance tests.
- B. The option to dispute the validity of the Department's test results is waived if the paired t-testing is not performed. Refer to this Section, Article 3.8.

END OF SECTION

SECTION 02821

CHAIN LINK FENCING AND GATES

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Chain link fencing and gates.

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete

1.3 REFERENCES

- A. AASHTO M 111: Zinc (Hot Dip Galvanized) Coatings on Iron and Steel Products
- B. AASHTO M 181: Chain Link Fence
- C. AASHTO M 232: Zinc Coating (Hot Dip) on Iron and Steel Hardware
- D. AASHTO M 280: Standard Specification for Metallic-Coated (Carbon) Steel Barbed Wire
- E. ASTM A 121: Metallic-Coated Carbon Steel Barbed Wire
- F. ASTM A 194: Carbon and Alloy Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
- G. ASTM A 392: Zinc-Coated Steel Chain-Link Fence Fabric
- H. ASTM F 436: Hardened Steel Washers
- I. ASTM A 491: Aluminum-Coated Steel Chain-Link Fence Fabric
- J. ASTM A 563: Carbon and Alloy Steel Nuts
- K. ASTM F 668: Polyvinyl Chloride (PVC) and Other Organic Polymer-Coated Steel Chain-Link Fence Fabric

- L. ASTM C 1107: Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
- M. ASTM F 1043: Strength and Protective Coatings on Steel Industrial Chain Link Fence Framework
- N. ASTM F 1083: Pipe, Steel, Hot-Dipped Zinc-Coated (Galvanized) Welded, for Fence Structures
- O. ASTM F 1554: Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength
- P. AWS D1.1 Structural Welding Code

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS Not Used

PART 2 PRODUCTS

2.1 GENERAL

- A. Concrete
 - 1. Class B Concrete – Refer to Section 03055.
- B. Non-shrink Grout
 - 1. Refer to ASTM C 1107.
- C. Anchor Bolt Assembly
 - 1. Anchor Bolts – Refer to ASTM F 1554.
 - 2. Nuts – Refer to ASTM A 563 or ASTM 194.
 - 3. Washers – Refer to ASTM F 436.
- D. Base Plate
 - 1. AASHTO M 270, Grade 36.
 - 2. Weld post to base plate according to AWS D1.1.
 - 3. Galvanize after fabrication according to AASHTO M 111.

2.2 POSTS, CAPS, RAILS, AND COUPLINGS

- A. Pipe Posts and Rails
 - 1. Schedule 40, hot-dip galvanized coated pipe.
 - a. Refer to ASTM F 1043 IA and ASTM F 1083
 - 2. High Strength, galvanized coated pipe.
 - a. Refer to ASTM F 1043 IC

- B. Fittings
 - 1. Malleable cast iron or pressed steel coated.
 - a. Refer to AASHTO M 232.
- C. Caps
 - 1. Equip all pipe posts with a galvanized steel or malleable iron weather-resistant cap that fits securely over the posts, with an apron around the outside of the post.
 - a. Refer to AASHTO M 232.
 - 2. Provide cap to permit passage of top rail when top rail is used.

2.3 CHAIN LINK FABRIC

- A. Provide either Type I zinc-coated steel or Type II aluminum-coated steel fence fabric as specified in AASHTO M 181, ASTM A 392, and ASTM A 491.
- B. Provide a polyvinyl chloride (PVC) coating when shown.
 - 1. Refer to ASTM F 668
- C. Use 0.148 inch diameter wire for fence fabric 6 ft or higher and 0.120 inch diameter wire for fabric less than 6 ft high.
- D. Provide 0.177 inch diameter spiral material for tension wires.
- E. Tie fence fabric to supporting members with wire of the same diameter as the fence fabric wire.

2.4 BARBED WIRE

- A. Provide zinc-coated barbed wire when zinc-coated fence is used as specified in AASHTO M 280.
- B. Use 0.099 inch diameter barbed wire with 0.080 inch diameter 4-point barbs on 5 inch centers.
- C. Provide aluminum coated barbed wire when aluminum coated fence is used as specified in ASTM A 121.
- D. Provide a support arm for barbed wire on top of a chain link fence that supports a 200 lb vertical load at the end of the arm without causing permanent deflection.

2.5 GATES

- A. Fabricate gate posts and frames of the sizes according to FG Series Standard Drawings.
 - 1. Fasten gate frame corners together with pressed steel or malleable iron corner ells, riveted or welded as shown.
 - 2. Galvanize welded steel gate frames after fabrication.
 - a. Refer to AASHTO M 111.
 - 3. Do not use closed cells that will prohibit dipping into galvanizing tanks.
- B. Follow the same standards for chain link fence fabric for covering the gate frames as for other fence fabric.
- C. Furnish each gate with the appropriate hinges, latch, and drop-bar locking device.

PART 3 EXECUTION

3.1 INSTALL POSTS

- A. Install according to FG Series Standard Drawings.
- B. Do not exceed the following spacing requirements when placing posts:

<u>Radii of Curve</u>	<u>Maximum Post Spacing</u>
Tangent or 500 ft	10 ft
200 ft to 500 ft	8 ft
100 ft to 200 ft	6 ft
0 ft to 100 ft	5 ft
- C. Install brace posts at maximum 500 ft intervals and at angle points of 30 degrees or greater.
- D. Set posts in concrete bases.
 - 1. Place concrete at least 6 inches below each post.
 - 2. Construct at least 12 inch diameter bases for end posts, pull posts, corner posts, gate posts, and line posts.

3.2 INSTALL FENCE FABRIC

- A. Locate bottom of fence fabric on the roadway side of posts unless otherwise specified.
 - 1. Place fabric approximately 1 inch above the ground.
 - 2. Maintain a straight grade between posts by excavating high points of the ground.
 - 3. Fill depression in the natural ground to within 1 inch of the bottom of fence.
- B. Stretch the fabric taut and securely fasten to fence posts.
 - 1. Use stretch bars and metal bands to fasten fence fabric to terminal, gate, corner, and pull posts.
 - a. Space metal bands at 1 ft intervals along the post.
 - 2. Cut the fabric at corner and pull posts.
 - 3. Fasten fabric to line posts with tie wires or metal bands at 14 inch intervals.
 - 4. Attach the top edge of fabric to the top rail or tension cable with wire ties at approximately 24 inch intervals.
 - 5. Attach the bottom edge of the fabric to the bottom tension wire with wire ties spaced at 24 inch intervals.

3.3 INSTALL GATES

- A. Install single gate or double gate as shown. Install plumb, level, and secure for full opening without interference.
- B. Install ground-set items in concrete for anchorage as shown in the standard drawing or as recommended by the manufacturer. Adjust hardware for smooth operation.
- C. Set gate openings according to manufacturer's dimensions.
- D. Fabric description numbers:
 - 1. First number indicates height.
 - 2. Second number indicates width of fabric opening.

END OF SECTION

SECTION 02825
CATTLE GUARD

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Cattle Guard fabrication and installation.
- B. Removal of existing cattle guard.
- C. Cast-in-place concrete aprons.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill
- B. Section 02075: Geotextiles
- C. Section 02221: Remove Structure and Obstruction
- D. Section 02317: Structural Excavation and Backfill
- E. Section 03055: Portland Cement Concrete
- F. Section 03056: Self-Consolidating Concrete (SCC)
- G. Section 03211: Reinforcing Steel and Welded Wire
- H. Section 03390: Concrete Curing
- I. Section 03575: Flowable Fill

1.3 REFERENCES

- A. AASHTO M 111: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
- B. AASHTO M 251: Plain and Laminated Elastomeric Bridge Bearings
- C. AASHTO M 252: Corrugated Polyethylene Drainage Pipe

- D. AWS D1.1 Structural Welding Code – Steel
- E. Precast Concrete Institute (PCI) Design Handbook
- F. Precast Concrete Institute (PCI) Tolerance Manual for Precast and Prestressed Concrete Construction, MNL-135.
- G. UDOT Structural Steel Construction Manual

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS Not Used

PART 2 PRODUCTS

2.1 GENERAL

- A. Refer to SW Series Standard Drawings.

2.2 MATERIALS

- A. Concrete
 - 1. Class AA(AE). Refer to Section 03055.
 - 2. Self-Consolidating Concrete – SCC. Refer to Section 03056.
- B. Reinforcing Steel
 - 1. Use coated reinforcing steel. Refer to Section 03211.
- C. Lifting Devices
 - 1. Use lifting devices that can support the required vertical and horizontal forces with the applicable safety factors according to the Component Handling and Erection Bracing requirements in the PCI Design Handbook.
 - 2. Galvanize steel elements. Refer to AASHTO M111.
- D. Flowable Fill
 - 1. Refer to Section 03575.
- E. Drainage Geotextile
 - 1. Refer to Section 02075

- F. Free Draining Granular Backfill
 - 1. Refer to Section 02056.
- G. Corrugated Polyethylene Pipe
 - 1. Refer to AASHTO M 252.
- H. Bearing Pad Strips
 - 1. Use 50 durometer hardness (Shore A), Grade 3 elastomer.
 - 2. Refer to AASHTO M 251.

2.3 TRAFFIC GRILL UNIT AND WELDED END GUARD UNIT FABRICATION

- A. Fabricate according to the UDOT Structural Steel Construction Manual and AWS D1.1
- B. Hot-dip galvanize after fabrication. Refer to AASHTO M 111.

2.4 PRECAST CONCRETE CATTLE GUARD UNIT FABRICATION

- A. Use a UDOT Certified Concrete Precaster.
- B. Limit fabrication tolerance for length to: $+1/8$ inch to $-3/8$ inch or as indicated in the Certified Precaster's plant quality control manual, whichever is more stringent.
 - 1. Use the tolerances specified for Ribbed Wall Panel in the PCI Tolerance Manual, MNL-135, where no cattle guard specific tolerances are specified in the Certified Precaster's plant quality control manual.
- C. Permanently mark each precast unit with date of casting, supplier identification, and identifiers to facilitate field placement.
 - 1. Stamp markings in fresh concrete.
- D. Curing
 - 1. Refer to Section 03390.
- E. Predrill holes for apron doveled bars when aprons are shown.
 - 1. Use template to mark drilling locations to prevent drill from damaging stud anchors.

PART 3 EXECUTION

3.1 GENERAL

- A. Structural excavation and backfill. Refer to Section 02317.
- B. Reinstall removed fence as required.

3.2 SUBGRADE PREPARATION

- A. Place flowable fill as shown.
 - 1. Allow flowable fill to become firm and unyielding before placing free draining granular backfill.
- B. Place free draining granular backfill as shown
 - 1. Contour to distribute roadway grade break among adjacent joints between precast cattle guard units.
 - 2. Do not allow gap between precast cattle guard units at roadway surface to exceed $\frac{3}{4}$ inch.

3.3 CAST-IN-PLACE APRON PLACEMENT

- A. Dowel and epoxy reinforcing steel as shown. Refer to Section 03211.
- B. Place and finish concrete
 - 1. Dampen subgrade just before concrete placement
 - 2. Hand methods of strike-off and consolidation are permitted.
 - 3. Finish the surface smooth with a concrete finishing float.
 - a. Do not add water to the surface of the concrete
 - b. Remove form marks and irregularities.
 - 4. Round edges to a $\frac{1}{2}$ inch radius.
 - 5. Brush exposed surfaces to a transverse broom finish.
- C. Cure concrete. Refer to Section 03390.

END OF SECTION

SECTION 02841

W-BEAM GUARDRAIL

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. W-Beam guardrail, post and blocks, W-Beam and Thrie-Beam guardrail transition element, W-Beam guardrail anchors, W-beam buried in back slope terminals, W-Beam curved guardrail elements, W-Beam guardrail delineation and certification requirements for installation of W-Beam guardrail systems.

1.2 RELATED SECTIONS

- A. Section 02842: Delineators
- B. Section 02890: Retroreflective Sheeting
- C. Section 03055: Portland Cement Concrete
- D. Section 03211: Reinforcing Steel and Welded Wire
- E. Section 06055: Timber and Timber Treatment

1.3 REFERENCES

- A. AASHTO M 111: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
- B. AASHTO M 180: Corrugated Sheet Steel Beams for Highway Guardrail
- C. AASHTO M 232: Zinc Coating (Hot-Dip) on Iron and Steel Hardware
- D. AASHTO M 235: Epoxy Resin Adhesives
- E. AASHTO M 270: Carbon and High Strength Low Alloy Structural Steel, Shapes, Plates and Bars and Quenched-and Tempered Alloy Structural Steel Plates for Bridges
- F. ASTM A 500: Cold Formed Welded and Seamless Carbon Steel Structure tubing in Rounds and Shapes

- G. ASTM A 563: Carbon and Alloy Steel Nuts
- H. ASTM F 436: Hardened Steel Washers
- I. ASTM F 3125: Heavy Hex Structural Bolts

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Certificate of compliance for all products for information.
- B. Guardrail Training Certificate for information – Refer to this Section, Article 1.6 B.

1.6 CERTIFIED GUARDRAIL INSTALLER

- A. Responsibilities
 - 1. Must be present at each location within the project and be an active participant during the installation and maintenance repair of guardrail or elements of a W-Beam Guardrail System.
- B. Qualifications
 - 1. Attend the Department's Guardrail Installation Training Course (GRIT) and pass the UDOT Guardrail Training Final Examination.
 - a. American Traffic Safety Services Association (ATSSA) Guardrail Installation Training may be used as an alternate to the Department's GRIT.

PART 2 PRODUCTS

2.1 W-BEAM GUARDRAIL AND HARDWARE

- A. W-Beam and Thrie-Beam Rail – Refer to BA Series Standard Drawings.
 - 1. Minimum galvanizing requirement: Refer to AASHTO M 180, Type 1, Class A.
- B. Bottom W-Beam rail or steel rub rail – Refer to BA Series Standard Drawings.
 - 1. W-Beam – Refer to requirements of this Section.
 - 2. Channel rub rail – Refer to AASHTO M 160 and AASHTO M 270.
 - a. Galvanize according to AASHTO M 111 after punching and cutting is complete.
 - b. Epoxy – Refer to AASHTO M 235.

- C. Hardware – Refer to BA Series Standard Drawings.
 - 1. Manufacturer – Refer to AASHTO M 180.
 - 2. Coatings – Refer to AASHTO M 232 or M 298.

2.2 W-BEAM GUARDRAIL POSTS AND OFFSET BLOCKS

- A. Steel post – Refer to AASHTO M 270 and AASHTO M 160.
 - 1. Galvanize according to AASHTO M 111 after punching and cutting is complete.
- B. Wood Post and wood blocks – Refer to Section 06055.
- C. Composite or plastic offset blocks for steel post installations.
 - 1. Certify according to MASH test requirements.
- D. Solid Rock Formation Post Backfill.
 - 1. Free draining two fractured face granular backfill meeting Table 1 and at 90% following AASHTO T335.

Table 1

Sieve Size	Percent Passing
1½ inch	90-100
1 inch	20-55
¾ inch	0-15
⅜ inch	0-5

2.3 TRANSITION ELEMENTS

- A. Thrie-Beam Terminal Connector and Terminal Connector Plate Hardware.
 - 1. Structural hex bolts conforming to ASTM F 3125, Grade A 325
 - 2. Hardened washers and heavy hex nuts according to ASTM F 436 and A 563 respectively.
 - 3. Galvanize according to AASHTO M 111.

2.4 ANCHOR ELEMENTS

- A. Refer to the requirements of this Section for W-Beam guardrail, posts, blocks, and hardware.
- B. Anchor Plate – Refer to AASHTO M 270.
 - 1. Galvanize according to AASHTO M 111 after punching, drilling, and cutting is complete.

- C. Bearing Plate – Refer to AASHTO M 270.
 - 1. Galvanize according to AASHTO M 111 after punching, drilling, and cutting is complete.
- D. Anchor Post Foundation Tube and Plate.
 - 1. Tube – Refer to ASTM A 500.
 - 2. Plate – Refer to AASHTO M 270.
 - 3. Galvanize according to AASHTO M 111 after punching, drilling, cutting, and welding is complete.
- E. Strut and yolk assembly – Refer to ASTM A36.
 - 1. Galvanize according to AASHTO M 111 after punching, drilling, and cutting is complete.

2.5 W-BEAM BURIED-IN-BACKSLOPE TERMINALS

- A. Refer to BA Series Standard Drawings for Buried-In-Backslope Terminal Anchor options.
 - 1. Steel Post with plate – Refer to AASHTO M 270 and AASHTO M 160. Galvanize according to AASHTO M 111 after punching, drilling, cutting, and welding is complete.
 - 2. Concrete Block – Use Class A(AE) concrete. Refer to Section 03055. Refer to Section 03211 for coated reinforcing steel requirement.

2.6 W-BEAM CURVED GUARDRAIL

- A. Refer to plan set for radius requirements.
- B. Shop bend curves.
- C. Refer to this Section for W-Beam guardrail and anchor system.
- D. Controlled Release Terminal Post (CRT) breakaway post according to wood post and block requirements of this Section.
 - 1. Drill two 3½ inch holes center of post, 29 and 45 inches from top of post.

2.7 W-BEAM GUARDRAIL LONG SPAN

- A. W-Beam – Refer to requirements of this Section.

- B. CRT breakaway post – Refer to wood post and block requirements of this Section.
 - 1. Drill two 3½ inch holes center of post.
 - a. 32 and 47¾ inches from top of post.

2.8 W-BEAM MEDIAN BARRIER

- A. Refer to requirements of this section for W-Beam guardrail, posts, blocks, rub rail channel, and hardware.

2.9 BARRIER DELINEATION

- A. Sheeting – Refer to Section 02890.
- B. Hardware – Refer to GW Series Standard Drawings.

PART 3 EXECUTION

3.1 PREPARATION

- A. Site
 - 1. Do not install guardrail until final grading has been verified by the Engineer.
 - 2. Protect work area when removing guardrail barriers, guardrail end treatment, or crash cushions until the barriers, guardrail end treatment, or crash cushion are reconstructed or the hazard is mitigated.

3.2 POSTS AND BLOCKS

- A. Drill required holes in posts and blocks according to BA Series Standard Drawings before installation.
 - 1. Field drilling post is acceptable when installing the bottom rail.
 - a. Coat field drilled steel post with a field-applied cold zinc material.
- B. Drive post if satisfactory results are obtained without damaging the post.
 - 1. Do not drive posts through asphalt.
 - 2. Refer to BA 4D Series Standard Drawings.
- C. Excavate postholes when not driven.
 - 1. Use existing excavated post hole material as backfill.
 - 2. Compact backfill material into bottom of hole if hole is over - excavated.

3. Compact existing backfill material around post to the satisfaction of the Engineer.

3.3 RAIL ELEMENTS

- A. Drill or punch required holes according to BA Series Standard Drawings before installation.
 1. Do not field drill holes in rail element.

3.4 DELINEATION

- A. Install delineation according to BA and GW Standard Drawings.

3.5 RAISING EXISTING W-BEAM GUARDRAIL

- A. Raise W-beam guardrail to maintain the minimum rail height according to BA Series Standard Drawings.
 1. Only W-beam guardrail with two or three-hole post system is eligible to be raised.
 - a. Raise the two or three-hole post systems rail element only if the rail element is not currently set in the highest hole.
 2. W-beam guardrail with single-hole post system cannot be raised. Do not lift posts in order to raise rail.
- B. Remove existing rail element and blocks.
- C. Reinstall rail elements and blocks on existing two or three-hole post system using the next hole up on the two or three-hole system to meet the height requirements according to BA Series Standard Drawings.
 1. Do not drill new holes in posts.

END OF SECTION

SECTION 02842

DELINEATORS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Delineators, Type I and Type II.
- B. Culvert and maintenance markers.

1.2 RELATED SECTIONS

- A. Section 02765: Pavement Marking Paint
- B. Section 02890: Retroreflective Sheeting

1.3 REFERENCES

- A. AASHTO M 111: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
- B. ASTM A 1011: Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability
- C. ASTM D 638: Tensile Properties of Plastics
- D. ASTM D 4956: Standard Specification for Retroreflective Sheeting for Traffic Control
- E. ASTM G 23: Operating Light-Exposure Apparatus (Carbon-Arc Type) With and Without Water for Exposure of Nonmetallic Materials

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS **Not Used**

PART 2 PRODUCTS

2.1 STEEL POSTS

- A. Supply and galvanize posts as specified. Refer to ASTM A 1011 and AASHTO M 111.
- B. Use flanged, channel-shaped steel posts that weight 2 lb/ft.
- C. Make all cuts before galvanizing posts. Refer to GW Series Standard Drawings.

2.2 FLEXIBLE POSTS

- A. Free of burns, discoloration, contamination, and other defects.
- B. Remains flexible at temperatures from -5 degrees F to +140 degrees F.
- C. Capable of being driven into an earth shoulder with or without a pilot hole.
- D. Tensile strength of 1,100 psi. Refer to ASTM D 638.

2.3 QUALITY CONTROL – FLEXIBLE POST TESTING

- A. Meet Cold Bend Test
 - 1. Subject two posts to a temperature of -10 degrees F, $\pm$ 5 degrees F for at least four hours.
 - 2. Immediately bend each post four times through a 90 degree angle around a 2 inch mandrel.
 - 3. Each post must return to its original straight configuration within a $\pm$ 10 degree angle within five minutes at the end of the four bends.
 - 4. Any cracking or significant loss of rigidity are grounds for failure.
- B. Meet Hot Bend Test
 - 1. Subject two posts to a temperature of 100 degrees F $\pm$ 5 degrees F for at least four hours.
 - 2. Satisfy all bending and physical requirements specified in the Cold Bend Test.
- C. Meet Impact Resistance Test
 - 1. Subject post to impacts by a typical sedan as follows:
 - a. Three hits - 0 degree angle at 0 degrees F.
 - b. Three hits - 0 degree angle at 100 degrees F.
 - c. Ten hits - 0 degree angle at 35 mph.
 - d. Five hits - 15 degree angle at 55 mph.

2. Acceptable Results
 - a. Installed post remain intact, securely anchored, and within ± 10 degrees of vertical orientation.
 - b. Installed post shows minimal signs of cracking or loss of rigidity.
 - c. Installed post retains at least 50 percent of its reflective sheeting.
 - d. Impact vehicle suffers little or no damage during the impact test.
- D. Exposure
 1. Expose the specimens for 500 hours in a carbon arc-type apparatus according to ASTM G 23, Method 1.
 2. Acceptable results:
 - a. Exposure does not result in delamination, distress, or discoloration.
 - b. Sheeting is not removable from the specimens without damage.
 - c. Post is resistant to ultraviolet light, ozone, hydrocarbons, and other weathering.

2.4 SHEETING

- A. Refer to Section 02890.

2.5 PAINT

- A. Refer to Section 02765.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Posts – Visibly free of bends or twists both before and after installation.

END OF SECTION

SECTION 02843

CRASH CUSHIONS AND BARRIER END TREATMENTS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Crash cushions with concrete pads, and barrier end treatments.
- B. Crash cushion and barrier end treatment markers.

1.2 RELATED SECTION

- A. Section 02890: Retroreflective Sheeting
- B. Section 03055: Portland Cement Concrete

1.3 REFERENCES

- A. UDOT Guidelines for Crash Cushions and Barrier End Treatments

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Manufacturer certified installer certification for information.
- B. Letter of certification for each system and each location affirming it is installed according to Department's and the manufacturer's specifications, for information.
 - 1. Use Form C-112 available on the Department's Web site. Refer to <http://www.udot.utah.gov/go/standardsreferences/>.
- C. Shop and installation drawings for each system type installed.

1.6 CERTIFIED INSTALLER

- A. Use manufacturer certified installer to perform the installation.

PART 2 PRODUCTS

2.1 CRASH CUSHIONS AND BARRIER END TREATMENTS

- A. Select from the approved products list within the UDOT Guidelines for Crash Cushions and Barrier End Treatments.
 - 1. Refer to the UDOT Guidelines for Crash Cushions and Barrier End Treatments, for specific uses and requirements for each approved system type. Refer to <http://www.udot.utah.gov/go/standardsreferences>.
 - 2. Refer to CC Series Standard Drawings for crash cushion or end treatment grading and marker requirements.

- B. Crash Cushions
 - 1. Type A, B, and D.
 - a. Supply system with an adequate width as shown.
 - b. Include additional fender panels, posts and hardware to gain the required width for hazard protection greater than the manufactured width of a system.
 - c. Supply system for the required speed according to UDOT Guidelines for Crash Cushions and Barrier End Treatments,
 - d. Galvanize steel parts according to manufacturer's requirements.
 - e. Supply transition element for the approach of opposing traffic when system is installed with bi-directional traffic and the back of system is within the maximum required clear zone.
 - 1) Two transition elements required when system is installed with W-beam median barrier.
 - f. Supply crash cushion markers. Refer to CC Series Standard Drawings.
 - 2. Type C
 - a. Galvanize steel parts according to manufacturer's requirements.
 - b. Supply double-sided W-beam transition element when system is installed in conjunction with concrete barrier. Refer to BA 4 Series Standard Drawings.
 - c. Supply crash cushion markers. Refer to CC Series Standard Drawings.

- C. Barrier End Treatments
 - 1. Type G and H
 - a. Supply post option as described in UDOT Guidelines for Crash Cushion and Barrier End Treatments, edition.

- b. Supply system with 12½ ft galvanized W-beam rail elements according to manufacturer's requirements.
- c. Supply manufacturer approved impact head and hardware.
- d. Galvanize steel parts according to manufacturer's requirements.
- e. Supply crash cushion markers. Refer to CC Series Standard Drawings.

2.2 CRASH CUSHION MARKERS

- A. Marker plate and self-adhesive sheeting requirements:
 - 1. Refer to CC Series Standard Drawings.
 - 2. Construct marker plate 18 x 18 inch using 0.032 gauge aluminum with appropriate object marker sheeting.
 - a. Drill a $\frac{7}{16}$ inch hole in each corner of plate.
 - b. Sheeting – Refer to Section 02890.
 - 1) Date placement specified in Section 02890 is not required.
 - 3. Use of self-adhesive sheeting without marker plate. Sheeting – Refer to Section 02890.
 - a. Date placement specified in Section 02890 is not required.
 - b. Use appropriate sheeting type for the substrate that sheeting is placed on.
 - c. Place a minimum of 324 in², typically 18 x 18 inch, on the approach end of crash cushion or barrier end treatment.
 - d. Use of manufacturer's supplied object marker sheeting is acceptable when sheet type and amount meet or exceed requirements
- B. Marker post 60 inch long and 2 inch outside diameter OD using black polyethylene material
- C. Flexible marker post 36 inch long and 2 ½ inch outside diameter OD using white or yellow polyethylene material, minimum 0.110 inch tube wall thickness.

2.3 CONCRETE

- A. AA(AE) – Refer to 03055

PART 3 EXECUTION

3.1 INSTALLATION

- A. Install crash cushion or barrier end treatment after final grading according to the CC Series Standard Drawings has been verified by Engineer.
 - 1. Construct concrete pad according to system requirements.
 - a. Crash Cushion Types A, B, and D require concrete pad constructed to manufacturer's requirements when installed in a permanent application.
 - b. Crash Cushion Type C and End Treatment Types G, and H.
 - 1) Create block out hole by forming, saw cutting, or other similar method required when installing ground mounted system into asphalt or concrete.
 - 2. Compact backfill material around posts and foundation tubes with mechanical effort to match the density of the surrounding soil.
 - 3. Install marker post. Refer to CC Series Standard Drawings.
 - a. Close top of marker post.
 - b. Drill three $\frac{7}{16}$ inch mounting holes.
 - c. Apply three 4 inch bands of yellow sheeting. Refer to Section 02890.
 - 4. Install flexible marker post. Refer to CC Series Standard Drawings.
 - a. Close top of marker post.
 - b. Apply two 4 inch bands of yellow sheeting. Refer to Section 02890.
 - c. Use a mechanical surface mounting system and install according to manufacturer's requirements.
 - 1) Use a mounting system such that the marker post can be removed and replaced without disturbing the mounting base.
- B. Install crash cushion and barrier end treatments according to:
 - 1. CC Series Standard Drawings and UDOT Guidelines for Crash Cushion and Barrier End Treatments.
 - 2. Manufacturer's specifications and recommendations.

END OF SECTION

SECTION 02844

CONCRETE BARRIER

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Fabricating and placing precast concrete barriers
- B. Constructing cast-in-place concrete barriers

1.2 RELATED SECTIONS

- A. Section 02075: Geotextiles
- B. Section 02842: Delineators
- C. Section 03055: Portland Cement Concrete
- D. Section 03056: Self-Consolidating Concrete (SCC)
- E. Section 03152: Concrete Joint Control
- F. Section 03211: Reinforcing Steel and Welded Wire
- G. Section 03390: Concrete Curing
- H. Section 03392: Penetrating Concrete Sealer
- I. Section 09981: Concrete Coating
- J. Section 13553: ITS Conduit

1.3 REFERENCES

- A. AASHTO M 111: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
- B. AASHTO M 180: Corrugated Sheet Steel Beams for Highway Guardrail
- C. AASHTO M 235: Epoxy Resin Adhesives
- D. ASTM A 36: Carbon Structural Steel

- E. ASTM A 449: Hex Cap Screws, Bolts and Studs, Steel, Heat Treated, 120/105/90 ksi Minimum Tensile Strength, General Use
- F. ASTM A 563: Carbon and Alloy Steel Nuts
- G. ASTM A 572: High-Strength Low-Alloy Columbium-Vanadium Structural Steel
- H. ASTM C 578: Rigid, Cellular Polystyrene Thermal Insulation
- I. ASTM C 1315: Liquid Membrane-Forming Compounds Having Special Properties for Curing and Sealing Concrete
- J. ASTM D 1621: Compressive Properties of Rigid Cellular Plastics
- K. ASTM D 1777: Thickness of Textile Materials
- L. ASTM D 6364: Determining Short-Term Compression Behavior of Geosynthetics
- M. ASTM F 436: Hardened Steel Washers
- N. ASTM F 1554: Anchor Bolts, Steel, 36, 55 and 105-ksi Yield Strength
- O. ASTM F 3125: High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength
- P. AWS D1.5: Bridge Welding Code
- Q. UDOT Quality Management Plan

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Manufacturer's product data sheets for review for the following:
 - 1. Barrier seal
 - 2. Composite drainage material
 - 3. Rigid plastic foam
 - 4. Anchor rods
 - 5. Epoxy
 - 6. Include recommended product installation instructions for each product for information.

1.6 ACCEPTANCE

- A. Remove and replace barrier section that does not meet the barrier smoothness requirements. Refer to this section, Article 3.3 paragraph D.

PART 2 PRODUCTS

2.1 CONCRETE

- A. Class AA(AE) – Refer to Section 03055.
- B. SCC – Refer to Section 03056.

2.2 STRUCTURAL STEEL

- A. Constant Slope Concrete Barrier:
 - 1. Connection pin, connection loop, and stabilization pin.
 - a. Refer to ASTM A 36.
 - b. Galvanize after fabrication according to AASHTO M 111.
- B. F-Shape Concrete Barrier:
 - 1. Connection pin
 - a. ASTM A 449, Type 1
 - b. Plate washer
 - 1) ASTM A 572, Grade 50
 - c. Galvanize after fabrication according to AASHTO M 111.
 - 2. Connection loop
 - a. ASTM A 36
 - b. Galvanize after fabrication according to AASHTO M 111.
 - 3. Stabilization pin and plate washer
 - a. ASTM A 36.
 - b. Galvanize after fabrication according to AASHTO M 111.
- C. Welding
 - 1. Refer to AWS D1.5.

2.3 REINFORCING STEEL

- A. Coated according to Section 03211.
- B. Use of non-coated reinforcing steel allowed for precast F-Shape barrier sections constructed for use in work zone applications.

2.4 BARRIER SEAL

- A. Polyester polyurethane open-cell foam 100 percent impregnated with asphalt.
- B. Foam Unit Weight Requirements
 - 1. Before impregnation 68 lbs/yd³ to 85 lbs/yd³
 - 2. After impregnation 252 lbs/yd³ to 270 lbs/yd³
- C. Impregnated asphalt foam returns to 95 percent of its original volume when compressed to 25 percent of its volume and released.
- D. Impregnated asphalt foam remains stable at temperatures ranging from -40 degrees F to +150 degrees F.

2.5 PRECAST F-SHAPE CONCRETE BARRIER

- A. Use a Department pre-qualified supplier.
- B. Refer to BA Series Standard Drawings for reinforcing steel.
- C. Mark each barrier with 1½ inch numbers indicating the date of casting and identification number supplied by the inspector.
 - 1. Mark “WORK ZONE ONLY” if barrier uses uncoated reinforcement.
 - 2. Impress ¼ inch deep into the top center of the barrier.
- D. Prevent cracking or damage during handling and storage of precast units.
 - 1. Replace units with cracks greater than 0.007 inch or damaged precast units.
- E. Cure in accordance with Section 03390.

2.6 BARRIER DELINEATION

- A. Sheeting – Refer to Section 02842.
- B. Hardware – Refer to GW Series Standard Drawings.

2.7 SURFACE SEALING MATERIAL

- A. Penetrating concrete sealer – Refer to Section 03392.
- B. Concrete coating – Refer to Section 09981.
 - 1. Use only when colored concrete coating is specified.

2.8 EXTRUSION AND SLIP FORM MACHINES FOR CAST-IN-PLACE CONSTANT SLOPE BARRIER

- A. Use equipment capable of vertical adjustment to the grade line while in forward motion.
- B. Use equipment with an attached grade line gauge or pointer to make a continual comparison with the barrier being placed and the offset guideline.
- C. Use equipment capable of accommodating pavement to barrier reinforcing bars.

2.9 CAST-IN-PLACE CONSTANT SLOPE CONCRETE BARRIER

- A. Electrical/ITS Conduits and Junction Boxes.
 - 1. Size and quantity according to contract drawings.
 - 2. Refer to Section 13553.
- B. Expansion Joints
 - 1. Use preformed joint filler – Refer to Section 03152.

2.10 COMPOSITE DRAINAGE MATERIAL

- A. Use a two-layer geocomposite sheet drain consisting of drainage geotextile bonded to one side of a three-dimensional drainage core.
 - 1. Drainage geotextile – Refer to Section 02075.
 - 2. Drainage core:
 - a. High strength polystyrene or polypropylene.
 - b. Minimum compressive strength – 15,000 lbs/ft².
 - 1) Refer to ASTM D 1621 or ASTM D 6364.
 - c. Minimum thickness – 0.40 inches.
 - 1) Refer to ASTM D 1777.
 - d. Capable of multidirectional flow.

2.11 RIGID PLASTIC FOAM

- A. Preformed, extruded cellular polystyrene thermal insulation material that has a water absorption property of 0.3 or less – Refer to ASTM C 578.

2.12 FREE STANDING BARRIER, F-SHAPE TO CAST-IN-PLACE BARRIER TRANSITION

- A. Refer to BA Series Standard Drawings.

- B. Thrie-Beam Rail:
 - 1. Minimum galvanizing requirement conforming to AASHTO M 180, Type 1, Class A.
- C. Thrie-Beam Terminal Connector Hardware:
 - 1. Structural hex bolts conforming to ASTM F 3125, Grade A 325.
 - 2. Hardened washers and heavy hex nuts according to ASTM F 436 and A 563 respectively.
 - 3. Galvanize according to AASHTO M 111.
- D. Transition Cap
 - 1. Refer to ASTM A 36.
 - 2. Galvanize according to AASHTO M 111.
 - 3. Connection
 - a. ½ inch diameter, threaded anchor rods conforming to ASTM F 1554, Grade 55.
 - b. Hardened washers and heavy hex nuts according to ASTM F 436 and A 563 respectively
 - c. Epoxy – Refer to AASHTO M 235 Type IV.
- E. Field Side Strap – Refer to ASTM A 36.
 - 1. Galvanize according to AASHTO M 111.

PART 3 EXECUTION

3.1 PREPARATION

- A. Site Considerations
 - 1. Protect work area when removing traffic barriers and crash cushions until the barriers and crash cushion are reconstructed or the hazard is mitigated.
 - 2. Concrete Barrier:
 - a. Complete grading requirements and place any required paved surfaces before installing barrier.
 - 1) Refer to BA Series Standard Drawings.

3.2 PRECAST CONCRETE BARRIER – 32 INCH F- SHAPE

- A. Installation includes moving, stockpiling, and placing all barriers.
- B. Place barrier seal between each barrier unit so that enough pressure is exerted on the sealing material to form and maintain a permanent bond.
- C. Refer to BA Series Standard Drawings.

3.3 CAST-IN-PLACE CONSTANT SLOPE CONCRETE BARRIER – 42 INCH AND 54 INCH

- A. Refer to BA Series Standard Drawings.
- B. Do not use precast mortar blocks to support the reinforcing steel.
- C. Constant Slope Barrier Placed by Extrusion or Slip Form
 1. Provide an offset guideline for the extrusion or slip form machine to maintain the predetermined grade.
 2. Feed concrete to the extrusion or slip form machine at a uniform rate.
 3. Operate machine uniformly restraining forward motion.
 - a. Produce well-compacted, dense concrete with consistency that maintains the shape of the barrier without support.
 - b. Produce a well-compacted mass of concrete free from surface pits larger than 1 inch in diameter and requiring no further finishing.
 4. Saw or cut joints before applying curing compound.
 5. Give the surface a final soft brush finish with strokes parallel to the line of barriers before applying curing compound.
 6. Do not finish with a brush application of grout.
- D. Cast-in-Place Constant Slope Barrier Smoothness:
 1. Verify top of barrier during concrete placement using a 12 ft straightedge.
 - a. Top of barrier must not vary more than ½ inch, measured from top of surface of barrier to bottom surface of straight edge throughout barrier run.
 2. Verify sloped face of barrier during concrete placement using a 4 ft straightedge.
 - a. Surface of traffic face must not vary more than 1 inch, measured from face of barrier to bottom surface of straight edge throughout barrier run.
- E. Mark barrier at beginning, end, and 1,000 ft intervals with 1½ inch numbers indicating the date of casting.
 1. Impress ¼ inch deep into the front face of barrier, 6 inches below the top.
- F. Curing – Refer to Section 03390.

- G. Seal concrete surfaces with a penetrating concrete sealer or a concrete coating.
 - 1. Penetrating concrete sealer
 - a. Use when a colored concrete coating is not specified.
 - b. Not required when curing compound meets ASTM C 1315.
 - c. Refer to Section 03392.
 - 2. Concrete Coating
 - a. Use when a colored concrete coating is specified.
 - b. Refer to Section 09981.

3.4 DELINEATION HARDWARE

- A. Concrete Barrier – Attach L Barrier Reflector. Refer to GW Series Standard Drawings.
- B. Attachment Location – Refer to BA Series Standard Drawings.
- C. Application – Refer to GW Series Standard Drawings.

END OF SECTION

SECTION 02845

HIGH TENSION CABLE BARRIER

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. High tension cable barrier system.

1.2 RELATED SECTIONS

- A. Section 01554: Traffic Control
- B. Section 02056: Embankment, Borrow, and Backfill
- C. Section 02890: Retroreflective Sheeting
- D. Section 03055: Portland Cement Concrete
- E. Section 03211: Reinforcing Steel and Welded Wire

1.3 REFERENCES

- A. AASHTO M 30: Metallic-coated Steel Wire Rope and Fittings for Highway Guardrail
- B. AASHTO M 268: Flat and Vertical Retroreflective Sheeting for Traffic Control
- C. AASHTO Manual for Assessing Safety Hardware (MASH 2016)
- D. ASTM A 36: Carbon Structural Steel
- E. ASTM A 123: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
- F. ASTM A 709: Carbon Structural Steel
- G. ASTM A 1011: Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

- H. ISO 12066-2002: Steel Wire Ropes – Determination of the Actual Modulus of Elasticity

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Installation details, parts list, and contacts for information.
1. Furnish the following before physical completion:
 - a. System installation details and parts list.
 - b. List and contact information for repair part suppliers.
 - c. List of Utah-based, manufacturer-trained installers.
- B. Certificate of Compliance for Wire Rope for information–
1. Furnish a separate certification from the wire rope manufacturer stating it meets the cable barrier manufacturer's requirements if the wire rope is an out-sourced product of the cable barrier system manufacturer.
- C. Herbicide submittals for review.
1. Product data sheets and application instructions
 2. Safety data sheets
 3. License issued by the Utah Department of Agriculture for right-of-way application
- D. Assembly Drawings and Manuals to be submitted at first training session for information.
1. Provide assembly drawings and manuals.
 - a. Training Materials
 - 1) Installation manuals.
 - 2) Maintenance manuals.
 - 3) Materials deemed necessary to conduct training for proper installation and maintenance of cable barrier system.
 - b. Assembly drawings for the installation of the following:
 - 1) Cable Barrier Terminal
 - 2) Typical installation of line posts and cable
- E. Tension log for information.
1. Include manufacturer's recommended tension chart.

1.6 TRAINING AND LITERATURE

- A. Provide installation and maintenance training and certification.
 - 1. Manufacturer supplying the high tension cable barrier system is responsible to perform training.
 - a. Provide one training session no more than 14 days before installing the system to the following:
 - 1) Contractor (Prime)
 - 2) Installation Contractor (Subcontractor)
 - 3) Engineer and designees
 - b. Provide one training session before physical completion and invite the following:
 - 1) Region Maintenance Engineer and designee
 - 2) Region Operations Engineer and designee
 - 3) District Engineer and designee
 - 4) Area Supervisor and designee
 - 5) Local Maintenance Station personnel
 - 6) Engineer for Maintenance (Complex) and designee
 - 7) Representative from the Division of Traffic and Safety
 - 8) FHWA-Utah Division representative

PART 2 PRODUCTS

2.1 HIGH TENSION CABLE BARRIER

- A. Provide 4 wire rope High Tension Cable Barrier system with the following requirements:
 - 1. System meeting MASH 2016 Test Level 3 and certified by the Department for crashworthiness on a 6H:1V or flatter slope.
 - 2. Capable of roadside or median installation.
 - 3. Maximum deflection of 9 ft under MASH 2016 TL-3 conditions with 10 ft post spacing.
 - 4. Capable of post and cable separation during impact.
 - 5. Produce no flying debris capable of vehicle windshield intrusion.
- B. Cable Barrier MASH Gating Terminal
 - 1. Cable Barrier Gating Terminal – MASH 2016 Approved
 - a. Use a terminal required by the manufacturer that will maintain cable tension after a designed impact and certified by the Department meeting Test Level 3 crashworthiness requirements
- C. Provide hardware and miscellaneous items associated with cable barrier and anchor system.

- D. High Tension Cable Gauge capable of measuring cable tension for Department repairs.

2.2 MATERIALS

- A. Wire Rope – Galvanized wire rope $\frac{3}{4}$ inch 3 x 7, construct according to AASHTO M 30 Type I, Class A coating, with a Modified Breaking Strength Minimum equal to 39,000 lb.
 - 1. Pre-stretch wire rope during manufacturing to exhibit a minimum wire rope modulus of elasticity of 19,000,000 psi according to ISO 12076-2002, with no bedding or pre-stretching of the rope permitted in testing.
 - 2. Factory swaged ends on each end of typical cable length.
 - a. Each cable must have a left hand thread on one end and a right hand thread on the other end, 1 inch in diameter. Ends able to accept a closed body style turnbuckle.
 - b. Provide galvanized hardware.
- B. Hardware and Miscellaneous Items
 - 1. Meet manufacturer's requirements for hardware and miscellaneous items as outlined in the manufacturer's specifications for the installation of the cable barrier system. Items to include but not limited to the following:
 - a. Anchor and terminal fittings
 - 1) 1 inch diameter fittings required
 - b. Turnbuckles, rigging screws, field splice connections
 - 1) 1 inch diameter fittings required
 - c. Parts used to separate and hold cable barrier at designed height.
- C. Line post, sleeve sockets – 3 ft, 5 ft, and 10 ft post spacings.
 - 1. Line posts
 - a. Meet all manufacturers' specifications.
 - b. Posts meets the requirements of ASTM A 1011 or ASTM A 36.
 - c. Galvanize according to ASTM A 123, after fabrication.
 - d. Post is able to hold the wire ropes at the design height.
 - 2. Line post sleeve
 - a. Steel sleeves to manufacturers' specifications.
 - b. Steel sleeves constructed using material according to ASTM A 1011 or ASTM A 36.
 - c. Galvanize according to ASTM A 123, after fabrication.

- D. Cast-in-place foundations.
 - 1. Concrete class AA(AE) – Refer to Section 03055
 - 2. Reinforcing Steel
 - a. Coated. Refer to Section 03211
- E. Cable Barrier MASH Gating Terminal
 - 1. Meet manufacturer's specifications.
 - a. Supply specialty posts and hardware.
 - 2. Cast-in-place concrete foundation.
 - 3. Object marker delineation using AASHTO M 268 Type IX retroreflective sheeting.
 - 4. Construct marker post 60 inch long and 2 inch OD using black polyethylene material.
 - a. Close top of marker post
 - b. Drill three $\frac{7}{16}$ inch mounting holes.
 - c. Apply three 4 inch bands of yellow sheeting. Refer to Section 02890.
- F. Herbicide
 - 1. Bare-ground or total vegetation grade.
- G. Delineation
 - 1. Sheeting Requirements
 - a. Refer to Section 02890
 - b. Application
 - 1) Line Post – Refer to this Section, Article 3.4 paragraph A.
- H. High Tension Cable Gauge
 - 1. Furnish gauge capable of measuring cable tension during initial installation.

PART 3 EXECUTION

3.1 PREPARATION

- A. Site Considerations
 - 1. Begin installation of cable barrier post foundations and terminals after final grading is complete as shown. Clear approach areas of any fixed objects or hazards.

2. Apply a bare-ground treatment to 2 ft on each side of the cable system.
 - a. Apply after cable barrier foundations have been installed and the excess material has been removed or graded into surrounding area.
 - b. Follow product-labeling requirements for selected product.

3.2 CAST-IN-PLACE CONCRETE FOUNDATIONS REQUIREMENTS

- A. Do not over excavate hole or use a tubular concrete form for foundation casting.
- B. Install manufacturers' required reinforcing steel as shown.
- C. Install post sleeve no greater than ½ inch above finished grade of foundation.
- D. Fill the excavated hole with concrete. Refer to BA 5 Series Standard Drawings.
 1. Align post sleeve to within ¾ inch alignment from the posts on either side.
 - a. Sleeve alignment can be adjusted to meet placement requirements on a horizontal curve.
 - b. Meet manufacturer's requirements if supplied system uses offset posts.
 2. Install foundation in such a manner that the post will stand vertically and be no greater than ½ inch from plumb.
- E. Allow concrete to cure a minimum of seven days and until it achieves 4,000 psi before installing cable.

3.3 POST AND CABLE INSTALLATION

- A. Install posts according to manufacturer's height requirements to provide proper cable height.
 1. Install o-excluder cap.
- B. Install cable according to manufacturer's requirements.
- C. Tension immediately after initial installation to manufacturer's requirements.
 1. Recheck and adjust tension 5 days, 10 days, and 15 days after initial tensioning.
 2. Maintain tension log showing time, date, location, ambient temperature, and final tension reading signed by the person performing the tension reading.

3.4 DELINEATION

A. Sheeting

1. Install appropriate sheeting on:
 - a. First line post
 - b. Last line post
 - c. Intermediate line posts spaced no more than 50 ft from nearest post with sheeting.
2. Sheeting color – White or yellow, color to correspond with the adjacent edge line.
3. Install so that sheeting is on approach side of nearest lane of traffic.
 - a. Medians width less than 25 ft and barrier system within 15 ft of any travel lane.
 - 1) Install a minimum 10 in² on approach side of nearest lane of traffic.
 - a) Install a minimum 10 in² on approach side of nearest lane of traffic.
 - b) Install a minimum 8 in² on backside side of nearest lane of traffic.
 - 2) Medians width greater than 25 ft.
 - a) Install a minimum 10 in² on approach side of nearest lane of traffic.

END OF SECTION

END OF SECTION

SECTION 02890

RETROREFLECTIVE SHEETING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Retroreflective sheeting materials for traffic signs and traffic control devices.

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES

- A. ASTM B 209: Aluminum and Aluminum-Alloy Sheet and Plate
- B. ASTM D 4280: Extended Life Type, Nonplowable, Raised Retroreflective Pavement Markers
- C. ASTM D 4956: Retroreflective Sheeting for Traffic Control
- D. Code of Federal Regulations (CFR)
- E. Construction of Roads and Bridges on Federal Highway Projects FP-03
- F. SDDCTEA Pamphlet 55-14: Traffic Engineering for Better Signs and Markings

1.4 DEFINITIONS

- A. Retroreflective Sheeting
 - 1. Sheeting – The retroreflective or non-reflective material that comprises the background, legend (word messages and symbols), and border.
 - 2. Sheeting Components – The matched component products required for the manufacture of highway signs consist of the sheeting, cutout letters and borders, adhesives, inks, and overlay films.
 - 3. Permanent Traffic Signs - Include sign installations that are in their final configuration and that are expected to have a multi-year life. Examples include freeway guide signs, regulatory signs, warning signs, barrier markers, crash cushion markers, freeway crossover markers and delineation.

4. Work Zone Standard Signs – MUTCD and Department standard application signs including but not limited to Road Work Ahead, Work Zone Speed Limit, Flagger Symbols, Business Access, and Regulatory signing within the work zone.

5. Work Zone Project Specific Signs – Signs that have legends specific to the project and that cannot be reused on a future project. Examples include: “Alder Street closed from 1st to 2nd Avenue – use Birch Street” or project notification signs. These are typically used for only one construction season.
6. Flexible Work Zone Devices – Include such devices as roll up signs, cones, tall cones, and flags with retroreflective sheeting.
7. Work Zone Channelization Devices – Include such devices as drums, vertical panels, barricades, tubular markers, and pavement marking tabs.

1.5 SUBMITTALS

- A. Authorized Products List (APL) Compliance Form for information.

PART 2 PRODUCTS

2.1 MATERIALS

- A. Retroreflective Sheeting
 1. Meet ASTM D 4956 Standard Specification for Retroreflective Sheeting for Traffic Control.
- B. Non-reflective Sheeting – As specified and according to the recommendation of the retroreflective sheeting manufacturer.
- C. Use matched component cutout legends, symbols, and borders.
- D. Use only acrylic EC film to achieve color. Do not use vinyl EC film to achieve color.

2.2 PERMANENT APPLICATIONS

- A. Permanent Traffic Signs and Traffic Control Devices
 1. Meet or exceed the minimum requirements of ASTM D 4956 Type XI.
 2. Use fluorescent sheeting for orange, yellow and yellow-green.

- B. Raised Pavement Marker/Median Reflector
 - 1. Meet or exceed the minimum retroreflectivity requirements of ASTM D 4280.

2.3 WORK ZONE APPLICATIONS

- A. Work Zone Signing
 - 1. Standard signs
 - a. Meet or exceed the minimum requirements of ASTM D 4956 Type XI.
 - b. Use fluorescent retroreflective sheeting for orange, yellow, and yellow-green.
 - 2. Project specific signs
 - a. Meet or exceed the minimum requirements of ASTM D 4956 Type III high intensity prismatic sheeting.
 - b. Use fluorescent retroreflective sheeting for orange, yellow, and yellow-green.
 - 3. Traffic incident management area signs
 - a. Use fluorescent pink.
- B. Flexible Work Zone Devices
 - 1. Meet or exceed the minimum requirements of ASTM D 4956 Type VI.
- C. Channelization Devices
 - 1. Vertical panels, barricade Types I, II, and III, and directional indicator barricades.
 - a. Meet or exceed the minimum requirements of ASTM D 4956 Type XI.
 - b. Use of standard orange acceptable.
 - 2. Meet or exceed the minimum requirements of ASTM D 4956 Type V for temporary raised pavement markers.
 - 3. Meet or exceed the minimum requirements of ASTM D 4956 Type IV for all other channelization devices.
 - 4. Use fluorescent retroreflective sheeting for orange and yellow.
 - 5. Do not obscure retroreflective sheeting with paint or other materials.

2.4 MANUFACTURER WARRANTY

- A. Failure of the sheeting inks or overlay films provided, sold, or recommended for use by the sheeting supplier, constitutes a failure of the entire sign and replacement of the sign under the manufacturer's warranty is required.

PART 3 EXECUTION Not Used

END OF SECTION

SECTION 02891
TRAFFIC SIGNS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Ground mounted roadside traffic signs, including supports and foundations.
- B. Traffic signs for overhead sign structure including the frame, brackets, and hardware.
- C. Removal of existing sign panel frame, attachments, lighting and hardware from overhead sign structure.

1.2 RELATED SECTIONS

- A. Section 01721: Survey
- B. Section 02317: Structural Excavation
- C. Section 02890: Retroreflective Sheeting
- D. Section 02893: Overhead Sign/VMS Structure
- E. Section 02894: Precast Roadside Sign Foundations
- F. Section 03055: Portland Cement Concrete
- G. Section 03211: Reinforcing Steel and Welded Wire

1.3 REFERENCES

- A. AASHTO M 232: Zinc Coating (Hot Dip) on Iron and Steel Hardware
- B. AISI 1008: Carbon Steel
- C. AISI 1038: Carbon Steel
- D. AISI 1045: Carbon Steel

- E. AMS 6378 D: Steel Bar
- F. ANSI B18-21-1: Helical Spring Lock Washer
- G. ANSI 360: Cast Aluminum
- H. ASME B18.2.1: Square, Hex, Heavy Hex and Askew Head Bolts
- I. ASTM A 36: Carbon Structural Steel
- J. ASTM A 53: Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
- K. ASTM A 123: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
- L. ASTM A 135: Electric-Resistance-Welded Steel Pipe
- M. ASTM A 153: Zinc Coating (Hot-Dip) on Iron and Steel Hardware
- N. ASTM A 314 Stainless Steel Billets and Bars for Forging
- O. ASTM A 449: Hex Cap Screws, Bolts and Studs, Steel, Heat Treated, 120/105/90 ksi Minimum Tensile Strength, General Use
- P. ASTM A 500: Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
- Q. ASTM A 502 Rivets Steel Structural
- R. ASTM A 513: Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing
- S. ASTM A 526: Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process, Commercial Quality
- T. ASTM A 536: Ductile Iron Castings
- U. ASTM A 653: Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
- V. ASTM A 787: Electric-Resistance-Welded Metallic-Coated Carbon Steel Mechanical Tubing

- W. ASTM B 85: Aluminum-Alloy Die Castings
- X. ASTM B 209: Aluminum and Aluminum-Alloy Sheet and Plate
- Y. ASTM B 221: Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
- Z. ASTM B 833: Zinc and Zinc Alloy Wire for Thermal Spraying (Metallizing) for the Corrosion Protection of Steel
- AA. ASTM F 1554: Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength
- BB. ASTM F 3125: High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength
- CC. American Plywood Association (APA) Product Standard

1.4 DEFINITIONS

- A. Flexible Sign Base – A sign mounting system that is designed to allow the sign panel and post to bend and return to an upright position.
- B. Traffic Sign Components
 - 1. Panel – Fully assembled panel type defined here:
 - a. Type
 - 1) A: Retroreflective sheeting on sheet aluminum or plastic.
 - 2) PW: Retroreflective sheeting on plywood-backed sheet aluminum.
 - b. Legend
 - 1) Non-reflective legend and border.
 - 2) Retroreflective legend and border.
 - 2. Panel Overlay – Attaching new panels to all or part of an existing panel.
 - 3. Panel Replacement – Removing the existing panel and attaching a new panel to the frame.
 - 4. Sign – An assembly comprised of panel, panel with frame, or panel with “Z” bar.
 - 5. Size – Width x Height.
 - 6. Substrate – The base aluminum or plastic material, to which the background sheeting is attached.

1.5 SUBMITTALS

- A. Manufacturer's product data sheets for information.
- B. Overhead Sign. Refer to Section 02893 for the following
 - 1. Shop drawings for review.
 - 2. Material submittals for information.
 - 3. Bolting Procedures for review.
 - 4. Welding submittals for information.
 - 5. Construction documentation for information.

1.6 SIGN CODES

- | | | |
|----|-------------------|----|
| A. | New Sign | N |
| B. | Relocation | R |
| C. | Removal | X |
| D. | Panel Replacement | PR |
| E. | Panel Overlay | PO |

PART 2 PRODUCTS

2.1 MATERIALS

- A. Substrate Aluminum – 0.080 or 0.125 inch thick as follows. Refer to ASTM B 209 alloy 6061-T6, or 5052-H38.
 - 1. Use 0.125 for signs installed on frame and Gore signs.
 - 2. Use 0.080 for all other signs.
- B. Substrate Plastic
 - 1. 0.125 inch thick
 - 2. Use only on signs mounted on flexible sign bases.
- C. Plywood Backing – Use APA product standard PS 1-83, Group One: ½ inch thick.
 - 1. 30/30, high density BB exterior (Douglas Fir) B Grade.
 - 2. Plugged-core (Douglas Fir) with ½ inch maximum gaps.
- D. Tubular steel sign mounting hardware
 - 1. Aluminum Z bar refer to ASTM B 221.

2. Pipe clamp casting refer to ASTM B 85 or aluminum alloy ANSI 360.0
 3. Galvanize all non-aluminum components according to ASTM A 135.
 4. U-Bolt use american standard regular semi-finished hex nuts and lock washers according to ASME B18.2.1.
- E. Freeway ground mount sign frame
1. Galvanize frame according to ASTM 123.
 2. Galvanize hardware except for sign facing hardware.
 3. Sign facing hardware options
 - a. Fasteners
 - 1) 18-8 round head ¼ inch stainless steel fastener refer to ASTM A 314 Class 304.
 - 2) Stainless steel washer
 - 3) Nylon washer
 - 4) Stainless steel nut
 - b. Pop rivets that conform to ASTM A 502 grade 1.
- F. Bases
1. Small sign tubular steel post base (B1) (socket system)
 - a. Tubular Socket
 - 1) 12 GA galvanized steel ASTM A 787.
 - b. Wedge
 - 1) 11 GA galvanized steel ASTM A 526
 - 2) G-90 coating.
 2. Round post slipbase sign base (B3)
 - a. Ground stub
 - 1) Galvanized steel ASTM A 536
 - 2) Hot dip galvanized after fabrication.
 - b. Slipbase casting
 - 1) Galvanized steel ASTM A 536.
 - c. Bolt retaining plate
 - 1) 30 Gauge Steel ASTM A 563
 - 2) G 90 coating.
 - d. Dog and Cap screws
 - 1) zinc plated
 3. Surface mounted tubular steel sign base (B4A)
 - a. Base plate
 - 1) Refer to ASTM A 36 plate steel.
 - 2) Galvanize according to ASTM A 123 after fabrication.
 - b. Pipe stub
 - 1) Refer to ASTM A 53 GBR schedule 40.
 - 2) Galvanize according to ASTM A 123 after fabrication.
 - c. All hardware galvanized or zinc plated

4. Side mounted tubular steel sign base (B4B)
 - a. Base casting
 - 1) Refer to ASTM A 536 65-45-12.
 - b. All hardware galvanized or zinc plated.
5. Concrete barrier pivot sign base (B5D)
 - a. Refer to ASTM A 36 steel.
 - b. Galvanize according to ASTM A 123 after fabrication.
 - c. Hardware requirements
 - 1) Anchor rods - ASTM A 449 or ASTM F 1554, grade 55.
 - 2) $\frac{3}{4}$ inch hex bolt - ASTM F3125, grade A 325.
 - 3) $\frac{3}{8}$ inch hex bolt-SAE grade 5.
 - 4) Use galvanized hardware.
6. Freeway sign base (B6A, B6B, B6C)
 - a. Refer to ASTM A 53 grade B for Pipe.
 - b. Refer to ASTM A 36 for S and W shapes and plates.
 - c. Use bolts, nuts, and washers conforming to ASTM F 3125 grade A325.
7. Multi-directional breakaway base (B7 Series)
 - a. Hardware requirements (Use bolts, nuts, and washers conforming to ASTM F 3125 grade A325.)
 - 1) Bracket aluminum 60601-T6.
 - 2) Special Bolt
 - a) Refer to ASTM A 449.
 - b) Galvanize according to ASTM A 153/B 695.
 - 3) Coupling
 - a) Refer to AMS 6378 D.
 - b) Galvanize according to ASTM A 153 and polyester coat.
 - 4) Hinge plate
 - a) Refer to AISI 4130 Steel
 - b) Galvanize according to ASTM A 153
 - 5) Anchor (B7A, B7D, B7E)
 - a) Refer to AISI 1038 rod
 - b) Refer to AISI 1008 coil.
 - 6) Anchor (B7B, B7C)
 - a) Refer to AISI 1045 rod
 - b) Refer to AISI 1008 coil.
 - 7) Bolts for bracket and hinge assembly
 - a) Refer to ASTM A 325.
 - b) Galvanize according to ASTM A 153.
 - 8) Lock washers for bracket and hinge assembly
 - a) Refer to ANSI B18-21-1.
 - b) Galvanize according to ASTM A 153.

- 9) Nuts for bracket and hinge assembly
 - a) Refer to ASTM A 563 grade DH.
 - b) Galvanize according to ASTM A 153.
- 10) Nuts for hinge hardware assembly
 - a) Refer to ASTM A 325
 - b) Galvanize according to ASTM A 153
- 11) Shims
 - a) Galvanized sheet steel.
- 12) Cap-screw for bracket assembly (B7B, B7C)
 - a) Refer to ASTM A 307
 - b) Galvanize according to ASTM A 153

G. Posts, "T" and "U" brackets, extensions, and hardware

- 1. Posts (P2, P3, P4, P5)
 - a. Refer to ASTM A 513 or ASTM A 787.
 - b. Galvanize according to ASTM A 653 or ASTM A 123.
- 2. Posts (P3.0, P3.5, P4.0, P4.5, P4.5-80)
 - a. Refer to ASTM A 53 type E steel pipe.
 - b. Galvanize according to AASHTO M 232.
- 3. Square Posts ($4 \times 4 \times \frac{3}{16}$, $4 \times 4 \times \frac{1}{4}$)
 - a. Refer to ASTM A 500 Grade B steel square tube.
 - b. Galvanize according to AASHTO M 232.
- 3. "T" and "U" Bracket
 - a. Galvanize according to ASTM A 653 or ASTM A 123.
 - b. Coat weld joints with zinc rich paint.
- 4. 90 degree post extension
 - a. Refer to ASTM A 787 AWG fabricated from galvanized sheet steel 0.083 inch thick ASTM A 653 SS grade G90.
 - b. Weld seam using zinc wire according to ASTM B 833.
 - c. Treat tube outside diameter with chromate conversion solution and protect with clear acrylic film.
- 5. Posts (Pipe Standard Weight) for freeway signs
 - a. Use structural steel conforming to ASTM A 53 Grade B Steel
 - b. Galvanize according to ASTM A 123.
- 6. S Section and W Section steel posts
 - a. Use structural steel conforming to ASTM A 36.
 - b. Galvanize according to ASTM A 123.

H. Retroreflective and non-reflective sheeting – Refer to Section 02890.

I. Foundation for ground mounted signs

- 1. Precast foundation – Refer to Section 02894.
- 2. Concrete – For B1, and B3 bases use class A(AE) or B(AE), for all other bases use class A(AE), refer to Section 03055.

- a. A commercially prepackaged air-entrained mix that is designed to meet or exceed a 28 day minimum compressive strength of 4,000 psi may be used as a substitute for applications requiring Classes A(AE) or B(AE) concrete.
 - 1) Proportion the mix according to the manufacturer's instructions to achieve a strength of 4000 psi at 28 days.
 - 2) Mix thoroughly before placing.
 - 3) Consolidate the mix in-place by rodding or vibrating.
- 3. Reinforcing steel – Refer to Section 03211

PART 3 EXECUTION

3.1 FABRICATION

- A. Panel
 - 1. Do not reverse screen sign larger than 7 ft²/color.
 - 2. Establish proper orientation of the sheeting for all traffic signs.
 - a. Verify cutout legends, symbols, and borders have the same sheeting orientation as background sheeting.
 - 3. Attach sheeting and sheeting components to aluminum substrate.
 - 4. Attach aluminum substrate to plywood backing when required. 5. Permanent sign and sheeting identification
 - a. Affix to each sign a 1 inch high, two-digit number representing the fabrication year to the front lower left corner of each sign.
 - 1) Use numbers of contrasting color to sheeting color.
 - 2) Represent the year 2012 as “12”, and so forth.
 - b. Affix to each sign a 2 inch x 2 inch impermeable, non-fading weather-resistant, self-adhesive label.
 - 1) Attach label where it will not obscure sheeting.
 - 2) The label contains the month and year of manufacture, Contractor’s name, and type of sheeting.
- B. Ground Mounted Signs
 - 1. Fabricate base, post, extensions, frames, and brackets according to SN drawings and as shown.
 - 2. Saw cut or shear plates. No flame cutting is allowed.

- C. Overhead Signs
 - 1. Fabricate frame and brackets as shown for the structure.

3.2 INSTALLATION – GROUND MOUNTED SIGNS

- A. Excavate – Refer to Section 02317.
- B. Construct sign post foundations with concrete according to dimensions.
- C. Compact backfill to a density equal to surrounding materials.
- D. Establish proper elevation and orientation of all signs, components, and determine proper sign post lengths as dictated by construction slopes. Refer to SN Series Standard Drawings.
- E. Allow for 7 day cure time before installing sign post in B1 base.

3.3 INSTALLATION – OVERHEAD SIGN

- A. Sign Panel Replacement
 - 1. Provide all new steel components and hardware necessary for the placement of new traffic sign panels on an existing overhead sign structure. This includes, but is not limited to the sign frame, brackets, and U-bolts.
- B. Establish proper elevation and orientation of all signs as shown.
 - 1. Coordinate the placement of all signs with the Engineer before installation.
- C. Attach sign panel to the structure as shown and according to Section 02893.
 - 1. Do not weld to overhead sign structure.

3.4 RELOCATE EXISTING GROUND MOUNTED SIGN

- A. Remove existing foundations to at least 6 inch below the ground line and backfill.
- B. Provide and install new posts and accessories as required.
- C. Retrofit as required to meet current standards.

3.5 RELOCATE EXISTING MILEPOST SIGN

- A. Survey location of existing milepost signs as specified in Section 01721 before removal.

- B. Milepost sign may be removed and stored to facilitate project work.
- C. Comply with other requirements listed in this Section, Article 3.4.
- D. Reestablish milepost sign at location according to Section 01721.

3.6 REMOVE EXISTING GROUND MOUNTED SIGN

- A. Remove foundations to at least 6 inch below the ground line and backfill.

3.7 REMOVE OVERHEAD SIGN

- A. Remove overhead sign attached to bridge structure as shown.
 - 1. Remove sign panel, frame, brackets and all attachments to the structure.
 - 2. Cut or grind element flush to the concrete surface that are embedded or epoxied to the deck, parapet or girder.
 - a. Apply galvanization coating over exposed cut surfaces according to Section 02893.
 - b. Notify the Engineer of any damage that occurred during removal to the parapet, deck or girder.
- B. Remove overhead sign attached to overhead sign structure as shown.
 - 1. Remove existing sign panel, frame, attachments and hardware.
 - a. Do not damage or modify any portion of the structure that is to remain.
 - 2. Remove lighting from the structure
 - a. Remove all lighting elements from the sign and structure.
 - b. Remove power from the structure to at least the first junction box. Terminate power according to local and state laws and guidelines.

3.8 LIMITATIONS

- A. Do not remove a sign that is being replaced until the new sign is placed and uncovered.
- B. Cover signs that require temporary covering with an opaque material.
 - 1. Secure at the rear of the sign so that the sign is not damaged.
 - 2. Maintain covering until covering or sign is removed.

END OF SECTION

SECTION 02894

PRECAST ROADSIDE SIGN FOUNDATIONS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Precast concrete foundations for roadside traffic signs, non-overhead.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill
- B. Section 03055: Portland Cement Concrete
- C. Section 03211: Reinforcing Steel and Welded Wire

1.3 REFERENCES

- A. AASHTO M 111: Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
- B. AASHTO M 291: Carbon and Alloy Steel Nuts
- C. AASHTO M 292: Carbon and Alloy Steel Nuts for Bolts for High-Pressure or High-Temperature Service or Both
- D. AASHTO M 293: Hardened Steel Washers
- E. ASTM A 36: Carbon Structural Steel
- F. ASTM F 1554: Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength.
- G. UDOT Quality Management Plan

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Working Drawings
 - 1. Detailed shop drawings of lifting methods and devices for review.
 - a. Provide the seal of a Professional Engineer (PE) or

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Professional Structural Engineer (SE) licensed in the State of Utah.

- b. Include supporting engineering calculations.

PART 2 PRODUCTS

2.1 CONCRETE

- A. Class AA(AE) – Refer to Section 03055.

2.2 REINFORCING STEEL

- A. Refer to Section 03211.

2.3 ANCHOR BOLTS, NUTS, AND WASHERS

- A. Use bolts, nuts, and washers displaying the manufacturer's markings.
- B. Anchor Bolts – Refer to ASTM F 1554.
- C. Nuts
 - 1. Refer to AASHTO M 291 or AASHTO M 292.
 - 2. Use heat-treated Grades DH and 2H.
- D. Washers – Refer to AASHTO M 293.
- E. Galvanize according to AASHTO M 111.

2.4 ANCHOR BOLT TEMPLATES

- A. Refer to ASTM A 36.
- B. Galvanize according to AASHTO M 111.

2.5 SHAFT FABRICATION

- A. Use a UDOT Certified Concrete Precaster or a pre-qualified project site caster for concrete products according to the Department Quality Management Plan: Precast-Prestressed Concrete Structures.
- B. Obtain a minimum compressive strength of 500 psi before stripping the forms.
- C. Permanently mark the top of the shaft with date of casting and

supplier identification.

1. Stamp markings in fresh concrete.

2.6 DRILLING EQUIPMENT

A. Capable of:

1. Drilling holes to the required diameter, location, alignment, and depth in the type of materials present at the shaft locations.
2. Installing and removing casing.

2.7 FREE DRAINING GRANULAR BACKFILL

A. Refer to Section 02056.

PART 3 EXECUTION

3.1 DRILLING

A. Drilling Holes

1. Drill straight, vertical holes to the depth shown or as directed by the Engineer.
2. Drill holes within 3 inches of plan position.
3. Remove all loose material from the bottom of the drilled holes before placing the shaft.
4. Do not use water or slurry for drilling operations.

B. Casing

1. Furnish and place casing when required to prevent the drilled hole from caving and any time groundwater is encountered.
2. Remove casing during or after the backfill is placed.

C. Uncased Holes

1. Drill uncased, dry, non-caving holes in a continuous operation.

D. Obtain Engineer's approval to place shaft under water.

3.2 SHAFT INSTALLATION

A. Place and compact free draining granular backfill in the bottom of the drilled hole to the proper elevation before placing the shaft.

B. Place free draining granular backfill up to the top of the shaft.

1. Compact by hand using a maximum of 12-inch lifts.

END OF SECTION

SECTION 02911

HYDRAULIC EROSION CONTROL PRODUCTS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Hydraulic Erosion Control Products (HECP) used to reduce soil erosion on disturbed areas of a construction project.

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES

- A. ASTM D 2974: Determining the Water (Moisture) Content, Ash Content, and Organic Material of Peat and Other Organic Soils
- B. ASTM D 6566: Measuring Mass per Unit Area of Turf Reinforcement Mats
- C. ASTM D 6567: Measuring the Light Penetration of a Erosion Control Product (ECP)
- D. ASTM D 7322: Determination of Erosion Control Product (ECP) Ability to Encourage Seed Germination and Plant Growth Under Bench-Scale Conditions
- E. ASTM D 7367: Determining Water Holding Capacity of Fiber Mulches for Hydraulic Planting
- F. ASTM D 8297: Determination of Erosion Control Products (ECP) Performance in Protecting Slopes from Sequential Rainfall-Induced Erosion Using a Tilted Bed Slope
- G. ASTM D 8298: Determination of Erosion Control Products (ECP) Performance in Protecting Slopes from Continuous Rainfall-Induced Erosion Using a Tilted Bed Slope

1.4 DEFINITIONS

- A. Hydraulic Erosion Control Products (HECP) – Biodegradable wood, wheat, straw or other natural fibrous material manufactured through thermal or mechanical process and generally pre-packaged with binding agents which when mixed with water is applied hydraulically as a slurry to reduce soil erosion and aid in vegetation establishment.

1.5 SUBMITTALS

- A. Authorized Products List (APL) Compliance Form for information.
- B. HECP work plan for review, include the following:
 - 1. Description of the measuring technique.
 - 2. Calculated area to receive each Type of HECP
 - 3. Product name, application rate and total quantity of each Type of HECP to be used. Refer to this Section, Paragraph 3.2 C.
 - 4. Product name, application rate and total quantity of Tackifier for HECP Type 1, if required

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver material and products in original ultra violet (UV) and weather-resistant factory labeled packages showing the name of the manufacturer and product description.

PART 2 PRODUCTS

2.1 HYDRAULIC EROSION CONTROL PRODUCTS (HECP)

- A. Furnish HECP according to Table 1:

Table 1

Requirements for Hydraulic Erosion Control Products				
Physical Properties	Test Method	HECP Type 1	HECP Type 2	HECP Type 3
Min. Organic Matter	ASTM D 2974	90%	90%	90%
Min. Mass per Unit Area	ASTM D 6566	8 oz/yd ²	11 oz/yd ²	11.5 oz/yd ²
Min. Ground Cover	ASTM D 6567	90%	95%	98%
Min. Water Holding Capacity	ASTM D 7367	400%	600%	1000%
Colored/Dyed	Observed	Yes	Yes	Yes
Performance Properties				
Min. Functional Longevity	Manufacturer	3 months	6 months	12 months
C Factor	ASTM D 8297* or ASTM D 8298*	≤ 0.30	≤ 0.05	≤ 0.01
Min. Vegetation Establishment	ASTM D 7322	300%	400%	500%

* Type 1 Test, see ASTM, Table 4.

2.2 GENERAL REQUIREMENTS

- A. All HECP must contain some type or combination of non-toxic additives such as tackifiers, flocculants, polymers, co-polymers or other binding agents as part of manufacturer packaging to satisfy requirements in Table 1 except as follows.
 - 1. HECP Type 1 products which do not contain a pre-mixed tackifier require a tackifier to be field mixed into the HECP.
- B. HECP shall not contain any germination or growth inhibiting factors.

2.3 WATER

- A. Free of dirt, silt and other detrimental matter.

2.4 TACKIFIER FOR HECP TYPE 1

- A. General
 - 1. Nonflammable
 - 2. Nontoxic to aquatic organisms.
 - 3. Functional for a minimum of 180 days.

- B. Guar-based tackifier
 - 1. Derived from ground endosperm of guar seeds (*Cyamopsis tetragonolobus*) and treated with dispersant agents.
 - 2. Required for HECF Type 1 products without tackifier at a rate of 50 lb per acre.
- C. Psyllium-based tackifier
 - 1. Manufactured from the finely ground muciloid coating of *Plantago ovata* or *Plantago ispaghula* seeds.
 - 2. Will hydrate when mixed with water to form a slurry and produces a firm, resilient and rewettable membrane on the soil surface after application.
 - 3. Required for HECF Type 1 products without tackifier at a rate of 100 lb per acre.
- D. Starch-based tackifier
 - 1. Typically derived from corn or potatoes in a non-ionic, cold-water soluble (pre-gelatinized) granular form.
 - 2. Required for HECF Type 1 products without tackifier at a rate of 150 lb per acre.

PART 3 EXECUTION

3.1 PREPARATION

- A. Complete required grading and topsoil placement in designated areas before applying HECF.
- B. Measure the area to be stabilized with HECF to calculate the quantity of HECF required.
 - 1. Coordinate with the Engineer to determine acceptability of the measuring technique to be used before measuring the areas to be stabilized.
- C. Time restrictions of HECF application:
 - 1. Apply HECF within 24 hours after seeding.
 - 2. Apply when soil is dry and rain is not forecasted to occur within 24-48 hours after HECF application.

3.2 PRE-APPLICATION VERIFICATION TEST

- A. Apply HECF to 100 yd² test area in the presence of the Engineer.
 - 1. Demonstrate to the Engineer's satisfaction compliance with the Manufacturer's installation requirements.

3.3 HECP APPLICATION

- A. Use HECP components pre-packed by the manufacturer.
- B. Mix water and HECP according to the manufacturer's recommendation.
- C. Apply HECP according to Table 2, unless manufacturer's recommended rate is higher than the rate specified.

Table 2

HECP Application (Slope Gradient and Application Rates)		
HECP	Max. Slope Gradient (H:V)	Minimum Rate (lb per acre)
Type 1	3:1	2,500
Type 2	2:1	3,500
Type 3	1:1	4,000

- D. Apply a uniform layer of HECP slurry from a minimum of two opposing directions to meet manufacturer's recommendation for installed thickness and ground coverage.
 - 1. Verification test area may be used to visually verify quality of HECP application.
 - 2. Remove overspray before the HECP slurry dries.

END OF SECTION

SECTION 02912

TOPSOIL

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Stripping, stockpiling, and placement of salvaged topsoil.
- B. Furnishing, stockpiling, and placement of contractor furnished topsoil.

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. AASHTO T 194: Determination of Organic Matter in Soils by Wet Combustion
- B. North American Proficiency Testing (NAPT) Program
- C. Soil, Plant and Water Reference Methods for the Western Region (WREP)

1.4 DEFINITIONS

- A. Subsoils – from E, B or C soil horizon layers including heavy clay and coarse sand.
- B. Topsoil – The upper layer of soil, usually referred to as the “A” soil horizon, which is a visually distinct, dark layer of soil considered most suitable for seed germination and the support of vegetative growth.

1.5 SUBMITTALS

- A. Contractor furnished topsoil test results for review.
 - 1. Provide test results for all contractor furnished topsoil sources from an independent soil testing laboratory that meets the following:
 - a. Current participant in NAPT Program.

PART 2 PRODUCTS

2.1 SALVAGED TOPSOIL

- A. Existing topsoil that is stripped and excavated from on site and then stockpiled and spread onto completed areas and excludes the following:
1. Subsoils.
 2. Rock greater than 2 inch in any dimension.
 3. Materials from the pavement section.
 4. Construction waste.
 5. Petroleum byproducts.
 6. Refuse or other manmade materials.
 7. Roots greater than 2 inch in diameter or 12 inch in length.
 8. Noxious weed plants.

2.2 CONTRACTOR FURNISHED TOPSOIL

- A. Provide contractor furnished topsoil that meets the requirements in Table 1.
1. Contractor furnished topsoil may be accepted if any submitted test results for N, P, K, or Fe that are less than the acceptable range in Table 1, as follows:
 - a. Apply Nitrogen, Phosphorus, and Potassium (NPK) fertilizer at the rates indicated on the label to the quantity and location of topsoil as shown.

Table 1

Contractor Furnished Topsoil Requirements		
Parameters	Test Method	Acceptable Range
Soil pH	WREP S-1.10 or S-2.20	5.5 to 8.0
Electrical Conductivity (EC)	WREP S-1.20 or S-2.30	< 4 ds/m
Sodium Adsorption Ratio (SAR)	WREP S-1.60	< 5
Organic Matter	AASHTO T 194	> 1%
N (Nitrate-nitrogen)	WREP S-4.10	> 20 ppm
P (Phosphorous)	WREP S-4.10	> 15 ppm
K (Potassium)	WREP S-4.10	> 150 ppm
Fe (Iron)	WREP S-6.10	> 10 ppm
Sand	WREP S-14.10	< 70%
Silt	WREP S-14.10	< 70%
Clay	WREP S-14.10	< 30%
Percentage of Rock Fragments (≥ 3/8 inch in diameter)		< 5%

- B. Provide contractor furnished topsoil free from:
 - 1. Subsoils
 - 2. Rock larger than 1.5 inch in any dimension.
 - 3. Trash, litter and refuse

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

- A. Do not strip, stockpile, spread, or place topsoil that is covered by snow, frozen, too saturated, or otherwise in a condition detrimental to the work.
- B. Complete final grading, allow trenches to settle, and complete surface preparation by clearing the area to receive topsoil of all trash, debris, weeds, and rock 2 inch or larger and any of objectionable material before placing topsoil.
 - 1. Scarify to a 6 inch depth any areas such as stockpiles, staging areas and haul roads using vertical tillage, deep soil ripping or other method.

3.2 STRIP AND STOCKPILE SALVAGED TOPSOIL

- A. Strip only after site clearing and grubbing is completed and before excavation of subsoil areas begins.
 - 1. Coordinate depths, locations, and quantities with the COR.
- B. Store in windrows or stockpiles.
 - 1. Do not damage or impact any existing landscape plants.
 - 2. Grade to minimize erosion on and adjacent to the stockpiles.
 - 3. Provide appropriate temporary erosion controls as necessary.

3.3 TOPSOIL PLACEMENT

- A. Place and spread the topsoil to a uniform thickness of 4 inch unless otherwise shown.
 - 1. Break clods and lumps with harrows, disks, or other appropriate equipment to provide a uniform textured soil.
 - 2. Remove and dispose of clods, stones, stumps, roots, and other objects larger than 2 inch.
- B. Place topsoil to establish final grades adjacent to paved surfaces including walks, curbs or other hard surfaces as follows:
 - 1. 1 inch below in areas receiving seed or turf seed.
 - 2. 1½ inch below in areas receiving turf sod.

- C. Place topsoil on sloped areas as follows:
1. On slopes flatter than 3:1, disc or harrow the placed topsoil along the contour or cat-track the placed topsoil with tracked equipment perpendicular to the contours to create continuous cleat tracks that run parallel with the contours.
 2. On slopes 3:1 and steeper, cat-track the placed topsoil with tracked equipment perpendicular to the contours to create continuous cleat tracks that run parallel with the contours.
- D. Provide a suitable topsoil surface that meet the following conditions:
1. Lightly compacted to a firm surface with a friable or crumbly soil texture.
 - a. Re-work any topsoil surface that is over-compacted, hard, or crusted by cat-tracking, discing, or harrowing the topsoil surface.
 2. Free from noxious weeds and other undesirable plants that have germinated and are actively growing.
 - a. Remove plants growing in topsoil by hand or mechanical methods, or by appropriate chemical herbicide application.
 3. Free from soil erosion.
 4. Maintain suitable topsoil surface conditions until seed, turf seed or turf sod is completed.

END OF SECTION

SECTION 02922

SEED, TURF SEED, AND TURF SOD

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Seed, turf seed, and turf sod requirements and application.
- B. Surface preparation.

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. Utah Fertilizer Act, Utah Code, Title 4, Chapter 13
- B. Utah Seed Act, Utah Code, Title 4, Chapter 16

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Manufacturer's directions on drill calibration seven calendar days before seeding, for information. Refer to this Section, Article 3.3.
- B. Seed substitution request, for review, if required. Refer to this Section, Article 2.1
- C. Quantity of seed required, for review. Include at least the following:
 - 1. Description of the measuring technique of the area to receive seed.
 - 2. Calculated area to receive drill seed, broadcast seed, turf seed, or turf sod.
 - 3. Calculation of total quantity of seed, turf seed or turf sod to be purchased.
- D. The following documentation from seed supplier, for information:
 - 1. Letter of Certification documenting the purchase and mixing of project seed schedule, including any authorized seed substitutions. Include at least the following:
 - a. Common name of each seed.
 - b. Lot number of each seed
 - c. Purity percent of each seed

- d. Germination percent of each seed
- e. Dormant or hard seed percent of each seed
- f. Bulk lb and percent of each seed.
- g. Pure Live Seed (PLS) lb of each seed.
- h. Total combined bulk lb of project seed schedule.
- i. Total combined PLS lb of project seed schedule.
- j. Total number of bags and bulk lb per bag delivered.
- 2. Certified Copy of Seed Analysis for each seed, including any authorized seed substitutions. Include at least the following on seed certification reports and labels:
 - a. Botanical name and common name, including variety.
 - b. Name of seed testing laboratory
 - c. Origin and Lot number
 - d. Purity percent
 - e. Inert matter percent
 - f. Other crop percent
 - g. Weed seed percent
 - h. Noxious Weed percent
 - i. Germination percent
 - j. Dormant or hard seed percent
 - k. Total Viability percent
 - l. PLS percent
 - m. Last seed germination test date (month and year)
- E. Certification that turf sod is nursery grown and contains grass species shown, for review.
- F. Documentation indicating the date and time turf sod was cut at the nursery, for information.
- G. Fertilizer labels, for information.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Notify COR seven calendar days before purchasing and mixing project seed schedule shown or as modified by authorized seed substitution request.
- B. Deliver seed or turf seed to project in original seed supplier bags showing analysis of seed mixture, net weight, and date and location of packaging.
- C. Cut turf sod from nursery no more than 24 hours before installing.

- D. Deliver fertilizer in original manufacturer packaging bearing labels including brand name, weight and guaranteed analysis.
 - 1. Store in a weatherproof location.

PART 2 PRODUCTS

2.1 SEED AND TURF SEED

- A. Must conform to Utah Seed Act.
- B. Obtain seed from lots that have been tested by a state certified seed testing laboratory such as Association of Seed Analyst (AOSA) or Society of Commercial Seed Technologists (SCST).
 - 1. Seed germination test older than 18 months for grass seed and older than 12 months for shrub or tree seed are not acceptable.
- C. Do not use wet, moldy, or otherwise damaged seed.
- D. Seed Substitutions.
 - 1. Coordinate with the seed supplier and submit recommended substitution alternatives and revised project seed schedule to the COR if seed substitutions to project seed schedule are required.

2.2 TURF SOD

- A. Healthy and well-rooted nursery grown turf sod comprised of grass species shown and free of weeds.
- B. Machine cut in straight, uniform strips or rolls, cut at a depth between $\frac{3}{4}$ inch and 1 inch.

2.3 FERTILIZER FOR TURF SEED OR TURF SOD

- A. Nitrogen, Phosphorus, and Potassium (NPK) fertilizer:
 - 1. Must conform to Utah Fertilizer Act, be uniform in composition, granular, dry, and free flowing.
 - 2. Apply at the rates indicated on the label.

PART 3 EXECUTION

3.1 PREPARATION

- A. Measure the area to receive seed, turf seed, or turf sod before ordering from the supplier.
- B. Coordinate with COR within 14 calendar days before installation seed, turf seed or turf sod begins to verify that the following suitable topsoil surface conditions are met:
 - 1. Lightly compacted to a firm surface with a friable or crumbly soil texture.
 - a. Re-work any topsoil surface that is over-compacted, hard, or crusted topsoil surface must be reworked such as cat-tracking, discing, or harrowing the topsoil surface.
 - 2. Free from noxious weeds and other undesirable plants that have germinated and are actively growing.
 - a. Remove plants growing in topsoil by hand or mechanical methods, or by appropriate chemical herbicide application.
 - 3. Free from soil erosion.
 - a. Regrade and repair any surface, rill, or gulley erosion.
- C. Turf Seed Preparation.
 - 1. Coordinate with Engineer to establish an appropriate irrigation operating schedule.
 - 2. Apply NPK fertilizer at the rates indicated on the label and mix thoroughly into upper 1 inch of topsoil.
 - a. Do not apply fertilizer and seed at the same time in the same machine.
- D. Turf Sod Preparation.
 - 1. Coordinate with COR to establish an appropriate irrigation operating schedule.
 - 2. Apply NPK fertilizer at the rates indicated on the label and mix thoroughly into upper 1 inch of topsoil.

3.2 SEED – GENERAL

- A. Notify the COR seven calendar days before seeding.
- B. Apply seed at the rate shown.
 - 1. Drill seed and broadcast seed are applied at different rates.

- C. Install drill seed and broadcast seed within the seeding window assigned to ground elevation specified in Table 1.
1. Do not begin seeding activities if topsoil is covered in snow, frozen or saturated.
 2. Postpone seeding until the following year if the seeding is not completed within the given window.
 3. A late winter exception to the seeding window may be obtained from the Engineer if suitable weather and soil conditions exist.

Table 1

Seeding Window	
Elevation	Date
Below 4,000 ft	October 15 to December 31
4,000 to 6,000 ft	September 15 to November 30
Above 6,000 ft	September 1 to November 15

- D. Install rolled and hydraulic erosion control products within 24 hours after seed or turf seed application.

3.3 DRILL SEED METHOD

- A. Use drill seed on slopes 3:1 and flatter.
- B. Use a minimum-till drill, no-till drill, rangeland drill seeder or equivalent drill seeding equipment designed to plant a variety of seed types into rugged soils with minimal soil disturbance and equipped with the following:
1. Depth band
 2. Seed box agitator
 3. Seed metering device
 4. Furrow opener
 5. Seed-to-soil packing devices such as packer wheels, imprinters or drag chains
- C. Follow the drill manufacturer's directions in the presence of the Engineer.
1. Calibrate the drill to apply seed at the rate shown.
- D. Space drill rows a maximum of 8 inch apart.

- E. Set depth bands to drill seeds to a ½ inch depth.
- F. Drill parallel to the contours.
- G. Maintain the drill at the calibrated setting throughout the seeding operation.
- H. Allow the furrows that are created by the drill to remain.

3.4 BROADCAST SEED METHOD

- A. Use the broadcast method of seeding under the following conditions:
 - 1. Slopes steeper than 3:1.
 - 2. Slopes 3:1 and flatter where the area to be seeded is inaccessible to drill.
 - 3. The area to be seeded is not large enough to justify using a drill.
 - 4. Rocky surface conditions.
 - 5. Other areas allowed by Engineer.
- B. Do not mix seed as part of a Hydraulic Erosion Control Product (HECP) application.
- C. Broadcast the seed onto the topsoil using either of the following:
 - 1. Mechanical, cyclone or hand seeder.
 - 2. Hydraulic hydroseeder machine which is capable of continuous agitation and application of the seed, water, and up to 300 lb of a cellulose fiber mulch (allowed only as application tracer) slurry mixture per acre.
- D. Incorporate the seed into the topsoil using either of the following:
 - 1. Cat-track seeded area with tracked equipment perpendicular to the contours to create continuous cleat tracks that run parallel with the contours.
 - 2. Drag or rake the seed into the topsoil to a depth between ¼ to ½ inch.
- E. Coordinate with COR to be on site and observe the first 100 yd² of broadcast seed.t broadcast seed during windy weather.

3.5 TURF SEED PLACEMENT

- A. Apply turf seed after seedbed preparation. Refer to this Section, Article 3.1.

- B. Verify good seed to soil contact by either lightly dragging, raking or rolling seeded areas using a minimum 24 inch wide turf roller half filled with water.
- C. Lightly water and program the irrigation system to maintain a moist seedbed.
- D. Rope off newly seeded areas along walkways using bright plastic ribbon tape attached to stakes placed frequently enough to delineate the newly seeded area.

3.6 TURF SOD PLACEMENT

- A. Place turf sod with all edges and joints tightly butted.
 - 1. Do not stretch or overlap turf sod.
 - 2. Keep length seams in a straight line.
 - 3. Lay turf sod with staggered joints.
 - 4. Trim off excess material along the edges.
- B. Roll turf sod immediately after placing using a minimum 24 inch wide turf roller half filled with water.
 - 1. Re-roll if high spots remain.
 - 2. Lift turf sod and place sufficient topsoil to fill any depressions that remain.
 - 3. Thoroughly water with a fine spray to a depth sufficient that the underside of the new turf sod and soil immediately below the turf sod are thoroughly wet.

END OF SECTION

SECTION 02961

ROTOMILLING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Rotomilling of the existing bituminous surface at the location and to the depth shown.

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES **Not Used**

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS **Not Used**

PART 2 PRODUCTS

2.1 EQUIPMENT

- A. Use power-operated track propelled planing machine or grinder:
 - 1. Capable of milling to cross slope as shown.
 - 2. Self-propelled with sufficient power, traction, and stability to maintain accurate depth of cut.
 - 3. Maximum of $\frac{5}{8}$ inch between the cutting teeth on the mandrel.
 - 4. Use a minimum 12 ft mill head on interstates and freeways and a minimum 7 ft mill head on all other roads or as directed by the Engineer.
- B. Use cleaning equipment capable of sweeping, picking up millings, and cleaning the vertical edge to clean up after milling operation.

PART 3 EXECUTION

3.1 ROTOMILLING

- A. Contractor Quality Control:
 - 1. Measure and record rotomilling depths, taking two random measurements every 1,000 ft of each pass of the milling machine.
 - 2. Require an adjustment to the depth of the milling operation, up to an additional $\frac{3}{8}$ inch of depth, to remove unacceptable material or to improve ride.
- B. Rotomill existing bituminous pavement surface to the width and depth shown not to exceed an additional $\frac{3}{8}$ inch of plan depth, measured from original surface to the top of the high point of the milled surface.
 - 1. Produce a uniform pavement surface texture.
 - 2. Adjust the speed of the milling operation to obtain continuous horizontal grind marks on the roadway surface without a skipping pattern.
 - 3. Limit the difference between the high and low points of the milled surface to $\frac{1}{4}$ inch.
- C. Load the millings from milling operation into a truck in one operation.
- D. Millings not designated for use become the property of the contractor.
 - 1. Remove and dispose of millings promptly as specified and according to applicable rules and regulations.
 - 2. Do not place millings on the job site except as described or as approved by the Engineer.

3.2 CLEAN AND REPAIR

- A. Remove and clean all millings from the surface daily. Control dust created by the cutting action. Clean rotomilled surface after milling operation and before opening to traffic.
- B. Verify all milled surfaces are sturdy and sound.
 - 1. Remove any material that is loose or not well bonded.

END OF SECTION

SECTION 02963

PROFILE ROTOMILLING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Rotomilling of the existing bituminous surface at the location and to the profile and cross-slope shown.

1.2 RELATED SECTIONS

- A. Section 02741: Asphalt Mix

1.3 REFERENCES **Not Used**

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Submit for information the record of profile grades.

PART 2 PRODUCTS

2.1 MATERIALS

- A. Use the milled material obtained from the rotomill process passing a 2 inch sieve as optional use of Reclaimed Asphalt Pavement (RAP). Refer to Section 02741.

2.2 EQUIPMENT

- A. Use power-operated track propelled planing machine or grinder:
 - 1. Capable of milling to the plan cross slope.
 - 2. Self-propelled with sufficient power, traction, and stability to maintain accurate depth of cut.
 - 3. Maximum of $\frac{5}{8}$ inch between the cutting teeth on the mandrel.
- B. Use appropriate cleaning equipment capable of sweeping, picking up millings, and cleaning the vertical edge to clean up after milling operation.

PART 3 EXECUTION

3.1 PREPARATION

- A. Measure and record profile grades during rotomilling operation:
 - 1. Use a means that provides the required accuracy.
- B. Achieve a rotomilled depth in the field of plan depth not to exceed an additional ¼ inch, measured from original surface to the top of the ridge.
- C. Rotomill existing bituminous pavement surface to a depth below final profile and cross slope grade as shown.
- D. Clean rotomilled surface after milling operation and before opening to traffic.
- E. Remove and replace or repair damage by the Contractor's operation outside the widths and depths as shown.
 - 1. Repair damage to traffic due to loose material on milled surface.
- F. Use a wire guide or other approved grade control methods to control the profile grade and cross slope of the rotomill.
- G. Load the reclaimed material from milling operation into a truck in one operation.
 - 1. Remove and clean all millings from the surface daily.
 - 2. Control dust created by the cutting action.
- H. Dispose of the milled material as indicated by the COR.

3.2 CLEAN AND REPAIR

- A. Remove and clean all millings from the surface daily.
 - 1. Control dust created by the cutting action.
 - 2. Clean rotomilled surface after milling operation and before opening to traffic.
- B. Verify all milled surfaces are sturdy and sound.
 - 1. Remove any material that is loose or not well bonded.
- C. Dispose of the milled material.

END OF SECTION

SECTION 02981

GRINDING PAVEMENT

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Grinding pavements.

1.2 RELATED SECTIONS

- A. Section 02701: Pavement Smoothness

1.3 REFERENCES **Not Used**

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS **Not Used**

PART 2 PRODUCTS

2.1 CONCRETE GRINDING EQUIPMENT

- A. Provide and operate equipment utilizing diamond blades mounted on a self-propelled machine designed for grinding and texturing pavement.
- B. Do not use equipment that causes damage to the transverse or longitudinal joints.

2.2 ASPHALT BUMP GRINDING EQUIPMENT

- A. Use one of the following pieces of equipment:
 - 1. Equipment utilizing diamond blades mounted on a self-propelled machine designed for grinding and texturing pavement.
 - 2. A mobile, self-contained machine utilizing diamond tipped teeth specifically designed to improve longitudinal roadway smoothness with the following characteristics:
 - a. Independent dual parallel averaging arms with a minimum length of 25 ft to hold and control the drum height
 - b. A minimum 8 ft wide cutting drum tapered on the outer edges to eliminate a vertical edge

- c. Capable of reading common profiler data in real time and compensating for the roughness identified.
- 3. Equipment utilizing a hydraulically driven asphalt bump grinding attachment with the following characteristics:
 - a. A rotatable housing with swivel casters capable of rotating right to left and forward and backward.
 - b. An adjustable averaging ski with a minimum width of 1 ft and a maximum length of 12 ft.
 - c. A manual device for setting depth.
 - d. An internal water system with the ability to flow water through the center of the drum.
 - e. A double hit cutting drum designed specifically for planing/shaving asphalt to close tolerances with a minimum of two wraps of flighting with a flat 0.75 inch carbide or diamond bit on each wrap with 0.05 inch overlap.
- B. Do not use micro milling, fine milling or standard milling machines.

2.3 VACUUMING EQUIPMENT

- A. Use vacuuming equipment to remove residue and excess water.

PART 3 EXECUTION

3.1 GRIND

- A. Provide a uniform finished texture.
- B. Perform grinding in a longitudinal direction.
 - 1. Begin and end grinding at lines normal to the pavement centerline.
- C. Do not cause damage to the underlying surface of the pavement.
- D. Provide uniform transverse slope of the pavement, to maintain cross slope drainage, with no depressions or misalignment of slope greater than 0.25 inch in 10 ft when tested with a 10 ft straightedge.
- E. Do not grind structures.
- F. All residue from the grinding process becomes property and responsibility of the Contractor.

3.2 CONCRETE GRINDING

- A. Grind until the surface of both sides of the transverse joints are in the same plane and smoothness requirements are met.
- B. Provide a uniform finished texture with resultant surface in a parallel, corduroy-type texture consisting of grooves between 0.090 and 0.150 inch wide.
 - 1. Create a distance between the grooves of between 0.060 and 0.13 inch.
 - 2. Make peaks of the ridges approximately $\frac{1}{16}$ inch higher than the bottom of the grooves

3.3 SMOOTHNESS

- A. Refer to Section 02701.

END OF SECTION

SECTION 02983

GRINDING FOR GROOVED-IN PAVEMENT MARKINGS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Pavement grinding for the installation of grooved-in pavement markings.

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. ASTM D 4285: Indicating Oil or Water in Compressed Air

1.4 DEFINITIONS

- A. Pavement Markings - Are both Pavement Messages and Pavement Striping.
- B. Pavement Messages – Markings generally placed perpendicular to and across the flow of traffic such as word, symbol, and arrow markings; stop, yield, and crosswalk lines; speed measurement markings; parking space markings; markings on shoulder such as chevron markings, and others.
- C. Pavement Striping – Lines placed parallel and adjacent to the flow of traffic such as lane lines, center lines, edge lines, channelizing lines and others.

1.5 SUBMITTALS

- A. Pavement marking manufacturer's recommendations for groove installation and cleaning for information.

PART 2 PRODUCTS

2.1 EQUIPMENT

- A. Provide a machine that:
 - 1. Is capable of grinding or cutting to the required depth and cross section.

2. Is capable of providing a surface profile that meets pavement marking manufacturer's recommendations.
 3. Has sufficient power, traction, and stability to maintain an accurate depth of cut.
- B. Walk-behind machines are not allowed for pavement striping.
- C. Use appropriate cleaning equipment capable of sweeping, blowing out and picking up millings during and after grinding operations.

PART 3 EXECUTION

3.1 PREPARATION

- A. Establish horizontal control line for grinding by placing control points as needed.

3.2 GRINDING

- A. Grind pavement for pavement striping and in pavement message areas to specified depth, width, and length.
1. Maintain 1 inch to 3 inch between the groove edge and the Portland Cement Concrete Pavement (PCCP) longitudinal joint or place the groove within 2 inch of the control line on Asphalt pavements, or as shown.
 2. Do not grind bridge decks and approach slabs for pavement markings.
- B. Groove Depth
1. Epoxy Paint
 - a. Minimum depth of 100 mils and a maximum depth of 120 mils.
 2. Waterborne Paint
 - a. Minimum depth of 60 mils and a maximum depth of 80 mils.
 3. Preformed Thermoplastic
 - a. All single layer preformed thermoplastic messages and markings: minimum depth of 110 mils and a maximum depth of 200 mils.
 4. Tape
 - a. Minimum depth of 160 mils and a maximum depth of 200 mils.

- C. Groove Width
 - 1. Grind at full groove depth a maximum of 1 inch wider than the width of the pavement striping.
 - 2. Grind at full groove depth a minimum of 1 inch beyond outer pavement message dimensions.
- D. Groove Length
 - 1. Grind at full groove depth a maximum of 2 inches longer than the length of the pavement striping.
 - 2. Grind at full groove depth a minimum of 1 inch beyond outer pavement message dimensions.

3.3 CLEANING

- A. Remove millings and clean the surface as work progresses.
 - 1. Milled material is the property of the Contractor.
 - 2. Use an air compressor capable of producing a minimum of 185 cfm and 120 psi to remove fine dust from the groove.
 - a. Use compressed air free of oil and water.
 - 1) Refer to ASTM D 4285.
 - 3. Follow any additional manufacturer's recommendations for groove cleaning.

END OF SECTION

SECTION 03055

PORTLAND CEMENT CONCRETE

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Portland Cement Concrete.

1.2 RELATED SECTIONS

- A. Section 03390: Concrete Curing

1.3 REFERENCES

- A. AASHTO M 6: Fine Aggregate for Hydraulic Cement Concrete
- B. AASHTO M 80: Coarse Aggregate for Hydraulic Cement Concrete
- C. AASHTO M 85: Portland Cement
- D. AASHTO M 154: Air-Entraining Admixtures for Concrete
- E. AASHTO M 157: Ready-Mixed Concrete
- F. AASHTO M 194: Chemical Admixtures for Concrete
- G. AASHTO M 295: Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
- H. AASHTO M 307: Silica Fume Used in Cementitious Mixtures
- I. AASHTO T 160: Length Change of Hardened Hydraulic Cement Mortar and Concrete
- J. AASHTO T 325: Estimating the Strength of Concrete in Transportation Construction by Maturity Tests
- K. AASHTO T 358: Surface Resistivity Indication of Concrete's Ability to Resist Chloride Ion Penetration
- L. ASTM C 150: Portland Cement

- M. ASTM C 595: Blended Hydraulic Cements
- N. ASTM C 1157: Hydraulic Cement
- O. ASTM C 1567: Determining the Potential Alkali-Silica Reactivity of Combinations of Cementitious Materials and Aggregate (Accelerated Mortar-Bar Method)
- P. ASTM C 1602: Mixing Water Used in the Production of Hydraulic Cement Concrete
- Q. American Concrete Institute (ACI) Manual of Concrete Practice
- R. UDOT Materials Manual of Instruction
- S. UDOT Minimum Sampling and Testing Requirements
- T. UDOT Quality Management Plan
- U. AASHTO T 121: Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete
- V. ASTM C 1116: Fiber-Reinforced Concrete
- W. ASTM C 1609: Flexural Performance of Fiber-Reinforced Concrete (Using Beam With Third-Point Loading)
- X. ICC Evaluation Service (ICC-ES) AC308: Concrete with Synthetic Fibers

1.4 DEFINITIONS

- A. Cold Weather Protection Period: The required time during which the concrete is maintained at or above a specific temperature to prevent freezing of the concrete and to provide the necessary strength development for structural safety.
- B. Fibrillated Microfiber – Fibrillated synthetic fibers with diameters or equivalent diameters less than 0.012 inch.
- C. Macrosynthetic Fiber – Synthetic fibers with diameters or equivalent diameters greater than 0.012 inch.

1.5 SUBMITTALS

- A. Mix design for all AAA, AA, and A concrete classes to be used for approval.
1. The Government approves concrete mix designs based on trial batch test results or on project history.
 2. Include at least the following:
 - a. The proposed mix design.
 - b. Target slump value.
 - c. Trial batch test results.
 - d. Test results verifying that coarse and fine aggregates meet this Section, Article 2.2, paragraph B.
 - e. Test results for the proposed mix design for potential reactivity of coarse and fine aggregates according to UDOT Quality Management Plan 506: Ready-Mix Concrete.
 - f. Test results demonstrating the ability of the combinations of cementitious materials and aggregates to control the reactivity when using potentially reactive aggregates in a mix design.
 - g. Written plan for admixtures. Refer to this Section, Article 2.2, paragraph D.
 - h. Well-graded combined aggregate gradation for the mix design when used.
 - 1) Provide targets for each required sieve (listed in Tables 5 and 6) for control and acceptance.
 - 2) Submit the coarseness factor, 0.45 power chart, percentage retained (8-18 gradation chart) or a combination of methodologies.
 - 3) Identify the aggregate size and number of component stockpiles.
 - 4) Provide gradations for each component stockpile and the target percentages of each stockpile used to achieve the total combined gradation.
 - i. Test results establishing a density (unit weight) target of freshly mixed concrete according to AASHTO T 121 when using Class AA(LSF) and AA(ES) concrete.
- B. Mix design, manufacturer's product data, or manufacturer's labeling for Class B concrete for approval.
- C. Cold Weather Concreting Plan and Hot Weather Concreting Plan for review.
1. Include the following:
 - a. Detailed procedures for the placement, protection, curing, and temperature monitoring of concrete during cold and hot weather.

- b. Procedures to be implemented upon abrupt changes in weather conditions or equipment failures.
 - c. Refer to this Section, Article 3.1, paragraph D for cold weather concreting requirements and Article 3.1, paragraph E for hot weather concreting requirements.
- 2. Allow the COR 14 calendar days to review the plans.
 - a. The Government may grant an increase in contract time when this review and approval time is exceeded.
 - b. This review period applies each time the plans are submitted.
- 3. Do not begin cold weather concreting before the Cold Weather Concreting Plan is approved.
- 4. Do not begin hot weather concreting before the Hot Weather Concreting Plan is approved.
- 5. Not required for precast concrete members provided by prequalified suppliers. Refer to this Section, Article 3.1, subparagraph D1 for cold weather. Refer to this Section, Article 3.1 paragraph E3 for hot weather.

1.6 ACCEPTANCE

- A. Acceptance for strength, air entrainment, and slump is according to UDOT Minimum Sampling and Testing Requirements.
 - 1. The following exceptions apply when using Class AA(LSF) and AA(ES) concrete mixes:
 - a. Slump tests are suspended.
 - b. Test fresh concrete density (unit weight) according to AASHTO T 121 at the same frequency as Air Content and Concrete Temperature.
 - 1) Batch fails if the unit weight of the fresh concrete in the field varies more than $\pm 5 \text{ lb/ft}^3$ from the target density for fresh concrete established by the mix design.
 - c. Reject batch if water/cementitious ratio exceeds the water/cementitious ratio established in the trial batch.
- B. The Department may accept the item at a reduced price when concrete is below specified strength and does not have a separate strength pay factor.
 - 1. The pay factor will be applied to the quantity of the pay item that is represented by the strength tests that fall below a specified strength.
 - 2. Government will calculate the pay factor using Table 1 based on 28 day compressive strength.

Table 1

Price Reduction for 28 Day Compressive Strength		
AAA(AE), AA(LSF), AA(LS), AA(ES), AA(P) Concrete Classes		AA(AE), A(AE) Concrete Classes
PSI below Specified Strength	Pay Factor	Pay Factor
1-100	0.95	1.0
101-200	0.90	0.95
201-300	0.85	0.85
301-400	0.80	0.80
More than 400	Reject	Reject

3. The COR may accept a “reject” lot based on an engineering analysis. The Government applies a 0.50 pay factor if a reject lot is allowed to remain in-place.

PART 2 PRODUCTS

2.1 CONCRETE CLASSES AND MIX REQUIREMENTS

- A. Use only concrete mixes that have a Department approved mix design.
1. Refer to the requirements in Table 2.

Table 2

Concrete Classes and Mix Requirements							
Class	Coarse Aggregate Size	Maximum Water / Cementitious Ratio	Maximum Percent Shrinkage at 28 days AASHTO T 160	Chloride Ion Penetration AASHTO T 358 Table 1	Air Content Percent (%) *	Mix Design Compressive Strength f'_{cr} (psi)	28 Day Minimum Compressive Strength f'_c (psi) **
AAA(AE)	1" to No. 4 ¾" to No. 4	0.40	N/A	N/A	5.0 - 7.5	6,200 or $f'_c + 1200$	5,000 or as shown
AA(LSF)	1" to No. 4 ¾" to No. 4	0.42	0.035	Low to Negligible	5.0 - 7.5	5,200	4,000
AA(LS)	1" to No. 4 ¾" to No. 4	0.40	0.035	Low to Negligible	5.0 - 7.5	5,200	4,000
AA(P)	2" to No. 4 1½" to No. 4 1" to No. 4	0.44	0.042	N/A	4.0 - 7.0 4.5 - 7.5 5.0 - 7.5	5,200	4,000
AA(ES)***	1½" to No. 4 1" to No. 4 ¾" to No. 4	0.42	0.035	Low to Negligible	4.5 - 7.5 5.0 - 7.5 5.0 - 7.5	5,200	4,000

Table 2 Continued							
Class	Coarse Aggregate Size	Maximum Water / Cementitious Ratio	Maximum Percent Shrinkage at 28 days AASHTO T 160	Chloride Ion Penetration AASHTO T 358 Table 1	Air Content Percent (%) *	Mix Design Compressive Strength f'_{cr} (psi)	28 Day Minimum Compressive Strength f'_c (psi) **
AA(AE)	2" to No. 4 1½" to No. 4 1" to No. 4 ¾" to No. 4	0.44	N/A	N/A	4.0 - 7.0 4.5 - 7.5 5.0 - 7.5 5.0 - 7.5	5,200	4,000
A	1½" to No. 4 1" to No. 4 ¾" to No. 4	0.53 0.53 0.48	N/A	N/A	N/A	3,900	3,000
A(AE)	1½" to No. 4 1" to No. 4 ¾" to No. 4	0.53 0.53 0.48	N/A	N/A	4.5 - 7.5	3,900	3,000
B or B(AE)		0.62	N/A	N/A	N/A 3.0 - 6.0	3,250	2,500

Table 2 Notes:

* Values listed represent in-place air content. Make necessary adjustments for impacts to air content due to placement.

** For f'_c over 4,000 psi, design and proportion mixes according to ACI Manual of Concrete Practice 301: Specifications for Concrete and project specific criteria. Use air content percent in Table 2 for these mixes according to the class specified and the coarse aggregate size.

*** For Class AA(ES), achieve at least 3,000 psi at 24 hr.

Acronym Definitions:

AE = air-entrained

P = pavement

LSF = low shrinkage with fiber

ES = early strength

LS = low shrinkage

- B. Maximum nominal size of coarse aggregate:
1. Not larger than $\frac{1}{5}$ the narrowest dimension between sides of forms.
 2. Not larger than $\frac{1}{3}$ the depth of slabs.
 3. Not larger than $\frac{3}{4}$ the minimum clear distance between reinforcing bars or between bars and forms, whichever is less.
- C. Do not exceed water/cementitious ratio.
1. Calculate the water/cementitious ratio (w/c) by weight according to the following formula:

$$w = \frac{\text{Water}}{\text{Cement} + \text{Pozzolan}} c$$

- D. Do not exceed 30 percent total pozzolan in any mix unless approved or otherwise specified.
- E. Use 94 lb additional cementitious material per cubic yard to the amounts determined in the mix design for concrete deposited in water.
- F. Slump tolerance
 - 1. Establish the target slump by mix design trial batch.
 - 2. The target slump tolerance is the acceptable variation from the maximum target slump.
 - 3. Do not exceed a 9 inch slump.

Table 3

Target Slump Tolerance (inch)		
	Target Slump	
	3 inch or less	More than 3 inch
Plus tolerance	0	0
Minus tolerance	1 ½ inch	2 ½ inch

- G. Class AA(LSF) and AA(ES) concrete mixes require the following in addition to the requirements in Table 2:
 - 1. Synthetic Fiber Reinforcement according to this Section, paragraph 2.2F.
 - a. Provide fibrillated microfibers and macrosynthetic fibers
 - 2. A well-graded combined aggregate gradation according to this Section, paragraph 2.2 B3.

2.2 MATERIALS

- A. Cement
 - 1. Use Type II Portland Cement or equivalent according to Table 4 unless otherwise specified. Type III Portland Cement or equivalent may be used for precast items.
 - 2. Blended Hydraulic Cement
 - a. Blended hydraulic cement substituted for Portland Cement:
 - 1) Use ASTM C 1567 to verify that expansion is less than 0.1 percent 14 days after the zero reading.
 - 2) Refer to the equivalent cements listed in Table 4.
 - b. Do not exceed 30 percent total pozzolan limit when adding fly ash to a blended hydraulic cement.
 - 1) Submit documentation of the total pozzolan content with the mix design.

Table 4

Portland Cement/Blended Hydraulic Cement Equivalencies		
AASHTO M 85 Equivalent Alkalies 0.80 max percent	ASTM C 595	ASTM C 1157
*Type I	IP, IL, IT	GU
Type II	IP(MS), IT(MS)	MS
Type III	-	HE
*Type V	IP(HS), IT(HS)	HS

*Use only when specified

3. Do not mix cements originating from different sources.
4. Do not use air-entrained cement.

B. Aggregate

1. Coarse Aggregate
 - a. Use coarse aggregate that meets AASHTO M 80 physical properties. Use one of the gradations in Table 5.
 - b. Do not exceed percentages of deleterious substances as specified in AASHTO M 80, Table 2, for Class A aggregates.

Table 5

Coarse Aggregate Gradations - Percent Passing (by weight)									
Aggregate Size (inches or sieve size)	2½	2	1½	1	¾	½	⅜	No. 4	No. 200
2 to No. 4	100	95-100		35-70		10-30		0-5	0-1
1½ to No. 4		100	95-100		35-70		10-30	0-5	0-1
1 to No. 4			100	95-100		25-60		0-10	0-1
¾ to No. 4				100	90-100		20-55	0-10	0-1

2. Fine Aggregate
 - a. Use fine aggregate that meets AASHTO M 6 physical properties. Use the gradation in Table 6.
 - b. Do not exceed percentages of deleterious substances as specified in AASHTO M 6, Table 2, for class A aggregates, using option "b" for material finer than the No. 200 sieve.

Table 6

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Fine Aggregate Gradation	
Sieve Size	Percent Passing (by weight)
$\frac{3}{8}$ inch	100
No. 4	95 to 100
Table 6 Continued	
Sieve Size	Percent Passing (by weight)
No. 16	45 to 80
No. 50	10 to 30
No. 100	2 to 10
No. 200	0 to 3.0

3. A well-graded combined aggregate gradation may replace the gradation requirements in Tables 5 and 6.
 - a. Proportion combined aggregates using any combination of the 0.45 power chart, the 8-18 percent-retained and the Coarseness Factor charts in the UDOT Materials Manual of Instruction, Section 975: Guidelines for Well-Graded Combined Aggregate Gradations.
 - 1) Determine a combined gradation for the mix design. Provide targets for each sieve size ($\frac{3}{4}$ ", $\frac{1}{2}$ ", $\frac{3}{8}$ ", #4, #8, #16, #30, #50, #100)
 - 2) Provide a combined gradation within the tolerances of Table 7 and with less than 3 percent passing the #200 sieve.
 - 3) Maintain gradations within zone II of the coarseness factor chart.

Table 7

Tolerances for a Well Graded Combined Aggregate Gradation	
Sieve Size	Percent Passing Allowable Tolerance
$\frac{3}{4}$ ", $\frac{1}{2}$ ", $\frac{3}{8}$ "	± 10%
#4, #8	± 5%
#16, #30, #50	± 4%
#100	± 3%

- C. Water
 1. Use potable water or water that meets ASTM C 1602, including Table 2.
- D. Admixtures
 1. Do not use calcium chloride.
 2. Air Entrainment according to AASHTO M 154, including Section 5.
 3. Water Reducing Agents

- a. Refer to AASHTO M 194.
 - b. High Range Water Reducer (HRWR) – Submit a written plan for approval with the trial batch that details ingredients, production methods, handling, and placing.
 - 4. Accelerators – Refer to AASHTO M 194.
 - 5. Set Retarding and Hydration Stabilizing Admixtures – Refer to AASHTO M 194.
 - a. Establish and inform the COR of the effective life of the set-retarding or stabilizing admixture by trial batch if admixtures are required due to haul times exceeding the time limitations in this Section, Article 3.1, paragraph A.
 - b. Do not exceed manufacturer's recommendations for the use of the set retarding admixture.
 - c. Do not re-dose the concrete with additional set retarding admixture.
 - d. Add admixture at the batch plant at the time of initial batching operations.
 - e. Show on batch tickets the amount of admixture used.
 - f. Time of placement is established by the trial batch and supersedes the requirements in this Section, Article 3.1, paragraph A.
 - 6. Site-added air-entrainment – Refer to AASHTO M 154.
 - a. Limit the use of site-added air-entraining agents to one addition per load, regardless of quantity.
 - b. Use pre-measured admixtures.
 - c. Record amount used on batch ticket.
 - d. Rotate the drum at least 30 revolutions at the mixing speed recommended by the manufacturer.
- E. Pozzolan
- 1. Fly Ash
 - a. Class F according to AASHTO M 295 except Table 2.
 - 1) Loss on Ignition (LOI) Not to exceed 3 percent.
 - 2) Allowable CaO content Not to exceed 15 percent.
 - 3) Label the storage silo for fly ash to distinguish it from cement.
 - 4) Use different size unloading hoses and fittings for cement and fly ash.
 - 2. Natural Pozzolan (Class N)
 - a. Refer to AASHTO M 295.
 - b. May use instead of fly ash provided that the expansion does not exceed 0.1 percent. Refer to ASTM C 1567.
 - 3. Silica Fume
 - a. Refer to AASHTO M 307.

- F. Synthetic Fiber Reinforcement
1. Fibrillated microfiber
 - a. Use fibrillated polypropylene fibers at 2 lb/yd³ of concrete mix.
 2. Macrosynthetic fiber
 - a. Use 4 lb/yd³ of concrete mix.
 - b. Provide a minimum flexural strength ratio ($R_{e,3}$) of 25 percent when tested according to ASTM C 1609.
 3. Store the fibers in a dry, covered area, free of contamination.
 4. Evaluate trial batches to verify workability of the concrete.
 5. Conform to ASTM C 1116, Type III and the requirements of ICC-ES AC32 Section 3.1.1 (plastic shrinkage reinforcement) and Section 3.1.2 (shrinkage and temperature reinforcement).
 6. Do not introduce fibers at the same time as the cement is being introduced.
 7. Mix for at least five minutes after the addition of the fibers.

2.3 MIX DESIGN

- A. Design mixes to meet the requirements of this Section and project specific criteria.
- B. The Contractor assumes responsibility for the compatibility of admixtures with the mix design and their potential effects on concrete properties.
- C. Design the cementitious system to mitigate potential alkali-aggregate reactivity.
 1. Use at least 20 percent pozzolan by weight of the total cementitious system.
- D. Obtain approval from the COR for the project specific application of an approved mix design.

2.4 TRIAL BATCHES

- A. Use the same components in the trial batches that will be used in the project.
 1. Accelerators and site-added air-entrainment can be incorporated in the trial batch but are not required.
- B. Use UDOT certified TTQP Concrete and Concrete Strength Testing personnel to perform trial batches and strength tests.
- C. The Government or its certified representative may witness the trial batch.

- D. Mix concrete trial batches according to the UDOT Materials Manual of Instruction 974: Guidelines for Portland Cement Concrete Mix Design Trial Batches.
- E. Use a UDOT qualified laboratory to verify trial batch compressive and flexural strength testing.

2.5 AGGREGATE STOCKPILES

- A. Construct stockpile platforms so that subgrades are prevented from intruding into aggregates.
- B. Build stockpiles at least two days before use.
- C. Provide separate stockpiles for coarse and fine aggregates.
- D. Construct stockpiles to minimize segregation of aggregates
- E. Allow washed aggregates to drain to uniform moisture content before use (12 hours minimum).

2.6 BATCH MATERIALS

- A. Batch Tolerances. Refer to AASHTO M 157.
 - 1. Cementitious Material : ± 1 percent of the required mass
 - 2. Aggregate: ± 2 percent of the required mass
 - 3. Total Water: ± 3 percent of the required mix amount
- B. Truck-Mixed Concrete (Dry-Batch)
 - 1. Do not load trucks in excess of their rated mixing capacity, 63 percent of the drum gross volume, or less than 2 yd³.
 - 2. The truck rating plate must be readable.

PART 3 EXECUTION

3.1 LIMITATIONS

- A. Timing – Deliver, place, and consolidate concrete as follows unless otherwise specified:
 - 1. Within 90 minutes of batching when the air temperature is below 80 degrees F.
 - 2. Within 75 minutes of batching when the air temperature is between 80 and 85 degrees F.
 - 3. Within 60 minutes of batching when the air temperature is

above 85 degrees F.

- B. Concrete Temperature – Place concrete when the concrete temperature is between 50 and 90 degrees F unless otherwise specified.
- C. Pumping and Conveying Equipment
 - 1. Do not use equipment or a combination of equipment and the configuration of that equipment that causes a loss of entrained air content that exceeds one-half of the range of air content allowed by specification.
 - a. Replace, reconfigure, or repair equipment that does not meet this requirement.
 - 2. Contractor is responsible to verify and monitor air loss.
- D. Cold Weather – Comply with the following when placing, finishing, curing, and protecting concrete exposed to cold weather during the protection period. Cold weather applies when the temperature is forecast to fall below 35 degrees F during the protection period.
 - 1. Provide necessary cold weather protection for placing, finishing, curing and protecting in-place concrete such as covers, insulation, and heat.
 - a. Follow the authorized Cold Weather Concreting Plan when placing cast-in-place concrete.
 - b. Follow the prequalified supplier's approved Quality Control Plan when fabricating precast concrete members.
 - 2. Concrete materials
 - a. Do not use chemical anti-freeze additives in the concrete. This does not apply to normal accelerators. Refer to AASHTO M 194.
 - b. Remove and replace concrete damaged by frost action at no additional cost to the Department.
 - c. Do not use material containing frost or lumps.
 - 3. Determine the concrete compressive strength by one of the following methods:
 - a. Field cured cylinders cured and protected the same as the concrete being protected.
 - b. Maturity method. Refer to AASHTO T 325.
 - 4. Maintain the temperature of the concrete at or above 50 degrees F during and after placement until the end of the protection period.
 - a. Measure the specified concrete temperature at the concrete surface. Use surface thermometers insulated from the surrounding air.

- 5. Placing concrete
 - a. Do not place concrete during adverse weather including rain, snow, and high winds without adequate protection approved by the ~~Engineer~~COR.
 - b. Do not proceed with the placement of concrete if the temperature of all contact surfaces, including reinforcement, is less than 36 degrees F or greater than 95 degrees F.
 - c. Cease placement operations when the ambient temperature is 40 degrees F and decreasing unless adequate precautions are taken according to the approved Cold Weather Concreting Plan.
- 6. Protection of in-place concrete
 - a. Maintain the concrete above 50 degrees F during placement and until the end of the protection period.
 - 1) The protection period is the time required for the concrete to reach a compressive strength of at least 3,500 psi.
 - 2) Extend the duration of the protection period at least 24 hr beyond the termination of the cure before exposing the concrete to freezing temperatures when curing by the water method. Refer to Section 03390.
 - b. Comply with the following when heating is required.
 - 1) Adequately vent combustion-type heaters that produce carbon monoxide.
 - 2) Position heaters and ducts so the hot dry air does not cause areas of the concrete surface to overheat or dry.
 - 3) Keep concrete surfaces moist to avoid excessive loss of moisture from the concrete when applying external heat.
- 7. Termination of protection
 - a. Limit the drop in temperature of concrete surfaces to 40 degrees F during any 24 hour period when removing cold weather protection until the surface temperature of the concrete reaches that of the ambient air temperature.
- E. Hot Weather – Comply with the following when placing, finishing, curing, and protecting concrete exposed to hot weather during the protection period.
 - 1. Hot weather limitations apply at any time of the year when a combination of high ambient temperature, high concrete temperature, low relative humidity, and high wind speed have the potential to impair the quality of freshly mixed or hardened

- concrete by accelerating the rate of moisture loss and the rate of cement hydration, or otherwise causes detrimental results.
2. Monitor site conditions, including air temperature, relative humidity, and wind speed, to assess the need for evaporation control measures.
 - a. Begin monitoring no later than 1 hour before beginning concrete placing operations.
 - b. Continue to monitor site conditions at intervals of 20 minutes or less until required curing procedures are applied.
 3. Provide necessary hot weather protection.
 - a. Follow the approved Hot Weather Concreting Plan when placing cast-in-place concrete.
 - b. Follow the prequalified supplier's approved Quality Control Plan when fabricating precast concrete members.
 - c. Initiate evaporation control measures when concrete and air temperatures, relative humidity of the air, and wind speed have the capacity to evaporate free water from the fresh concrete surface at a rate equal to or greater than 0.2 lb/ft²/hr.
 - 1) Determine the evaporation rate of surface moisture using the NRMCA Nomograph in Appendix B of ACI 305.1.
 4. Cool all surfaces that will come in contact with the concrete to below 95 degrees F

3.2 CYLINDER STORAGE DEVICE

- A. Provide and maintain cylinder storage device.
 1. Maintain cylinders at a temperature range of 60 degrees F to 80 degrees F for the initial 16 hour curing period.
 2. Do not move the cylinders during this period.
 3. Equip the storage device with an automatic 24 hour temperature recorder that continuously records on a time/temperature chart with an accuracy of ± 1 degree F.
 4. Have the storage device available at the point of placement at least 24 hours before placement.
 5. Stop placement of concrete if the storage device is not provided or cannot accommodate the required number of test cylinders. Cylinder strength results may not be disputed if storage devices are not provided.
 6. Use water containing hydrated lime if water is to be in contact with cylinders.
 7. The COR may require a 24 hour test run to determine the

storage device capability to maintain and record temperature.

END OF SECTION

SECTION 03056

SELF-CONSOLIDATING CONCRETE (SCC)

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Self-consolidating concrete.

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete

1.3 REFERENCES

- A. AASHTO M 6: Fine Aggregate for Portland Cement Concrete
- B. AASHTO M 80: Coarse Aggregate for Portland Cement Concrete
- C. AASHTO T 347: Slump Flow of Self-Consolidating Concrete (SCC)
- D. AASHTO T 345: Passing Ability of Self-Consolidating Concrete (SCC) by J-Ring
- E. AASHTO T 351: Visual Stability Index (VSI) of Self-Consolidating Concrete (SCC)
- F. ASTM C 494: Chemical Admixtures for Concrete
- G. ASTM C 1602: Mixing Water Used in the Production of Hydraulic Cement Concrete
- H. ASTM C 1610: Static Segregation of Self-Consolidating Concrete Using Column Technique
- I. ASTM C 1712: Rapid Assessment of Static Segregation Resistance of Self-Consolidating Concrete Using Penetration Test
- J. American Concrete Institute (ACI) Standards
- K. UDOT Materials Manual of Instruction

- L. UDOT Minimum Sampling and Testing Requirements
- M. UDOT Quality Management Plans

1.4 DEFINITIONS

- A. Self-Consolidating Concrete (SCC) – A highly flowable non-segregating concrete mix that spreads into place, and is able to flow and fill all corners of the formwork, even in the presence of congested reinforcement by means of its own mass with no mechanical vibration.

PART 2 PRODUCTS

1.1 CONCRETE CLASSES AND MIX REQUIREMENTS

- A. Refer to Section 03055, except as modified in this Section.
- B. Slump Flow – 20 to 30 inches. Refer to AASHTO T 347.
- C. Rapid Assessment of Static Segregation less than or equal to 15 mm. Refer to ASTM C 1712.
- D. Static Segregation – less than 10 percent. Refer to ASTM C 1610.
- E. Passing Ability of Self-Consolidating Concrete (SCC) by J-Ring
Refer to AASHTO T 345
- F. Do not exceed water/cementitious ratio of 0.40.

1.2 MATERIALS

- A. General
 - 1. Refer to Section 03055, except as modified in this Section.
- B. Aggregate
 - 1. Use Coarse Aggregate according to AASHTO M 80 physical properties and the combined aggregate gradation requirements of Table 1.
 - a. Do not exceed percentages of deleterious substances as shown in AASHTO M 80, Table 2, for Class A aggregates.

2. Use Fine Aggregate according to AASHTO M 6 physical properties and the combined aggregate gradation requirements of Table 1.
 - a. Do not exceed percentages of deleterious substances according to AASHTO M 6 for class A aggregates, using class B for material finer than the No. 200 sieve.

Table 1

Combined Aggregate Gradations Percent Passing by Dry Weight of Aggregate		
Sieve Size	$\frac{3}{4}$ inch Operating Bands	$\frac{1}{2}$ inch Operating Bands
$\frac{3}{4}$ inch	95 – 100	–
$\frac{1}{2}$ inch	65 – 95	95 – 100
$\frac{3}{8}$ inch	58 – 83	65 – 95
No. 4	35 – 65	50 – 80
No. 8	25 – 50	30 – 60
No. 16	15 – 35	20 – 45
No. 30	10 – 35	12 – 35
No. 50	5 – 20	5 – 20
No. 100	1 – 12	2 – 12
No. 200	0 – 5	0 – 5

- C. Water
 1. Use potable water or water according to ASTM C 1602, including Table 2.
- D. Admixtures
 1. Viscosity Modifying Admixtures (VMA) according to ASTM C 494, Type S.
 - a. Do not exceed manufacturer's recommendations for the use of the viscosity modifying admixture.
 - b. Show the amount of admixture used on batch tickets.
 - c. Do not allow site-added VMA

PART 3 EXECUTION

1.1 GENERAL

- A. Refer to Section 03055

END OF SECTION

SECTION 03058

RAPID STRENGTH CONCRETE (RSC)

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Rapid Strength Concrete (RSC)

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete

1.3 REFERENCES

- A. AASHTO M 154: Air-Entraining Admixtures for Concrete
- B. AASHTO M 157: Ready-Mixed Concrete
- C. AASHTO T 22: Compressive Strength of Cylindrical Concrete Specimens
- D. AASHTO T 97: Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading)
- E. ASTM C 157: Length Change of Hardened Hydraulic-Cement Mortar and Concrete
- F. ASTM C 309: Liquid Membrane-Forming Compounds for Curing Concrete
- G. ASTM C 1074: Estimating Concrete Strength by the Maturity Method
- H. ASTM C 1600: Rapid Hardening Hydraulic Cement
- I. ASTM C 1602: Mixing Water Used in the Production of Hydraulic Cement Concrete
- J. UDOT Materials Manual of Instruction
- K. UDOT Minimum Sampling and Testing Requirements

1.4 DEFINITIONS

- A. Cold Weather Protection Period – The required time during which the concrete is maintained at or above a specific temperature to prevent freezing of the concrete and to provide the necessary strength development for structural safety.
- B. Production Day – A 24 hour period in which RSC is being placed

1.5 SUBMITTALS

- A. Mix Design to be used for approval
 - 1. The Department approves concrete mix designs based on trial batch test results.
 - 2. Include at least the following:
 - a. The proposed mix design using the same format and units used by the volumetric mixer printout.
 - b. Trial batch test results.
 - c. Test results verifying that coarse and fine aggregates meet Section 03055, Article 2.2, paragraph B.
 - d. Written plan for admixtures
 - 3. Submit a new mix design for approval, whenever the source of any mix component changes.
 - 4. Mix designs are valid for one year from date of approval.
- B. Cold Weather Concreting Plan and Hot Weather Concreting Plan for review.
 - 1. Include the following:
 - a. Detailed procedures for monitoring and maintaining temperatures of all mix components including water, aggregates, and all other admixtures.
 - b. Detailed procedures for the mixing, placement, protection, curing, and temperature monitoring of concrete
 - c. Procedures to be implemented upon abrupt changes in weather conditions or equipment failures.
 - d. Refer to this Section, Article 3.3, paragraph D for cold weather concreting requirements and Article 3.3, paragraph E for hot weather concreting requirements.
- C. Volumetric mixer daily production data in electronic or printed media at the end of each production day, including the following:
 - 1. Weight of cement per revolution count
 - 2. Weight of each aggregate size per revolution count
 - 3. Gate openings for each aggregate size.
 - 4. Weight of water added to the concrete per revolution count
 - 5. Moisture content of each aggregate size

Rapid Strength Concrete (RSC)

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6. Individual volume of admixtures per revolution count
7. Time of Day
8. Day of week
9. Production start and stop times
10. Volumetric mixer identification
11. Specific mix design of concrete being produced
12. Source of the individual aggregate sizes
13. Source, brand, and type of cement
14. Source, brand, and type of individual admixtures
15. Name and signature of the operator

1.6 ACCEPTANCE

- A. General acceptance will be based on a comparison of the approved mix design and the printed ticket.
 1. Verify correct amount of air entraining admixture is added and adjust mix proportions as necessary
 2. Verify water/cement ratio meets the requirements of Table 1 and adjust mix proportions as necessary
- B. Acceptance sampling and testing of material is according to UDOT Minimum Sampling and Testing Requirements
- C. The Government rejects the portion of a lot represented by a strength test that is more than 500 psi below the 28 Day Minimum Compressive Strength $f'c$ (psi) in Table 1.
 1. The COR may accept reject material based on the materials dispute resolution process.
 - a. The Government applies a 0.50 pay factor to the pay item for the quantity represented if reject material is allowed to remain in-place.
 2. The Government will monitor the strength test results and may unapprove the mix design based on low strength test results.

2

PART 2 PRODUCTS

2.1 CONCRETE MIX REQUIREMENTS

- A. Use only concrete mixes that have a UDOT approved mix design.
 1. Refer to the requirements in Table 1.

Table 1

Property	Test method	Requirements
Air Entrainment	AASHTO M 154 including Section 5	5.0%- 7.5%

3 Day Flexural Strength	AASHTO T 97	550 psi minimum
28 Day Flexural Strength	AASHTO T 97	650 psi minimum
4 hour Compressive strength	AASHTO T 22	4000 psi minimum
Mix Design Compressive Strength f'_{cr} (psi)	AASHTO T 22	5200 psi minimum
28 Day Compressive Strength f'_c (psi)	AASHTO T 22	4000 minimum
4 hour Drying shrinkage	ASTM C 157	0.04 maximum
28 Day Drying shrinkage	ASTM C 157	0.04 maximum
Water/Cement Ratio by weight		0.38 - 0.44

2.2 MATERIALS

- A. Cement
 - 1. Use RSC meeting the requirements of ASTM C1600.
 - 2. Do not use pozzolans.
- B. Aggregate
 - 1. Meet the requirements of Section 03055, Article 2.2, paragraph B
 - a. Use a $\frac{3}{4}$ inch coarse aggregate gradation or no course aggregate.
- C. Water
 - 1. Use potable water or water that meets ASTM C 1602, including Table 2.
- D. Admixtures
 - 1. Meet the requirements of Section 03055, Article 2.2, paragraph D.
- E. Curing Compound
 - 1. Refer to ASTM C 309, Type 2, Class B.

2.3 MIX DESIGN

- A. Design mixes to meet the requirements of this Section and project specific criteria.
- B. The Contractor assumes responsibility for the compatibility of admixtures with the mix design and their potential effects on concrete properties.
- C. Obtain approval from the COR for the project specific application of an approved mix design.

2.4 TRIAL BATCHES

- A. Perform 1.5 hour, 2 hour, 2.5 hour, 3 hour, 3.5 hour, 4 hour, 6 hour, 24 hour and 28 day compressive strength tests – Refer to AASHTO T 22
 - 1. Develop maturity meter curves, Nurse-Saul method, for use in opening to traffic – Refer to ASTM C 1074.
- B. Perform 4 hour, 24 hour, 7 day and 28 day shrinkage tests – Refer to ASTM C 157
- C. Use the same components in the trial batches to be used in production.
- D. Use Department Certified TTQP Concrete and Concrete Strength Testing personnel to perform trial batches and strength tests.
- E. The Department or its representative may witness the trial batch.
- F. Use a Department qualified laboratory to verify trial batch compressive and flexural strength testing.
- G. Mix concrete trial batches using the volumetric mixer truck to be used in production if a volumetric truck is used in production; otherwise, mix concrete trial batches according to the UDOT Materials Manual of Instruction 974; Guidelines for Portland Cement Concrete Mix Design Trial Batches.

2.5 EQUIPMENT

- A. Volumetric Mixer Truck
 - 1. The volumetric mixer truck is required to have a current UDOT certification and calibrated annually.

- B. Maturity Meters
 - 1. Produce computer generated printout of the data logger.
 - 2. Provide sensor probes.

PART 3 EXECUTION

3.1 CONCRETE MIXING

- A. Volumetric truck mixing
 - 1. Aggregates must be dried and drained to a stable moisture content prior to loading.
 - 2. Free moisture content of the fine aggregate must not exceed 8 percent of its saturated surface-dry weight.
 - 3. Do not use ice to cool directly. Ice must be melted before entering the mixer if ice is used to cool the mix water.
 - 4. Proportion and charge the cement into the volumetric mixer such that there is no variance of the required quantity due to conditions such as wind or accumulation of equipment.
 - 5. Do not mix more material in the volumetric mixer than will allow complete mixing.
 - 6. Do not add water after discharge.
 - 7. Do not apply finishing aids to the concrete surface.
- B. Batch plant mixing
 - 1. Batch Tolerances. Refer to AASHTO M 157.
 - a. Cementitious Material: ± 1 percent of the required mass.
 - b. Aggregate: ± 2 percent of the required mass.
 - c. Total Water: ± 3 percent of the required mix amount.
 - 2. Limit loads to 6.5 cu yd.
- C. Truck Washout
 - 1. Provide, at jobsite, a washout bin of sufficient size to contain and properly dispose of any wash water and associated concrete residue.

3.2 CURING

- A. Apply concrete curing compound to the entire concrete surface immediately after surface finishing of individual placements.
 - 1. Apply at an approximate application rate of 200 ft²/gal.
 - 2. Prevent damage to the pavement surface texture.
- B. Follow any additional cement manufacturer curing requirements.

- C. Refer to Section 03055 for cylinder storage requirements.
 - 1. Do not move concrete test cylinders from placement area for at least 4 hours.
 - 2. Remove concrete test cylinders from job site immediately before opening for traffic or at 4 hours, whichever is later.

3.3 PRE-ACTIVITY MEETING

- A. Conduct pre-activity meeting with Department representative, contractor, and RSC supplier before any concrete placement and discuss the following:
 - 1. Concrete curing.
 - 2. Concrete protection.
 - 3. Washout bin staging.
 - 4. Handling of concrete cylinders.
 - 5. Placement of maturity meters.
 - 6. Opening to traffic.

3.4 LIMITATIONS

- A. Do not open to traffic until material reaches a compressive strength of 3000 psi based on maturity meter.
 - 1. Place the maturity meter in the last placement of the production day.
 - 2. Concrete compressive strength of 3500 psi is required in cold weather.
- B. Concrete Temperature – Place concrete when the concrete temperature is between 50 and 80 degrees F, taken immediately after discharge of chute, unless otherwise specified.
- C. Cold Weather – Comply with the following when the ambient temperature is forecast to drop below 40 degrees F before the concrete reaches a compressive strength of 3500 psi as determined by the maturity method.
 - 1. Follow the authorized cold weather plan.
 - 2. Provide necessary cold weather protection for mixing, placing, finishing, curing, and protecting in-place concrete.
 - 3. Blanket concrete immediately after curing compound is applied.
 - 4. Do not use chemical anti-freeze additives in the concrete.
 - 5. Maintain the temperature of the concrete above 50 degrees F during mixing, placing, and curing.
 - 6. Do not proceed with placement of concrete if the temperature of all contact surfaces, including reinforcement, is less than 36 degrees F.

- D. Hot Weather – Comply with the following when mixing, placing, finishing, curing, and protecting concrete exposed to hot weather until the concrete reaches a compressive strength of 3000 psi as determined by the maturity method.
1. Hot weather limitations apply at any time of the year when a combination of high ambient temperature, high concrete temperature, low relative humidity, and high wind speed have the potential to impair the quality of the concrete by accelerating the rate of moisture loss and rate of cement hydration, or otherwise causes detrimental results.
 2. Follow the authorized hot weather plan.
 3. Do not proceed with the placement when the temperatures of all contact surfaces, including reinforcement, are greater than 95 degrees F.
 4. Cool all surfaces that will come in contact with the concrete to below 95 degrees F.

END OF SECTION

SECTION 03152

CONCRETE JOINT CONTROL

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Joint Filler and Joint Sealer
- B. Backer Rods and Waterstops

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. AASHTO M 153: Preformed Sponge Rubber and Cork Expansion Joint Fillers for Concrete Paving and Structural Construction
- B. AASHTO M 213: Preformed Expansion Joint Fillers for Concrete Paving and Structural Construction (Non-extruding and Resilient Bituminous Types)
- C. AASHTO LRFD Bridge Construction Specifications
- D. ASTM C 309: Liquid Membrane-Forming Compounds for Curing Concrete
- E. ASTM C 509: Elastomeric Cellular Preformed Gasket and Sealing Material
- F. ASTM C 639: Rheological (Flow) Properties of Elastomeric Sealants
- G. ASTM C 661: Indentation Hardness of Elastomeric-Type Sealants by Means of a Durometer
- H. ASTM C 679: Tack-Free Time of Elastomeric Sealants
- I. ASTM C 719: Adhesion and Cohesion of Elastomeric Joint Sealants Under Cyclic Movement (Hockman Cycle)
- J. ASTM C 793: Effects of Laboratory Accelerated Weathering on Elastomeric Joint Sealants
- K. ASTM C 794: Adhesion-in-Peel of Elastomeric Joint Sealants

- L. ASTM C 920: Elastomeric Joint Sealants
- M. ASTM D 412: Vulcanized Rubber and Thermoplastic Rubbers and Thermoplastic Elastomers-Tension
- N. ASTM D 792: Density and Specific Gravity (Relative Density) of Plastics by Displacement
- O. ASTM D 1621: Compressive Properties of Rigid Cellular Plastics
- P. ASTM D 1622: Apparent Density of Rigid Cellular Plastics
- Q. ASTM D 1623: Tensile and Tensile Adhesion Properties of Rigid Cellular Plastics
- R. ASTM D 6690: Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements
- S. Federal Specifications
- T. Military Specifications (MIL)

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Manufacturer's product data sheets and recommended installation instructions.

PART 2 PRODUCTS

2.1 HOT POURED JOINT SEALANT

- A. Refer to ASTM D 6690, Type II for general requirements, physical properties, packing, marking, and sampling.

2.2 PREMOLDED JOINT FILLERS

- A. Refer to AASHTO M 153
- B. Refer to AASHTO M 213.

2.3 SILICONE JOINT SEALER

- A. Silicone joint sealer and special category for self-leveling.
 - 1. Both made of low-modulus silicone specifically formulated to seal portland cement concrete pavement joints.
- B. Furnish in a one-part, non-acid curing formulation. Refer to ASTM C 309.
- C. Meet the requirements in Table 1 for Silicone joint sealer and Table 2 for Self-leveling silicone joint sealer.

Table 1

Requirements (Silicone Joint Sealer)		
Description	Requirement	Test Method
Tensile Stress –150 percent max elongation, 7-day cure at 77 ± 3 degrees F and 45-55 percent relative humidity (rh).	45 psi maximum	ASTM D 412 (DIE C)
Flow	0.3 inch maximum	MIL-S-8802
Extrusion Rate 100 degrees to 0 degrees F	0.2 - 0.6 lbs/min	MIL-S-8802
Tack-Free Time	20 - 75 minutes	MIL-S-8802
Specific Gravity	1.01 - 1.51	ASTM D 792
Durometer Hardness, shore A – cured 7 days at 77 ± 3 degrees F and 45-55 percent relative humidity (rh).	10 - 25	ASTM C 661
Shelf Life	6 month minimum from date of shipment from plant or point of manufacture.	
Ozone and Ultraviolet (UV) Resistance	No chalking, cracking, or bond loss.	ASTM C 793
Movement capability and adhesion. Magnitude of cycles movement is appropriate for sealant category, cure 7 days in air 77 ± 3 degrees F then 7 days in water 77 ± 3 degrees F.	No Failures	ASTM C 719

Table 2

Requirements for Self-Leveling (Silicone Joint Sealer)		
Description	Requirement	Test Method
Rheological Properties – Type I	Pass	ASTM C 639
Elongation, percent minimum	800 %	ASTM D 412 DIE C, Mod.
Tensile Stress at 150 percent Elongation	40 psi maximum	ASTM D412
Accelerated Weathering	No chalking, cracking, or bond loss	ASTM C793
Shelf Life	Six month min. from date of shipment from plant or point of manufacture.	
Durometer Hardness, Shore OO – cured 14 days at 77 ± 3 degrees F and 45 - 55 percent, rh	40-80	ASTM C661
Movement capability and adhesion. Magnitude of cycles movement is appropriate for sealant category, cure 14 days in air 77 ±3 degrees F, and then 7 days in water 77 ±3 degrees F	No failures	ASTM C719

- D. Sealant must be delivered in the manufacturer's original sealed container, displaying the lot number, expiration date of the shelf-life warranty, and the sealer trade name.

2.4 BACKER ROD

- A. Use closed-cell, polyethylene-foam rods that meet the requirements in Table 3.

Table 3

Backer Rod Requirements and Test Methods		
Description	Requirement	Test Method
Diameter	Joint width + 1/8 inch	
Density	2 lbs/ft ³	ASTM D 1622
Tensile Strength	25 psi	ASTM D 1623
Absorption	0.5 % by volume	ASTM C 509
Compression	25 percent at 8 psi	ASTM D 1621
Deflection		ASTM D 1621

2.5 JOINT SEALER (STRUCTURES)

- A. Cold-applied, gun-grade, polyurethane base material that cures under field conditions to form a rubber-like, non-sag, elastomeric joint seal as specified in Federal Specifications TT-S-00230 C, Type II, Class A and ASTM C 920, Type S or M, Grade NS, Class 35 or higher, with at least the following Use: T, M, I.
- B. Use material that bonds tightly to the sides of the concrete groove and exhibits the physical properties in Table 4 when cured and tested after 21 days at 73 degrees F.

Table 4

Physical Properties of Joint Sealer (Structures) and Test Methods		
Description	Requirement	Test Method
Modulus of Elasticity at 100 percent Elongation	65 psi minimum	ASTM D 412
Hardness (Shore A)	30 minimum	ASTM C 661
Elongation (at break)	300 % minimum	ASTM D 412
Tensile Strength	175 psi minimum	ASTM D 412
Adhesion-in-Peel	20 lbs/linear inch (pli)	ASTM C 794
Service Temperature Range	-20 °F to 180 °F	TT-S-00230 C
Final Cure	5 to 8 days	ASTM C 679
Color	Gray	

2.6 WATERSTOPS

- A. Refer to AASHTO LRFD Bridge Construction Specifications, Section 8.9.2.6

PART 3 EXECUTION

3.1 RELIEF JOINT SEALING

- A. Form or saw cut joint as shown.
- B. Cleaning and Drying
 - 1. Asphalt joint
 - a. Clean 6 inches on both sides of the joint of foreign matter and loosened particles with a hot compressed air heat lance immediately before sealing the joint.

- b. Adequate cleaning is determined by surface darkening at least 12 inches wide, centered on the joint.
 - 2. Concrete joint
 - a. Clean joint and concrete surface by sand blasting before applying the sealant.
 - b. Remove sand left by sandblasting from the joint and surrounding surfaces using compressed air.
- C. Use a backer rod compatible with the sealant, when shown.
- D. Apply hot poured joint sealant as shown.

END OF SECTION

SECTION 03211

REINFORCING STEEL AND WELDED WIRE

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Reinforcing steel, steel welded wire reinforcement, dowelled anchors, T-headed bars, mechanical couplers, and grouted splice couplers.
- B. Coating for reinforcing steel, steel welded wire reinforcement, and dowelled anchors.

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES

- A. AASHTO M 31: Deformed and Plain Carbon and Low-Alloy Steel Bars for Concrete Reinforcement
- B. AASHTO M 336: Steel Wire and Welded Wire, Plain and Deformed, for Concrete Reinforcement
- C. AASHTO T 106: Compressive Strength of Hydraulic Cement Mortar (Using 50-mm or 2-in Cube Specimens)
- D. ASTM A 108: Steel Bar, Carbon and Alloy, Cold-Finished
- E. ASTM A 641: Zinc-Coated (Galvanized) Carbon Steel Wire
- F. ASTM A 493: Stainless Steel Wire and Wire Rods for Cold Heading and Cold Forging
- G. ASTM A 706: Deformed and Plain Low-Alloy Steel Bars for Concrete Reinforcement
- H. ASTM A 767: Zinc-Coated (Galvanized) Steel Bars for Concrete Reinforcement
- I. ASTM A 775: Epoxy-Coated Steel Reinforcing Bars
- J. ASTM A 884: Epoxy-Coated Steel Wire and Welded Wire Reinforcement

- K. ASTM A 934: Epoxy-Coated Prefabricated Steel Reinforcing Bars
- L. ASTM A 955: Deformed and Plain Stainless-Steel Bars for Concrete Reinforcement
- M. ASTM A 970: Headed Steel Bars for Concrete Reinforcement
- N. ASTM A 1060: Zinc-Coated (Galvanized) Steel Welded Wire Reinforcement, Plain and Deformed, for Concrete
- O. ASTM D 3963: Fabrication and Jobsite Handling of Epoxy-Coated Steel Reinforcing Bars
- P. ASTM E 3121: Field Testing of Anchors in Concrete or Masonry
- Q. ACI 355.4: Qualification of Post-Installed Adhesive Anchors in Concrete
- R. American Welding Society (AWS) Standards
- S. Concrete Reinforcing Steel Institute (CRSI) Manual of Standard Practice

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Working Drawings
 - 1. Detailed shop drawings for review of the following:
 - a. Field bending procedure if required. Provide the seal of a Professional Engineer (PE) or Professional Structural Engineer (SE) licensed in the State of Utah.
 - b. Mechanical butt splice shop drawings when proposed details differ from the plans and specifications.
 - 1) Show number and location of mechanical butt splices.
 - 2) Provide two samples of mechanical butt splices and test to destruction in the presence of the Engineer.
- B. Material Submittals
 - 1. Certificate of Compliance from the manufacturer for the following products.
 - a. Reinforcing steel
 - b. Grouted splice couplers

2. Continuous butt welded reinforcing hoops.
 - a. Manufacturer's Quality Control (QC) procedures for the hoop fabrication for review. Include the following as a minimum:
 - 1) The pre-production procedures for the qualification of material and equipment.
 - 2) The methods and frequencies for performing QC procedures during production.
 - 3) The calibration procedures and calibration frequency for all equipment.
 - 4) The welding procedure specification (WPS) for resistance welding.
 - 5) The method for identifying and tracking lots.
 - b. Two samples of welded splices for verification testing.
3. Grouted Splice Couplers
 - a. Independent test report for review confirming coupler compliance for each supplied coupler size with the following:
 - 1) Develop 150 percent of the specified yield strength of the connected bar.
 - 2) Determine by testing the amount of time and grout compressive strength required to provide 100 percent of the specified minimum yield strength of the attached reinforcing bar. Use this value to determine when to release bracing.
 - 3) Use the same grout in the testing that will be used in the construction.
 - 4) Requirements for the grout including required strength gain to develop the specified minimum yield strength of the connected reinforcing bar.
4. Post-installed adhesive anchor information for review
 - a. Product data indicating product characteristics, storage and handling requirements and limitations
 - 1) The manufacturer's product data must document that the adhesive used is compatible with the type and extent of the anchor coating or corrosion resistant material used.
 - b. Manufacturer's Printed Installation Instructions (MPII)
 - c. Current ICC-ES Evaluation Report in the name of the proposed adhesive anchor system to be used
 - d. Material Safety Data Sheets
 - e. List of locations where post-installed adhesive anchors will be installed in an overhead or upwardly inclined orientation

- C. Continuous resistance butt welded hoop welder certifications for information.
 - 1. Include welder test reports for each operator, process, and position as required by AWS specifications.
 - 2. Include a letter that states the certified welders have been using the process without an interruption of more than six months since being certified.
- D. Grouted splice coupler quality assurance qualifications for information
 - 1. Grouted splice coupler installer qualifications
 - a. Provide names and state experience or training of personnel responsible for installation of the grouted splice couplers.
 - 1) Include a certificate or statement of training with the names of the trained individuals signed by the manufacturer's representative
 - 2. Grouted splice coupler manufacturer's technical support representative qualifications. Include at least the following:
 - a. Company name
 - b. Name and phone number of the technical support representative
 - c. List of projects using the submitted grouted splice coupler with at least two years of satisfactory performance. List the following for each project:
 - 1) Project name
 - 2) Bridge location (state routes and bridge identifiers)
 - 3) Scope of work
 - 4) Products used
- E. Post-installed adhesive anchor installer qualifications for information when anchors are to be installed in an overhead or upwardly inclined orientation.
 - 1. Provide current certification for qualified adhesive anchor installer according to the ACI Adhesive Anchor Installer Certification program, or equivalent.

1.6 QUALITY ASSURANCE

- A. Grouted splice coupler manufacturer's technical support representative
 - 1. Provide a technical support representative onsite during preparation and grouting of the grouted splice coupler

- B. Grouted splice coupler installers
 - 1. Installers must be familiar with installation and grouting of splice couplers and that have completed at least two successful projects in the last two years.
 - a. Train new personnel within three months of installation by a manufacturer's technical representative as an acceptable substitution for the experience.

PART 2 PRODUCTS

2.1 REINFORCING STEEL

- A. Deformed or plain carbon steel bars
 - 1. Refer to AASHTO M 31, Grade 60.
- B. Deformed or plain low-alloy steel bars
 - 1. Refer to ASTM A 706, Grade 60.
- C. Deformed or plain stainless steel bars
 - 1. Refer to ASTM A 955, Type XM-28.

2.2 WIRE AND WIRE REINFORCEMENT

- A. Refer to AASHTO M 336 for cold drawn steel wire.
- B. Refer to AASHTO M 366 for steel welded wire reinforcement.

2.3 T-HEADED BARS

- A. Use T-headed bars consisting of deformed rebar with steel plates friction welded to one end of the rebar. Friction welding conforms to the authorized quality control manual and AWS C6.2, Friction Welding of Metals.
 - 1. Headed Bars that meet the requirements of ASTM A 970 may be substituted.
- B. Use deformed rebar according to ASTM A 706, Grade 60.
- C. Cut plate heads for T-headed bars from flats of hot-rolled steel according to ASTM A 108.

2.4 COATINGS

- A. Epoxy Coating
 - 1. Reinforcing steel
 - a. Refer to ASTM A 775 or ASTM A 934.
 - 2. Wire and wire reinforcement
 - a. Refer to ASTM A 884.
- B. Galvanized Coating
 - 1. Reinforcing steel
 - a. Refer to ASTM A 767.
 - 2. Wire and wire reinforcement.
 - a. Coat welded wire after fabrication.
 - b. Refer to ASTM A 1060.
- C. Coat bars as described.
 - 1. Maintain epoxy coating thickness between 8 and 12 mils.
 - 2. Maintain galvanized coating thickness according to ASTM A 767.
 - 3. Coat bars after bending unless the fabricator can show that satisfactory results can be obtained by coating before bending.
 - 4. Do not use bent bars with visible cracks or damage in the coating.
- D. Coating Repair Materials
 - 1. Meet requirements of ASTM D 3963 for repair of epoxy coatings.
 - 2. Use Inorganic Zinc-Rich Paint with 65 to 69 percent metallic zinc by weight, or greater than 92 percent metallic zinc by weight in dry film for repair of galvanized coatings.

2.5 DOWELLED ANCHORS

- A. Post-Installed Adhesive Anchors
 - 1. Use adhesive anchoring system satisfying the assessment criteria of ACI 355.4.
 - a. Meet anchor category 1 or 2 as documented in the ICC-ES evaluation report for the selected anchor system.
 - 1) Anchor category 1 corresponds to a strength reduction factor with no supplemental reinforcement of $\phi_d = 0.65$.
 - 2) Anchor category 2 corresponds to a strength reduction factor with no supplemental reinforcement of $\phi_d = 0.55$.
 - 2. Use adhesive anchoring system with qualifications for use in seismic loading and overhead applications.

3. Use galvanized anchors in adhesive anchoring system unless ICC-ES report explicitly states compatibility with the coating or corrosion resistant materials used for the anchor.
 4. Meet the following minimum characteristic bond stresses at or above the maximum short term temperature shown regardless of anchor diameter:
 - a. τ_{cr} : 475 psi
 - b. τ_{uncr} : 840 psi
 - c. Provide the minimum characteristic bond stresses for a maximum short term temperature of at least 160 degrees F, if not shown.
 5. Supply the adhesive anchor product as a complete system including:
 - a. New adhesive cartridge
 - b. Clean mixing nozzle
 - c. Extension tube
 - d. Dispensing gun
 - e. Piston plug
 - f. Manufacturer recommended supplies for cleaning the drilled hole
- B. Use reinforcing steel, bolts, and anchors as shown.

2.6 BAR SUPPORTS AND TIE WIRE

- A. Provide epoxy coated, galvanized, plastic coated, or plastic bar supports and tie wire that meet the following requirements when not in contact with stainless steel:
1. Meet the requirements of Table 1.
 2. Remove contaminants that affect the adhesion of the coating to the wire.
 3. Use an electrostatic spray method, fluidized bed, or flocking to apply an epoxy coating.
 4. Apply plastic coating by spraying, dipping, or using as a powder.
 5. Provide galvanized coating according to ASTM A 641
 6. Use patching material according to the manufacturer's recommendation to repair damaged coating.
 - a. Use patching material that is compatible with the coating and that is inert in concrete.
 - b. Do not repair hanger marks on the coated bar supports that result from the coating application process. Hanger marks are not considered damaged coating.
 7. Use 16 gauge coated tie wire.
 - a. Do not use galvanized tie wire with epoxy coated reinforcing steel.

- B. Precast concrete block bar supports that conform to the following:
 - 1. Provide minimum 28-day compressive strength of 2,500 psi.
 - 2. Use three inch thick supports with sides ranging from 4 to 6 inch with a minimum ground contact area of 24 in².
- C. Provide bar supports and tie wires for use with stainless steel bars that meet the following:
 - 1. Meet the requirements of Table 1.
 - 2. Provide bar supports that are plastic or stainless steel conforming to the requirements of ASTM A 493, Type 316.
 - 3. Provide tie wires that are plastic or stainless steel conforming to the requirements of ASTM A 493, Type 316, annealed.

2.7 MECHANICAL SPLICE COUPLER

- A. Elastic capacity couplers:
 - 1. Use reinforcing steel mechanical splice couplers capable of developing in tension 125 percent of the specified yield strength of the reinforcing steel bar.
- B. Ultimate strength couplers:
 - 1. Use where shown.
 - 2. For Grade 60 reinforcing steel bars
 - a. Use reinforcing steel mechanical splice couplers capable of transferring at least 95 ksi in tension between the spliced reinforcing steel bars.
- C. Use coated couplers with the same type of coating as the reinforcing steel being spliced.
- D. Use stainless steel splice coupler with stainless steel reinforcement.

2.8 GROUTED SPLICE COUPLER

- A. Use grouted splice couplers to join precast elements as shown.
 - 1. Provide couplers that use cementitious grout placed inside a steel casting. Grout is part of the proprietary system and is provided by the coupler manufacturer.
- B. Use one of the following grouted splice coupler manufacturers according to the requirements of this Section. Use one of the following providers:
 - 1. NMB Splice Sleeve
Splice Sleeve North America, Inc.
38777 West Six Mile Road, Suite 205
Livonia, MI 48152
Reinforcing Steel and Welded Wire

2. Sleeve-Lock Grout Sleeve System
Dayton Superior Corporation
1125 Byers Road
Miamisburg, OH 45342
 3. Lenton Interlok
Pantair USA
34600 Solon Road
Solon, OH 44139
- C. Use grouted splice couplers that provide 150 percent of the specified yield strength of the connected bar.
- D. Use grout supplied by the manufacturer of the coupler and that matches the certified test report for the coupler.
- E. Use the same coating system as used for the reinforcing steel.
1. Use grouted splice couplers that join the reinforcing steel without removal of the epoxy coating on the spliced bar when using epoxy coated reinforcing steel.

2.9 CONTINUOUS RESISTANCE BUTT WELDED HOOPS

- A. Weld only reinforcing steel conforming to ASTM A 706 as shown.
1. Use resistance butt welded splices for continuous hoops.
- B. Refer to AWS D1.4: Structural Welding Code - Reinforcing Steel.
- C. Change welding procedures to reflect chemical composition of the steel.
1. Welders must have correct mill test report (chemical analysis) from the heat in which the steel was made.
- D. Apply coating after all welding has been completed.

2.10 FABRICATION

- A. Use Department Prequalified Suppliers for all reinforcing steel products.
- B. Bend reinforcement to the shapes as shown. Refer to CRSI Manual of Standard Practice.
- C. Do not heat the bars during the bending operations.

PART 3 EXECUTION

3.1 DELIVERY, STORAGE, AND HANDLING

- A. Protect the bars and the coating during handling and storage.
 - 1. Use systems with padded contact areas when handling epoxy coated bars.
 - 2. Pad all bundling bands for epoxy coated bars.
 - 3. Lift all bundles with strong-back, multiple supports, or a platform bridge.
 - 4. Do not drop or drag bars.
- B. Store bars above the ground surface on wooden or padded supports.
 - 1. Place timbers between bundles when stacking is necessary.
 - 2. Space the supports close enough to prevent sags in the bundles.
- C. Cover epoxy coated reinforcing steel with an opaque covering upon delivery to the project site.
 - 1. Protect epoxy coated reinforcing steel that has been partially embedded in concrete or placed in formwork.
 - a. Cover with an opaque covering before 30 days exposure to sunlight.
 - 2. Place the opaque coverings to provide air circulation and prevent condensation on the reinforcing steel.
- D. Ship, handle, and store stainless reinforcing steel so it does not come in contact with carbon steel.
 - 1. Cover stainless reinforcing steel with tarps during outdoor storage.
 - 2. Separate bundles of stainless reinforcing steel from other types of reinforcing steel with wooden spacers.
 - 3. Store stainless reinforcing steel on wooden supports off the ground or floor.
- E. Store post-installed adhesive anchor adhesive in a cool, dry and dark place before use, according to the MPPII.

3.2 COATING REPAIR

- A. General
 - 1. Follow product manufacturer recommendations for repairs.
 - 2. Clean and dry repair area.
 - 3. Protect nearby surfaces from overspray.
 - 4. Allow at least 45 minutes of drying time before encasing repaired reinforcing bar in concrete.

- B. Epoxy Coatings
 - 1. Do not apply repair materials at temperatures below 50 degrees F except where application instructions explicitly state lower temperatures are allowed.
 - 2. Apply multiple coats when using spray applied coating repair materials.
 - a. Allow a few minutes between coats to avoid drips and runs.
- C. Limits for Repair
 - 1. Repair minor damage to coatings and cut reinforcing steel bar ends with specified coating repair materials before placement.
 - a. Refer to this Section, Article 3.2.
 - 2. Do not use bars with moderate damage to coatings within any one foot length of the reinforcing steel bar or as determined by the Engineer.
 - a. Cut ends are not considered damage for this purpose.
 - 3. Replace installed reinforcing steel bars that have significant damage to coatings within any one foot length of the reinforcing steel bar or as determined by the COR.

3.3 PLACEMENT

- A. Maintain a clean surface keeping all reinforcing steel free from loose mill scale, loose or thick rust, dirt, paint, oil, or grease.
- B. Place and secure reinforcing steel in its designated position.
 - 1. Placing Tolerances
 - a. Decks or members 10 inches or less in thickness
 - 1) Cover: $-\frac{1}{8}$ inch, $+\frac{1}{4}$ inch.
 - 2) Longitudinal spacing for individual bars: ± 1 inch.
 - a) Clear spacing between bars: not less than the greater of $1\frac{1}{2}$ inches, $1\frac{1}{2}$ bar diameters, and $1\frac{1}{2}$ times the maximum aggregate size.
 - 3) Average spacing for 10 bars: $+\frac{1}{16}$ inch.
 - a) Do not use tolerance to decrease number of bars or increase bar spacing.
 - b. Members 10 to 20 inches in thickness
 - 1) Cover: $\pm \frac{1}{4}$ inch.
 - 2) Longitudinal spacing for individual bars, stirrups, or ties: ± 1 inch.
 - a) Clear spacing between bars: not less than the greater of $1\frac{1}{2}$ inches, $1\frac{1}{2}$ bar diameters, and $1\frac{1}{2}$ times the maximum aggregate size.

- 3) Average spacing for 10 bars: $+1/16$ inch.
 - a) Do not use tolerance to decrease number of bars or increase bar spacing.
 - c. Members greater than 20 inches in thickness
 - 1) Cover: $-1/4$ inch, $+1/2$ inch.
 - 2) Spacing for stirrups or ties: ± 3 inches.
 - a) Clear spacing between bars: not less than the greater of $1\frac{1}{2}$ inches, $1\frac{1}{2}$ bar diameters, and $1\frac{1}{2}$ times the maximum aggregate size.
 - 3) Longitudinal bar spacing ± 3 inches.
 - a) Clear spacing between bars: not less than the greater of $1\frac{1}{2}$ inches, $1\frac{1}{2}$ bar diameters, and $1\frac{1}{2}$ times the maximum aggregate size.
 - 4) Average spacing for 20 bars: $+1/4$ inch.
 - a) Do not use tolerance to decrease number of bars or increase bar spacing.
 - d. Length of bar laps -1 inch
 - e. Embedment length -1 inch
- C. Tie bars together with ties at intersections except when spacing is less than 9 inches in each direction, in which case tie at alternate intersections.
 - 1. Tie bundled bars together at not more than 6 ft centers.
- D. Maintain the required distance from the forms and between layers of reinforcing steel with prefabricated chairs, ties, hangers, or other devices.
- E. Do not tack weld reinforcing bars in place.
- F. Overlap at least one panel of welded wire reinforcement sheets to each other and fasten at the ends and edges.
- G. Support reinforcing steel for concrete "T" beams, pier caps, approach slabs, and deck slabs on metal chairs or bar supports according to this Section, Article 2.6.
- H. Space chairs for supporting the top steel and bolsters for supporting the bottom steel not more than 4 ft on center of the bar in each direction.
- I. Tie deck steel to beams or forms at regular intervals of not more than 5 ft on center along the beams to prevent steel movement during concrete placement.

- J. Support reinforcing steel for slabs on grade on metal chairs attached to a sand plate or use precast concrete block supports according to this Section, Article 2.6.
 - 1. Precast concrete block supports may only be used to support the bottom mat of bars.
 - a. Do not use blocks that are cracked or damaged.
- K. Place stainless steel reinforcing steel so that it does not come in contact with carbon steel.
 - 1. Do not tie stainless steel to uncoated or coated carbon steel reinforcing steel, galvanized attachments, or galvanized conduits.
 - a. Maintain at least 1 inch clearance between the metals using nylon or polyethylene spacers when stainless steel reinforcing or dowels must be near coated or uncoated reinforcing, or galvanized metals.
 - 1) Bind using nylon cable ties.
 - 2) Maintain at least 1 inch clearance unless insufficient space exists.
 - a) Either bar may be sleeved with a $\frac{1}{8}$ inch minimum thick insulator material, such as polyethylene, nylon or rubber tube, extending at least 1 inch in either direction past the point of closest contact between the two dissimilar bars.
 - b) Sleeves are not allowed for bars that run parallel to each other.
- L. Do not place concrete until the Engineer has verified the reinforcing steel placement and fastening.

3.4 FIELD CUTTING

- A. Saw or shear coated bars that are specified to be cut in the field.
 - 1. Do not flame cut.
- B. Repair the coating at the sawed or sheared end using the specified patching or repair material.

3.5 SPLICING

- A. Furnish all reinforcing steel in the lengths shown.
- B. Do not splice bars except where shown.

- C. Stagger splices as far as possible.
- D. Place and tie lapped splices to maintain the clearance to the surface of the concrete shown.
- E. Do not allow lap splices in vertical column reinforcing bars unless shown.
- F. Do not lap splice No. 14 and No. 18 bars.
 - 1. Use mechanical splice couplers.
- G. Use mechanical splice couplers when shown.
 - 1. Follow the manufacturer's published recommendations for equipment and splicing procedures.

3.6 FIELD BENDING

- A. Do not field bend reinforcing steel unless shown.
- B. Follow the authorized field bending procedures.
- C. Use methods that do not damage coatings.
- D. Do not heat the bars during the bending operations.
- E. Do not bend bars partially embedded in concrete except as shown or pre-approved by the COR.
 - 1. Do not field straighten or re-bend fabricated bent bars.

3.7 INSTALLATION OF DOWELLED ANCHORS

- A. Install doweled anchors according to the authorized Manufacturer's Printed Installation Instructions, applicable specifications, and as shown.
- B. Post-Installed Adhesive Anchors
 - 1. Do not use post-installed adhesive anchors in sustained tension applications.
 - 2. Installation
 - a. The qualified anchor installer is responsible to perform all cleaning and installation operations for post-installed adhesive anchors for overhead or upwardly inclined anchors.

- b. Drilling holes
 - 1) Locate existing reinforcing steel using a non-destructive method before drilling.
 - a) Notify the Engineer of location conflicts between existing reinforcing steel and new adhesive anchors
 - 2) Use a rotary impact drill.
 - 3) Do not use a lubricant.
 - 4) Drill hole to the diameter and depth described in the MPII.
 - 5) Blow debris and dust out the hole with compressed air.
 - 6) Brush the full depth of the hole.
 - 7) Blow dust out of the brushed hole with compressed air.
 - 8) Do not install dowelled anchors until the holes are verified by the Engineer.
 - c. Installing anchors
 - 1) Install the epoxy from the bottom of the hole outward.
 - 2) Twist the anchor as it is being inserted to prevent the formation of air bubbles
 - 3) Secure anchors to prevent displacement while adhesive cures.
 - 4) Follow MPII for set times
 - d. Limitations
 - 1) Do not install anchors until the concrete has cured at least 21 days or until concrete achieves 100 percent of the specified 28 day minimum compressive strength based on field cured cylinders.
 - 2) Do not install anchors unless concrete temperature is expected to be at least 50 degrees F for the three hours immediately after installation.
 - 3) Do not install anchors if concrete temperature is over 100 degrees F.
3. Proof test post-installed adhesive anchors as specified and when shown, in the presence of the Engineer and as directed by the Engineer:
- a. Test anchors installed in an overhead or upwardly inclined position.
 - 1) Test 25 percent of the first 40 anchors installed and 10 percent of anchors installed thereafter unless shown otherwise.
 - b. Allow anchor adhesives to cure at least 48 hours before proof testing.

- c. Tension test according to ASTM E 3121.
 - 1) Test in the confined condition.
 - 2) Tension anchors to 80 percent of the anchor yield strength.
 - 3) Hold tension at the specified load for at least 30 seconds.
 - a) Measurable displacement of anchor is a failure condition regardless of final tension.
 - 4) Notify the Engineer if concrete cracking occurs in the vicinity of the anchor after loading.

3.8 CONNECTIONS USING GROUTED SPLICE COUPLERS

- A. Use grouted splice coupler installers according to the submitted grouted splice coupler installer qualifications.
- B. Remove and clean all debris from the joints before grout application.
- C. Keep bonding surfaces free from laitance, dirt, dust, paint, grease, oil, or any contaminants other than water.
- D. Embed rebar anchor dowels to the minimum coupler embedment required by the manufacturer.
- E. Saturate Surface Dry (SSD) all concrete surfaces in the joint before connecting the elements.
- F. Use shims to verify that the reinforcing extensions are within the manufacturers recommended tolerance.
- G. Maintain a minimum grout and sleeve temperature of 50 degrees F until the temporary bracing is removed.
- H. Mix structural grout and coupler grout just before use according to the manufacturer's instructions.
- I. Follow the manufacturer's recommendations for coupler installation and grouting.
- J. Monitor the grouting operation to verify that all sleeves have been filled.
- K. Protect sleeves from vibration, shock, or other excessive movement until temporary bracing is removed.

- L. Conform to the following when installing couplers above a horizontal joint:
1. Determine the thickness of shims to provide the specified elevation within tolerance.
 2. Follow non-shrink grout manufacturer's recommendations for mixing, joint surface preparation, and application.
 3. Place non-shrink grout on the interface between the two elements being joined before setting the element.
 - a. Crown the thickness of the grout toward the center of the joint so that the grout can be displaced outward as the element is lowered onto the joint.
 - b. Prevent the grout from entering the coupler above elements by using grout dams or seals.
 4. Set the element in place.
 - a. Engage all couplers in the joint.
 - b. Allow the grout to seep out of the joint.
 5. Trowel off excess grout to form a neat joint once the element is set, plumbed, and aligned.
 - a. Pack grout into any voids around the joint perimeter.
 6. Flush out the coupler with clean potable water.
 7. Mix the special coupler grout according to the manufacturer's recommendations for methods and proportions of mix and water.
 8. Make four sets of three 2 inch cube specimens for testing.
 - a. Cure the specimens according to AASHTO T 106.
 - b. Test one set of cubes for compressive strength to determine when to release bracing. Refer to this Section, Article 1.5 paragraph B3a2.
 - c. Test one set of cubes at 28 days for acceptance.
 - d. Store extra sets for longer term testing if necessary.
 - e. Use a Department qualified laboratory to take the samples and perform the tests.
 9. Pump the coupler grout into the coupler that is cast into the element.
 - a. Start from the lower port.
 - b. Pump until the grout is flowing freely from the upper port.
 - c. Cap the upper port first and then remove the nozzle to cap the lower port.
 10. Cure the joint according to the grout manufacturer's recommendations.
- M. Conform to the following when installing couplers below a horizontal joint:
1. Determine shim thickness to provide the specified elevation within tolerance.

2. Before setting the element:
 - a. Mix the coupler grout paying strict attention to the manufacturer's recommendations for methods and proportions of mix and water.
 - b. Clean debris from the interior using compressed air.
 - 1) Remove any rain water using a vacuum that can remove water from the confined space in the coupler.
 - c. Place the coupler grout into the coupler by pouring or pumping.
 - d. Place grout on the interface between the two elements being joined.
 - 1) Crown the thickness of the grout toward the center of the joint so that the grout can be displaced outward as the element is lowered onto the joint.
 3. Trowel off excess grout to form a neat joint once the element is set, plumbed, and aligned.
 - a. Pack grout into any voids around the joint perimeter.
- N. Conform to the following when installing couplers in vertical joints (horizontal bar/coupler connection):
1. Establish a method to provide the specified elevations, alignment, and spacing within tolerance.
 2. Use washers or seals to prevent mixing the joint grout and the coupler grout.
 3. Apply epoxy adhesive to the interface between the two elements being joined.
 4. Set the element in place.
 - a. Engage all couplers in the joint.
 5. Flush out the couplers with clean potable water once the element is set, plumbed, and aligned.
 6. Mix the coupler grout paying strict attention to the manufacturer's recommendations for methods and proportions of mix and water.
 7. Pump the coupler grout into the coupler that is cast into the element.
 - a. Start from the port closest to the joint.
 - b. Pump until the grout is flowing freely from the other port.
 - c. Cap the port farthest from the joint first and then remove the nozzle to cap the other port.
 8. Form the edges of the joint and place grout into the joint.
 9. Cure the joint according to the grout manufacturer's recommendations.

3.9 FIELD QUALITY CONTROL

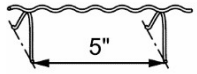
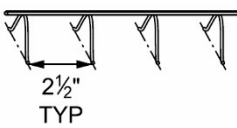
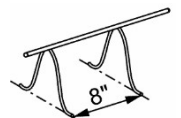
- A. Inspect coated bars for damage to the coating after the bars are in place and immediately before concrete placement.

Reinforcing Steel and Welded Wire

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Table 1

Bar Supports							
Types and Sizes				Minimum Wire Sizes ² and Geometry			
Symbol	Bar Support Illustration	Type of Support	Standard Sizes	Nominal Height	Carbon Steel		Geometry
					Top	Legs	
SB ¹		Slab Bolster	¾, 1, 1½, and 2 inch heights in 5 ft and 10 ft lengths	All	4 ga. Corrugated	6 ga.	Legs Spaced 5 inches on Center, Vertical Corrugations Spaced 1 inch on Center (See Note 3)
BB ¹		Beam Bolster	1, 1½, and 2 inch; over 2 inch to 5 inch heights in increments of ¼ inch lengths of 5 ft.	Up to 1½ inch incl.	7 ga.	7 ga.	Legs Spaced 2½ inches on Center (See Note 3)
				Over 1½ inch to 2 inches incl.	7 ga.	7 ga.	
				Over 2 inches to 3½ inches incl.	4 ga.	4 ga.	
				Over 3½ inch	4 ga.	4 ga.	
BC		Individual Bar Chair	¾, 1, 1½, and 1¾ inch heights	All	-----	7 ga.	(See Note 3)
JC		Joist Chair	4, 5, and 6 inch widths and ¾, 1, and 1½ inch heights	All	-----	6 ga.	(See Note 3)
HC or HPC*	 * SAND PLATE NEED NOT BE COATED	Individual High Chair	2 inch to 15 inch heights in increments of ¼ inch.	2 inches to 3½ inches incl.	-----	4 ga.	Legs at 20 degree or less with vertical. When height exceeds 12 inches, legs are reinforced with welded crosswires or encircling wires (See Note 4)
				Over 3½ inches to 5 inches incl.	-----	4 ga.	
				Over 5 inches to 9 inches incl.	-----	2 ga.	
				Over 9 inches to 15 inches incl.	-----	0 ga.	
CHC		Continuous High Chair	Same as HC in 5 ft and 10 ft lengths	2 inches to 3½ inches incl.	2 ga.	4 ga.	Legs at 20 degree or less with vertical. All legs 8¼ inches on center maximum, with leg within 4 inches of end of chair, and spread between legs not less than 50 percent of nominal height. (See Note 5)
				Over 3½ inches to 5 inches incl.	2 ga.	4 ga.	
				Over 5 inches to 9 inches incl.	2 ga.	2 ga.	
				Over 9 inches to 15 inches incl.	2 ga.	0 ga.	

Notes and Bar Supports Table, see next page.

Notes:

1. Provide top wire on continuous supports, not otherwise designated as corrugated, which may be straight or corrugated at the option of the manufacturer.
2. Provide minimum wire sizes that are American steel and wire gauges.
3. Provide adequate stability against overturning. The leg spread measured between points of support on the minor axis must be at least 70 percent of the nominal height.
4. Provide adequate stability against overturning. The leg spread measured between points of support on the minor axis must be at least 55 percent of the nominal height.
5. Provide adequate stability against overturning and adequate load capacity. The leg spread measured between points of support on the minor axis must not exceed the minimum and maximum percentages of the nominal height as shown.

Table 2

Support Axis		
Nominal Height (inches)	Distance Between Supports as a Percent of Nominal Height	
	Minimum	Maximum
Under 4	70	No Limit
4	70	95
6	65	90
8	60	85
10	55	80
12	50	75
Over 12	50	75

END OF SECTION

SECTION 03390

CONCRETE CURING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Concrete curing materials and methods.
- B. This section does not apply to cast-in-place Portland Cement Concrete Pavement and Lean Concrete Base Course.
- C. This section does not apply to dry cast precast concrete members, except for precast concrete box and three-sided culvert structures.

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete
- B. Section 03310: Structural Concrete

1.3 REFERENCES

- A. ASTM C 309: Liquid Membrane-Forming Compounds for Curing Concrete
- B. ASTM C 1315: Liquid Membrane-Forming Compounds Having Special Properties for Curing and Sealing Concrete
- C. UDOT Quality Management Plans

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Manufacturer's product data sheets and recommended installation instructions.

PART 2 PRODUCTS

2.1 CURING COMPOUND

- A. Refer to this Section, Part 3, Tables 1 and 2.
- B. Limit Volatile Organic Compounds (VOC) content to 350 grams / liter maximum.

PART 3 EXECUTION

3.1 CURING MEASURES

- A. Prevent drying of exposed concrete surfaces after placing concrete and until applying the selected curing method.
 - 1. Keep exposed concrete surfaces moist by fogging if the surfaces begin to dry before the selected curing method can be applied.
- B. Cure newly placed concrete to prevent loss of water by one or more of the following methods according to this Section.
 - 1. Forms-in-Place Method (FIPM)
 - a. Cure formed surfaces of concrete by retaining the forms in place without loosening for the specified curing period.
 - b. Complete the cure using one of the methods specified for the element when forms are removed before completing the specified curing period.
 - 2. Water Method (WM)
 - a. Keep concrete surfaces continuously wet by ponding, spraying, or covering with materials that are continuously and thoroughly wet.
 - 1) Acceptable materials for covering concrete surfaces include cotton mats, multiple layers of burlap, or other materials that retain water.
 - 2) Secure the cover materials to prevent wind or other forces from removing them.
 - 3) Keep the cover materials saturated throughout the curing period.
 - b. Do not erode or damage the finish.
 - c. Prevent excess water from impacting traffic on in-service roadways.

3. Liquid Membrane-Curing Compound Method (LMCCM)
 - a. Thoroughly seal exposed concrete surfaces with a liquid membrane-curing compound immediately after finishing operations are completed for exposed concrete surfaces or immediately after forms are removed and necessary patching is complete when forms are removed before the end of the curing period.
 - b. Apply liquid membrane-curing compound uniformly at the rates specified in Tables 1 and 2.
 - c. Immediately repair damage to the curing compound film during the specified curing period by re-spraying.
4. Waterproof Cover Method (WCM)
 - a. Exposed concrete surfaces must be wet before installing cover.
 - b. Cover exposed concrete surfaces with a material that prevents moisture loss from the concrete.
 - 1) Do not use materials that have lost their waterproof qualities.
 - 2) Secure cover to prevent displacement by wind.
 - c. Use this method only when the covering can be secured adequately to prevent moisture loss.
 - d. Maintain the air temperature beneath the cover at not less than 50 degrees F.
 - 1) The use of insulated blankets is permitted.
5. Steam or Radiant-Heat Method (SRHM)
 - a. Use only for precast concrete members manufactured in prequalified plants.
 - b. Use a complete steam or radiant heat curing system that includes 24 hour temperature control and monitoring devices.
 - 1) Use temperature recording devices as necessary to verify that temperatures are uniform throughout the enclosure and within the limits specified.
 - c. Steam Heat Curing System
 - 1) Use a suitable enclosure to contain live steam and minimize moisture and heat losses.
 - 2) Use low-pressure and saturated steam.
 - 3) Maintain 90 to 100 percent relative humidity in the curing enclosure.
 - 4) Do not apply heat directly on the concrete or cause localized high temperatures.
 - d. Radiant Heat Curing System
 - 1) Apply heat by means of pipes circulating steam, hot oil, or hot water, or by electric heating elements.
 - 2) Use a suitable enclosure to contain the heat.

- 3) Minimize moisture loss by covering exposed concrete surfaces with plastic sheeting.
- e. Waiting Period
 - 1) Do not apply the initial application of heat before the initial set of the concrete except to maintain the minimum temperature within the curing enclosure.
 - 2) Maintain the temperature within the curing enclosure at not less than 50 degrees F.
 - a) Live steam or radiant heat may be used to maintain the curing enclosure at the proper minimum temperature.
 - b) Keep the concrete wet during this period.
- f. Curing Period
 - 1) Increase the temperature within the concrete during the initial application of heat at an average rate not to exceed 40 degrees F per hour until the curing temperature is reached.
 - 2) Do not exceed a concrete temperature of 160 degrees F when applying heat.
 - 3) Maintain the concrete temperature at between 50 degrees F and 160 degrees F.
 - 4) Maintain the curing temperature until the concrete achieves the specified strength for terminating the curing.
 - 5) Decrease the concrete temperature at a rate not to exceed 40 degrees F per hour until reaching a temperature of not more than 20 degrees F above the air temperature to which the concrete will be exposed when discontinuing heat.
- g. Transfer the stressing force to the concrete immediately after discontinuing steam curing or radiant heat curing for prestressed members.

C. Prevent exposed concrete surfaces from drying when transitioning between curing methods.

D. Fogging

- 1. Use fogging when necessary to prevent drying of exposed concrete surfaces.
- 2. Use fogging equipment with compressed air misters that atomize the water and produce a very fine mist and not a spray.
 - a. Use equipment that allows for adjusting the rate of fogging depending on the conditions that are present.
 - b. Maintain misters at least 5 ft above the concrete surface and aimed in a direction not lower than horizontal.

- c. Do not use fogging to apply excess water to the concrete surface to aid finishing.
 - d. Do not affect the water/cement ratio of the concrete.
 - e. Discontinue fogging when a fine coating of water or sheen is visible on the concrete surface.
 - 3. Do not damage the concrete surface.
- E. Follow the hot and cold weather limitations according to Section 03055.

3.2 CURING COMPOUND APPLICATION

- A. Comply with the following when applying liquid membrane-curing compound to structural elements in bridges, box culverts, headwalls, retaining walls, concrete drainage structures, and concrete slope protection.
- 1. Do not use curing compounds on surfaces that require a rubbed finish or on surfaces of construction joints against which new concrete will be cast, unless approved.
 - a. Completely remove the curing compound before casting new concrete against the surface when curing compound is allowed.
 - 2. Do not use curing compounds on architectural surfaces that require a concrete coating or penetrating concrete sealer and where removal will diminish the texture.
- B. Preparation
- 1. Verify concrete surfaces are ready for curing.
 - a. Complete all patching and surface finishing before applying curing compound.
 - 2. Prepare concrete surfaces and apply curing compound according to product manufacturer's recommendations. Refer to Tables 1 and 2 for application rates.
 - 3. Keep surfaces moist until the curing compound is applied.

3.3 CURE CAST-IN-PLACE CONCRETE

- A. Cure all formed surfaces using the FIPM.
- B. Cure exposed surfaces of newly placed cast-in-place concrete according to the curing methods and curing periods in Table 1.
- 1. Determine the concrete compressive strength using field cured cylinders cured the same as the concrete member when compressive strength is used to determine the curing period. The curing period is the specified number of consecutive days when compressive strength is not used to determine the curing period.

Table 1

Cast-in-Place Concrete Curing Requirements				
Element	Curing Methods for Exposed Surfaces ¹	Curing Period ²	Curing Compound	
			Type	Application Rate
Bridge Decks and Approach Slabs	LMCCM and WM	14 days	ASTM C309, Type I D, Class A	Manufacturer's recommended rate
Closure Pours in Bridge Decks and Approach Slabs	LMCCM and WM	7 days and f'c ³	ASTM C309, Type I D, Class A	Manufacturer's recommended rate
Other Bridge Elements (superstructure, substructure, and foundation elements)	- LMCCM or - WM or - WCM	0.70 f'c or 7 days ³	ASTM C309, Type I D, Class A	100 ft ² /gal
Box Culverts (including wingwalls, and aprons), Headwalls, Retaining Walls, Concrete Drainage Structures, Sign Structure Foundations	- LMCCM or - WM or - WCM	0.70 f'c or 7 days	ASTM C309, Type I D, Class A	100 ft ² /gal
Concrete barrier	- LMCCM or - WM or - WCM	0.70 f'c or 7 days	ASTM C309, Type I D, Class A or ASTM C1315, Type 1, Class A ⁴	100 ft ² /gal
Curbs, gutters, flatwork, sidewalks, driveways, concrete slope protection, and other concrete items not specified	LMCCM	7 days	ASTM C309, Type I D, Class A	100 ft ² /gal
Notes: ¹ Use FIPM for all formed surfaces. Specified curing methods apply to exposed concrete surfaces and any formed surfaces where the forms are removed before the curing period ends. ² f'c = specified 28 day minimum compressive strength. ³ The curing period for bridge elements that use high early strength concrete may be reduced to the greater of 3 days and the time required to achieve the specified 28 day minimum compressive strength. (Refer to Section 03310 for when the use of high early strength concrete is permitted.) ⁴ Use a curing compound meeting ASTM C309 when removal is required.				

- C. Concrete Barrier
 - 1. Broom clean the formed surfaces of the barrier after removing forms.
 - 2. Apply curing compound to exposed concrete surfaces immediately after finishing operations are completed.

3.4 CURE PRECAST CONCRETE

- A. Cure all formed surfaces using the FIPM.
- B. Cure exposed surfaces of newly placed precast concrete according to the curing methods and curing periods in Table 2.
 - 1. Determine the concrete compressive strength using field cured cylinders cured the same as the concrete member when compressive strength is used to determine the curing period. The curing period is the specified number of consecutive days when compressive strength is not used to determine the curing period.

Table 2

Precast Concrete Curing Requirements				
Element	Curing Methods for Exposed Surfaces ¹	Curing Period ²	Curing Compound	
			Type	Application Rate
Precast Concrete Deck Panels (full depth), Precast Approach Slabs, (includes parapets when cast concurrent with precast deck and approach slab panels)	• LMCCM and WM	14 days	ASTM C309, Type I D, Class A	Manufacturer's recommended rate
Precast Substructure Elements, Partial Depth Precast Deck Panels (non-prestressed)	• WM or • SRHM	0.7 f'c or 7 days	Not used	Not used
Prestressed Concrete Members (includes all pretensioned concrete members where pretensioning is required in the plans)	• SRHM or • WM or • WCM	Specified release strength (f'ci)	Not used	Not used
Precast Noise Walls, Precast Retaining/Noise Walls, MSE Retaining Wall Panels	• SRHM or • WM or • WCM	0.70 f'c or 7 days	Not used	Not used
Precast Box Culvert Structures and Precast Three-Sided Culvert Structures (wet cast and dry cast)	• SRHM or • WM or • WCM • LMCCM	0.70 f'c or 7 days	ASTM C309, Type I D, Class A	100 ft ² /gal
Wet Cast Concrete Drainage Structures (such as manholes, grade rings, catch basin grade sections, pipe end sections, precast inlets and boxes)	• SRHM or • WM or • WCM • LMCCM	0.50 f'c or 7 days	ASTM C309, Type I D, Class A	100 ft ² /gal
Modular Block (wet cast)	• WM or • WCM or • LMCCM	0.70 f'c or 7 days	ASTM C309, Type I D, Class A	100 ft ² /gal

Table 2 (Continued)

Precast Concrete Curing Requirements				
Element	Curing Methods for Exposed Surfaces ¹	Curing Period ²	Curing Compound	
			Type	Application Rate
Concrete Barrier	LMCCM	7 days and until certified according to QMP ³	ASTM C1315, Type 1, Class A	100 ft ² /gal
PCC Pavement Panels	LMCCM and WM	14 days	ASTM C309, Type I D, Class A	Manufacturer's recommended rate
Table 2 Notes: ¹ Use FIPM for all formed surfaces. Specified curing methods apply to exposed concrete surfaces and any formed surfaces where the forms are removed before the curing period ends. ² f'c = specified 28 day minimum compressive strength. ³ QMP = UDOT Quality Management Plan: Precast-Prestressed Concrete Structures.				

- C. Precast Concrete Deck Panels (full depth), and Precast Approach Slabs
1. Refer to this Section, Article 3.3, paragraph C.
- D. Precast Concrete Barrier
1. Broom clean the formed surfaces of the barrier after removing forms.
 2. Apply curing compound to exposed concrete surfaces immediately after finishing operations are completed.

END OF SECTION

SECTION 03575
FLOWABLE FILL

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Flowable fill.

1.2 RELATED SECTIONS

- A. Section 03055: Portland Cement Concrete

1.3 REFERENCES

- A. AASHTO M 194: Chemical Admixture for Concrete
- B. ASTM D 4832: Preparation and Testing of Controlled Low Strength Material (CLSM) Test

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Batch Proportions
 - 1. Seven days before placement.
- B. Trial Batch
 - 1. Certified test results or conduct laboratory trial batch to verify strength before placement.

PART 2 PRODUCTS

2.1 MATERIALS

- A. Cement – Refer to Section 03055.
- B. Pozzolan – Refer to Section 03055.
- C. Sand.

- D. Coarse aggregate – Determine a suitable aggregate size and gradation for the intended application.
- E. Admixtures
 - 1. Water reducers and set accelerators. Refer to AASHTO M 194.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Combine materials to meet the requirements for strength and constructability as required. Determine strength from trial batches at 28 days.
 - 1. Minimum strength – 50 psi according to ASTM D 4832.
 - 2. Maximum strength – 150 psi according to ASTM D 4832.
 - 3. Slump: 5 inches to 10 inches.

END OF SECTION

SECTION 06055

TIMBER AND TIMBER TREATMENT

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Timber for signs, guardrails, and posts, and the preservative treatment for timber.

1.2 RELATED SECTIONS Not Used

1.3 REFERENCES

- A. AASHTO M 133: Preservatives and Pressure Treatment Processes for Timber
- B. AASHTO M 168: Wood Products
- C. Southern Pine Inspection Bureau (SPIB) Standard Grading Rules
- D. Western Wood Products Association (WWPA) Standard Grading Rules

1.4 DEFINITIONS Not Used

1.5 SUBMITTALS

- A. Manufacturer's product data sheets and recommended installation instructions.

PART 2 PRODUCTS

2.1 STRUCTURAL TIMBER AND LUMBER

- A. Refer to AASHTO M 168.
- B. Treat, if required, using preservative. Refer to AASHTO M 133.

2.2 POSTS

- A. Pressure treated wood posts that comply with the current WWPA Standard Grading Rules or SPIB Grading Rules.
- B. Temporary Use Sign Posts
 - 1. Use only one type of wood species on any one project.
 - 2. S4S, 80 percent FOHC, Douglas Fir-Larch, Grade No. 2 or better.
 - 3. Hem-fir, Lodgepole Pine, Ponderosa Pine, or Southern Yellow Pine, Grade No. 1 or better.
- C. Guardrail Post
 - 1. Use only one species of wood on any one project.
 - 2. Surfaced or rough-sawn posts and offset blocks.
 - 3. Douglas Fir-Larch, Hem-Fir, Lodgepole Pine, Ponderosa Pine, or Southern Yellow Pine.
 - 4. Grade No. 1 or better.
- D. Steel Post with Wood Block
 - 1. Routed wood offset block of Douglas Fir or Southern Yellow Pine.
 - 2. Surfaced or rough sawn blocks.
 - 3. Grade No. 1.

2.3 TREATMENT

- A. Meet requirements of AASHTO M 133.

PART 3 EXECUTION Not Used

END OF SECTION

SECTION 09981

CONCRETE COATING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Water repellent and concrete stain applied to concrete surface.

1.2 RELATED SECTIONS **Not Used**

1.3 REFERENCES

- A. ASTM D 968: Abrasion Resistance of Organic Coatings by Falling Abrasive
- B. ASTM D 3960: Volatile Organic Compound (VOC) Content of Paints and Related Coatings
- C. ASTM D 4285: Indicating Oil or Water in Compressed Air
- D. ASTM D 4541: Pull-Off Strength of Coatings Using Portable Adhesion Testers
- E. ASTM D 7234: Pull-Off Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Concrete Coating System
 - 1. Manufacturer's product data sheets for review.
 - a. Include recommended installation instructions for each product used for information.
 - 2. Letter of compatibility under the concrete stain manufacturer's letterhead, for review, indicating that the concrete stain to be used are compatible with the water repellents to be used.
 - 3. Three 8½ inch by 11 inch samples of each concrete stain color required on the project for review before ordering concrete coating materials.

PART 2 PRODUCTS

2.1 CONCRETE COATING SYSTEM

- A. Water Repellent
 - 1. Use Alkoxy Silane (98 percent or greater silane) as a base coat.
 - a. Meet VOC content of 350 g/L or less for a reactive penetrating sealer. Refer to ASTM D 3960.
- B. Concrete Stain
 - 1. Use a 100 percent acrylic emulsion stain that meets the requirements of Table 1.
 - 2. Provide the colors as described.

Table 1

PROPERTIES OF CURED CONCRETE STAIN		
Property	Value	Method
Abrasion Resistance, min. liters per mil	30	ASTM D 968
Adhesion to Concrete	Substrate failure	ASTM D 4541 or ASTM D 7234
VOC content, max. g/L	250	ASTM D 3960

2.2 EQUIPMENT

- A. Sandblaster/Air Compressor
 - 1. Air free of water and oil according to ASTM D 4285.
- B. Pressure Washer
 - 1. At least 3000 psi but no more than 5000 psi.

PART 3 EXECUTION

3.1 GENERAL

- A. Use only one concrete coating system on an individual element.

- B. Furnish, prepare, apply, cure, and store materials according to the product manufacturer's directions and as specified in this Section.
 - 1. Do not allow material to freeze
 - 2. Deliver concrete coating materials to the project site in sealed containers bearing the manufacturer's original labels with the brand, color, and type clearly marked on each container.
- C. Omit water repellent when applying concrete stain over graffiti.
- D. Weather Limitations
 - 1. Apply when the outside air temperature is between 45 and 90 degrees F and will remain between 45 and 90 degrees F per the manufacturer's recommended cure time.
 - 2. Do not apply within 24 hours of a rain event or pressure washing, nor if a rain event is forecasted within 8 hours.

3.2 PREPARATION BEFORE WATER REPELLENT APPLICATION

- A. Cure concrete for 28 days before water repellent application.
 - 1. Rapid setting patch materials do not need to cure for 28 days but must meet moisture content levels specified in this Section.
- B. Completely remove curing compounds, oil, grease, dirt, and any other surface contaminants from concrete surfaces.
 - 1. Use one of the following methods:
 - a. Pressure washing
 - b. Shotblasting
 - c. Sandblasting
- C. Remove dust or debris using compressed air immediately before each application of the product.
- D. Measure moisture content in the presence of the Engineer with an electronic concrete moisture meter at three or more locations along the area to be sealed.
 - 1. Measure recently patched concrete areas separately.
 - 2. Do not apply water repellent if any measurements exceed 4.0 percent.

3.3 WATER REPELLENT APPLICATION

- A. Apply according to the manufacturer's recommendations, except for the following.
 - 1. Apply each coat at an application rate of 1 gal penetrating concrete sealer to 250 sq ft.

- B. Apply two separate coats of penetrating concrete sealer.
 - 1. Apply second coat while the first coat is still wet to the touch.

3.4 PREPARATION BEFORE CONCRETE STAIN APPLICATION

- A. Allow water repellent to cure for at least 5 hours.
- B. Pressure wash surface and allow area to dry for 24 hours and complete any additional surface preparation requirements per manufacturer's recommendations.
- C. Stir the coating materials thoroughly before application.

3.5 CONCRETE STAIN APPLICATION

- A. Apply according to the manufacturer's recommendations.
- B. Apply two coats of the concrete stain according to the manufacturer's recommendations, but not less than the following rates.
 - 1. Apply the first coat evenly at a minimum application rate of 1 gal concrete stain to 350 sq ft working in one direction.
 - 2. Cure the first coat of the concrete stain at least 4 hours before the second coat of the concrete stain is applied.
 - 3. Apply the second coat evenly at a minimum application rate of 1 gal concrete stain to 350 sq ft working in the opposite direction of the first coat.

3.6 PROTECTION

- A. Prevent sandblasting material, coating products and debris from falling into waterways, pedestrian areas, traffic areas, or onto railroad tracks.
- B. Protect surrounding surfaces from overspray.

END OF SECTION

SECTION 13431

PRECAST CONCRETE FIBER OPTIC AND UTILITY VAULT

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Precast concrete fiber optic and utility vaults, and maintenance markers.
- B. Type IV-CV, and Type V-CV Precast Concrete Fiber Optic and Utility Vaults.

1.2 RELATED SECTIONS

- A. Section 02056: Embankment, Borrow, and Backfill
- B. Section 02221: Remove Structure and Obstruction
- C. Section 03575: Flowable Fill
- D. Section 13553: ITS Conduit

1.3 REFERENCES

- A. AASHTO LRFD Bridge Design Specifications
- B. USDA Rural Utilities Service (RUS)

1.4 DEFINITIONS **Not Used**

1.5 SUBMITTALS

- A. Manufacturer's shop drawings.
 - 1. Shop drawings must bear the seal of a Licensed Professional Engineer and will not deviate from the functional dimensions shown.
- B. Manufacturer's product specifications for information
- C. Manufacturer's warranty provisions

PART 2 PRODUCTS

2.1 FILL

- A. Free Draining Granular Backfill – Refer to Section 02056.
- B. Granular Backfill Borrow – Refer to Section 02056.
- C. Flowable Fill – Refer to Section 03575.
- D. Hand-mix grout
 - 1. Minimum strength – 50 psi
 - 2. Maximum strength – 150 psi
 - 3. Slump – 5 inches to 10 inches

2.2 FIBER OPTIC VAULTS AND COVERS

- A. Fiber vaults – pre-cast concrete. Refer to AT Series Standard Drawings for dimensions of vault types.
- B. Provide fiber vaults that are adjustable upwards to meet a new finished grade by the addition of manufacturer-provided risers, collars, or grade rings.
- C. Provide fiber vaults that have the cover frames cast separately from the vault structure and are removable and replaceable.
- D. Furnish an internal heavy-duty non-metallic cable storage rack system on two opposite fiber vault walls.
 - 1. Furnish and install two stanchions on each rack wall.
 - a. Fasten to the fiber vault wall according to the manufacturer's instructions.
 - b. Stanchions are a minimum of 36 inches for Type IV-CV fiber vault racks.
 - c. Stanchions are a minimum of 45 inches For Type V-CV fiber vault racks.
 - 2. Use cable rack arms that are appropriate for the amount of slack cable being installed but not be less than 4 rack arms in a Type IV vault and 5 rack arms in a Type V vault, capable of supporting a minimum of 250 lb.
- E. Supply Type IV-CV Fiber Optic and Utility Vaults with a hinged metal lid for unpaved areas or a heavy duty round manhole-style lid in paved areas.
 - 1. Supply Type V-CV vaults with a manhole-style lid only.

2. Fiber vaults with hinged metal lids will be rated for AASHTO LRFD HL – 93 loading according to AASHTO LRFD Bridge Design Specifications.
 3. Fiber vaults with round manhole style lids will be rated for AASHTO LRFD HL- 93 loading according to AASHTO LRFD Bridge Design Specifications.
- F. Provide hinged metal lids that have double-doors (split) and double hinges.
1. Provide a lid that operates freely over 180 degrees.
 2. Provide a lid with drop handles for easy opening.
 3. Provide a neoprene gasket between the hinged lid and the vault frame.
- G. Provide lids with a grounding lug with $\frac{1}{2}$ - 13 NC female threads on the underside of the manhole and hinged metal lids.
- H. Provide hinged and manhole lids with Penta-Head vandal resistant stainless steel bolts with washer.
1. All bolt holes must be self-draining.
- I. Furnish hinged metal lids that have a non-skid diamond pattern or similar surface that has a raised pattern of at least $\frac{3}{32}$ inch high.
- J. Provide lids for all vaults that are marked, "UDOT FIBER OPTIC" using a minimum 3 inch high letters with $\frac{1}{8}$ inch line thickness.
1. Form the letters by engraving, casting, stamping, or with a precise weld bead.
- K. Manhole Lid Access Points – recessed steel pull slots or holes to allow removal or opening of the manhole with a hook or lever.

2.3 MAINTENANCE MARKERS

- A. Furnish and install Utility Marker Posts for each vault location. Refer to AT Series Standard Drawings.

2.4 LOCATE BALL OR DISK Not Used

PART 3 EXECUTION

3.1 BACKFILL

- A. Place 12 inches of Free Draining Granular Backfill under vaults.
- B. Hand tamp Granular Backfill Borrow or approved native soil around the vault lid on unpaved surfaces.
 - 1. Match the top 6 inches to the composition, density, and elevation of the surrounding surface.

3.2 FIBER OPTIC AND UTILITY VAULT

- A. Install according to manufacturer's recommendations and AT Series Standard Drawings
- B. Core drill holes to match conduit entry where required without damaging the vault. Core drill holes at thin wall sections only. Do not "knock out" thin wall sections. Use grout to create a complete seal between conduit and the structure wall. Finish grout smooth and flush with the interior wall.
 - 1. Make core drilled holes in vault not more than ¼ inches larger than conduit diameter.
 - 2. Seal conduit and micro-duct ends inside all vaults with at least 2 inch thick duct seal after cables are installed.
 - 3. Seal vacant conduit and micro-ducts with a manufactured conduit plug and attach detectable pull tape. Refer to Section 13553.
- C. Install vault plum and level and grade backfill flush with vault.
- D. Do not locate fiber vaults on slopes or low lying locations with poor drainage.
- E. Do not install vaults within the traveled way or shoulders unless shown.

- F. Conduit in fiber vaults
 - 1. Install conduits into thin wall section only.
 - 2. Extend PVC conduit 2 inches, HDPE conduit 6 inches, micro-duct oversheath 6 inches and individual micro-ducts 2 ft beyond the inside wall of the vault.
 - 3. Align ITS conduit ends by color at each side of the box.
 - 4. Install bushings on all conduits before cable installation according to Section 13553.
 - 5. Refer to AT Series Standard Drawings.
- G. Remove concrete sidewalk or other surfaces.
 - 1. Refer to Section 02221.
 - 2. Replace with in-kind materials to match the existing grade, texture, and color of concrete or other surface.
- H. Encase conduit in flowable fill or hand-mix grout where the conduit enters the fiber optic vault.

3.3 RESTORATION

- A. Restore areas disturbed or damaged during the installation of all vaults.

END OF SECTION

End of 2026 Standard Specifications

For Road and Bridge Construction

Standards

Coordinator

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Coordinator

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